Common Security Advisory Framework Version 2.1

Committee Specification 01

09 November 2026

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OASIS Common Security Advisory Framework (CSAF) TC

Chairs

Justin Murphy (justin.murphy@mail.cisa.dhs.gov), DHS Cybersecurity and Infrastructure Security Agency (CISA)
Omar Santos (osantos@cisco.com), Cisco Systems
Stefan Hagen (stefan@hagen.link), Individual

Editors

Stefan Hagen (stefan@hagen.link), Individual
Thomas Schmidt (thomas.schmidt@bsi.bund.de), Federal Office for Information Security (BSI) Germany

Additional artifacts

This prose specification is one component of a Work Product that also includes:

- Aggregator JSON schema: https://docs.oasis-open.org/csaf/csaf/v2.1/cs01/schema/aggregator.json.
  Latest stage: https://docs.oasis-open.org/csaf/csaf/v2.1/schema/aggregator.json.
- CSAF JSON schema: https://docs.oasis-open.org/csaf/csaf/v2.1/cs01/schema/csaf.json.
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- Extension Content JSON schema: https://docs.oasis-open.org/csaf/csaf/v2.1/cs01/schema/extension-content.json.
  Latest stage: https://docs.oasis-open.org/csaf/csaf/v2.1/schema/extension-content.json.
- Extension Metadata JSON schema: https://docs.oasis-open.org/csaf/csaf/v2.1/cs01/schema/extension-metadata.json.
  Latest stage: https://docs.oasis-open.org/csaf/csaf/v2.1/schema/extension-metadata.json.
- Extension Metaschema JSON schema: https://docs.oasis-open.org/csaf/csaf/v2.1/cs01/schema/extension-metaschema.json.
  Latest stage: https://docs.oasis-open.org/csaf/csaf/v2.1/schema/extension-metaschema.json.
- Meta JSON schema: https://docs.oasis-open.org/csaf/csaf/v2.1/cs01/schema/meta.json.
  Latest stage: https://docs.oasis-open.org/csaf/csaf/v2.1/schema/meta.json.
- Provider JSON schema: https://docs.oasis-open.org/csaf/csaf/v2.1/cs01/schema/provider.json.
  Latest stage: https://docs.oasis-open.org/csaf/csaf/v2.1/schema/provider.json.

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Related work

This specification replaces or supersedes:

- *Common Security Advisory Framework Version 2.0*. Edited by Langley Rock, Stefan Hagen, and Thomas Schmidt. 18 November 2022. OASIS Standard. https://docs.oasis-open.org/csaf/csaf/v2.0/os/csaf-v2.0-os.html. Latest stage: https://docs.oasis-open.org/csaf/csaf/v2.0/csaf-v2.0.html.

Declared JSON namespaces:

- https://docs.oasis-open.org/csaf/csaf/v2.1/schema/aggregator.json
- https://docs.oasis-open.org/csaf/csaf/v2.1/schema/csaf.json
- https://docs.oasis-open.org/csaf/csaf/v2.1/schema/extension-content.json
- https://docs.oasis-open.org/csaf/csaf/v2.1/schema/extension-metadata.json
- https://docs.oasis-open.org/csaf/csaf/v2.1/schema/extension-metaschema.json
- https://docs.oasis-open.org/csaf/csaf/v2.1/schema/meta.json
- https://docs.oasis-open.org/csaf/csaf/v2.1/schema/provider.json

Abstract

The Common Security Advisory Framework (CSAF) Version 2.1 is the definitive reference for the language which supports creation, update, and interoperable exchange of security advisories as structured information on products, vulnerabilities and the status of impact and remediation among interested parties.

Status

This document was last revised or approved by the membership of OASIS on the above date. The level of approval is also listed above. Check the "Latest stage" location noted above for possible later revisions of this document. Any other numbered Versions and other technical work produced by the Technical Committee (TC) are listed at https://www.oasis-open.org/committees/tc_home.php?wg_abbrev=csaf#technical.

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Citation format

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1. Introduction

1.1 IPR Policy

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1.2 Terminology

The key words "MUST", "MUST NOT", "REQUIRED", "SHALL", "SHALL NOT", "SHOULD", "SHOULD NOT", "RECOMMENDED", "NOT RECOMMENDED", "MAY", and "OPTIONAL" in this document are to be interpreted as described in BCP 14 [RFC2119] and [RFC8174] when, and only when, they appear in all capitals, as shown here.

For purposes of this document, the following terms and definitions apply:

Advisory

reporting item that describes a condition present in an artifact and that requires action by the consumers.

Advisory Document

artifact in which an analysis tool reports a result.

Advisory Management System

software system that consumes the documents produced by analysis tools, produces advisories that enable engineering and operating organizations to assess the quality of these software artifacts at a point in time, and performs functions such as filing security advisories and displaying information about individual advisories. **Note**: An Advisory Management System can interact with a document viewer to display information about individual advisories.

Advisory Matching

process of determining whether two advisories are targeting the same products and conditions.

Artifact

sequence of bytes addressable via a URI. *Examples*: A physical file in a file system such as a source file, an object file, a configuration file or a data file; a specific version of a file in a version control system; a database table accessed via an HTTP request; an arbitrary stream of bytes returned from an HTTP request, a product URL, a common product enumeration value.

Coordinated Vulnerability Disclosure (CVD)

A disclosure process that includes coordination.

Coordination

set of activities including identifying and engaging stakeholders, mediating, communicating and other planning in support of vulnerability disclosure (conforming to [ISOIEC29147]).

Critical Extension

An extension that if an instance of it is included in a CSAF Document causes at least one mandatory test to fail.

CSAF 2.0 to CSAF 2.1 Converter

A CSAF Producer which takes a CSAF 2.0 Document as input and converts it into a valid CSAF 2.1 Document.

CSAF Additional Test

A test that is not yet defined in section 6.

CSAF Asset Matching System

program that connects to or is an asset database and is able to manage CSAF Documents as required by CSAF Management System as well as matching them to assets of the asset database.

CSAF Basic Validator

A program that reads a document and checks it against the JSON schema and performs mandatory tests.

CSAF Consumer

program that reads and interprets a CSAF Document.

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CSAF Content Management System

program that is able to create, review and manage CSAF Documents and is able to preview their details as required by CSAF Viewer.

CSAF Converter

CSAF Producer that transforms the output of an analysis tool from its native output format into the CSAF format.

CSAF Core

All parts of CSAF except for the parts covering extensions.

CSAF Direct Producer

analysis tool which acts as a CSAF Producer.

CSAF Document

security advisory text document in the format defined by this document.

CSAF Downloader

A program that retrieves CSAF Documents in an automated fashion.

CSAF Extended Validator

A CSAF Basic Validator that additionally performs recommended tests.

CSAF Extension

A specified JSON object conveying additional information different from the content that can be conveyed with the CSAF Core elements.

CSAF Extension Additional Test

A test whose execution depends on the presence of the specifying CSAF Extension that provides additional checks in the context of the CSAF Extension or the CSAF Document the extension is embedded in.

CSAF Extension Bundle

A of compilation of machine-readable artifacts related to a single CSAF Extension.

CSAF Extension Collection

A set of multiple CSAF Extension Packages.

CSAF Extension Overlay Test

A test whose execution depends on the presence of the specifying CSAF Extension that extends or replaces a test standardized in this specification or a CSAF Additional Test.

CSAF Extension Package

A of compilation of all artifacts related to a single CSAF Extension.

CSAF Extension Schema

A JSON schema specifying the content and properties of a CSAF Extension.

CSAF Extension Specification

The specification of a single CSAF Extension and related material.

CSAF Extension Test

A test that is either a CSAF Extension Overlay Test or a CSAF Extension Additional Test.

CSAF Full Validator

A CSAF Extended Validator that additionally performs informative tests.

CSAF Library

A library that implements CSAF data capabilities.

CSAF Library with Basic Validation

A CSAF Library that also satisfies the conformance target “CSAF Basic Validator”.

CSAF Library with Extended Validation

A CSAF Library that also satisfies the conformance target “CSAF Extended Validator”.

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CSAF Library with Full Validation

A CSAF Library that also satisfies the conformance target “CSAF Full Validator”.

CSAF Management System

program that is able to manage CSAF Documents and is able to display their details as required by CSAF Viewer.

CSAF Modifier

CSAF Post-Processor which takes a CSAF Document as input and modifies the structure or values of properties. The output is a valid CSAF Document.

CSAF Post-Processor

CSAF Producer that transforms an existing CSAF Document into a new CSAF Document, for example, by removing or redacting elements according to sharing policies.

CSAF Producer

program that emits output in the CSAF format.

CSAF RVISC ID Updater

A CSAF Post-Processor that updates vulnerability IDs in a given CSAF based on the entries in RVISC.

CSAF SBOM Matching System

A program that connects to or is an SBOM database and is able to manage CSAF Documents as required by CSAF Management System as well as matching them to SBOM components of the SBOM database.

CSAF Superseder

A CSAF Post-Processor that transforms a given CSAF into a superseded one.

CSAF Translator

CSAF Post-Processor which takes a CSAF Document as input and translates values of properties into another language. The output is a valid CSAF Document.

CSAF Viewer

CSAF Consumer that reads a CSAF Document, displays a list of the results it contains, and allows an end user to view each result in the context of the artifact in which it occurs.

CSAF Withdrawer

A CSAF Post-Processor that transforms a given CSAF into a withdrawn one.

CVRF CSAF Converter

CSAF Producer which takes a CVRF document as input and converts it into a valid CSAF Document.

Disclosure

act of initially providing vulnerability information to a party that was not believed to be previously aware (conforming to [ISOIEC29147]).

Document

output file produced by an analysis tool, which enumerates the results produced by the tool.

Driver

tool component containing an analysis tool’s or converter’s primary executable, which controls the tool’s or converter’s execution, and which in the case of an analysis tool typically defines a set of analysis rules.

Embedded Link

syntactic construct which enables a message string to refer to a location mentioned in the document.

Empty Array

array that contains no elements, and so has a length of zero.

Empty Object

object that contains no properties.

Empty String

string that contains no characters, and so has a length of zero.

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(End) User
person who uses the information in a document to investigate, triage, or resolve results.

Engineering System
software analysis environment within which analysis tools execute. **Note**: An engineering system might include a build system, a source control system, a result management system, a bug tracking system, a test execution system, and so on.

Extension
tool component other than the driver (for example, a plugin, a configuration file, or a taxonomy).

External Property File
file containing the values of one or more externalized properties.

Externalizable Property
property that can be contained in an external property file.

Externalized Property
property stored outside of the CSAF Document to which it logically belongs.

False Positive
result which an end user decides does not actually represent a problem.

Filter
refine a list by selecting entries that match given criteria.

Fingerprint
stable value that can be used by a result management system to uniquely identify a result over time, even if a relevant artifact is modified.

Formatted Message
message string which contains formatting information such as Markdown formatting characters.

Fully Qualified Logical Name
string that fully identifies the programmatic construct specified by a logical location, typically by means of a hierarchical identifier.

Hierarchical String
string in the format <component>{/<component>}*.

Issuing Party
entity that releases a CSAF document, independent of its role according to the distribution system. It is usually the same one as described by the value of $.document.publisher in that CSAF document.

Line
contiguous sequence of characters, starting either at the beginning of an artifact or immediately after a newline sequence, and ending at and including the nearest subsequent newline sequence, if one is present, or else extending to the end of the artifact.

Line (Number)
1-based index of a line within a file. **Note**: Abbreviated to "line" when there is no danger of ambiguity with "line" in the sense of a sequence of characters.

Localizable
subject to being translated from one natural language to another.

Message String
human-readable string that conveys information relevant to an element in a CSAF Document.

Nested Artifact
artifact that is contained within another artifact.

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Newline Sequence

sequence of one or more characters representing the end of a line of text. **Note**: Some systems represent a newline sequence with a single newline character; others represent it as a carriage return character followed by a newline character.

Notification

reporting item that describes a condition encountered by a tool during its execution.

Opaque

neither human-readable nor machine-parsable into constituent parts.

Parent (Artifact)

artifact which contains one or more nested artifacts.

Plain Text Message

message string which does not contain any formatting information.

Plugin

tool component that defines additional rules.

Policy

set of rule configurations that specify how results that violate the rules defined by a particular tool component are to be treated.

Problem

result which indicates a condition that has the potential to detract from the quality of the program. *Examples*: A security vulnerability, a deviation from contractual or legal requirements.

Product

is any deliverable (e.g. software, hardware, specification, or service) which can be referred to with a name. This applies regardless of the origin, the license model, or the mode of distribution of the deliverable.

Property

attribute of an object consisting of a name and a value associated with the name.

Redactable Property

property that potentially contains sensitive information that a CSAF Direct Producer or a CSAF Post-Processor might wish to redact.

Reporting Item

unit of output produced by a tool, either a result or a notification.

Reporting Configuration

the subset of reporting metadata that a tool can configure at runtime, before performing its scan. *Examples*: severity level, rank

Repository

container for a related set of files in a version control system.

Search

compile a list of entries that match given criteria.

Taxonomy

classification of analysis results into a set of categories.

Tag

string that conveys additional information about the CSAF Document element to which it applies.

Text Artifact

artifact considered as a sequence of characters organized into lines and columns.

Text Region

region representing a contiguous range of zero or more characters in a text artifact.

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Tool Component
component of an analysis tool or converter, either its driver or an extension, consisting of one or more files.

Top-Level Artifact
artifact which is not contained within any other artifact.

Translation
rendering of a tool component's localizable strings into another language.

Triage
decide whether a result indicates a problem that needs to be corrected.

User
see end user.

VCS
version control system.

Vendor
the community, individual, or organization that created or maintains a product (including open source software and hardware providers).

VEX
Vulnerability Exploitability eXchange - enables a supplier or other party to assert whether or not a particular product is affected by a specific vulnerability, especially helpful in efficiently consuming SBOM data.

Viewer
see CSAF Viewer.

Vulnerability
functional behavior of a product or service that violates an implicit or explicit security policy (conforming to [ISOIEC29147]).

White Space
code point used to improve text readability or token separation as defined in section 12.2 of [ECMA-262].

XML
eXtensible Markup Language - the format used by the predecessors of this standard, namely CVRF 1.1 and CVRF 1.2.

1.3 Normative References

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1.5 Typographical Conventions

Keywords defined by this specification use this monospaced font.

> Normative source code uses this paragraph style.

The information models of the CSAF schemata are illustrated in the prose in a generic YAML format in the shape of outlines. In contrast to extracts of JSON snippets in prior versions of this specification such YAML snippets can be annotated and validated both as being well-formed and matching the schema part. These outlines are directly extractable per JSONPaths (see [RFC9535]) from the corresponding CSAF schema as documented in the normative format of this specification. The types used are the general Mapping (JSON object), Sequence (JSON array), and String types. The latter may be further constrained to specific facets like for example enumerations noted as String.Enum. General instances (like a CSAF document) are noted as such in angle brackets to emphasize the topology. Items of sequences are noted as YAML sequence items and annotated per comments naming the kind of sequence item they represent.

JSONPath is also used within the standard as syntax for specifying paths or referring to elements relevant to the current context. They can have multiple results when being applied to a JSON instance or schema. However, when referring to a schema with one of the keywords $ref or $dynamicRef the value is a URI-Reference (cf. section 8.2.3 of [JSON-Schema-Core]). To represent paths which contain the branch recursion below $.product_tree.branches[*] the descendant segment syntax $.product_tree..branches[*] is used. Implementations need to ensure that only matching paths with the intended type are selected. For example, this could be done by checking that the property names that are omitted through the .. syntax along the path between product_tree and branches are just branches. Including but not limited to paths containing x_extensions could introduce arbitrary other property names.

Some sections of this specification are illustrated with non-normative examples introduced with “Example” or “Examples” like so:

Example 1:

> Informative examples also use this paragraph style but preceded by the text "Example(s)".

All examples in this document are informative only.

All other text is normative unless otherwise labeled e.g. like the following informative comment:

> This is a pure informative comment that may be present, because the information conveyed is deemed useful advice or common pitfalls learned from implementer or operator experience and often given including the rationale.

This document adheres to the Modern Language Association (MLA) style guidelines for formatting titles and terms.

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2. Design Considerations

The Common Security Advisory Framework (CSAF) is a language to exchange Security Advisories formulated in JSON.

The term Security Advisory as used in this document describes any notification of security issues in products of and by providers. Anyone providing a product is considered in this document as a vendor, i.e. developers or maintainers of information system products or services. This includes all authoritative product vendors, Product Security Incident Response Teams (PSIRTs), and product resellers and distributors, including authoritative vendor partners. A security issue is not necessarily constrained to a problem statement, the focus of the term is on the security aspect impacting (or not impacting) specific product-platform-version combinations. Information on presence or absence of workarounds is also considered part of the security issue. This document is the definitive reference for the language elements of CSAF version 2.1. The encompassing JSON schema file noted in the Additional Artifacts section of the title page SHALL be taken as normative in the case a gap or an inconsistency in this explanatory document becomes evident. The following presentation in this section is grouped by topical area, and is not simply derivative documentation from the schema document itself. The information contained aims to be more descriptive and complete. Where applicable, common conventions are stated and known common issues in usage are pointed out informatively to support implementers of document producers and consumers alike.

This minimal required information set does not provide any useful information on products, vulnerabilities, or security advisories. Thus, any real-world Security Advisory will carry additional information as specified in section 3 Schema elements.

Care has been taken, to design the containers for product and vulnerability information to support fine-grained mapping of security advisories onto product and vulnerability and minimize data duplication through referencing. The display of the elements representing Product Tree and Vulnerability information has been placed in the sections named accordingly.

2.1 Construction Principles

A Security Advisory represented by a CSAF document is created through the application of multiple schemas during a process of collaboration between multiple stakeholders. The distinct and in part complex schemas are orchestrated, put into context and connected with each other serving the goal of providing actionable, structured, and validated information targeting a high degree of automation.

The format chosen is JSON schema (cf. [JSON-Schema-Core], [JSON-Schema-Validation], [JSON-Hyper-Schema]) which allows validation and delegation to sub schema providers. The latter aligns well with separation of concerns and shares the format family of information interchange utilized by the providers of product and vulnerability information which migrated from XML to JSON since the creation of CSAF CVRF version 1.2, the pre-predecessor of this specification.

The acronym CSAF, “Common Security Advisory Framework”, stands for the target of concerted mitigation and remediation accomplishment.

Technically, the use of JSON schema allows validation and proof of model conformance (through established schema based validation) of the declared information inside CSAF documents.

The CSAF schema structures its derived documents into three main classes of the information conveyed:

1. The frame, aggregation, and reference information of the document
2. Product information considered relevant by the creator
3. Vulnerability information and its relation to the products declared in 2.

Wherever possible repetition of data has been replaced by linkage through ID elements. Consistency on the content level thus is in the responsibility of the producer of such documents, to link e.g. vulnerability information to the matching product.

A dictionary like presentation of all defined schema elements is given in the section 3. Any expected relations to other elements (linkage) is described there. This linking relies on setting attribute values accordingly (mostly guided by industry best practice and conventions) and thus implies, that any deep validation on a semantic level (e.g. does the CWE match the described vulnerability) is to be ensured by the producer and consumer of CSAF documents. It is out of scope for this specification.

Proven and intended usage patterns from practice are given where possible.

Delegation to industry best practices technologies is used in referencing schemas for:

- Classification for Document Distribution
  - Traffic Light Protocol (TLP)
    - Default Definition: https://www.first.org/tlp/

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- Exploit Prediction
  - Exploit Prediction Scoring System (EPSS) [EPSS]
- Package Data
  - Package-URL (PURL) [ECMA-427]
- Platform Data
  - Common Platform Enumeration (CPE) Version 2.3 [CPE23-N]
- Vulnerability Categorization
  - Stakeholder-Specific Vulnerability Categorization [SSVC]
    - JSON Schema Reference: https://certcc.github.io/SSVC/data/schema/v2/SelectionList_2_0_0.schema.json
- Vulnerability Classification
  - Common Weakness Enumeration (CWE) [CWE]
    - CWE List: https://cwe.mitre.org/data/index.html
- Vulnerability Scoring
  - Common Vulnerability Scoring System (CVSS) Version 4.0 [CVSS40]
    - JSON Schema Reference: https://www.first.org/cvss/cvss-v4.0.json
  - Common Vulnerability Scoring System (CVSS) Version 3.1 [CVSS31]
    - JSON Schema Reference: https://www.first.org/cvss/cvss-v3.1.json
  - Common Vulnerability Scoring System (CVSS) Version 3.0 [CVSS30]
    - JSON Schema Reference: https://www.first.org/cvss/cvss-v3.0.json
  - Common Vulnerability Scoring System (CVSS) Version 2.0 [CVSS2]
    - JSON Schema Reference: https://www.first.org/cvss/cvss-v2.0.json

Even though not all - especially the referenced - JSON schemas prohibit specifically additional properties and custom keywords, it is strongly recommended not to use them. Suggestions for new fields SHOULD be made through issues in the TC's GitHub. The JSON schemas defined in this standard do not allow the use of additional properties and custom keywords.

> The standardized fields allow for scalability across different issuing parties and dramatically reduce the human effort and need for dedicated parsers as well as other tools on the side of the consuming parties.

Sole exception are the dedicated extension points (x_extensions) which allow the usage of extension defined outside this specification. To still ensure interoperability and guide tools how to deal with this data a CSAF Extension Content Schema and CSAF Extension Metaschema has been defined and SHALL be adhered to (cf. section 2.4). Document issuers are advised to always balance pros and cons regarding the use of extension.

Section 4 defined profiles that are used to ensure a common understanding of which fields are required in a given use case. Additional conventions are stated in section 5. The tests given in section 6 support CSAF producers and consumers to verify rules from the specification which can not be tested by the schema. Section 7 states how to distribute and where to find CSAF documents. Safety, Security and Data Protection are considered in section 8. Finally, a set of conformance targets describes tools in the ecosystem.

2.2 Format Validation

The JSON schema 2020-12 dialect per default uses the format keyword just as annotation. To be able to ensure that the format constraints are validated as intended, the following metaschema is defined.

```
{
  "$schema": "https://json-schema.org/draft/2020-12/schema",
  "$id": "https://docs.oasis-open.org/csaf/csaf/v2.1/schema/meta.json",
  "$dynamicAnchor": "meta",
  "$vocabulary": {
    "https://json-schema.org/draft/2020-12/vocab/core": true,
    "https://json-schema.org/draft/2020-12/vocab/applicator": true,
    "https://json-schema.org/draft/2020-12/vocab/unevaluated": true,
    "https://json-schema.org/draft/2020-12/vocab/validation": true,
    "https://json-schema.org/draft/2020-12/vocab/meta-data": true,
    "https://json-schema.org/draft/2020-12/vocab/format-assertion": true,
    "https://json-schema.org/draft/2020-12/vocab/content": true
  },
  "allOf": [
```

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```
    { "$ref": "https://json-schema.org/draft/2020-12/meta/core" },
    { "$ref": "https://json-schema.org/draft/2020-12/meta/applicator" },
    { "$ref": "https://json-schema.org/draft/2020-12/meta/unevaluated" },
    { "$ref": "https://json-schema.org/draft/2020-12/meta/validation" },
    { "$ref": "https://json-schema.org/draft/2020-12/meta/meta-data" },
    { "$ref": "https://json-schema.org/draft/2020-12/meta/format-assertion" },
    { "$ref": "https://json-schema.org/draft/2020-12/meta/content" }
  ]
}
```

All vocabularies that can be used are defined in this metaschema as JSON schema 2020-12 dialect has the rule of non-inheritability of vocabularies.

It is then consequently used in all JSON schemas defined in this standard and replaces the reference to the JSON schema 2020-12.

```
{
  "$schema": "https://docs.oasis-open.org/csaf/csaf/v2.1/schema/meta.json",
  // ...
}
```

The format validation is enforced by setting the corresponding vocabulary as required.

If a library used to parse, modify or create CSAF content is unable to deal with this meta schema, it could reach the objective by interpreting the schema as JSON schema 2020-12 dialect and enforcing the format validation via its implementation.

2.3 Date and Time

This standard uses the date-time format as defined in JSON Schema Draft 2020-12 Section 7.3.1. In accordance with [RFC3339] and [ISO8601-1], the following rules apply:

- The letter T separating the date and time SHALL be upper case.
- The separator between date and time SHALL be the letter T.
- The letter Z indicating the timezone UTC SHALL be upper case.
- Fractions of seconds are allowed as specified in the standards mention above with the full stop (.) as separator.
- Empty timezones SHALL NOT be used.
- The ABNF of RFC 3339, section 5.6 applies.

In contrast to the aforementioned standards, leap seconds SHALL NOT be used.

While a full support of [RFC3339] would be preferred, significant challenges have been mentioned by implementers as most libraries are lacking the support for leap seconds. To ensure interoperability, the decision was made to prohibit leap seconds.

2.4 Extensions

This standard allows for extensions to the standardized schema.

The following rules apply:

- An extension SHALL NOT occur in any other place than specified.
- An extension SHALL satisfy the Conformance Target “CSAF Extension”.
- The schema specifying the content and properties of the CSAF Extension SHALL satisfy the Conformance Target “CSAF Extension Schema”.
- For official and registered extensions a CSAF Extension Package SHALL be provided.
- CSAF Extensions SHOULD NOT provide CSAF Extension Overlay Tests for tests in the preset extensions.

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2.4.1 Classes

There three classes of extensions:

1. Official extensions: These CSAF Extensions are specified by the OASIS CSAF TC and ensure interoperability.

   > Those extensions are used to backport upcoming features or integrate new features at the OASIS CSAF TC's discretion.

2. Registered extensions: These CSAF Extensions are specified by a named party, publicly available, well-documented and registered with the OASIS CSAF TC. Collisions are avoided through being listed in the register.

   > Those extension can be used for conveying additional information to a broader community and making it available for tool integration. The OASIS CSAF TC may or may not consider to replace such extension with a registered one or take it as input for a future version of CSAF.

3. Experimental extensions: These CSAF Extension can be used for tests and experiments. They SHOULD NOT be used in production.

2.4.2 Lists

The OASIS CSAF TC maintains:

- a list of official CSAF Extensions and their CSAF Extension Packages,
- a list of registered CSAF Extensions and their CSAF Extension Packages,
- a list of deprecated extensions, and
- a list of deny-listed extensions.

Deprecated extensions can still be used but support for them is removed in near future. If avoidable, they SHOULD NOT be used. Deny-listed extensions SHALL NOT be used. The lists of deprecated extensions and deny-listed extensions MAY contain extensions that do not fulfill the conformance target CSAF Extension.

The list MAY contain additional examples.

2.4.3 Metaschema

The CSAF Extension Metaschema SHALL be used to validate a CSAF Extension Schema.

2.4.4 Content Schema

An extension SHALL contain exactly the following elements: CSAF Extension Schema (`$schema`), Extension Category (`category`), Critical (`critical`) and Content (`content`).

```
$schema: String
category: String.Enum
content: Mapping
critical: Boolean
```

> The CSAF Extension Content Schema works an interface definition. It is used in the CSAF Schema and defines a common structure for all CSAF Extensions.

2.4.4.1 Content Schema Property - Schema

CSAF Extension Schema (`$schema`) of value type CSAF Extension Content `$schema` Type (`content_schema_t`) contains the URL of the CSAF Extension JSON schema which the JSON object promises to be valid for. This SHOULD also be the location where the JSON schema can be retrieved. The value SHOULD match the `$id` of the JSON schema that defines the extension. The URL SHOULD contain a human-readable name for the extension before the version string. The versioning SHALL use [SemVer]. URLs using a domain mentioned in [RFC2606] SHALL be used according to their defined purpose.

Examples 1:

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```
https://www.example.com/some-path/to-a-csaf-extension/schema/our-awesome-metric_1.0.0.json
https://oil-and-gas.isac.example/.well-known/csaf/extensions/schema/product-safety_17.41.0.json
```

2.4.4.2 Content Schema Property - Category

Extension Category (category) of value type string and enum holds the category of the extension content. Valid enum values are:

```
essential
significant
supplementary
```

The value essential indicates, that the content provided through this extension is crucial to understand of the CSAF document this extension is included in. CSAF consumers and CSAF validators SHALL warn if they process a CSAF Document including such extension instance and do not have the extension in question already implemented.

The value significant indicates, that the content provided through this extension is highly relevant and significantly aids in understanding the overall content of the CSAF document. CSAF consumers and CSAF validators SHOULD warn if they process a CSAF Document including such an extension instance and do not have the extension in question already implemented.

The value supplementary indicates, that the content provided through this extension provides additional information. CSAF consumers and CSAF validators MAY warn if they process a CSAF Document including such an extension instance and do not have the extension in question already implemented.

2.4.4.3 Content Schema Property - Content

Content (content) of value type object contains the additional information in its properties. A Content object has at least 1 property. The property names (JSON keys) can be chosen freely; they SHOULD characterize the information given in its value.

2.4.4.4 Content Schema Property - Critical

Critical (critical) of value type boolean determines whether using the extension would fail a mandatory test. The default value for this is false.

For any failing test, a CSAF Extension Test SHALL be provided. A CSAF Tool that encounters a critical extension it does not have implemented SHALL fail processing of the CSAF document, unless the user explicitly forces it to continue.

2.4.5 Metadata

The CSAF Extension Metadata is a representation of metadata for a CSAF Extension. It is used to provide information for automated tools on certain requirements and incompatibilities the CSAF Extension has.

Examples 1:

```
https://www.example.com/some-path/to-a-csaf-extension/metadata/our-awesome-metric_1.0.0.json
https://oil-and-gas.isac.example/.well-known/csaf/extensions/metadata/product-safety_17.41.0.json
```

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3. Schema Elements

The CSAF schema describes how to represent security advisory information as a JSON document.

The CSAF schema Version 2.1 builds on the JSON Schema draft 2020-12 rules extended by the format validation enforcement (see 2.2).

```
"$schema": "https://docs.oasis-open.org/csaf/csaf/v2.1/schema/meta.json"
```

The schema identifier is:

```
"$id": "https://docs.oasis-open.org/csaf/csaf/v2.1/schema/csaf.json"
```

The further documentation of the schema is organized via Definitions and Properties.

- Definitions provide types that extend the JSON schema model
- Properties use these types to support assembling security advisories

Types and properties together provide the vocabulary for the domain specific language supporting security advisories.

The two mandatory properties are $schema and document. The three additional properties, product_tree, vulnerabilities, and x_extensions are optional.

3.1 Definitions

The definitions ($defs) introduce the following domain specific types into the CSAF language: Acknowledgments (acknowledgments_t), Action ID (action_id_t), Branches (branches_t), Contact (contact_t), Document-local Vuln ID (dl_vuln_id_t), Entity Group ID (entity_group_id_t), Entity ID (entity_id_t), Entity Refs (entity_refs_t), Extensions (extensions_t), Full Product Name (full_product_name_t), Language (lang_t), Notes (notes_t), Product Group ID (product_group_id_t), Product Groups (product_groups_t), Product ID (product_id_t), Products (products_t), References (references_t), Subpath (subpath_t) and Version (version_t).

```
$defs:
  acknowledgments_t: Sequence
  action_id_t: String.Pattern
  branches_t: Sequence
  contact_t: Mapping
  dl_vuln_id_t: String.Pattern
  entity_group_id_t: String.Pattern
  entity_id_t: String.Pattern
  entity_refs_t: Sequence
  extensions_t: Sequence
  full_product_name_t: Mapping
  lang_t: String.Pattern
  notes_t: Sequence
  product_group_id_t: String
  product_groups_t: Sequence
  product_id_t: String
  products_t: Sequence
  references_t: Sequence
  subpath_t: Mapping
  version_t: String.Pattern
```

3.1.1 Acknowledgments Type

List of Acknowledgments (acknowledgments_t) type instances of value type array with 1 or more elements contain a list of Acknowledgment elements.

```
$defs:
  acknowledgments_t: Sequence
  # ...
```

The value type of Acknowledgment is object with at least 1 and at most four properties. Every such element acknowledges contributions by describing those that contributed. The properties are: names, organization, summary, and urls.

```
$defs:
  acknowledgments_t:
  - # <acknowledgment-instance>:
    names: Sequence
    organization: String
    summary: String
```

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```
  urls: Sequence
# ...
```

3.1.1.1 Acknowledgments Type - Names

List of acknowledged names (names) of value type array with 1 or more items holds the names of contributors being recognized. Every such item of value type string with 1 or more characters represents the name of the contributor and contains the name of a single contributor being recognized.

Examples 1:

```
Albert Einstein
Johann Sebastian Bach
```

3.1.1.2 Acknowledgments Type - Organization

The contributing organization (organization) of value type string with 1 or more characters and holds the name of the contributing organization being recognized.

Examples 1:

```
CISA
Google Project Zero
Talos
```

3.1.1.3 Acknowledgments Type - Summary

Summary of the acknowledgment (summary) of value type string with 1 or more characters SHOULD represent any contextual details the document producers wish to make known about the acknowledgment or acknowledged parties.

Example 1:

```
First analysis of Coordinated Multi-Stream Attack (CMSA)
```

3.1.1.4 Acknowledgments Type - URLs

List of URLs (urls) of acknowledgment is a container (value type array) for 1 or more string of type URL that specifies a list of URLs or location of the reference to be acknowledged. Any URL of acknowledgment contains the URL or location of the reference to be acknowledged. Value type is string with format URI (uri).

3.1.1.5 Acknowledgments Type - Example

Example 1:

```
"acknowledgments": [
  {
    "names": [
      "Johann Sebastian Bach",
      "Georg Philipp Telemann",
      "Georg Friedrich Händel"
    ],
    "organization": "Baroque composers",
    "summary": "wonderful music"
  },
  {
    "organization": "CISA",
    "summary": "coordination efforts",
    "urls": [
      "https://cisa.gov"
    ]
  },
  {
    "organization": "BSI",
    "summary": "assistance in coordination"
  },
  {
    "names": [
      "Antonio Vivaldi"
    ],
    "summary": "influencing other composers"
  }
],
```

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The above SHOULD lead to the following outcome in a human-readable advisory:

> We thank the following parties for their efforts:
>
> - Johann Sebastian Bach, Georg Philipp Telemann, Georg Friedrich Händel from Baroque composers for wonderful music
> - CISA for coordination efforts (see: https://cisa.gov)
> - BSI for assistance in coordination
> - Antonio Vivaldi for influencing other composers

3.1.2 Action ID Type

The Action ID Type (action_id_t) of value type string with 1 or more characters is a reference token for action instances. It SHALL conform to pattern (regular expression):

```
^[^\\s\\-_\\.](.*[^\\s\\-_\\.])?$
```

The value contains a token required to identify an action uniquely in the context of the current document so that it can be referred to from other parts in the document.

```
$defs:
  # ...
  action_id_t: String.Pattern
  # ...
```

Examples 1:

```
1
AID-0001
Some ID
```

3.1.3 Branches Type

List of branches (branches_t) with value type array contains 1 or more branch elements as children of the current element.

```
$defs:
  # ...
  branches_t: Sequence
  # ...
```

Every Branch holds exactly 3 properties and is a part of the hierarchical structure of the product tree. The properties name and category are mandatory. In addition, the object contains either a branches or a product property.

```
$defs:
  # ...
  branches_t:
  - # <branch-instance>:
    branches: $defs.branches_t
    category: String.Enum
    name: String
    product: $defs.full_product_name_t
  # ...
```

> branches_t supports building a hierarchical structure of products that allows to indicate the relationship of products to each other and enables grouping for simpler referencing. As an example, the structure MAY use the following levels: vendor -> product_family -> product_name -> product_version. It is recommended to use the hierarchical structure of vendor -> product_name -> product_version whenever possible to support the identification and matching of products on the consumer side.

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3.1.3.1 Branches Type - Branches

List of branches (branches) has the value type branches_t.

3.1.3.2 Branches Type - Category

Category of the branch (category) of value type string and enum describes the characteristics of the labeled branch. Valid enum values are:

```
architecture
host_name
language
patch_level
platform
product_family
product_name
product_version
product_version_range
service_pack
specification
vendor
```

The value architecture indicates the architecture for which the product is intended.

> This usually identifies the CPU instruction-set architecture. See also example [2 (of section 3.1.3.3)] in section 3.1.3.3.

The value host_name indicates the host name of a system/service.

The value language indicates the language of the product.

The value patch_level indicates the patch level of the product.

The value platform indicates the (CPU) platform for which the product is intended.

> As the same architecture is used in different platforms, this allows a differentiation. See also example [3 (of section 3.1.3.3)] in section 3.1.3.3.

The value product_family indicates the product family that the product falls into.

The value product_name indicates the name of the product.

The value product_version indicates exactly a single version of the product. The value of the adjacent name property can be numeric or some other descriptor. However, it SHALL NOT contain version ranges of any kind.

> It is recommended to enumerate versions wherever possible. Nevertheless, the TC understands that this is sometimes impossible. To reflect that in the specification and aid in automatic processing of CSAF documents the value product_version_range was introduced. See next section for details.

The value product_version_range indicates a range of versions for the product. The value of the adjacent name property SHALL NOT be used to convey a single version.

> In most cases, product version ranges hinder the use of product identification helpers as they need to identify a product completely. In such case, the product identification and also therefore matching, independent whether that is against SBOMs or assets, solely relies on the categorized strings. This contradicts the goal of using unique product identifiers which would resolve the naming and identification issue for products and are a next major step towards more automation and providing only relevant information to users. Therefore, the use of product version ranges should be avoided, whenever possible.

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The value service_pack indicates the service pack of the product.

The value specification indicates the specification such as a standard, best common practice, etc.

The value vendor indicates the name of the vendor or manufacturer that makes the product.

3.1.3.3 Branches Type - Name

Name of the branch (name) of value type string with 1 or more characters contains the canonical descriptor or 'friendly name' of the branch.

Examples 1:

```
10
365
Microsoft
Office
PCS 7
SIMATIC
Siemens
Windows
```

A leading v or V in the value of name SHOULD only exist for the categories product_version or product_version_range if it is part of the product version as given by the vendor.

Examples 2 (for name when using architecture):

```
AArch32
AArch64
AMD64
ARMv7
ARMv8-R
ARMv9.7-A
IA-64
MISP32
m68k
ppc64el
RV64
s390
SPARC
x86
```

Examples 3 (for name when using platform):

```
AMD
ARM
Intel
```

3.1.3.3.1 Branches Type - Name under Product Version

If adjacent property category has the value product_version, the value of name SHALL NOT contain version ranges of any kind.

Examples 1 (for name when using product_version):

```
10
17.4
v3
```

> The product_version is the easiest way for users to determine whether their version is meant (provided that the given ancestors in the product tree matched): If both version strings are the same, it is a match - otherwise not. Therefore, it is always recommended to enumerate product versions instead of providing version ranges.

Examples 2 (for name when using product_version which are invalid):

```
8.0.0 - 8.0.1
8.1.5 and later
<= 2
prior to 4.2
```

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```
All versions < V3.0.29
V3.0, V4.0, V4.1, V4.2
```

All the examples above contain some kind of a version range and are therefore invalid under the category product_version.

3.1.3.3.2 Branches Type - Name under Product Version Range

If adjacent property category has the value product_version_range, the value of name SHALL contain version ranges. The value of SHALL obey to exactly one of the following options:

1. VErsion Range Specifier (VERS)

   VERS is an ongoing community effort to address the problem of version ranges. Its draft specification is available at [VERS].

   VERS SHALL be used in its canonical form. To convey the term "all versions", the special string vers:all/* SHALL be used.

   According to the interpretation used here, the canonical form requires that the VERS is normalized.

   Examples 1 (for name *when using* product_version_range *with VERS):*

   ```
   vers:gem/>=2.2.0|!=2.2.1|<2.3.0
   vers:npm/1.2.3|>=2.0.0|<5.0.0
   vers:pypi/0.0.0|0.0.1|0.0.2|0.0.3|1.0|2.0pre1
   vers:tomee/>=8.0.0-M1|<=8.0.1
   ```

   Through the definitions of the VERS specification a user can compute whether a given version is in a given range.

2. VERS-like Specifier (vls)

   This option uses only the constraint part from the VERS specification; special requirements regarding VERS state above apply. It SHALL NOT have the scheme nor the type part. Neither * nor a constraint denoting a single version is valid. Even though 1.0.0|1.1.0 is considered valid vls, it is RECOMMENDED to enumerate the versions instead. It is a fallback option and SHOULD NOT be used unless really necessary.

   The reason for that is, that it is nearly impossible for tools to reliably determine whether a given version is in the range or not. Also, different tools might come to different results regarding this question. This includes, but is not limited to, tests 6.2.49, 6.2.50.2, 6.3.19.2.

   Tools MAY support this on best effort basis.

   Consequently, situations might occur where the result is not determinable by that tool which is an admissible scenario and valuable information for the user.

   Examples 2 (for name *when using* product_version_range *with vls):*

   ```
   <=2
   <4.2
   <V3.0.29
   >=8.1.5
   ```

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```
>10.9a|!=10.9c|!=10.9f|<=10.9k
<2024-4-pabc0019|>2024-10-pefd0010|<2024-12-pjkl2010|>2024-12-pjkl5010|<=2025-1-pghi1001
```

3.1.3.4 Branches Type - Product

Product (product) has the value type Full Product Name (full_product_name_t).

3.1.4 Contact Type

Contact Type (contact_t) of value type object with 1 or more properties contains information on how to contact the party. The properties are Contact Details (details), Email (email), Public OpenPGP Key URL (public_openpgp_key_url), and Contact URL (url). If the property public_openpgp_key_url is set, the email SHALL be set as well.

```
$defs:
  # ...
  contact_t:
    details: String
    email: String.EMAIL
    public_openpgp_key_url: String.URI
    url: String.URI
  # ...
```

Contact details (details) of value type string with 1 or more characters contains details regarding ways to reach the party, e.g. through web sites, phone numbers, and postal mail addresses.

Example 1:

```
Example Company can be reached at tel:+493023125232,
or via our website at https://www.example.com/contact.
```

Email (email) of value type string of 6 or more characters with format email contains the email address that can be used to reach the party.

Examples 2:

```
productcert@example.net
psirt@example.com
reporter@securityresearcher.example
vulnerability@coordinator.example
```

Public OpenPGP Key URL (public_openpgp_key_url) has value type string of 11 or more characters with format uri and pattern (regular expression):

```
^https:\\/\\/
```

Public OpenPGP Key URL contains a URL pointing to a public OpenPGP key valid for the email of party provided in the sibling property email.

It is desired that the OpenPGP Key contains the same email address in its user ID as given through the property email. However, due to data protection and operation concerns neither a user ID in the OpenPGP key nor an exact match to the value of email is enforced by this standard. The use of aliases is permitted. The issuing party is responsible for ensuring the usability of the key provided.

The URL MAY point to a location that redirects. Redirects SHALL fulfill the same requirements as specified in 7.1.6. The content delivered SHALL be a valid OpenPGP key allowing encryption as ASCII armored file with the matching content type. See [RFC4880] and [RFC3156] for more details.

Examples 3:

```
https://coordinator.example/.well-known/openpgpkey/hu/nxdcs8npc6mn3xyfpcbiqhcu9s357r5m?l=vulnerability
https://example.net/.well-known/openpgpkey/hu/euwmpyfh4rzf8ymbqhjjhrirgib4dyfs?l=productcert
https://openpgpkey.securityresearcher.example/.well-known/openpgpkey/securityresearcher.example/hu/enudbakzkbdym3ymwjy9pcxztka75f73?l=reporter
https://psirt.example.com/security/openpgp/latest
```

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Contact URL (url) has value type string of 11 or more characters with format uri and pattern (regular expression):

```
^https:\\/\\/
```

Contact URL contains a URL that can be used to reach the party.

Examples 4:

```
https://www.example.com/psirt/contact
https://www.example.net/psirt/report-a-vulnerability--expert-form
https://www.example.org/.well-known/security.txt
```

3.1.5 Document-local Vuln ID Type

The Document-local Vuln ID Type (dl_vuln_id_t) of value type string with 6 or more characters is a document-local reference token for vulnerability instances. It SHALL conform to pattern (regular expression):

```
^VULN-[0-9A-Za-z._-]+$
```

The value contains a token required to identify a vulnerability uniquely in the context of the current document so that it can be referred to from other parts in the document. It SHOULD NOT to be used to refer to a vulnerability from outside of the CSAF document.

```
$defs:
  # ...
  dl_vuln_id_t: String.Pattern
  # ...
```

Examples 1:

```
VULN-0001
VULN-CVE-1900-0001
```

3.1.6 Entity Group ID Type

The Entity Group ID Type (entity_group_id_t) of value type string with 6 or more characters is a reference token for entity group instances. It SHALL conform to pattern (regular expression):

```
^EGID-[0-9A-Za-z][0-9A-Za-z._-]*$
```

The value contains a token required to identify a group of entities uniquely in the context of the current document so that it can be referred to from other parts in the document.

```
$defs:
  # ...
  entity_group_id_t: String.Pattern
  # ...
```

Examples 1:

```
EGID-0001
EGID-CERTs
EGID-Coordinators
EGID-vendors
```

3.1.7 Entity ID Type

The Entity ID Type (entity_id_t) of value type string with 5 or more characters is a reference token for entity instances. It SHALL conform to pattern (regular expression):

```
^EID-[0-9A-Za-z][0-9A-Za-z._-]*$
```

The value contains a token required to identify an entity uniquely in the context of the current document so that it can be referred to from other parts in the document.

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```
$defs:
  # ...
  entity_id_t: String.Pattern
  # ...
```

Examples 1:

```
  EID-0001
  EID-CERT
  EID-Example_Company
  EID-org.example
```

3.1.8 Entity Refs Type

List of entity references (entity_refs_t) of value type array with 1 or more unique items (a set) specifies a list of entity_ids or entity_group_ids to give context to the parent item.

```
$defs:
  # ...
  entity_refs_t: Sequence
  # ...
```

Value type of every such entity reference item is any of Entity Group ID Type (entity_group_id_t) or Entity ID Type (entity_id_t).

```
$defs:
  # ...
  entity_refs_t:
  - # <entity-ref-instance>:
    # !AnyOf<
    - $defs.entity_group_id_t
    - $defs.entity_id_t
    #>
  # ...
```

This type allows to reference Entity IDs directly and indirectly via an Entity Group ID within the same data structure.

> The usage of the anyOf schema constraint allows for a faster evaluation. However, such construct needs to ensure that the schemas have no overlap.

3.1.9 Extensions Type

List of extensions (extensions_t) of value type array with 1 or more unique items (a set) contains a list of extension elements for the current context.

```
$defs:
  # ...
  extensions_t: Sequence
  # ...
```

Each item SHALL comply with the rules for CSAF Extensions and validate against the CSAF Extension Content schema as well as the schema given through its $schema property.

3.1.10 Full Product Name Type

Full Product Name (full_product_name_t) of value type object specifies information about the product and assigns the product ID. The properties name and product_id are required. The property product_identification_helper is optional.

```
$defs:
  # ...
  full_product_name_t:
    name: String
    product_id: $defs.product_id_t
    product_identification_helper: Mapping
  # ...
```

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3.1.10.1 Full Product Name Type - Name

Textual description of the product (name) has value type string with 1 or more characters. The value SHOULD be the product's full canonical name, including version number and other attributes, as it would be used in a human-friendly document.

Examples 1:

```
Cisco AnyConnect Secure Mobility Client 2.3.185
Microsoft Host Integration Server 2006 Service Pack 1
```

3.1.10.2 Full Product Name Type - Product ID

Product ID (product_id) holds a value of type Product ID (product_id_t).

3.1.10.3 Full Product Name Type - Product Identification Helper

Helper to identify the product (product_identification_helper) of value type object provides in its properties at least one method which aids in identifying the product in an asset database. Of the given eight properties cpe, hashes, model_numbers, purls, sbom_urls, serial_numbers, skus, and x_generic_uris, 1 is mandatory.

```
$defs:
  # ...
  full_product_name_t:
    # ...
    product_identification_helper:
      cpe: String.Pattern
      hashes: Sequence
      model_numbers: Sequence
      purls: Sequence
      sbom_urls: Sequence
      serial_numbers: Sequence
      skus: Sequence
      x_generic_uris: Sequence
  # ...
```

A helper to identify the product SHALL identify the product described by the name in its entirety. A matching algorithm has to be able to rely on the product_identification_helper information during the matching without having to check e.g. the branches hierarchy leading to the product.

Only exception is the use of a CPE with a version part of *: A matching algorithm SHALL check whether a parent element along the path leading to the product_identification_helper exists, that has the category product_version_range. In such case, the corresponding value of the product version range SHALL be used as version information for the matching.

3.1.10.3.1 Full Product Name Type - Product Identification Helper - CPE

Common Platform Enumeration representation (cpe) has value type string of 5 or more characters with pattern (regular expression):

```
^((cpe:2\\.3:[aho\\*\\-](:((((\\?*|\\*?)([a-zA-Z0-9\\-\\._]|(\\\\[\\\\\\\\*\\?!\"#\\$%&'\\(\\)\\+,\\/:;<=>@\\[\\]\\^`\\{\\|\\}~]))+(\\?*|\\*?))|[\\*\\-])){5}(:(([a-zA-Z]{2,3}(-([a-zA-Z]{2}|[0-9]{3}))?)|[\\*\\-]))(:((((\\?*|\\*?)([a-zA-Z0-9\\-\\._]|(\\\\[\\\\\\\\*\\?!\"#\\$%&'\\(\\)\\+,\\/:;<=>@\\[\\]\\^`\\{\\|\\}~]))+(\\?*|\\*?))|[\\*\\-])){4})|([c][pP][eE]:\\/[AHOaho]?(:[A-Za-z0-9\\._\\-~%]*){0,6}))$
```

The Common Platform Enumeration (CPE) attribute refers to a method for naming platforms external to this specification. See [CPE23-N] for details.

Both, CPE 2.2 and CPE 2.3, are supported in CSAF.

3.1.10.3.2 Full Product Name Type - Product Identification Helper - Hashes

List of hashes (hashes) of value type array with 1 or more unique items contains a list of cryptographic hashes usable to identify files.

```
$defs:
  # ...
  full_product_name_t:
    # ...
    product_identification_helper:
      # ...
```

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```
    hashes: Sequence
    # ...
  # ...
```

Cryptographic hashes of value type object contains all information to identify a file based on its cryptographic hash values. Any cryptographic hashes object has the two mandatory properties file_hashes and filename.

```
$defs:
  # ...
  full_product_name_t:
    # ...
    product_identification_helper:
      # ...
      hashes:
        file_hashes: Sequence
        filename: String
      # ...
  # ...
```

List of file hashes (file_hashes) of value type array with 1 or more unique items contains a list of cryptographic hashes for this file.

```
$defs:
  # ...
  full_product_name_t:
    # ...
    product_identification_helper:
      # ...
      hashes:
        file_hashes: Sequence
        # ...
      # ...
  # ...
```

Each File hash of value type object contains one hash value and algorithm of the file to be identified. Any File hash object has the two mandatory properties algorithm and value.

```
$defs:
  # ...
  full_product_name_t:
    # ...
    product_identification_helper:
      # ...
      hashes:
        file_hashes:
        - # <file_hash-instance>:
          algorithm: String.Pattern
          value: String.Pattern
        # ...
      # ...
  # ...
```

The Algorithm of the cryptographic hash representation (algorithm) has value type string with 1 or more characters with pattern (regular expression):

```
^[0-9a-z][0-9a-z-]*$
```

The Algorithm of the cryptographic hash representation contains the name of the cryptographic hash algorithm used to calculate the value. The default value for algorithm is sha256. Secure cryptographic hash algorithms SHOULD be preferred.

Examples 1:

```
blake2b512
sha256
sha3-512
sha384
sha512
```

These values are derived from the currently supported digests in OpenSSL [OPENSSL]. Leading dashes were removed.

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The command openssl dgst -list (Version 3.4.0 from 2024-10-22) outputs the following:

```
Supported digests:
-blake2b512       -blake2s256       -md4
-md5              -md5-sha1         -mdc2
-ripemd           -ripemd160        -rmd160
-sha1             -sha224           -sha256
-sha3-224         -sha3-256         -sha3-384
-sha3-512         -sha384           -sha512
-sha512-224       -sha512-256       -shake128
-shake256         -sm3              -ssl3-md5
-ssl3-sha1        -whirlpool
```

The Value of the cryptographic hash representation (value) has value type string of 32 or more characters with pattern (regular expression):

```
^[0-9a-f]{32,}$
```

The Value of the cryptographic hash attribute contains the cryptographic hash value in lowercase hexadecimal representation.

Examples 2:

```
37df33cb7464da5c7f077f4d56a32bc84987ec1d85b234537c1c1a4d4fc8d09dc29e2e762cb5203677bf849a2855a0283710f1f5fe1d6ce8d5ac85c645d0fcb3
4775203615d9534a8bfca96a93dc8b461a489f69124a130d786b42204f3341cc
9ea4c8200113d49d26505da0e02e2f49055dc078d1ad7a419b32e291c7afebbb84badfbd46dec42883bea0b2a1fa697c
```

The Filename representation (filename) of value type string with 1 or more characters contains the name of the file which is identified by the hash values.

Examples 3:

```
WINWORD.EXE
msotadddin.dll
sudoers.so
```

If the value of the hash matches and the filename does not, a user SHOULD prefer the hash value. In such cases, the filename SHOULD be used as informational property.

3.1.10.3.3 Full Product Name Type - Product Identification Helper - Model Numbers

The list of models (model_numbers) of value type array with 1 or more unique items contains a list of model numbers.

A list of models SHOULD only be used if a certain range of model numbers with its corresponding software version is affected, or the model numbers change during update.

This can also be used to identify hardware. If necessary, the software, or any other related part, SHALL be bind to that via a product path.

```
$defs:
  # ...
  full_product_name_t:
    # ...
    product_identification_helper:
      # ...
      model_numbers: Sequence
      # ...
  # ...
```

Any given model number of value type string with at least 1 character represents a model number of the component to identify - possibly with placeholders.

The terms “model”, “model number” and “model variant” are mostly used synonymously. Often it is abbreviated as “MN”, “M/N” or “model no.”.

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If a part of a model number of the component to identify is given, it SHALL begin at the first and end at the last character position of the string representing the targeted component. The wildcard characters ? (for a single character) and * (for zero or more characters) signal exclusion of characters at these positions from matching. This applies also to the first character. An unescaped * SHALL be the only * wildcard in the string. As part of the model number, the special characters ?, * and \ SHALL be escaped with \.

> Note: A backslash SHALL be escaped itself in a JSON string.

Examples 1:

```
*-G109A/EU?
2024-*
6RA8096-4MV62-0AA0
6RA801?-??V62-0AA0
IC25T060ATCS05-0
```

3.1.10.3.4 Full Product Name Type - Product Identification Helper - PURLs

List of PURLs (purls) of value type array with 1 or more unique items contains a list of Package-URL (PURL) identifiers.

```
$defs:
  # ...
  full_product_name_t:
    # ...
    product_identification_helper:
      # ...
      purls: Sequence
      # ...
  # ...
```

A Package-URL representation has value type string of 7 or more characters with pattern (regular expression):

```
^pkg:[a-z][a-z0-9\\.\\-]*\\/.+
```

> The given pattern does not completely evaluate whether a PURL is valid according to the [ECMA-427] specification. It provides a more generic approach and general guidance to enable forward compatibility. CSAF uses only the canonical form of PURL to conform with section 3.3 of [RFC3986]. Therefore, URLs starting with pkg:// are considered invalid.

The PURL attribute refers to a method for reliably identifying and locating software packages external to this specification. See [ECMA-427] and [PURL] for details. Multiple PURLs can be specified to allow for identifiers to locate identical components in different locations.

If multiple PURLs are specified, they SHALL only differ in their qualifiers. Otherwise, separate product branches SHOULD be used to differentiate between the components.

3.1.10.3.5 Full Product Name Type - Product Identification Helper - SBOM URLs

The list of SBOM URLs (sbom_urls) of value type array with 1 or more unique items contains a list of URLs where SBOMs for this product can be retrieved.

> The SBOMs might differ in format or depth of detail. Currently supported formats are SPDX, CycloneDX, and SWID.

```
$defs:
  # ...
  full_product_name_t:
    # ...
    product_identification_helper:
      # ...
      sbom_urls: Sequence
```

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```
    # ...
  # ...
```

Any given SBOM URL of value type string with format uri contains a URL of one SBOM for this product.

Examples 1:

```
https://raw.githubusercontent.com/CycloneDX/bom-examples/master/SBOM/keycloak-10.0.2/bom.json
https://swinslow.net/spdx-examples/example4/main-bin-v2
```

3.1.10.3.6 Full Product Name Type - Product Identification Helper - Serial Numbers

The list of serial numbers (serial_numbers) of value type array with 1 or more unique items contains a list of serial numbers.

A list of serial numbers SHOULD only be used if a certain range of serial numbers with its corresponding software version is affected, or the serial numbers change during update.

```
$defs:
  # ...
  full_product_name_t:
    # ...
    product_identification_helper:
      # ...
      serial_numbers: Sequence
      # ...
  # ...
```

Any given serial number of value type string with at least 1 character represents a serial number of the component to identify - possibly with placeholders.

If a part of a serial number of the component to identify is given, it SHALL begin at the first and end at the last character position of the string representing the targeted component. The wildcard characters ? (for a single character) and * (for zero or more characters) signal exclusion of characters at these positions from matching. This applies also to the first character. An unescaped * SHALL be the only * wildcard in the string. As part of the serial number, the special characters ?, * and \ SHALL be escaped with \.

> Note: A backslash SHALL be escaped itself in a JSON string.

Examples 1:

```
*RF8R71YR???
11S45N0249Z1ZS9*
DSEP147100
L15-VM-???
L234.696.30.044.712
```

3.1.10.3.7 Full Product Name Type - Product Identification Helper - SKUs

The list of stock keeping units (skus) of value type array with 1 or more unique items contains a list of stock keeping units.

A list of stock keeping units SHOULD only be used if the list of product paths is used to decouple e.g. hardware from the software, or the stock keeping units change during update. In the latter case the remediations SHALL include the new stock keeping units or a description how it can be obtained.

> The use of the list of product paths in the first case is important. Otherwise, the end user is unable to identify which version (the affected or the not affected / fixed one) is used.

```
$defs:
  # ...
  full_product_name_t:
    # ...
    product_identification_helper:
```

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```
    # ...
    skus: Sequence
    # ...
  # ...
```

Any given stock keeping unit of value type string with at least 1 character represents a stock keeping unit (SKU) of the component to identify - possibly with placeholders.

> Sometimes this is also called “item number”, “article number” or “product number”.

If a part of a stock keeping unit of the component to identify is given, it SHALL begin at the first and end at the last character position of the string representing the targeted component. The wildcard characters ? (for a single character) and * (for zero or more characters) signal exclusion of characters at these positions from matching. This applies also to the first character. An unescaped * SHALL be the only * wildcard in the string. As part of the stock keeping unit, the special characters ?, * and \ SHALL be escaped with \.

> Note: A backslash SHALL be escaped itself in a JSON string.

Examples 1:

```
11????2
320196154130703
i800*
i8005-5.0
2.063.*.?:221-A
```

3.1.10.3.8 Full Product Name Type - Product Identification Helper - Generic URIs

List of generic URIs (x_generic_uris) of value type array with 1 or more unique items contains a list of identifiers which are either vendor-specific or derived from a standard not yet supported.

```
$defs:
  # ...
  full_product_name_t:
    # ...
    product_identification_helper:
      # ...
      x_generic_uris: Sequence
  # ...
```

Any such Generic URI item of value type object provides the two mandatory properties Namespace (namespace) and URI (uri).

```
$defs:
  # ...
  full_product_name_t:
    # ...
    product_identification_helper:
      # ...
      x_generic_uris:
      - # <x_generic_uri-instance>:
        namespace: String.URI
        uri: String.URI
  # ...
```

The namespace of the generic URI (namespace) of value type string with format uri refers to a URL which provides the name and knowledge about the specification used or is the namespace in which these values are valid.

The URI (uri) of value type string with format uri contains the identifier itself.

> These elements can be used to reference a specific component from an SBOM:

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Example 1 (linking a component from a CycloneDX SBOM using the bomlink mechanism):

```
"x_generic_uris": [
  {
    "namespace": "https://cyclonedx.org/capabilities/bomlink/",
    "uri": "urn:cdx:411dafd2-c29f-491a-97d7-e97de5bc2289/1#pkg:maven/org.jboss.logging/jboss-logging@3.4.1.Final?type=jar"
  }
]
```

Example 2 (linking a component from an SPDX SBOM):

```
"x_generic_uris": [
  {
    "namespace": "https://spdx.github.io/spdx-spec/latest/document-creation-information/#65-spdx-document-namespace-field",
    "uri": "https://swinslow.net/spdx-examples/example4/main-bin-v2#SPDXRef-libc"
  }
]
```

3.1.10.4 Full Product Name Type - Extensions

Product-level Extensions (x_extensions) of value type Extensions Type (extensions_t) contains list of extensions valid at the full product name element level of the CSAF document and associated with this full product name element.

```
$defs:
  # ...
  full_product_name_t:
    # ...
    x_extensions:  # $defs.extensions_t
    - # <x_extension-instance>:
      $schema: String
      category: String.Enum
      content: Mapping
      critical: Boolean
      # ...
  # ...
```

3.1.11 Language Type

Language type (lang_t) has value type string with pattern (regular expression):

```
^(([A-Za-z]{2,3}(-[A-Za-z]{3}(-[A-Za-z]{3}){0,2})?|[A-Za-z]{4,8})(-[A-Za-z]{4})?(-([A-Za-z]{2}|[0-9]{3}))?(-([A-Za-z0-9]{5,8}|[0-9][A-Za-z0-9]{3}))*(-[A-WY-Za-wy-z0-9](-[A-Za-z0-9]{2,8})+)*(-[Xx](-[A-Za-z0-9]{1,8})+)?|[Xx](-[A-Za-z0-9]{1,8})+|[Ii]-[Dd][Ee][Ff][Aa][Uu][Ll][Tt]|[Ii]-[Mm][Ii][Nn][Gg][Oo])$
```

The value identifies a language, corresponding to IETF BCP 47 / RFC 5646. See IETF language registry: https://www.iana.org/assignments/language-subtag-registry/language-subtag-registry

> The pattern given above is derived from the ABNF depicted in Figure 1 in [RFC5646]. CSAF skips those grandfathered language tags that are deprecated at the time of writing the specification. Even though the private use language tags are supported they should not be used to ensure readability across the ecosystem. It is recommended to follow the conventions for the capitalization of the subtags even though it is not mandatory as most users are used to that.

CSAF Producers SHALL follow the conventions for the capitalization of the subtags. CSAF validators SHALL treat subtags as case-insensitive unless stated otherwise in the test.

Examples 1:

```
de
en
fr
frc
jp
```

3.1.12 Notes Type

List of notes (notes_t) of value type array with 1 or more items of type Note contains notes which are specific to the current context.

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```
$defs:
  # ...
  notes_t: Sequence
  # ...
```

Value type of every such Note item is object with the mandatory properties category and text providing a place to put all manner of text blobs related to the current context. A Note object MAY provide the optional properties audience, group_ids, product_ids and title.

```
$defs:
  # ...
  notes_t:
  - # <note-instance>:
    audience: String
    category: String.Enum
    group_ids: $defs.product_groups_t
    product_ids: $defs.products_t
    text: String
    title: String
  # ...
```

Audience of note (audience) of value type string with 1 or more characters indicates who is intended to read it.

Examples 1:

```
all
executives
operational management and system administrators
safety engineers
```

Note category (category) of value type string and enum contains the information of what kind of note this is. Valid enum values are:

```
description
details
faq
general
legal_disclaimer
other
summary
```

The value description indicates the note is a description of something. The optional sibling property title MAY have more information in this case.

The value details indicates the note is a low-level detailed discussion. The optional sibling property title MAY have more information in this case.

The value faq indicates the note is a list of frequently asked questions.

The value general indicates the note is a general, high-level note. The optional sibling property title MAY have more information in this case.

The value legal_disclaimer indicates the note represents any possible legal discussion, including constraints, surrounding the document.

The value other indicates the note is something that doesn't fit the other categories. The optional sibling attribute title SHOULD have more information to indicate clearly what kind of note to expect in this case.

The value summary indicates the note is a summary of something. The optional sibling property title MAY have more information in this case.

Group IDs (group_ids) are of value type Product Groups (product_groups_t) and contain a list of Product Groups the current note item applies to.

Product IDs (product_ids) are of value type Products (products_t) and contain a list of Products the current note item applies to.

Note content (text) of value type string with 1 or more characters holds the content of the note. Content varies depending on type.

Title of note (title) of value type string with 1 or more characters provides a concise description of what is contained in the text of the note.

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Examples 2:

```
Details
Executive summary
Technical summary
Impact on safety systems
```

3.1.13 Product Group ID Type

The Product Group ID Type (product_group_id_t) of value type string with 1 or more characters is a reference token for product group instances. The value is a token required to identify a group of products so that it can be referred to from other parts in the document. There is no predefined or required format for the Product Group ID (product_group_id) as long as it uniquely identifies a product group in the context of the current document.

```
$defs:
  # ...
  product_group_id_t: String
  # ...
```

Examples 1:

```
CSAFGID-0001
CSAFGID-0002
CSAFGID-0020
```

Even though the standard does not require a specific format it is recommended to use different prefixes for the Product ID and the Product Group ID to support reading and parsing the document.

3.1.14 Product Groups Type

List of Product Group ID (product_groups_t) of value type array with 1 or more unique items (a set) of type Product Group ID (product_group_id_t) specifies a list of product_group_ids to give context to the parent item.

```
$defs:
  # ...
  product_groups_t: Sequence
  # ...
```

3.1.15 Product ID Type

The Product ID Type (product_id_t) of value type string with 1 or more characters is a reference token for product instances. The value is a token required to identify a full_product_name so that it can be referred to from other parts in the document. There is no predefined or required format for the Product ID (product_id) as long as it uniquely identifies a product in the context of the current document.

```
$defs:
  # ...
  product_id_t: String
  # ...
```

Examples 1:

```
CSAFPID-0004
CSAFPID-0008
```

Even though the standard does not require a specific format it is recommended to use different prefixes for the Product ID and the Product Group ID to support reading and parsing the document.

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3.1.16 Products Type

List of Product IDs (products_t) of value type array with 1 or more unique items (a set) of type Product ID (product_id_t) specifies a list of product_ids to give context to the parent item.

```
$defs:
  # ...
  products_t: Sequence
  # ...
```

3.1.17 References Type

List of references (references_t) of value type array with 1 or more items of type Reference holds a list of Reference objects.

```
$defs:
  # ...
  references_t: Sequence
  # ...
```

Value type of every such Reference item is object with the mandatory properties url and summary holding any reference to conferences, papers, advisories, and other resources that are related and considered related to either a surrounding part of or the entire document and to be of value to the document consumer. A reference object MAY provide the optional property category.

```
$defs:
  # ...
  references:
  - # <reference-instance>:
    category: String
    summary: String
    url: String.URI
  # ...
```

Category of reference (category) of value type string and enum indicates whether the reference points to the same document or vulnerability in focus (depending on scope) or to an external resource. Valid enum values are:

```
external
self
```

The default value for category is external.

The value external indicates, that this document is an external reference to a document or vulnerability in focus (depending on scope).

The value self indicates, that this document is a reference to this same document or vulnerability (also depending on scope).

> This includes links to documents with the same content but different file format (e.g. advisories as PDF or HTML).

Summary of the reference (summary) of value type string with 1 or more characters indicates what this reference refers to.

URL of reference (url) of value type string with format uri provides the URL for the reference.

3.1.18 Subpath Type

Subpath (subpath_t) of value type object contains the next node along the current path and its relationship to the previous node. The properties category and next_product_reference are required.

```
$defs:
  # ...
  subpath_t:
    category: String.Enum
    next_product_reference: $defs.product_id_t
  # ...
```

Relationship category (category) of value type string and enum defines the category of relationship between the previous item and the referenced next product. Valid enum values are:

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```
default_component_of
external_component_of
installed_on
installed_with
optional_component_of
```

The value default_component_of indicates that the entity labeled with one Product ID (e.g. CSAFPID-0001) is a default component of an entity with another Product ID (e.g. CSAFPID-0002). These Product IDs SHOULD NOT be identical to provide minimal redundancy.

The value external_component_of indicates that the entity labeled with one Product ID (e.g. CSAFPID-0001) is an external component of an entity with another Product ID (e.g. CSAFPID-0002). These Product IDs SHOULD NOT be identical to provide minimal redundancy.

The value installed_on indicates that the entity labeled with one Product ID (e.g. CSAFPID-0001) is installed on a platform entity with another Product ID (e.g. CSAFPID-0002). These Product IDs SHOULD NOT be identical to provide minimal redundancy.

The value installed_with indicates that the entity labeled with one Product ID (e.g. CSAFPID-0001) is installed alongside an entity with another Product ID (e.g. CSAFPID-0002). These Product IDs SHOULD NOT be identical to provide minimal redundancy.

The value optional_component_of indicates that the entity labeled with one Product ID (e.g. CSAFPID-0001) is an optional component of an entity with another Product ID (e.g. CSAFPID-0002). These Product IDs SHOULD NOT be identical to provide minimal redundancy.

Next product reference (next_product_reference) of value type Product ID (product_id_t) holds a Product ID that refers to the Full Product Name element, which is referenced as the second element of the relationship.

3.1.19 Version Type

The Version (version_t) type has value type string with pattern (regular expression):

```
^(0|[1-9][0-9]*)$|^((0|[1-9]\\d*)\\.(0|[1-9]\\d*)\\.(0|[1-9]\\d*)(?:-((?:0|[1-9]\\d*|\\d*[a-zA-Z-][0-9a-zA-Z-]*)(?:\\.(?:0|[1-9]\\d*|\\d*[a-zA-Z-][0-9a-zA-Z-]*))*))?(?:\\
+([0-9a-zA-Z-]+(?:\\.[0-9a-zA-Z-]+)*))?)$
```

The version specifies a version string to denote clearly the evolution of the content of the document. There are two options how it can be used:

- semantic versioning (preferred; according to the rules below)
- integer versioning

A CSAF document SHALL use only one versioning system.

Examples 1:

```
1
4
0.9.0
1.4.3
2.40.0+21AF26D3
```

3.1.19.1 Version Type - Integer Versioning

Integer versioning increments for each version where the $.document.tracking.status is final the version number by one. The regular expression for this type is:

```
^(0|[1-9][0-9]*)$
```

The following rules apply:

1. Once a versioned document has been released, the contents of that version SHALL NOT be modified. Any modifications SHALL be released as a new version.
2. Version zero (0) is for initial development before the initial_release_date. The document status SHALL be draft. Anything MAY change at any time. The document SHOULD NOT be considered stable.

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3. Version 1 defines the initial release to the specified target group. Each new version where $.document.tracking.status is final has a version number incremented by one.
4. Pre-release versions (document status draft) SHALL carry the new version number. Sole exception is before the initial release (see rule 2). The combination of document status draft and version 1 MAY be used to indicate that the content is unlikely to change.
5. Build metadata is never included in the version.
6. Precedence SHALL be determined by integer comparison.

3.1.19.2 Version Type - Semantic Versioning

Semantic versioning derived the rules from [SemVer]. The regular expression for this type is:

```
^((0|[1-9]\\d*)\\.(0|[1-9]\\d*)\\.(0|[1-9]\\d*)(?:-((?:0|[1-9]\\d*|\\d*[a-zA-Z-][0-9a-zA-Z-]*)(?:\\.(?:0|[1-9]\\d*|\\d*[a-zA-Z-][0-9a-zA-Z-]*))*))?(?:\\+([0-9a-zA-Z-]+(?:\\.[0-9a-zA-Z-]+)*))?)$
```

The goal of this structure is to provide additional information to the end user whether a new comparison with the asset database is needed. The "public API" in regard to CSAF is the CSAF document with its structure and content. This results in the following rules:

1. A normal version number SHALL take the form X.Y.Z where X, Y, and Z are non-negative integers, and SHALL NOT contain leading zeroes. X is the major version, Y is the minor version, and Z is the patch version. Each element SHALL increase numerically. For instance: 1.9.0 -> 1.10.0 -> 1.11.0.
2. Once a versioned document has been released, the contents of that version SHALL NOT be modified. Any modifications SHALL be released as a new version.
3. Major version zero (0.y.z) is for initial development before the initial_release_date. The document status SHALL be draft. Anything MAY change at any time. The document SHOULD NOT be considered stable. Changes which would increment the major version according to rule 7 are tracked in this stage with (0.y.z) by incrementing the minor version y instead. Changes that would increment the minor or patch version according to rule 6 or 5 are both tracked in this stage with (0.y.z) by incrementing the patch version z instead.
4. Version 1.0.0 defines the initial release to the specified target group. The way in which the version number is incremented after this release is dependent on the content and structure of the document and how it changes.
5. Patch version Z (x.y.Z | x > 0) SHALL be incremented if only backwards compatible bug fixes are introduced. A bug fix is defined as an internal change that fixes incorrect behavior.

   > In the context of the document this is the case e.g. for spelling mistakes.

6. Minor version Y (x.Y.z | x > 0) SHALL be incremented if the content of an existing element changes except for those which are covered through rule 7. It SHALL be incremented if substantial new information are introduced or new elements are provided. It MAY include patch level changes. Patch version SHALL be reset to 0 when minor version is incremented.
7. Major version X (X.y.z | X > 0) SHALL be incremented if a new comparison with the end user's asset database is required. This includes:
   - changes (adding, removing elements or modifying content) in $.product_tree or elements which contain $.product_tree in their path
   - adding or removing items of $.vulnerabilities
   - adding or removing elements in:
     - $.vulnerabilities[*].product_status.first_affected
     - $.vulnerabilities[*].product_status.known_affected
     - $.vulnerabilities[*].product_status.last_affected
   - removing elements from:
     - $.vulnerabilities[*].product_status.first_fixed
     - $.vulnerabilities[*].product_status.fixed
     - $.vulnerabilities[*].product_status.known_not_affected

   It MAY also include minor and patch level changes. Patch and minor version SHALL be reset to 0 when major version is incremented.

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8. A pre-release version (document status draft) MAY be denoted by appending a hyphen and a series of dot separated identifiers immediately following the patch version. Identifiers SHALL comprise only ASCII alphanumerics and hyphens [0-9A-Za-z-]. Identifiers SHALL NOT be empty. Numeric identifiers SHALL NOT include leading zeroes. Pre-release versions have a lower precedence than the associated normal version. A pre-release version indicates that the version is unstable and might not satisfy the intended compatibility requirements as denoted by its associated normal version.

   Examples 1:

   ```
   1.0.0-0.3.7
   1.0.0-alpha
   1.0.0-alpha.1
   1.0.0-x-y-z.--
   1.0.0-x.7.z.92
   ```

9. Pre-release SHALL NOT be included if $.document.tracking.status is final.

10. Build metadata MAY be denoted by appending a plus sign and a series of dot separated identifiers immediately following the patch or pre-release version. Identifiers SHALL comprise only ASCII alphanumerics and hyphens [0-9A-Za-z-]. Identifiers SHALL NOT be empty. Build metadata SHALL be ignored when determining version precedence. Thus two versions that differ only in the build metadata, have the same precedence.

    Examples 2:

    ```
    1.0.0+20130313144700
    1.0.0+21AF26D3----117B344092BD
    1.0.0-alpha+001
    1.0.0-beta+exp.sha.5114f85
    ```

11. Precedence refers to how versions are compared to each other when ordered.

    1. Precedence SHALL be calculated by separating the version into major, minor, patch and pre-release identifiers in that order (Build metadata does not figure into precedence).
    2. Precedence is determined by the first difference when comparing each of these identifiers from left to right as follows: Major, minor, and patch versions are always compared numerically.

       Example 3:

       ```
       1.0.0 < 2.0.0 < 2.1.0 < 2.1.1
       ```

    3. When major, minor, and patch are equal, a pre-release version has lower precedence than a normal version:

       Example 4:

       ```
       1.0.0-alpha < 1.0.0
       ```

    4. Precedence for two pre-release versions with the same major, minor, and patch version SHALL be determined by comparing each dot separated identifier from left to right until a difference is found as follows:
       1. Identifiers consisting of only digits are compared numerically.
       2. Identifiers with letters or hyphens are compared lexically in ASCII sort order.
       3. Numeric identifiers always have lower precedence than non-numeric identifiers.
       4. A larger set of pre-release fields has a higher precedence than a smaller set, if all of the preceding identifiers are equal.

       Example 5:

       ```
       1.0.0-alpha < 1.0.0-alpha.1 < 1.0.0-alpha.beta < 1.0.0-beta < 1.0.0-beta.2 < 1.0.0-beta.11 < 1.0.0-rc.1 < 1.0.0
       ```

Note, that the following values do no conform the semantic versioning described above.

Examples 6 (which are invalid):

```
1.16.13.14-Cor
1.0.0-x-y-z.–
```

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```
1.0.0+21AF26D3—-117B344092BD
2.5.20+3f93da6b+7cc
3.20.0-00
```

3.2 Properties

These final five subsections document the five properties of a CSAF document. The two mandatory properties $schema and document, as well as the optional properties product_tree, vulnerabilities, x_extensions in that order.

3.2.1 Schema Property

JSON schema ($schema) of value type string and const with format uri contains the URL of the CSAF JSON schema which the document promises to be valid for. The value for this const is:

```
https://docs.oasis-open.org/csaf/csaf/v2.1/schema/csaf.json
```

This value allows for tools to identify that a JSON document is meant to be valid against this schema. Tools can use that to support users by automatically checking whether the CSAF adheres to the JSON schema identified by this URL.

3.2.2 Document Property

Document level meta-data (document) of value type object with the six mandatory properties Category (category), CSAF Version (csaf_version), Distribution (distribution), Publisher (publisher), Title (title), and Tracking (tracking) captures the meta-data about this document describing a particular set of security advisories. In addition, the document object MAY provide the nine optional properties Acknowledgments (acknowledgments), Aggregate Severity (aggregate_severity), Involvement (involvement), Language (lang), License expression (license_expression), Notes (notes), References (references), Source Language (source_lang), and Document-level Extensions (x_extensions).

```
<csaf-instance>:
  document:
    acknowledgments: $defs.acknowledgments_t
    aggregate_severity: Mapping
    category: String.Pattern
    csaf_version: String.Const
    distribution: Mapping
    involvement: Mapping
    lang: $defs.lang_t
    license_expression: String
    notes: $defs.notes_t
    publisher: Mapping
    references: $defs.references_t
    source_lang: $defs.lang_t
    title: String
    tracking: Mapping
    x_extensions: $defs.extensions_t
```

3.2.2.1 Document Property - Acknowledgments

Document acknowledgments (acknowledgments) of value type Acknowledgments Type (acknowledgments_t) contains a list of acknowledgment elements associated with the whole document.

```
<csaf-instance>:
  document:
    acknowledgements:  # $defs.acknowledgments_t
    - # <acknowledgement-instance>:
      names: Sequence
      organization: String
      summary: String
      urls: Sequence
      # ...
    # ...
```

3.2.2.2 Document Property - Aggregate Severity

Aggregate severity (aggregate_severity) of value type object with the mandatory property text and the optional property namespace is a vehicle that is provided by the document producer to convey the urgency and criticality with which the one or more vulnerabilities reported should be addressed. It is a document-level metric and applied to the document as a whole — not

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any specific vulnerability. The range of values in this field is defined according to the document producer's policies and procedures.

```
<csaf-instance>:
  document:
    # ...
    aggregate_severity:
      namespace: String
      text: String
    # ...
```

The Namespace of aggregate severity (namespace) of value type string with format uri points to the namespace so referenced.

The Text of aggregate severity (text) of value type string with 1 or more characters provides a severity which is independent of - and in addition to - any other standard metric for determining the impact or severity of a given vulnerability (such as CVSS).

Examples 1:

```
Critical
Important
Moderate
```

3.2.2.3 Document Property - Category

Document category (category) has value type string of 1 or more characters with pattern (regular expression):

```
^[^\\s\\-_\\.](.*[^\\s\\-_\\.])?$
```

Document category defines a short canonical name, chosen by the document producer, which will inform the end user as to the category of document.

It is directly related to the profiles defined in section 4.

```
<csaf-instance>:
  document:
    # ...
    category: String.Pattern
    # ...
```

Examples 1:

```
csaf_base
csaf_security_advisory
csaf_vulnerability_report
csaf_vex
Example Company Security Notice
```

3.2.2.4 Document Property - CSAF Version

CSAF version (csaf_version) of value type string and const gives the version of the CSAF specification which the document was generated for. The value for this const is:

```
2.1
```

3.2.2.5 Document Property - Distribution

Rules for document sharing (distribution) of value type object with the mandatory property Traffic Light Protocol (TLP) (tlp) and the optional properties Sharing Group (sharing_group) and Text (text) describes any constraints on how this document might be shared.

```
<csaf-instance>:
  document:
    # ...
    distribution:
```

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```
    sharing_group: Mapping
    text: String
    tlp: Mapping
  # ...
```

If multiple values are present, the TLP information SHOULD be preferred as this aids in automation. The Sharing Group SHALL be interpreted as specification to the TLP information. Therefore, the Sharing Group MAY also be used to convey special TLP restrictions:

Examples 1:

```
E-ISAC members-only
Only releasable to European Energy sector
Releasable to NATO countries
```

> Note that for such restrictions the Sharing Group Name SHALL exist and all participants SHALL know the associated Sharing Group IDs to allow for automation.

3.2.2.5.1 Document Property - Distribution - Sharing Group

Sharing Group (sharing_group) of value type object with the mandatory property Sharing Group ID (id) and the optional property Sharing Group Name (name) contains information about the group this document is intended to be shared with.

```
<csaf-instance>:
  document:
    # ...
    distribution:
      sharing_group:
        id: String.Pattern
        name: String
      # ...
    # ...
```

Sharing Group ID (id) has value type string with format uuid and pattern (regular expression):

```
^(([0-9a-f]{8}-[0-9a-f]{4}-4[0-9a-f]{3}-[0-9a-f]{4}-[0-9a-f]{12})|([0]{8}-([0]{4}-){3}[0]{12})|([f]{8}-([f]{4}-){3}[f]{12}))$
```

Sharing Group ID provides the unique ID for the sharing group. This ID is intended to be globally unique and MAY also be used by different issuing parties to share CSAF data within a closed group, e.g. during a Multi-Party Coordinated Vulnerability Disclosure case.

> Note, that participants in such cases usually differ. Therefore, it is advised to use one ID per case. Otherwise, the consequences of adding or removing parties from a case and the implications to other cases have to be considered.

The ID SHOULD NOT change throughout different CSAF documents, if the same sharing group is addressed. It SHALL differ if a different sharing group is addressed.

> It is assumed that the ID is globally unique, if constructed according to the specification for UUID Version 4.

The ID SHALL be valid according to [RFC9562] and recorded in the 8-4-4-4-12 notation in lowercase. The ID SHALL be a UUID Version 4 for any closed sharing group, i.e. TLP:GREEN and above.

The following ID values SHOULD NOT be used unless there are technical reasons for them. Therefore, they are reserved for implementation-specific situations:

- A system MAY use the Max UUID for TLP:CLEAR CSAF documents.

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> For example, the system uses the UUID as an indication whether a user allowed to see the document. The security considerations from [RFC9562] should be reflected on.

- A system MAY use the Nil UUID for CSAF documents that SHALL NOT be shared.

> For example, the CSAF document is just being drafted and the accidental leakage should be prevented.

> Note, that both values do not indicate a closed sharing group.

A CSAF document with TLP:CLEAR SHOULD NOT contain a sharing group value and SHALL NOT contain any other value for the Sharing Group ID than Max UUID (ffffffff-ffff-ffff-ffff-ffffffffffff).

If an issuing party distributes multiple versions of a single CSAF document to different sharing groups, the rules for CSAF modifier (cf. section 9.1.8) regarding the generation of the value of $.document.tracking.id SHALL be applied. This implies that usually the sharing group ID is used as a prefix to the original $.document.tracking.id.

Sharing Group Name (name) of value type string with one or more characters contains a human-readable name for the sharing group.

The Sharing Group Name is optional and can be chosen freely by the entity establishing the sharing group. However, the following values are reserved for the conditions below:

- For the Max UUID, the value of name SHALL exist and be Public.
- For the Nil UUID, the value of name SHALL exist and be No sharing allowed.

3.2.2.5.2 Document Property - Distribution - Text

The Textual description (text) of value type string with 1 or more characters provides a textual description of additional constraints.

Examples 1:

```
Copyright 2024, Example Company, All Rights Reserved.
Distribute freely.
Share only on a need-to-know-basis only.
```

3.2.2.5.3 Document Property - Distribution - TLP

Traffic Light Protocol (TLP) (tlp) of value type object with the mandatory property Label (label) and the optional property URL (url) provides details about the TLP classification of the document.

```
<csaf-instance>:
  document:
    # ...
    distribution:
      # ...
      tlp:
        label: String
        url: String.URI
    # ...
```

The Label of TLP (label) with value type string and enum provides the TLP label of the document. Valid values of the enum are:

```
AMBER
AMBER+STRICT
CLEAR
GREEN
RED
```

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Note: In the TLP specification, there are only four labels. The part +STRICT is an extension to TLP:AMBER. To simplify the JSON structure, avoid additional business level tests and aid in parsing, consumption and processing, it is provided as a label to be selected instead of having a separate field.

The default value for label is CLEAR.

Note: This provides the suggested default value for anyone writing CSAF documents as the majority of those are intended to be publicly available.

The URL of TLP version (url) with value type string with format uri provides a URL where to find the textual description of the TLP version which is used in this document. The default value is the URL to the definition by FIRST:

```
https://www.first.org/tlp/
```

Examples 1:

```
https://www.us-cert.gov/tlp
https://www.bsi.bund.de/SharedDocs/Downloads/DE/BSI/TLP/merkblatt-tlp.pdf
```

3.2.2.6 Document Property - Involvement

Involvement (involvement) of value type object with the mandatory properties List of actions (actions) and List of entities (entities) contains the coordination record stating entities and actions between them. The optional property is List of entity groups (entity_groups).

```
<csaf-instance>:
  document:
    # ...
    involvement:
      actions: Sequence
      entities: Sequence
      entity_groups: Sequence
    # ...
```

3.2.2.6.1 Document Property - Involvement - Actions

List of actions (actions) of value type array with 1 or more unique items (a set) contains the timeline of actions.

```
<csaf-instance>:
  document:
    # ...
    involvement:
      actions: Sequence
      # ...
    # ...
```

Every Action item of value type object with the five mandatory properties List of acting entities (acting_entity_refs), Action ID (action_id), Action category (category), Date of the action (date), and Action status (status) contains details about a single event in the timeline. In addition, any Action item MAY expose the optional properties List of document-local vuln IDs dl_vuln_ids, Group IDs (group_ids), Product IDs (product_ids), List of receiving entities (receiving_entity_refs), List of referenced actions (referenced_action_ids), Summary of the action (summary), and Action status (status).

```
<csaf-instance>:
  document:
    # ...
    involvement:
      actions:
      - # <action-instance>:
        acting_entity_refs: $defs.entity_refs_id_t
        action_id: $defs.action_id_t
        category: String.Enum
        date: String.DateTime
        dl_vuln_ids: Sequence
```

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```
    group_ids: $defs.product_groups_t
    product_ids: $defs.products_t
    receiving_entity_refs: $defs.entity_refs_id_t
    referenced_action_ids: Sequence
    summary: String
    status: String.Enum
  # ...
# ...
```

List of acting entities (acting_entity_refs) of value type Entity Refs ID (entity_refs_id_t) contains a list of entities that act.

Action ID (action_id) of value type Action ID (action_id_t) contains the reference token for this action. Its value SHALL be unique across all action items.

Action category (category) of value type string and enum specifies the category which this action belongs to. Valid values are:

```
assessment
confirmation
coordination
discovery
dispute
embargo
exploitation
fix_deployment
fix_release
notification
observation
```

The value assessment indicates that the actor examines if the vulnerability is correct, how severe it is, or which products it applies to. An assessment action records that this examination took place, which entities took part in it, and when. Any resulting score or rating is given in $.vulnerabilities[*].metrics or $.vulnerabilities[*].threats, not in the action.

The value confirmation indicates that the actor confirms that it received or accepted the action that this action refers to (via referenced_action_ids).

The value coordination indicates that the parties coordinate their work.

The value discovery indicates that the actor found the vulnerability.

The value dispute indicates that the actor disputes the action that this action refers to (via referenced_action_ids). It can dispute that the action is correct, its severity, or that it occurred. It can also state that the actor does not accept the action.

The value embargo indicates that the actor and the receiving entities agree to not disclose the information before the intended disclosure date. The agreed date is the disclosure_date of each vulnerability that dl_vuln_ids names. An embargo action SHALL name a minimum of one party in receiving_entity_refs, and it SHOULD name the vulnerabilities under embargo in dl_vuln_ids. An agreement to delay disclosure is an embargo. Other joint work of the parties stays coordination. The entities in receiving_entity_refs are the parties that the agreement binds. They are not necessarily the members of $.document.distribution.sharing_group.

The value exploitation indicates that the actor used the vulnerability. With the status attempted, the action states that the actor tried to use the vulnerability and did not succeed.

The value fix_deployment indicates that the actor installed a fix.

The value fix_release indicates that the actor published a fix, or made a fix available.

The value notification indicates that the actor tells a different party about the vulnerability. To describe an action for which no suitable specific action category is available, the notification category SHOULD be used and any specific information SHOULD be described in the summary field of the action.

The value observation indicates that the actor observed that the vulnerability was used. The entities in acting_entity_refs are the entities that made the observation, and date is the date of the observation. An observation action MAY name an exploitation action in referenced_action_ids. That action gives the entity that used the vulnerability and the date of that use. The scope of an observation action that names an exploitation action SHOULD be equal to or narrower than the scope of that action, after the resolution of product groups. An entity that found the vulnerability itself records a discovery, not an observation.

A confirmation and a dispute are always about a different action. Each of them SHALL have a minimum of one value in referenced_action_ids, and each SHALL name a minimum of one party in receiving_entity_refs.

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For a confirmation, each entity in the acting_entity_refs set SHALL also be in the receiving_entity_refs set of the action that it refers to. Entity groups are resolved before this. Only a entity that received an action can confirm that action.

Date of the action (date) of value type string with format date-time contains the date when the action occurred.

List of document-local vuln IDs (dl_vuln_ids) of value type array with 1 or more unique items (a set) contains a list of local IDs referring to vulnerabilities within the same document that this action applies to. Every Document-local Vuln ID item of value type Document-local Vuln ID Type (dl_vuln_id_t) specifies a single vulnerability element the current action applies to.

Group IDs (group_ids) are of value type Product Groups (product_groups_t) and contain a list of Product Groups the current action item applies to.

Product IDs (product_ids) are of value type Products (products_t) and contain a list of Products the current action item applies to.

List of receiving entities (receiving_entity_refs) of value type Entity Refs ID (entity_refs_id_t) contains a list of entities that receive the action.

List of referenced actions (referenced_action_ids) of value type array with 1 or more unique items (a set) contains a list of other actions reference by this action. Every referenced action item of value type Action ID (action_id_t) refers to a single vulnerability.

Summary of the action (summary) of value type string with 1 or more characters contains information about the action.

Action status (status) of value type string and enum contains an observation about the action. Valid values are:

```
attempted
completed
deferred
discontinued
in_progress
outstanding
planned
```

The value attempted indicates that the actor tried to do the action, but it did not reach the recipient.

The value completed indicates that the actor completed the action.

The value deferred indicates that the actor decided not to act at this time. It will examine the decision again later.

The value discontinued indicates that the actor started the work and then stopped it before it was complete.

The value in_progress indicates that the actor is doing the work now.

The value outstanding indicates that a party waits for this action. The status gives an observation and tells you nothing about the work of the actor.

The value planned indicates that the a party waits for this action. The status gives an observation and tells you nothing about the work of the actor.

The commonly agreed order of these status values for a notification is:

1. outstanding
2. deferred
3. planned
4. attempted
5. in_progress
6. completed
7. completed referenced by a confirmation with status of completed

The difference between a completed notification and one that also is referenced by a completed confirmation can be an important distinction for the notifying entity. Most likely, the list of actions will not carry every status for each notification but will highlight the important steps in the process.

3.2.2.6.2 Document Property - Involvement - Entities

List of entities (entities) of value type array with 1 or more unique items (a set) contains a list of entities related.

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```
<csaf-instance>:
  document:
    # ...
    involvement:
      # ...
      entities: Sequence
      # ...
    # ...
```

Every Entity item of value type object with the two mandatory properties Entity category (category) and Entity ID (entity_id) contains information about a single entity. In addition, any Entity item MAY expose the optional properties Contact of the entity (contact) and Name of the entity (name).

```
<csaf-instance>:
  document:
    # ...
    involvement:
      # ...
      entities:
      - # <entity-instance>:
        category: String.Enum
        contact: $defs.contact_t
        entity_id: $defs.entity_id_t
        name: String
      # ...
    # ...
```

Entity category (category) of value type string and enum specifies the category of the party. Valid values are:

```
adversary
coordinator
discoverer
multiplier
other
public
reporting_authority
user
vendor
```

The value adversary indicates an entity that acts against the other parties. It may be identified or not identified.

The value coordinator indicates an entity that coordinates the disclosure between several parties, e.g. a CERT.

The value discoverer indicates an entity that finds vulnerabilities, for example a researcher.

It is assume that the discoverer also reports vulnerabilities.

The value multiplier indicates an entity that uses or distributes the data about a vulnerability.

The value other indicates an entity that has a role that the other values do not give.

The value public indicates the general public.

The value reporting_authority indicates an entity that receives or requires a report, e.g. a national CSIRT.

The value user indicates an entity that uses or operates a product mentioned.

The value vendor indicates an entity that develops or maintains of information system products or services. This includes all authoritative product vendors, product security incident response teams (PSIRTs), open source projects as well as product resellers and distributors, including authoritative vendor partners.

The category does not control the role: an entity can act in one action and receive in a different action. An adversary is permitted as a recipient, because a notification to an adversary records that the data went to that adversary.

Contact of the entity (contact) of value type Contact Type (contact_t) contains information on how to contact the entity.

```
<csaf-instance>:
  document:
```

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```
  # ...
  involvement:
    # ...
    entities:
    - # <entity-instance>:
      # ...
      contact:
        details: String
        email: String.EMAIL
        public_openpgp_key_url: String.URI
        url: String.URI
      # ...
    # ...
  # ...
```

Entity ID (entity_id) value type Entity ID (entity_id_t) contains an ID for the entity.

Contains an ID for the entity (name) of value type string with 1 or more characters contains the name of the entity.

3.2.2.6.3 Document Property - Involvement - Entity Groups

List of entity groups (entity_groups) of value type array with 1 or more unique items (a set) contains a list of entity groups.

```
<csaf-instance>:
  document:
    # ...
    involvement:
      # ...
      entity_groups: Sequence
    # ...
```

Every Entity group item of value type object with the three mandatory properties Entity Group ID (entity_group_id), List of Entity IDs (entity_ids), and Name of the entity group (name) defines a new logical group of entities that can then be referred to in other parts of the document to address a group of entities with a single identifier. In addition, any Entity group item MAY expose the optional property Summary of the entity group (summary).

```
<csaf-instance>:
  document:
    # ...
    involvement:
      # ...
      entity_groups:
      - # <entity_group-instance>:
        entity_group_id: $defs.entity_group_id_t
        entity_ids: Sequence
        name: String
        summary: String
    # ...
```

Entity Group ID (entity_group_id) value type Entity Group ID (entity_group_id_t) contains an ID for the entity group.

List of Entity IDs (entity_ids) of value type array with 2 or more unique items (a set) lists the Entity IDs of those entities which are known as one group in the document. Every Entity ID item of value type Entity ID (entity_id_t) specifies a single entity being part of the group.

Name of the entity group (name) of value type string with 1 or more characters contains a human-readable name for the entities grouped.

Summary of the entity group (summary) of value type string with 1 or more characters contains a human-readable summary stating the purpose of the group.

3.2.2.7 Document Property - Language

Document language (lang) of value type Language Type (lang_t) identifies the language used by this document, corresponding to IETF BCP 47 / RFC 5646.

3.2.2.8 Document Property - License Expression

License expression (license_expression) of value type string with 1 or more characters contains the SPDX license expression for the CSAF document. It SHALL NOT contain a license text. See Annex B of [SPDX31] for details. The DocumentRef part given in that ABNF SHALL NOT be used in CSAF. Any SPDX license identifier not from the official SPDX license identifier list

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SHALL contain a prefix of the form LicenseRef-<license-inventoring-entity>- where <license-inventoring-entity> is replaced with a unique name for the entity that provided the database this license identifier was found in. Unless otherwise previously established, the unique name SHOULD be a domain name. The same applies for AdditionRef- identifiers.

In addition, the following rules apply:

- License identification:
  1. The SPDX License List Version 3.26.0 or later SHALL be consulted to identify the appropriate SPDX license identifier or construct an expression based on a SPDX license identifier. Deprecated license identifiers SHOULD NOT be used. SPDX license identifiers that were deprecated before the version listed above SHALL NOT be used.

     > The list is available at [SPDX-L]. It includes also the exceptions.

  2. If the appropriate license identifier is not found in the SPDX License List or expression been possible to constructed, the license database AboutCode's "ScanCode LicenseDB" SHALL be consulted as a next step. License identifier from this database SHALL use the prefix LicenseRef-scancode- and addition expressions the prefix AdditionRef-scancode-. The construction of a license expression with such identifiers is also allowed.

     > The database is currently available at [SCANCODE-LICENSEDB].
     >
     > Note that at the time of this writing the AboutCode's "ScanCode LicenseDB" uses LicenseRef-scancode- for exceptions which is the format used in [SPDX231]. For the usage in CSAF, those values have to be converted.

  3. If the first two steps did not result in an appropriate license identifier or expression and no extant identifier was found, the issuing party SHALL create their own license identifier and necessary addition identifiers following the rules above. The identifier SHOULD contain the version number of the license respectively exception. The license text SHALL be made available through exactly one document note using the category legal_disclaimer. The title SHALL be License for English or an unspecified document language. For any other language, it SHOULD be the language specific translation of that term.
- License similarity:
  - Annex C of [SPDX31] applies.
- Case sensitivity: CSAF follows the recommendation of SPDX and enforces case sensitivity for SPDX license expressions in CSAF documents.

Examples 1:

```
CC-BY-4.0
LicenseRef-www.example.org-Example-CSAF-License-3.0
LicenseRef-scancode-public-domain
MIT OR any-OSI
```

When choosing a license expression, CSAF issuing parties SHALL consider to use a permissive license to enable the flow of information through the ecosystem.

> Restrictive licenses can prevent users to easily utilize tools that are hosted in the cloud as the tool providers cannot pre-import CSAF document but users need at least to activate the sources themselves. This is relevant for services and tools, regardless of whether they are commercial or not.

3.2.2.9 Document Property - Notes

Document notes (notes) of value type Notes Type (notes_t) holds notes associated with the whole document.

```
<csaf-instance>:
  document:
    # ...
    notes:  # $defs.notes_t
    - # <note-instance>:
```

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```
    audience: String
    category: String.Enum
    group_ids: $defs.product_groups_t
    product_ids: $defs.products_t
    text: String
    title: String
  # ...
```

The following combinations of category and title have a special meaning and SHALL be used as stated below:

Table 1: Requirements for combinations of category and title that have a special meaning.

category	title	content of text
description	Product Description	Contains a description of a product given in the product_tree in regard to field of application and core functionality. This SHOULD be bound to the corresponding product or product group.
general	AI Usage	Contains information on whether and how AI systems were used in discovering, analyzing, remediating, or documenting the content of the whole CSAF document, including the human oversight applied.
general	General Security Recommendations	Contains general advice and security recommendations that are related, generic and might be applicable independently of the content of the CSAF document.
legal_disclaimer	License	Contains the only license text of the document license.
summary	Summary	Contains a short summary of the content of the advisory.

If a note is specific to a product or product group it SHALL be bound via the group_ids respectively product_ids.

Example 1 (for text *when stating a legal disclaimer):*

```
This content is licensed under the [Creative Commons Attribution 4.0 International License](https://creativecommons.org/licenses/by/4.0/). If you distribute this content, or a modified version of it, you must provide attribution to Example Organization and provide a link to the original.
```

Example 2 (for text *when stating a legal disclaimer):*

```
THIS DOCUMENT IS PROVIDED ON AN \"AS IS\" BASIS AND DOES NOT IMPLY ANY KIND OF GUARANTEE OR WARRANTY, INCLUDING THE WARRANTIES OF MERCHANTABILITY OR FITNESS FOR A PARTICULAR USE. YOUR USE OF THE INFORMATION ON THE DOCUMENT OR MATERIALS LINKED FROM THE DOCUMENT IS AT YOUR OWN RISK. EXAMPLE COMPANY RESERVES THE RIGHT TO CHANGE OR UPDATE THIS DOCUMENT AT ANY TIME.
```

3.2.2.10 Document Property - Publisher

Publisher (publisher) of value type object with the mandatory properties Category (category), Name (name) and Namespace (namespace) provides information on the publishing entity. The two other optional properties are: contact and issuing_authority.

```
<csaf-instance>:
  document:
    # ...
    publisher:
      category: String
      contact: $defs.contact_t
      issuing_authority: String
      name: String
      namespace: String
    # ...
```

3.2.2.10.1 Document Property - Publisher - Category

The Category of publisher (category) of value type string and enum provides information about the category of publisher releasing the document. The valid values are:

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```
coordinator
discoverer
multiplier
other
translator
user
vendor
```

The value coordinator indicates individuals or organizations that manage a single vendor's response or multiple vendors' responses to a vulnerability, a security flaw, or an incident. This includes all Computer Emergency/Incident Response Teams (CERTs/CIRTs) or agents acting on the behalf of a researcher.

The value discoverer indicates individuals or organizations that find vulnerabilities or security weaknesses. This includes all manner of researchers.

The value multiplier indicates individuals or organizations that use existing CSAF documents or information that could be represented in CSAF, and create their own CSAF documents for distribution to a specific target audience. A single multiplier might have target audiences.

> For example, a National CSIRT might create different CSAF documents for the same vulnerability for critical infrastructure companies in different sectors, government agencies, non-critical industry, and the public based on information sharing agreements and threats to the target group.

The creation step can make use of a CSAF modifier that replaces metadata, e.g. the document publisher. Currently, this value includes multipliers, republishers, and forwarders.

The value translator indicates individuals or organizations that translate CSAF documents. This includes all manner of language translators, also those who work for the party issuing the original advisory.

The value other indicates a catchall for everyone else. Currently this includes editors, reviewers, and miscellaneous contributors.

The value user indicates anyone using a vendor's product.

The value vendor indicates developers or maintainers of information system products or services. This includes all authoritative product vendors, product security incident response teams (PSIRTs), open source projects as well as product resellers and distributors, including authoritative vendor partners.

3.2.2.10.2 Document Property - Publisher - Contact

Contact (contact) of value type Contact Type (contact_t) contains information on how to contact the publisher.

```
<csaf-instance>:
  document:
    # ...
    publisher:
      # ...
      contact:
        details: String
        email: String.EMAIL
        public_openpgp_key_url: String.URI
        url: String.URI
      # ...
    # ...
```

3.2.2.10.3 Document Property - Publisher - Issuing Authority

Issuing authority (issuing_authority) of value type string with 1 or more characters Provides information about the authority of the issuing party to release the document, in particular, the party's constituency and responsibilities or other obligations.

Examples 1:

```
Example Company PSIRT is responsible for vulnerability handling across all Example Company products and services.
Example CERT as designated CSIRT according to Directive (EU) 2022/2555 Article 12 section 1 acts as coordinator for the purposes of coordinated vulnerability disclosure.
Example ISAC publishes security information for products used by or relevant for its constituency enriched by a sector-specific assessment.
The Example Organization Security Team is responsible for vulnerability handling across all Example Organization offerings.
```

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3.2.2.10.4 Document Property - Publisher - Name

The Name of publisher (name) of value type string with 1 or more characters contains the name of the issuing party.

Examples 1:

```
BSI
Cisco PSIRT
Siemens ProductCERT
```

3.2.2.10.5 Document Property - Publisher - Namespace

The Namespace of publisher (namespace) of value type string with format uri contains a URL which is under control of the issuing party and can be used as a globally unique identifier for that issuing party. The URL SHALL be normalized.

An issuing party can choose any URL which fulfills the requirements state above. The URL MAY be dereferenceable. If an issuing party has chosen a URL, it SHOULD NOT change. Tools can make use of the combination of $.document.publisher.namespace and $.document.tracking.id as it identifies a CSAF document globally unique.

If an issuing party decides to change its Namespace it SHOULD reissue all CSAF documents with an incremented (patch) version which has no other changes than:

- the new publisher information
- the updated revision history
- the updated item in $.document.references[*] which points to the new version of the CSAF document
- an added item in $.document.references[*] which points to the previous version of the CSAF document (if the URL changed)

Examples 1:

```
https://csaf.io
https://www.example.com
```

3.2.2.11 Document Property - References

Document references (references) of value type References Type (references_t) holds a list of references associated with the whole document.

```
<csaf-instance>:
  document:
    # ...
    references:  # $defs.references_t
    - # <reference-instance>:
      category: String
      summary: String
      url: String.URI
    # ...
```

3.2.2.12 Document Property - Source Language

Source language (source_lang) of value type Language Type (lang_t) identifies if this copy of the document is a translation then the value of this property describes from which language this document was translated.

The property SHALL be present for any CSAF document with the value translator in $.document.publisher.category. The property SHALL NOT be present if the document was not translated.

If an issuing party publishes a CSAF document with the same content in more than one language, one of these documents SHOULD be deemed the “original”, the other ones SHOULD be considered translations from the “original”. The issuing party can retain its original publisher information including the category. However, other rules defined in the conformance clause “CSAF translator” SHOULD be applied.

3.2.2.13 Document Property - Title

Title of this document (title) of value type string with 1 or more characters SHOULD be a canonical name for the document, and sufficiently unique to distinguish it from similar documents.

Examples 1:

```
Cisco IPv6 Crafted Packet Denial of Service Vulnerability
Example Company Cross-Site-Scripting Vulnerability in Example Generator
```

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3.2.2.14 Document Property - Tracking

Tracking (tracking) of value type object with the six mandatory properties Current Release Date (current_release_date), Identifier (id), Initial Release Date (initial_release_date), Revision History (revision_history), Status (status), and Version (version) is a container designated to hold all management attributes necessary to track a CSAF document as a whole. The two optional additional properties are Aliases (aliases) and Generator (generator).

```
<csaf-instance>:
  document:
    # ...
    tracking:
      aliases: Sequence
      current_release_date: String.DateTime
      generator: Mapping
      id: String.Pattern
      initial_release_date: String.DateTime
      revision_history: Sequence
      status: String.Enum
      version: $defs.version_t
```

3.2.2.14.1 Document Property - Tracking - Aliases

Aliases (aliases) of value type array with 1 or more unique items (a set) representing Alternate Names contains a list of alternate names for the same document.

```
<csaf-instance>:
  document:
    # ...
    tracking:
      aliases: Sequence
      # ...
```

Every such Alternate Name of value type string with 1 or more characters specifies a non-empty string that represents a distinct optional alternative ID used to refer to the document.

Example 1:

```
  CVE-2019-12345
```

3.2.2.14.2 Document Property - Tracking - Current Release Date

Current release date (current_release_date) of value type string with format date-time holds the date when the current revision of this document was released.

3.2.2.14.3 Document Property - Tracking - Generator

Document Generator (generator) of value type object with mandatory property Engine (engine) and optional property Date (date) is a container to hold all elements related to the generation of the document. These items will reference when the document was actually created, including the date it was generated and the entity that generated it.

```
<csaf-instance>:
  document:
    tracking:
      # ...
      generator:
        date: String.DateTime
        engine: Mapping
      # ...
```

Date of document generation (date) of value type string with format date-time SHOULD be the current date that the document was generated. Because documents are often generated internally by a document producer and exist for a nonzero amount of time before being released, this field MAY be different from the Initial Release Date and Current Release Date.

Engine of document generation (engine) of value type object with mandatory property Engine name (name) and optional property Engine version (version) contains information about the engine that generated the CSAF document.

```
<csaf-instance>:
  document:
    tracking:
```

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```
# ...
generator:
  # ...
  engine:
    name: String
    version: String
# ...
```

Engine name (name) of value type string with 1 or more characters represents the name of the engine that generated the CSAF document.

Examples 1:

```
Red Hat rhsa-to-cvrf
Secvisogram
TVCE
```

Engine version (version) of value type string with 1 or more characters contains the version of the engine that generated the CSAF document.

> Although it is not formally required, the TC suggests to use a versioning which is compatible with Semantic Versioning as described in the external specification [SemVer]. This could help the end user to identify when CSAF consumers have to be updated.

Examples 2:

```
0.6.0
1.0.0-beta+exp.sha.a1c44f85
2
```

3.2.2.14.4 Document Property - Tracking - ID

Unique identifier for the document (id) has value type string of 1 or more characters with pattern (regular expression):

```
^[\\S](.*[\\S])?$
```

Unique identifier for the document holds the Identifier. It SHALL NOT start or end with a white space and SHALL NOT contain a newline sequence.

The ID is a simple label that provides for a wide range of numbering values, types, and schemes. Its value SHOULD be assigned and maintained by the original document issuing authority. It SHALL be unique for that organization.

Examples 1:

```
Example Company - 2019-YH3234
RHBA-2019:0024
cisco-sa-20190513-secureboot
```

This value is also used to determine the filename for the CSAF document (cf. section 5.1).

> The combination of $.document.publisher.namespace and $.document.tracking.id identifies a CSAF document globally unique. The combination of $.document.publisher.namespace, $.document.tracking.id, and $.document.tracking.version identifies that specific version of the CSAF document globally unique.

3.2.2.14.5 Document Property - Tracking - Initial Release Date

Initial release date (initial_release_date) of value type string with format date-time holds the date when this document was first released to the specified target group.

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> For TLP:CLEAR documents, this is usually the timestamp when the document was published. For TLP:GREEN and higher, this is the timestamp when it was first made available to the specific group. Note that the initial release date does not change after the initial release even if the document is later on released to a broader audience.

If the timestamp of the initial release date was set incorrectly, it SHALL be corrected. This change SHALL be tracked with a new entry in the revision history.

3.2.2.14.6 Document Property - Tracking - Revision History

The Revision History (revision_history) with value type array of 1 or more Revision History Entries holds one revision item for each version of the CSAF document, including the initial one.

```
<csaf-instance>:
  document:
    # ...
    tracking:
      # ...
      revision_history: Sequence
      # ...
```

Each Revision contains all the information elements required to track the evolution of a CSAF document. Revision History Entry items are of value type object with the three mandatory properties: Date (date), Number (number), and Summary (summary). In addition, a Revision MAY expose the optional property legacy_version.

```
<csaf-instance>:
  document:
    # ...
    tracking:
      # ...
      revision_history:
      - # <revision-instance>:
        date: String.DateTime
        legacy_version: String
        number: $defs.version_t
        summary: String
      # ...
```

The Date of the revision (date) of value type string with format date-time states the date of the revision entry.

Legacy version of the revision (legacy_version) of value type string with 1 or more characters contains the version string used in an existing document with the same content.

> This SHOULD be used to aid in the mapping between existing (human-readable) documents which might use a different version scheme and CSAF documents with the same content. It is recommended, to use the CSAF revision number to describe the revision history for any new human-readable equivalent.

The Number (number) has value type Version (version_t).

The Summary of the revision (summary) of value type string with 1 or more characters holds a single non-empty string representing a short description of the changes.

Each Revision item which has a number of 0 or 0.y.z SHALL be removed from the document if the document status is final. Versions of the document which are pre-release SHALL NOT have its own revision item. All changes SHALL be tracked in the item for the next release version. Build metadata SHOULD NOT be included in the number of any revision item.

Any issuing party using a CSAF document as basis and modifying it SHALL create a new revision history tracking the modified document. This applies especially to CSAF multiplier.

> The new revision history (and consequently the corresponding ensures current_release_date) that the CSAF document can be found and downloaded by CSAF downloaders that retrieve just those CSAF documents that have a newer current_release_date than the timestamp of their last visit at this source (incremental download).

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3.2.2.14.7 Document Property - Tracking - Status

Document status (status) of value type string and enum defines the draft status of the document. The value SHALL be one of the following:

```
draft
final
interim
```

The value draft indicates, that this is a pre-release, intended for issuing party's internal use only, or possibly used externally when the party is seeking feedback or indicating its intentions regarding a specific issue.

The value final indicates, that the issuing party asserts the content is unlikely to change. "Final" status is an indication only, and does not preclude updates. This SHOULD be used if the issuing party expects no, slow or few changes.

The value interim indicates, that the issuing party expects rapid updates. This SHOULD be used if the expected rate of release for this document is significant higher than for other documents. Once the rate slows down it SHALL be changed to final. This MAY be done in a patch version.

This is extremely useful for downstream vendors to constantly inform the end users about ongoing investigation. It can be used as an indication to pull the CSAF document more frequently.

3.2.2.14.8 Document Property - Tracking - Version

Version has the value type Version (version_t).

3.2.2.15 Document Property - Extensions

Document-level Extensions (x_extensions) of value type Extensions Type (extensions_t) contains list of extensions valid at the document property level of the CSAF document and associated with this document metadata.

```
<csaf-instance>:
  document:
    # ...
    x_extensions:  # $defs.extensions_t
    - # <x_extension-instance>:
      $schema: String
      category: String.Enum
      content: Mapping
      critical: Boolean
      # ...
```

3.2.3 Product Tree Property

Product Tree (product_tree) has value type object with 1 or more properties is a container for all fully qualified product names that can be referenced elsewhere in the document. The properties are Branches (branches), Full Product Names (full_product_names), Product Groups (product_groups), and Product Paths (product_paths).

```
<csaf-instance>:
  # ...
  product_tree:
    branches: $defs.branches_t
    full_product_names: Mapping
    product_groups: Sequence
    product_paths: Sequence
```

3.2.3.1 Product Tree Property - Branches

List of branches (branches) has the value type branches_t.

3.2.3.2 Product Tree Property - Full Product Names

List of full product names (full_product_names) of value type array with 1 or more items of type full_product_name_t contains a list of full product names.

3.2.3.3 Product Tree Property - Product Groups

List of product groups (product_groups) of value type array with 1 or more items of value type object contains a list of product groups.

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```
<csaf-instance>:
  # ...
  product_tree:
    # ...
    product_groups: Sequence
  # ...
```

The product group items are of value type object with the two mandatory properties Group ID (group_id) and Product IDs (product_ids) and the optional Summary (summary) property.

```
<csaf-instance>:
  # ...
  product_tree:
    # ...
    product_groups:
    - # <product_group-instance>:
      group_id: $defs.product_group_id_t
      product_ids: $defs.product_id_t
      summary: String
    # ...
  # ...
```

The summary of the product group (summary) of value type string with 1 or more characters gives a short, optional description of the group.

Examples 1:

```
Products supporting Modbus.
The x64 versions of the operating system.
```

Group ID (group_id) has value type Product Group ID (product_group_id_t). Its value SHALL be unique across all Product Group items within the document.

List of Product IDs (product_ids) of value type array with 2 or more unique items of value type Product ID (product_id_t) lists the product_ids of those products which are known as one group in the document.

3.2.3.4 Product Tree Property - Product Paths

List of product paths (product_paths) of value type array with 1 or more items contains a list of product paths.

```
<csaf-instance>:
  # ...
  product_tree:
    # ...
    product_paths: Sequence
  # ...
```

The Product path item is of value type object and has three mandatory properties: Beginning product reference (beginning_product_reference), Full Product Name (full_product_name), and Subpaths (subpaths). The Product path item establishes a path along existing full_product_name_t elements, allowing the document producer to define a path of multiple products that form a new full_product_name entry.

```
<csaf-instance>:
  # ...
  product_tree:
    # ...
    product_paths:
    - # <product_path-instance>:
      beginning_product_reference: $defs.product_id_t
      full_product_name: $defs.full_product_name_t
      subpaths: Sequence
    # ...
  # ...
```

The situation where a need for declaring a Product path arises, is given when a product is e.g. vulnerable only when installed together with another, or to describe operating system components.

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Beginning product reference (beginning_product_reference) of value type Product ID (product_id_t) holds a Product ID that refers to the Full Product Name element, which is the beginning node of the product path. It is also the product, the first node in subpaths links to.

> The product referenced in beginning_product_reference is usually the product that is more closely identified by the product path.

Full Product Name (full_product_name) of value type Full Product Name Type (full_product_name_t).

List of product subpaths (subpaths) of value type array with 1 or more items of type Subpath (subpath_t) contains an ordered list of product subpaths, each one relating to the path defined by all previous elements up to the beginning node of the product path.

```
<csaf-instance>:
  # ...
  product_tree:
    # ...
    product_paths:
    - # <product_path-instance>:
      # ...
      subpaths:
      - # <subpath-instance>: $defs.subpath_t
        category: String.Enum
        next_product_reference: $defs.product_id_t
      # ...
    # ...
  # ...
```

Example 1:

```
"product_tree": {
  "full_product_names": [
    {
      "product_id": "CSAFPID-908070601",
      "name": "Cisco AnyConnect Secure Mobility Client 4.9.04053"
    },
    {
      "product_id": "CSAFPID-908070602",
      "name": "Microsoft Windows"
    }
  ],
  "product_paths": [
    {
      "beginning_product_reference": "CSAFPID-908070601",
      "full_product_name": {
        "product_id": "CSAFPID-908070603",
        "name": "Cisco AnyConnect Secure Mobility Client 4.9.04053 installed on Microsoft Windows"
      },
      "subpaths": [
        {
          "category": "installed_on",
          "next_product_reference": "CSAFPID-908070602",
        }
      ]
    }
  ]
}
```

> The product Cisco AnyConnect Secure Mobility Client 4.9.04053" (Product ID: CSAFPID-908070601) and the product Microsoft Windows (Product ID: CSAFPID-908070602) form together a new product with the separate Product ID CSAFPID-908070603. The latter one can be used to refer to that combination in other parts of the CSAF document. In the preceding example, it might be the case that Cisco AnyConnect Secure Mobility Client 4.9.04053" is only vulnerable when installed on Microsoft Windows.

3.2.4 Vulnerabilities Property

Vulnerabilities (vulnerabilities) of value type array with 1 or more objects representing vulnerabilities by providing one or more properties represents a list of all relevant vulnerability information items.

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```
<csaf-instance>:
  # ...
  vulnerabilities: Sequence
```

The Vulnerability item of value type object with 1 or more properties is a container for the aggregation of all fields that are related to a single vulnerability in the document. Any vulnerability MAY provide the optional properties Acknowledgments (acknowledgments), Common Vulnerabilities and Exposures (CVE) (cve), Common Weakness Enumeration (CWE) (cwes), Disclosure Date (disclosure_date), Discovery Date (discovery_date), Document-local Vuln ID (dl_vuln_id), List of first known exploitation dates (first_known_exploitation_dates), Flags (flags), IDs (ids), Metrics (metrics), Notes (notes), Product Status (product_status), References (references), Remediations (remediations), Threats (threats), Title (title), and Vulnerability-level Extensions (x_extensions).

```
<csaf-instance>:
  # ...
  vulnerabilities:
  - # <vulnerability-instance>:
    acknowledgments: $defs.acknowledgments_t
    cve: String.Pattern
    cwes: Sequence
    disclosure_date: String.DateTime
    discovery_date: String.DateTime
    dl_vuln_id: $defs.dl_vuln_id_t
    first_known_exploitation_dates: Sequence
    flags: Sequence
    ids: Sequence
    metrics: Sequence
    notes: $defs.notes_t
    product_status: Mapping
    references: $defs.references_t
    remediations: Sequence
    threats: Sequence
    title: String
    x_extensions: $defs.extensions_t
    # ...
```

3.2.4.1 Vulnerabilities Property - Acknowledgments

Vulnerability acknowledgments (acknowledgments) of value type Acknowledgments Type (acknowledgments_t) contains a list of acknowledgment elements associated with this vulnerability item.

```
<csaf-instance>:
  # ...
  vulnerabilities:
  - # <vulnerability-instance>:
    acknowledgments:  # $defs.acknowledgments_t
    - # <acknowledgment-instance>:
      names: Sequence
      organization: String
      summary: String
      urls: Sequence
      # ...
    # ...
```

3.2.4.2 Vulnerabilities Property - CVE

CVE (cve) has value type string with pattern (regular expression):

```
^CVE-[0-9]{4}-[0-9]{4,}$
```

CVE holds the MITRE standard Common Vulnerabilities and Exposures (CVE) tracking number for the vulnerability.

3.2.4.3 Vulnerabilities Property - CWEs

List of CWEs (cwes) of value type array with 1 or more unique items (a set) of value type object contains a list of CWEs.

```
<csaf-instance>:
  # ...
  vulnerabilities:
  - # <vulnerability-instance>:
    # ...
    cwes: Sequence
    # ...
```

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If more than one CWE is specified, the most applicable weakness ID SHOULD be listed first. A CSAF viewer MAY decide to display just the first CWE item.

Every CWE item of value type object with the three mandatory properties Weakness ID (id), Weakness Name (name), CWE version (version) holds the MITRE standard Common Weakness Enumeration (CWE) for the weakness associated. For more information cf. [CWE].

```
<csaf-instance>:
  # ...
  vulnerabilities:
  - # <vulnerability-instance>:
    # ...
    cwes:
    - # <cwe-instance>:
      id: String.Pattern
      name: String.Pattern
      version: String.Pattern
    # ...
```

The Weakness ID (id) has value type string with pattern (regular expression):

```
^CWE-[1-9]\\d{0,5}$
```

It holds the ID for the weakness associated.

Examples 1:

```
CWE-22
CWE-352
CWE-79
```

The Weakness name (name) has value type string of 1 or more characters with pattern (regular expression):

```
^[^\\s\\-_\\.](.*[^\\s\\-_\\.])?$
```

The Weakness name holds the full name of the weakness as given in the CWE specification.

Examples 2:

```
Cross-Site Request Forgery (CSRF)
Improper Limitation of a Pathname to a Restricted Directory ('Path Traversal')
Improper Neutralization of Input During Web Page Generation ('Cross-site Scripting')
```

The CWE version (version) has value type string with pattern (regular expression):

```
^[1-9]\\d*\\.([0-9]|([1-9]\\d+))(\\.([0-9]|([1-9]\\d+)))?$
```

It holds the version string of the CWE specification this weakness was extracted from. When creating or modifying a CSAF document, the latest published version of the CWE specification SHOULD be used.

Examples 3:

```
"1.0",
"3.4.1",
"4.0",
"4.11",
"4.12"
```

3.2.4.4 Vulnerabilities Property - Disclosure Date

Disclosure date (disclosure_date) of value type string with format date-time holds the date and time the vulnerability was originally disclosed to the public.

For vulnerabilities not yet disclosed to the public, a disclosure date in the future SHOULD indicate the intended date for disclosure of the vulnerability. This is also sometimes called embargo date. As disclosure dates may change during a vulnerability disclosure process, an issuing party SHOULD produce an updated CSAF document to confirm that the

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vulnerability was in fact disclosed to the public at that time or update the disclosure_date with the new intended date in the future.

If a vulnerability was patched without stating that it is a vulnerability, the vulnerability is in the context of this specification not considered to be disclosed. As a consequence, the discovery_date needs to be earlier than or equal to the disclosure_date.

3.2.4.5 Vulnerabilities Property - Discovery Date

Discovery date (discovery_date) of value type string with format date-time holds the date and time the vulnerability was originally discovered.

3.2.4.6 Vulnerability Property - Document-local Vuln ID

Document-local Vuln ID (dl_vuln_id) of value type Document-local Vuln ID (dl_vuln_id_t) contains the local ID referring to this vulnerability from another part of the same document. Its value SHALL be unique across all vulnerability items within the document.

3.2.4.7 Vulnerabilities Property - First Known Exploitation Dates

List of first known exploitation dates (first_known_exploitation_dates) of value type array with 1 or more unique items (a set) contains a list of dates of first known exploitations.

```
<csaf-instance>:
  # ...
  vulnerabilities:
  - # <vulnerability-instance>:
    # ...
    first_known_exploitation_dates: Sequence
    # ...
```

Every First known exploitation date item of value type object with the two mandatory properties Date of the information (date) and Date of the exploitation (exploitation_date) holds at least 3 properties and contains information on when this vulnerability was first known to be exploited in the wild in the products specified. At least one of the optional elements Group IDs (group_ids) and Product IDs (product_ids) SHALL be present to state for which products or product groups this date is applicable.

> This information can be helpful to determine the risk of compromise. It can also be used to provide an indication for the time frame to be considered in a threat hunt for the exploitation this vulnerability.

```
<csaf-instance>:
  # ...
  vulnerabilities:
  - # <vulnerability-instance>:
    # ...
    first_known_exploitation_dates:
    - # <event-instance>:
      date: String.DateTime
      exploitation_date: String.DateTime
      group_ids: $defs.product_groups_t
      product_ids: $defs.products_t
    # ...
```

Date of the information (date) of value type string with format date-time contains the date when the information was last updated.

Date of the exploitation (exploitation_date) of value type string with format date-time contains the date when the exploitation happened.

> Different document issuers might have different knowledge about exploitations in the wild that happened. Therefore, the exploitation_date can differ.

Group IDs (group_ids) are of value type Product Groups (product_groups_t) and contain a list of Product Groups the current first known exploitation date item applies to.

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Product IDs (product_ids) are of value type Products (products_t) and contain a list of Products the current first known exploitation date item applies to.

3.2.4.8 Vulnerabilities Property - Flags

List of flags (flags) of value type array with 1 or more unique items (a set) of value type object contains a list of machine readable flags.

```
<csaf-instance>:
  # ...
  vulnerabilities:
  - # <vulnerability-instance>:
    # ...
    flags: Sequence
    # ...
```

Every Flag item of value type object with the mandatory property Label (label) contains product specific information in regard to this vulnerability as a single machine readable flag. For example, this could be a machine readable justification code why a product is not affected. At least one of the optional elements Group IDs (group_ids) and Product IDs (product_ids) SHALL be present to state for which products or product groups this flag is applicable.

> These flags enable the receiving party to automate the selection of actions to take.

In addition, any Flag item MAY provide the three optional properties Date (date), Group IDs (group_ids) and Product IDs (product_ids).

```
<csaf-instance>:
  # ...
  vulnerabilities:
  - # <vulnerability-instance>:
    # ...
    flags:
    - # <flag-instance>:
      date: String.DateTime
      group_ids: $defs.product_groups_t
      label: String.Enum
      product_ids: $defs.products_t
    # ...
```

Date of the flag (date) of value type string with format date-time contains the date when assessment was done or the flag was assigned.

Group IDs (group_ids) are of value type Product Groups (product_groups_t) and contain a list of Product Groups the current flag item applies to.

Label of the flag (label) of value type string and enum specifies the machine readable label. Valid enum values are:

```
component_not_present
inline_mitigations_already_exist
vulnerable_code_cannot_be_controlled_by_adversary
vulnerable_code_not_in_execute_path
vulnerable_code_not_present
```

The given values reflect the VEX not affected justifications. See [VEX-Justification] for more details. The values SHALL be used as follows:

- component_not_present: The software is not affected because the vulnerable component is not in the product.
- vulnerable_code_not_present: The product is not affected because the code underlying the vulnerability is not present in the product.

> Unlike component_not_present, the component in question is present, but for whatever reason (e.g. compiler options) the specific code causing the vulnerability is not present in the component.

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- vulnerable_code_cannot_be_controlled_by_adversary: The vulnerable component is present, and the component contains the vulnerable code. However, vulnerable code is used in such a way that an attacker cannot mount any anticipated attack.
- vulnerable_code_not_in_execute_path: The affected code is not reachable through the execution of the code, including non-anticipated states of the product.

  Components that are neither used nor executed by the product.

- inline_mitigations_already_exist: Built-in inline controls or mitigations prevent an adversary from leveraging the vulnerability.

Product IDs (product_ids) are of value type Products (products_t) and contain a list of Products the current flag item applies to.

3.2.4.9 Vulnerabilities Property - IDs

List of IDs (ids) of value type array with 1 or more unique ID items of value type object represents a list of unique labels or tracking IDs for the vulnerability (if such information exists).

```
<csaf-instance>:
  # ...
  vulnerabilities:
  - # <vulnerability-instance>:
    # ...
    ids: Sequence
    # ...
```

Every ID item of value type object with the two mandatory properties System Name (system_name) and Text (text) contains a single unique label or tracking ID for the vulnerability. In addition, any ID item MAY expose the optional properties Group IDs (group_ids) and Product IDs (product_ids).

```
<csaf-instance>:
  # ...
  vulnerabilities:
  - # <vulnerability-instance>:
    # ...
    ids:
    - # <id-instance>:
      group_ids: $defs.product_groups_t
      product_ids: $defs.products_t
      system_name: String
      text: String
    # ...
```

Group IDs (group_ids) are of value type Product Groups (product_groups_t) and contain a list of Product Groups the current ID item applies to.

This can be used to provide the information that this specific ID applies only to a specific set of product groups.

Product IDs (product_ids) are of value type Products (products_t) and contain a list of Products the current ID item applies to.

This can be used to provide the information that this specific ID applies only to a specific set of products.

System name (system_name) of value type string with 1 or more characters indicates the name of the vulnerability tracking or numbering system.

Examples 1:

```
Cisco Bug ID
GitHub Issue
https://github.com/oasis-tcs/csaf
```

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Text (text) of value type string with 1 or more characters is unique label or tracking ID for the vulnerability (if such information exists).

Examples 2:

```
CSCso66472
oasis-tcs/csaf#210
#1217
```

General examples may include an identifier from a vulnerability tracking system that is available to customers, such as:

- a Cisco bug ID,
- a GitHub Issue number,
- an ID from a Bugzilla system, or
- an ID from a public vulnerability database such as the X-Force Database.

The list of registered vulnerability id systems is available via [RVISC].

The ID MAY be a vendor-specific value but is not to be used to publish the CVE tracking numbers (MITRE standard Common Vulnerabilities and Exposures), as these are specified inside the dedicated CVE element.

3.2.4.10 Vulnerabilities Property - Metrics

List of metrics (metrics) of value type array with 1 or more unique items (a set) contains metric objects for the current vulnerability.

```
<csaf-instance>:
  # ...
  vulnerabilities:
  - # <vulnerability-instance>:
    # ...
    metrics: Sequence
    # ...
```

Every Metric item of value type object with the mandatory properties content and products and the optional property source contains all metadata about the metric including products it applies to and the source and the content itself.

```
<csaf-instance>:
  # ...
  vulnerabilities:
  - # <vulnerability-instance>:
    # ...
    metrics:
    - # <metric-instance>:
      content: Mapping
      products: $defs.products_t
      source: String.URI
    # ...
```

3.2.4.10.1 Vulnerabilities Property - Metrics - Content

Content (content) of value type object with the optional properties CVSS v2 (cvss_v2), CVSS v3 (cvss_v3), CVSS v4 (cvss_v4), EPSS (epss), Qualitative Severity Rating (qualitative_severity_rating), SSVC v2 (ssvc_v2), and Metrics-content-level Extensions (x_extensions) specifies information about (at least one) metric or score for the given products regarding the current vulnerability. A Content object has at least 1 property.

```
# &CERTCC-SSVC https://certcc.github.io/SSVC/data/schema/v2/
# &FIRST-CVSS https://www.first.org/cvss/
<csaf-instance>:
  # ...
  vulnerabilities:
  - # <vulnerability-instance>:
    # ...
    metrics:
    - # <metric-instance>:
```

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```
    content:
      cvss_v2: $ref.eval(concat( *FIRST-CVSS 'cvss-v2.0.json' ))
      cvss_v3:
        # !OneOf<
        - $ref.eval(concat( *FIRST-CVSS 'cvss-v3.0.json' ))
        - $ref.eval(concat( *FIRST-CVSS 'cvss-v3.1.json' ))
        # >
      cvss_v4: $ref.eval(concat( *FIRST-CVSS 'cvss-v4.0.json' ))
      epss: Mapping
      qualitative_severity_rating: String.Enum
      ssvc_v2:
        $ref.eval(concat(
          *CERTCC-SSVC 'Decision_Point_Value_Selection-2-0-0.schema.json'
        ))
      x_extensions: $defs.extensions_t
    # ...
  # ...
```

The property CVSS v2 (cvss_v2) holding a CVSS v2.0 value abiding by the schema at https://www.first.org/cvss/cvss-v2.0.json. See [CVSS2] for details.

The property CVSS v3 (cvss_v3) holding a CVSS v3.x value abiding by one of the schemas at https://www.first.org/cvss/cvss-v3.0.json or https://www.first.org/cvss/cvss-v3.1.json. See [CVSS30] respectively [CVSS31] for details.

The property CVSS v4 (cvss_v4) holding a CVSS v4.0 value abiding by the schema at https://www.first.org/cvss/cvss-v4.0.json. See [CVSS40] for details.

The property EPSS (epss) of value type object with the three mandatory properties Percentile (percentile), Probability (probability) and EPSS timestamp (timestamp) contains the EPSS data. See [EPSS] for details.

```
<csaf-instance>:
  # ...
  vulnerabilities:
  - # <vulnerability-instance>:
    # ...
    metrics:
    - # <metric-instance>:
      content:
        # ...
        epss:
          percentile: String.Pattern
          probability: String.Pattern
          timestamp: String.DateTime
        # ...
      # ...
    # ...
```

Percentile (percentile) has value type string with pattern (regular expression):

```
^(([0]\\.([0-9])+)|([1]\\.[0]+))$
```

The value contains the rank ordering of probabilities from highest to lowest.

Probability (probability) has value type string with pattern (regular expression):

```
^(([0]\\.([0-9])+)|([1]\\.[0]+))$
```

The value contains the likelihood that any exploitation activity for this Vulnerability is being observed in the 30 days following the given timestamp.

EPSS timestamp (timestamp) of value type string with format date-time holds the date and time the EPSS value was recorded.

The property Qualitative Severity Rating (qualitative_severity_rating) of value type string and enum contains an assessment of the severity of the vulnerability regarding the products on a qualitative scale. Valid enum values are:

```
critical
high
low
medium
none
```

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The value critical indicates that this vulnerability allows attackers to fully compromise a system or access sensitive data with little or no user interaction.

The value high indicates that this vulnerability can lead to significant impact such as unauthorized access or data loss, but usually require some conditions or user interaction.

The value low indicates that this vulnerability has a minimal impact and low likelihood of exploitation, usually causing minor issues or informational leaks.

The value medium indicates that this vulnerability can result in moderate impact like partial data exposure or denial of service, often needing specific circumstances.

The value none indicates that this flaw pose no security risk or impact like false positives or informational findings without real threat.

> The Qualitative Severity Rating is not a replacement for CVSS. It is intended to be used for sources that provide an assessment on a qualitative scale but no full CVSS vector.

CSAF consumer SHOULD give preference to CVSS if both, Qualitative Severity Rating and CVSS, are present. Issuing parties SHOULD consider using the SSVC decision point Provider Urgency from the cvss namespace to convey an additional assessment provided by a party.

The property SSVC v2 (ssvc_v2) holding an SSVC Selection List v2.0.0 value abiding by the schema at https://certcc.github.io/SSVC/data/schema/v2/SelectionList_2_0_0.schema.json. See [SSVC] for details.

The property Metrics-content-level Extensions (x_extensions) of value type Extensions Type (extensions_t) contains a list of extensions valid at the metrics-content-level of the CSAF document and associated with this metric element.

```
<csaf-instance>:
  # ...
  vulnerabilities:
  - # <vulnerability-instance>:
    # ...
    metrics:
    - # <metric-instance>:
      content:
        # ...
        x_extensions:  # $defs.extensions_t
        - # <x_extension-instance>:
          $schema: String
          category: String.Enum
          content: Mapping
          critical: Boolean
          # ...
      # ...
    # ...
  # ...
```

> This extension point can be used for metrics that are not supported in the CSAF standard (yet). For new versions of an official supported standard, the OASIS CSAF TC will consider providing an official extension to ensure interoperability.

3.2.4.10.2 Vulnerabilities Property - Metrics - Products

Product IDs (products) of value type products_t with 1 or more items indicates for which products the given content applies. A metric object SHOULD reflect the associated product's status (for example, a fixed product no longer contains a vulnerability and should have a CVSS score of 0, or simply no score listed; the known affected versions of that product can list the vulnerability score as it applies to them).

3.2.4.10.3 Vulnerabilities Property - Metrics - Source

Source (source) of value type string with format uri contains the URL of the source that originally determined the metric. If no source is given, then the metric was assigned by the document author.

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For example, this could point to the vendor advisory, discoverer blog post, a multiplier's assessment or other sources that provide metric information.

3.2.4.11 Vulnerabilities Property - Notes

Vulnerability notes (notes) of value type Notes Type (notes_t) holds notes associated with this vulnerability item.

```
<csaf-instance>:
  # ...
  vulnerabilities:
  - # <vulnerability-instance>:
    # ...
    notes:  # $defs.notes_t
    - # <note-instance>:
      audience: String
      category: String.Enum
      group_ids: $defs.product_groups_t
      product_ids: $defs.products_t
      text: String
      title: String
    # ...
  # ...
```

The following combinations of category and title have a special meaning and SHALL be used as stated below:

Table 2: Combinations of category and title with special meaning.

category	title	content of text
description	CVE Description	Contains the official and unchanged CVE description for this specific vulnerability.
description	Preconditions	Contains a description of the preconditions that have to be fulfilled to be able to exploit this specific vulnerability, e.g. user account or physical access.
general	AI Usage	Contains a information on whether and how AI systems were used in discovering, analyzing, remediating, or documenting this specific vulnerability, including the human oversight applied.
summary	Vulnerability Summary	Contains a summary of this specific vulnerability which is not the official CVE description.

If a note is specific to a product or product group it SHALL be bound via the group_ids respectively product_ids.

3.2.4.12 Vulnerabilities Property - Product Status

Product status (product_status) of value type object with 1 or more properties contains different lists of product_ids which provide details on the status of the referenced product related to the current vulnerability. The nine defined properties are First affected (first_affected), First fixed (first_fixed), Fixed (fixed), Known affected (known_affected), Known not affected (known_not_affected), Last affected (last_affected), Recommended (recommended), Under investigation (under_investigation) and Unknown (unknown) are all of value type Products (products_t).

```
<csaf-instance>:
  # ...
  vulnerabilities:
  - # <vulnerability-instance>:
    # ...
    product_status:
      first_affected: $defs.products_t
      first_fixed: $defs.products_t
      fixed: $defs.products_t
      known_affected: $defs.products_t
      known_not_affected: $defs.products_t
      last_affected: $defs.products_t
      recommended: $defs.products_t
      under_investigation: $defs.products_t
      unknown: $defs.products_t
    # ...
```

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First affected (first_affected) of value type Products (products_t) represents that these are the first versions of the releases known to be affected by the vulnerability.

First fixed (first_fixed) of value type Products (products_t) represents that these versions contain the first fix for the vulnerability but may not be the recommended fixed versions.

Fixed (fixed) of value type Products (products_t) represents that these versions contain a fix for the vulnerability but may not be the recommended fixed versions.

Known affected (known_affected) of value type Products (products_t) represents that these versions are known to be affected by the vulnerability. Actions are recommended to remediate or address this vulnerability.

> This could include for instance learning more about the vulnerability and context, and/or making a risk-based decision to patch or apply defense-in-depth measures. See $.vulnerabilities[*].remediations, $.vulnerabilities[*].notes and $.vulnerabilities[*].threats for more details.

Known not affected (known_not_affected) of value type Products (products_t) represents that these versions are known not to be affected by the vulnerability. No remediation is required regarding this vulnerability.

> This could for instance be because the code referenced in the vulnerability is not present, not exposed, compensating controls exist, or other factors. See $.vulnerabilities[*].flags and $.vulnerabilities[*].threats in category impact for more details.

Last affected (last_affected) of value type Products (products_t) represents that these are the last versions in a release train known to be affected by the vulnerability. Subsequently released versions would contain a fix for the vulnerability.

Recommended (recommended) of value type Products (products_t) represents that these versions have a fix for the vulnerability and are the vendor-recommended versions for fixing the vulnerability.

Under investigation (under_investigation) of value type Products (products_t) represents that it is not known yet whether these versions are or are not affected by the vulnerability. However, it is still under investigation - the result will be provided in a later release of the document.

Unknown (unknown) of value type Products (products_t) represents that it is not known whether these versions are or are not affected by the vulnerability. There is also no investigation and therefore the status might never be determined.

The individual properties form the following product status groups:

- Affected:

```
$.vulnerabilities[*].product_status.first_affected[*]
$.vulnerabilities[*].product_status.known_affected[*]
$.vulnerabilities[*].product_status.last_affected[*]
```

- Not affected:

```
$.vulnerabilities[*].product_status.known_not_affected[*]
```

- Fixed:

```
$.vulnerabilities[*].product_status.first_fixed[*]
$.vulnerabilities[*].product_status.fixed[*]
```

- Under investigation:

```
$.vulnerabilities[*].product_status.under_investigation[*]
```

- Unknown:

```
$.vulnerabilities[*].product_status.unknown[*]
```

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As the aforementioned product status groups contradict each other, the sets formed by the contradicting groups within one vulnerability item SHALL be pairwise disjoint.

> Note: An issuer might recommend ($.vulnerabilities[*].product_status.recommended) a product version from any group - also from the affected group, i.e. if it was discovered that fixed versions introduce a more severe vulnerability.

3.2.4.13 Vulnerabilities Property - References

Vulnerability references (references) of value type References Type (references_t) holds a list of references associated with this vulnerability item.

```
<csaf-instance>:
  # ...
  vulnerabilities:
  - # <vulnerability-instance>:
    # ...
    references:  # $defs.references_t
    - # <reference-instance>:
      category: String.Enum
      summary: String
      url: String.URI
    # ...
  # ...
```

3.2.4.14 Vulnerabilities Property - Remediations

List of remediations (remediations) of value type array with 1 or more Remediation items contains a list of remediations.

```
<csaf-instance>:
  # ...
  vulnerabilities:
  - # <vulnerability-instance>:
    # ...
    remediations: Sequence
    # ...
```

Every Remediation item of value type object with the two mandatory properties Category (category) and Details (details) specifies details on how to handle (and presumably, fix) a vulnerability. At least one of the optional elements Group IDs (group_ids) and Product IDs (product_ids) SHALL be present to state for which products or product groups this remediation is applicable.

In addition, any Remediation MAY expose the six optional properties Date (date), Entitlements (entitlements), Group IDs (group_ids), Product IDs (product_ids), Restart required (restart_required), and URL (url).

```
<csaf-instance>:
  # ...
  vulnerabilities:
  - # <vulnerability-instance>:
    # ...
    remediations:
    - # <remediation-instance>:
      category: String.Enum
      date: String.DateTime
      details: String
      entitlements: Sequence
      group_ids: $defs.product_groups_t
      product_ids: $defs.products_t
      restart_required: Mapping
      url: String.URI
    # ...
```

3.2.4.14.1 Vulnerabilities Property - Remediations - Category

Category of the remediation (category) of value type string and enum specifies the category which this remediation belongs to. Valid values are:

```
fix_planned
mitigation
no_fix_planned
none_available
```

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```
optional_patch
vendor_fix
workaround
```

The value workaround indicates that the remediation contains information about a configuration or specific deployment scenario that can be used to avoid exposure to the vulnerability. There MAY be none, one, or more workarounds available. This is typically the "first line of defense" against a new vulnerability before a mitigation or vendor fix has been issued or even discovered.

The value mitigation indicates that the remediation contains information about a configuration or deployment scenario that helps to reduce the risk of the vulnerability but that does not resolve the vulnerability on the affected product. Mitigations MAY include using devices or access controls external to the affected product. Mitigations MAY or MAY NOT be issued by the original author of the affected product, and they MAY or MAY NOT be officially sanctioned by the document producer.

The value vendor_fix indicates that the remediation contains information about an official fix that is issued by the original author of the affected product. Unless otherwise noted, it is assumed that this fix fully resolves the vulnerability.

The value optional_patch indicates that the remediation contains information about an patch that is issued by the original author of the affected product. Its application is not necessary, but might be desired by the user, e.g. to calm a security scanner by updating a dependency to a fixed version even though the dependency in the affected version was used in the product in a way that the product itself was not affected. Unless otherwise noted, it is assumed that this does not change the state regarding the vulnerability.

This is sometimes also referred to as a "regulatory compliance patch".

The value none_available indicates that there is currently no fix or other remediation available. The text in field details SHOULD contain details about why there is no fix or other remediation.

The value fix_planned indicates that there is a fix for the vulnerability planned but not yet ready. An issuing party might choose to use this category to announce that a fix is currently developed. The text in field details SHOULD contain details including a date when a customer can expect the fix to be ready and distributed.

The value no_fix_planned indicates that there is no fix for the vulnerability and it is not planned to provide one at any time. This is often the case when a product has been orphaned, declared end-of-life, or otherwise deprecated. The text in field details SHOULD contain details about why there will be no fix issued.

Some category values contradict each other and thus are mutually exclusive per product. Therefore, such a combination SHALL NOT be used in the list of remediations for the same product. This is independent from whether the product is referenced directly or indirectly through a product group. The following tables shows the allowed and prohibited combinations:

Table 3: Remediation Combinations

category value	workaround	mitigation	vendor_fix	optional_patch	none_available	fix_planned	no_fix_planned
workaround	allowed	allowed	allowed	prohibited	prohibited	allowed	allowed
mitigation	allowed	allowed	allowed	prohibited	prohibited	allowed	allowed
vendor_fix	allowed	allowed	allowed	prohibited	prohibited	prohibited	prohibited
optional_patch	prohibited	prohibited	prohibited	allowed	prohibited	prohibited	prohibited
none_available	prohibited	prohibited	prohibited	prohibited	allowed	prohibited	prohibited
fix_planned	allowed	allowed	prohibited	prohibited	prohibited	allowed	prohibited
no_fix_planned	allowed	allowed	prohibited	prohibited	prohibited	prohibited	allowed

Some category values contradict certain product status groups. Therefore, such a combination SHALL NOT exist in a vulnerability item for the same product. This is independent from whether the product is referenced directly or indirectly through a product group. The following tables shows the allowed, discouraged and prohibited combinations:

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Table 4: Product Status Remediation Category Combinations

category value	Affected	Not Affected	Fixed	Under Investigation	Unknown	Recommended
workaround	allowed	prohibited	prohibited	discouraged	discouraged	allowed
mitigation	allowed	prohibited	prohibited	discouraged	discouraged	allowed
vendor_fix	allowed	prohibited	prohibited	discouraged	discouraged	allowed
optional_patch	prohibited	allowed	discouraged	allowed	allowed	allowed
none_available	allowed	prohibited	prohibited	allowed	allowed	allowed
fix_planned	allowed	discouraged	prohibited	discouraged	discouraged	allowed
no_fix_planned	allowed	discouraged	prohibited	allowed	allowed	allowed

The following preference for combinations of remediation categories and product status groups is RECOMMENDED:

1. vendor_fix and Recommended
2. mitigation and Recommended
3. workaround and Recommended
4. optional_patch and Recommended
5. vendor_fix and Affected
6. mitigation and Affected
7. workaround and Affected
8. optional_patch and Under Investigation
9. optional_patch and Unknown
10. fix_planned and Recommended
11. fix_planned and Affected
12. optional_patch and Not Affected
13. none_available and Recommended
14. no_fix_planned and Recommended
15. none_available and Affected
16. none_available and Under Investigation
17. none_available and Unknown
18. no_fix_planned and Affected
19. no_fix_planned and Under Investigation
20. no_fix_planned and Unknown

The remaining discouraged combinations are appended at the end of the list:

1. optional_patch and Fixed
2. vendor_fix and Under Investigation
3. vendor_fix and Unknown
4. mitigation and Under Investigation
5. mitigation and Unknown
6. workaround and Under Investigation
7. workaround and Unknown
8. fix_planned and Under Investigation
9. fix_planned and Unknown
10. fix_planned and Not Affected
11. no_fix_planned and Not Affected

CSAF Viewers MAY sort the remediation items accordingly.

3.2.4.14.2 Vulnerabilities Property - Remediations - Date

Date of the remediation (date) of value type string with format date-time contains the date from which the remediation is available.

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3.2.4.14.3 Vulnerabilities Property - Remediations - Details

Details of the remediation (details) of value type string with 1 or more characters contains a thorough human-readable discussion of the remediation.

3.2.4.14.4 Vulnerabilities Property - Remediations - Entitlements

List of entitlements (entitlements) of value type array with 1 or more items of type Entitlement of the remediation as string with 1 or more characters contains a list of entitlements.

```
<csaf-instance>:
  # ...
  vulnerabilities:
  - # <vulnerability-instance>:
    # ...
    remediations:
    - # <remediation-instance>:
      # ...
      entitlements: Sequence
      # ...
    # ...
```

Every Entitlement of the remediation contains any possible vendor-defined constraints for obtaining fixed software or hardware that fully resolves the vulnerability.

3.2.4.14.5 Vulnerabilities Property - Remediations - Group IDs

Group IDs (group_ids) are of value type Product Groups (product_groups_t) and contain a list of Product Groups the current remediation item applies to.

3.2.4.14.6 Vulnerabilities Property - Remediations - Product IDs

Product IDs (product_ids) are of value type Products (products_t) and contain a list of Products the current remediation item applies to.

3.2.4.14.7 Vulnerabilities Property - Remediations - Restart Required

Restart required by remediation (restart_required) of value type object with the one mandatory property Category (category) and the optional property Details (details) provides information on the category of restart required by this remediation to become effective.

```
<csaf-instance>:
  # ...
  vulnerabilities:
  - # <vulnerability-instance>:
    # ...
    remediations:
    - # <remediation-instance>:
      # ...
      restart_required:
        category: String.Enum
        details: String
      # ...
    # ...
```

Category of restart (category) of value type string and enum specifies what category of restart is required by this remediation to become effective. Valid values are:

```
connected
dependencies
machine
none
parent
service
system
vulnerable_component
zone
```

The values SHALL be used as follows:

- none: No restart required.

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- vulnerable_component: Only the vulnerable component (as given by the elements of product_ids or group_ids in the current remediation item) needs to be restarted.
- service: The vulnerable component and the background service used by the vulnerable component need to be restarted.
- parent: The vulnerable component and its parent process need to be restarted. This could be the case if the parent process has no build-in way to restart the vulnerable component or process values / context is only given at the start of the parent process.
- dependencies: The vulnerable component and all components which require the vulnerable component to work need to be restarted. This could be the case e.g. for a core service of a software.
- connected: The vulnerable component and all components connected (via network or any type of inter-process communication) to the vulnerable component need to be restarted.
- machine: The machine on which the vulnerable component is installed on needs to be restarted. This is the value which SHOULD be used if an OS needs to be restarted. It is typically the case for OS upgrades.
- zone: The security zone in which the machine resides on which the vulnerable component is installed needs to be restarted. This value might be useful for a remediation if no patch is available. If the malware can be wiped out by restarting the infected machines but the infection spreads fast the controlled shutdown of all machines at the same time and restart afterwards can leave one with a clean system.
- system: The whole system which the machine resides on which the vulnerable component is installed needs to be restarted. This MAY include multiple security zones. This could be the case for a major system upgrade in an ICS system or a protocol change.

Additional restart information (details) of value type string with 1 or more characters provides additional information for the restart. This can include details on procedures, scope or impact.

3.2.4.14.8 Vulnerabilities Property - Remediations - URL

URL (url) of value type string with format uri contains the URL where to obtain the remediation.

3.2.4.15 Vulnerabilities Property - Threats

List of threats (threats) of value type array with 1 or more items of value type object contains information about a vulnerability that can change with time.

```
<csaf-instance>:
  # ...
  vulnerabilities:
  - # <vulnerability-instance>:
    # ...
    threats: Sequence
    # ...
  # ...
```

Every Threat item of value type object with the two mandatory properties Category (category) and Details (details) contains the vulnerability kinetic information. This information can change as the vulnerability ages and new information becomes available. In addition, any Threat item MAY expose the three optional properties Date (date), Group IDs (group_ids), and Product IDs (product_ids).

```
<csaf-instance>:
  # ...
  vulnerabilities:
  - # <vulnerability-instance>:
    # ...
    threats:
      category: String.Enum
      date: String.DateTime
      details: String
      group_ids: $defs.product_groups_t
      product_ids: $defs.products_t
    # ...
  # ...
```

Category of the threat (category) of value type string and enum categorizes the threat according to the rules of the specification. Valid values are:

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```
exploit_status
impact
target_set
```

The value exploit_status indicates that the details field contains a description of the degree to which an exploit for the vulnerability is known. This knowledge can range from information privately held among a very small group to an issue that has been described to the public at a major conference or is being widely exploited globally. For consistency and simplicity, this section can be a mirror image of the CVSS exploitMaturity (v4.0), respectively exploitCodeMaturity (v3.1 and v3.0) or exploitability (v2.0) metric. However, it can also contain a more contextual status, such as "Weaponized" or "Functioning Code".

The value impact indicates that the details field contains an assessment of the impact on the user or the target set if the vulnerability is successfully exploited or a description why it cannot be exploited. If applicable, for consistency and simplicity, this section can be a textual summary of the three CVSS impact metrics. These metrics measure how a vulnerability detracts from the three core security properties of an information system: Confidentiality, Integrity, and Availability.

The value target_set indicates that the details field contains a description of the currently known victim population in whatever terms are appropriate. Such terms MAY include: operating system platform, types of products, user segments, and geographic distribution.

Date of the threat (date) of value type string with format date-time contains the date when the assessment was done or the threat appeared.

Details of the threat (details) of value type string with 1 or more characters represents a thorough human-readable discussion of the threat.

Group IDs (group_ids) are of value type Product Groups (product_groups_t) and contain a list of Product Groups the current threat item applies to.

Product IDs (product_ids) are of value type Products (products_t) and contain a list of Products the current threat item applies to.

3.2.4.16 Vulnerabilities Property - Title

Title (title) has value type string with 1 or more characters and gives the document producer the ability to apply a canonical name or title to the vulnerability.

3.2.4.17 Vulnerabilities Property - Extensions

Vulnerability-level Extensions (x_extensions) of value type Extensions Type (extensions_t) contains a list of extensions valid at the vulnerability item level of the CSAF document and associated with this vulnerability element.

```
<csaf-instance>:
  # ...
  vulnerabilities:
  - # <vulnerability-instance>:
    # ...
    x_extensions:  # $defs.extensions_t
    - # <x_extension-instance>:
      $schema: String
      category: String.Enum
      content: Mapping
      critical: Boolean
      # ...
```

3.2.5 Extensions Property

Root-level Extensions (x_extensions) of value type Extensions Type (extensions_t) contains a list of extensions valid at the root-level of the CSAF document and associated with this CSAF document.

```
<csaf-instance>:
  # ...
  x_extensions:  # $defs.extensions_t
  - # <x_extension-instance>:
    $schema: String
    category: String.Enum
    content: Mapping
    critical: Boolean
    # ...
```

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4. Profiles

CSAF documents do not have many required fields as they can be used for different purposes. To ensure a common understanding of which fields are required in a given use case the standard defines profiles. Each subsection describes such a profile by describing necessary content for that specific use case and providing insights into its purpose. The value of $.document.category is used to identify a CSAF document's profile. The following rules apply:

1. Each CSAF document SHALL conform the **CSAF Base** profile.
2. Each profile extends the base profile "CSAF Base" - directly or indirect through another profile from the standard - by making additional fields from the standard mandatory. A profile can always add, but never subtract nor overwrite requirements defined in the profile it extends.
3. Any optional field from the standard can also be added to a CSAF document which conforms with a profile without breaking conformance with the profile. One and only exempt is when the profile requires not to have a certain set of fields.
4. Values of $.document.category starting with csaf_ are reserved for existing, past, upcoming and future profiles defined in the CSAF standard.
5. Values of $.document.category starting with csaf_deprecated_ are used for official profiles that are marked deprecated. Those profiles are mostly there to allow backwards compatibility, e.g. with older CSAF versions. Therefore, they SHOULD NOT be used for newly created CSAF documents.
6. Values of $.document.category that do not match any of the values defined in section 4 of this standard SHALL be validated against the "CSAF Base" profile.
7. Local or private profiles MAY exist and tools MAY choose to support them.
8. If an official profile and a private profile exists, tools SHALL validate against the official one from the standard.

4.1 Profile 1: CSAF Base

This profile defines the default required fields for any CSAF document. Therefore, it is a "catch all" for CSAF documents that do not satisfy any other profile. Furthermore, it is the foundation all other profiles are build on.

A CSAF document SHALL fulfill the following requirements to satisfy the profile "CSAF Base":

- The following elements SHALL exist and be valid:
  - $['$schema']
  - $.document.category
  - $.document.csaf_version
  - $.document.distribution.tlp.label
  - $.document.publisher.category
  - $.document.publisher.name
  - $.document.publisher.namespace
  - $.document.title
  - $.document.tracking.current_release_date
  - $.document.tracking.id
  - $.document.tracking.initial_release_date
  - $.document.tracking.revision_history[*].date
  - $.document.tracking.revision_history[*].number
  - $.document.tracking.revision_history[*].summary
  - $.document.tracking.status
  - $.document.tracking.version
- The value of $.document.category SHALL NOT be equal to or a close match for any value that is intended to only be used by another profile nor to the (case insensitive) name of any other profile from the standard. Such case insensitive matching does ignore any dash, hyphen, minus, white space, invisible and underscore characters. To explicitly select the use of this profile the value csaf_base SHOULD be used.

> Neither CSAF Security Advisory nor csaf security advisory are valid values for $.document.category.

An issuing party might choose to prepend its $.document.publisher.name to a value in $.document.category that would otherwise be intended to only be used by another profile, to state that the CSAF document does not use the profile associated with this value.

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In this case, the (case insensitive) string “CSAF” SHALL be removed from the value. This SHOULD be done if the issuing party is unable or unwilling to use the value csaf_base, e.g. due to legal or cooperate identity reasons.

> Both values Example Company Security Advisory and Example Company security_advisory in $.document.category are therefore valid categories for the profile “CSAF Base”. This is important to prepare forward compatibility as later versions of CSAF might add new profiles. Therefore, the values which can be used for the profile “CSAF Base” might change.

4.2 Profile 2: Security Incident Response

This profile SHOULD be used to provide a response to a security breach or incident. This MAY also be used to convey information about an incident that is unrelated to the issuing party’s own products or infrastructure.

> Example Company might use a CSAF document satisfying this profile to respond to a security incident at ACME Inc. and the implications on its own products and infrastructure.

A CSAF document SHALL fulfill the following requirements to satisfy the profile “Security Incident Response”:

- The following elements SHALL exist and be valid:
  - all elements required by the profile “CSAF Base”.
  - $.document.notes with at least one item which has a category of description, details, general or summary

    > Reasoning: Without at least one note item which contains information about response to the event referred to this doesn’t provide any useful information.

  - $.document.references with at least one item which has a category of external

    > The intended use for this field is to refer to one or more documents or websites which provides more details about the incident.

- The value of $.document.category SHALL be csaf_security_incident_response.

4.3 Profile 3: Informational Advisory

This profile SHOULD be used to provide information which are **not related to a vulnerability** but e.g. a misconfiguration.

A CSAF document SHALL fulfill the following requirements to satisfy the profile “Informational Advisory”:

- The following elements SHALL exist and be valid:
  - all elements required by the profile “CSAF Base”.
  - $.document.notes with at least one item which has a category of description, details, general or summary

    > Reasoning: Without at least one note item which contains information about the “issue” which is the topic of the advisory it is useless.

  - $.document.references with at least one item which has a category of external

    > The intended use for this field is to refer to one or more documents or websites which provide more details about the issue or its remediation (if possible). This could be a hardening guide, a manual, best practices or any other helpful information.

- The value of $.document.category SHALL be csaf_informational_advisory.

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- The element $.vulnerabilities SHALL NOT exist. If there is any information that would reside in the element $.vulnerabilities the CSAF document SHOULD use another profile, e.g. “Security Advisory”.

If the element $.product_tree exists, a user SHALL assume that all products mentioned are affected.

4.4 Profile 4: Security Advisory

This profile SHOULD be used to provide information which is related to vulnerabilities and corresponding remediations.

A CSAF document SHALL fulfill the following requirements to satisfy the profile “Security Advisory”:

- The following elements SHALL exist and be valid:
  - all elements required by the profile “CSAF Base”.
  - $.product_tree which lists all products referenced later on in the CSAF document regardless of their state.
  - $.vulnerabilities which lists all vulnerabilities.
  - at least one of
    - $.vulnerabilities[*].cve
    - $.vulnerabilities[*].ids
  - $.vulnerabilities[*].notes

    > Provides details about the vulnerability.

  - $.vulnerabilities[*].product_status

    > Lists each product’s status in regard to the vulnerability.

  - $.vulnerabilities[*].product_status.known_affected

    > Lists affected products in regard to the vulnerability.

  - For each product given in $.vulnerabilities[*].product_status.fixed, the corresponding affected version SHALL be given.

    > Corresponding versions are usually in the same branches element.

- The value of $.document.category SHALL be csaf_security_advisory.
- The following elements SHOULD exist:
  - $.vulnerabilities[*].product_status.fixed

    > Lists fixed products in regard to the vulnerability.

  - $.vulnerabilities[*].remediations

    > Lists for each affected product in regard to the vulnerability appropriate remediations.

4.5 Profile 5: VEX

This profile SHOULD be used to provide information of the “Vulnerability Exploitability eXchange”. The main purpose of the VEX format is to state that and why a certain product is, or is not, affected by a vulnerability. See [VEX] for details.

A CSAF document SHALL fulfill the following requirements to satisfy the profile “VEX”:

- The following elements SHALL exist and be valid:

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  - all elements required by the profile “CSAF Base”.
  - $.product_tree which lists all products referenced later on in the CSAF document regardless of their state.
  - $.vulnerabilities which lists all vulnerabilities.
  - at least one of
    - $.vulnerabilities[*].product_status.fixed
    - $.vulnerabilities[*].product_status.known_affected
    - $.vulnerabilities[*].product_status.known_not_affected
    - $.vulnerabilities[*].product_status.under_investigation
  - at least one of
    - $.vulnerabilities[*].cve
    - $.vulnerabilities[*].ids
  - $.vulnerabilities[*].notes

    Provides details about the vulnerability.

- For each item in
  - $.vulnerabilities[*].product_status.known_not_affected an impact statement SHALL exist as machine readable flag in $.vulnerabilities[*].flags or as human readable justification in $.vulnerabilities[*].threats. For the latter one, the category value for such a statement SHALL be impact and the details field SHALL contain a description why the vulnerability cannot be exploited.
  - $.vulnerabilities[*].product_status.known_affected additional product specific information SHALL be provided in $.vulnerabilities[*].remediations as an action statement. Optional, additional information MAY also be provide through $.vulnerabilities[*].notes and $.vulnerabilities[*].threats.

    The use of the categories no_fix_planned and none_available for an action statement is permitted.

  Even though Product status lists Product IDs, Product Group IDs can be used in the remediations and threats object. However, it SHALL be ensured that for each Product ID the required information according to its product status as stated in the two points above is available. This implies that all products with the status known_not_affected SHALL have an impact statement and all products with the status known_affected SHALL have additional product specific information regardless of whether that is referenced through the Product ID or a Product Group ID.

- The value of $.document.category SHALL be csaf_vex.

4.6 Profile 6: Deprecated Security Advisory

This profile MAY be used to provide information which is related to vulnerabilities and corresponding remediations, e.g. when converting CSAF documents from older CSAF versions or a human-readable format. It SHOULD NOT be used for newly created documents. The profile “Security Advisory” from section 4.4 SHOULD be used instead.

The definition of the profile “Deprecated Security Advisory” in CSAF 2.1 matches the definition of profile “Security Advisory” in CSAF 2.0.

A CSAF document SHALL fulfill the following requirements to satisfy the profile “Deprecated Security Advisory”:

- The following elements SHALL exist and be valid:
  - all elements required by the profile “CSAF Base”.
  - $.product_tree which lists all products referenced later on in the CSAF document regardless of their state.
  - $.vulnerabilities which lists all vulnerabilities.

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- $.vulnerabilities[*].notes

> Provides details about the vulnerability.

- $.vulnerabilities[*].product_status

> Lists each product's status in regard to the vulnerability.

- The value of $.document.category SHALL be csaf_deprecated_security_advisory.

4.7 Profile 7: Withdrawn

This profile SHALL be used for any CSAF document that is withdrawn. It SHALL NOT be used for any superseded document.

A CSAF document SHALL fulfill the following requirements to satisfy the profile "Withdrawn":

- The following elements SHALL exist and be valid:
  - all elements required by the profile "CSAF Base".
  - $.document.notes with exactly one item using the category description describing the original content and the reasons for the withdrawal

    > Other items, such as a legal disclaimer, may exist alongside the required one.

    The title SHALL be Reasoning for Withdrawal for English or an unspecified document language. For any other language, it SHOULD be the language specific translation of that term.
  - $.document.tracking.revision_history with at least 2 entries. Any previous items SHALL NOT be removed.

    > A CSAF document cannot be withdrawn during the initial release to its specified target group. In such case, the CSAF document should not be released at all. If it was shared previously in draft status, then the $.document.tracking.status is kept in draft.

- The value of $.document.category SHALL be csaf_withdrawn.
- The elements $.product_tree and $.vulnerabilities SHALL NOT exist.

The CSAF document MAY link to additional information through $.document.references.

4.8 Profile 8: Superseded

This profile SHALL be used for any CSAF document that is superseded. It SHALL NOT be used for any withdrawn document.

A CSAF document SHALL fulfill the following requirements to satisfy the profile "Superseded":

- The following elements SHALL exist and be valid:
  - all elements required by the profile "CSAF Base".
  - $.document.notes with exactly one item using the category description

    > Other items, such as a legal disclaimer, may exist alongside the required one.

    The title SHALL be Reasoning for Supersession for English or an unspecified document language. For any other language, it SHOULD be the language specific translation of that term.
  - $.document.tracking.revision_history with at least 2 entries. Any previous items SHALL NOT be removed.

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> A CSAF document cannot be superseded during the initial release to its specified target group. In such case, the CSAF document should not be released at all. If it was shared previously in draft status, then the $.document.tracking.status is kept in draft.

  - $.document.references containing at least one item with category external The summary SHALL start with Superseding Document for English or an unspecified document language. For any other language, it SHOULD be the language specific translation of that term.
- The value of $.document.category SHALL be csaf_superseded.
- The elements $.product_tree and $.vulnerabilities SHALL NOT exist.

4.9 Profile 9: Vulnerability Report

This profile SHALL be used to exchange a vulnerability report during the Coordinated Vulnerability Disclosure (CVD) process. Providing a structured mechanism for exchanging information throughout the coordinated vulnerability disclosure process among the participating parties it supports communication between discoverers, vendors, and, where applicable, coordinators. Information such as affected products, technical details, severity assessments, and supporting evidence can be exchanged privately using this profile to facilitate coordination. It MAY also be used to report previously partially or full-disclosed vulnerabilities to corresponding parties.

A CSAF document SHALL fulfill the following requirements to satisfy the profile "Vulnerability Report":

- The following elements SHALL exist and be valid:
  - All elements required by the profile "CSAF Base".
  - $.product_tree which lists all products referenced later on in the CSAF document regardless of their state.
  - $.vulnerabilities which lists all vulnerabilities to be reported.
  - at least one of
    - $.vulnerabilities[*].cve
    - $.vulnerabilities[*].ids

    Each reported vulnerability needs a unique identifier to track the vulnerability as well as avoid misunderstandings and confusion during the coordination. For any vulnerability that has a CVE assigned regardless of the state of the CVE, its CVE identifier SHALL be used. In any other case, the issuing party SHALL provide an identifier that uniquely identifies the vulnerability within its own organization. An entity SHOULD make use of its corresponding RVISC entry. If other participants during the coordination use different identifiers, they SHOULD be added to the report to provide a single place for such mapping.
  - $.vulnerabilities[*].notes with at least one item using one of the following combinations:
    - the category summary: The title SHALL be Vulnerability Summary for English or an unspecified document language. For any other language, it SHOULD be the language specific translation of that term.
      This entry SHALL be used to describe the vulnerability that requires coordination with participating parties.
    - the category description: The title SHALL be CVE Description for English or an unspecified document language. For any other language, it SHOULD be the language specific translation of that term.
      This entry SHALL be used to describe the vulnerability that requires coordination with participating parties.

    > Other items may exist alongside the required one.

  - $.vulnerabilities[*].product_status

    > Lists each product's status with respect to the reported vulnerability as determined by the discoverer, or whoever is the issuing party of this document.

  - $.vulnerabilities[*].product_status.known_affected

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Identifies the products that the issuing party has determined to be affected. This requirement ensures that a Vulnerability Report profile identifies at least one product to which the reported vulnerability is known to apply.

- The value of $.document.category SHALL be csaf_vulnerability_report.
- The following elements SHOULD exist:
  - $.document.acknowledgments

    Identifies the individuals or organizations that SHOULD receive public acknowledgment for the discovery. If omitted, it indicates that the discoverer does not wish to receive public acknowledgment following disclosure.

  - $.document.aggregate_severity

    Provides an overall severity assessment for the report.

  - $.vulnerabilities[*].metrics

    Provides vulnerability assessment metrics, such as CVSS, EPSS, SSVC, or other supported scoring systems.

  - $.vulnerabilities[*].references

    Identifies publicly available information relevant to the reported vulnerability, including references supporting evidence of known exploitation where applicable.

  - $.vulnerabilities[*].threats

    Describes the reported security impact or exploitation status.

  - $.vulnerabilities[*].title

    Provides a concise title for each reported vulnerability.

- The value of $.document.distribution.tlp.label SHOULD be TLP:AMBER to promote sharing on a need-to-know basis within the vendor's or coordinator's organization and, where appropriate, their clients. Additional information, such as sharing_group or text, MAY be provided to further clarify the issuing party's intended sharing expectations.
- If the issuing party is aware of active exploitation, that information SHOULD be communicated using the SSVC Exploitatation metric as Active or PoC. Evidence supporting known exploitation MAY be referenced using $.vulnerabilities[*].references to assist investigation by vendors and coordinators.
- The following elements MAY exist:
  - $.vulnerabilities[*].disclosure_date which SHOULD be used to communicate or coordinate an intended public disclosure date for the reported vulnerabilities.
  - $.document.involvement

    Records coordination activities between the parties involved in the disclosure of the vulnerabilities.

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5. Additional Conventions

This section provides additional rules for handling CSAF documents.

5.1 Filename

The following rules SHALL be applied to determine the filename for the CSAF document:

1. The value `$.document.tracking.id` SHALL be converted into lowercase.
2. Any character sequence which is not part of one of the following groups SHALL be replaced by a single underscore (`_`):
   - lower-case ASCII letters (0x61 - 0x7A)
   - digits (0x30 - 0x39)
   - special characters: `+` (0x2B), `-` (0x2D)

   > The regex `[^+\-a-z0-9]+` can be used to find a character sequence which has to be replaced by an underscore. However, it SHALL NOT be applied before completing the first step.
   >
   > Even though the underscore `_` (0x5F) is a valid character in the filename it is replaced to avoid situations where the conversion rule might lead to multiple consecutive underscores. As a result, a `$.document.tracking.id` with the value `2022_#01-A` is converted into `2022_01-a` instead of `2022_01-a`.
3. The file extension `.json` SHALL be appended.

Examples 1:

```
cisco-sa-20190513-secureboot.json
example_company_-_2019-yh3234.json
rhba-2019_0024.json
```

> It is currently considered best practice to indicate that a CSAF document is invalid by inserting `_invalid` into the filename in front of the file extension.

Examples 2:

```
cisco-sa-20190513-secureboot_invalid.json
example_company_-_2019-yh3234_invalid.json
rhba-2019_0024_invalid.json
```

5.2 Separation in Data Stream

If multiple CSAF documents are transported via a data stream in a sequence without requests inbetween, they SHALL be separated by the Record Separator in accordance with [RFC7464].

5.3 Sorting

The keys within a CSAF document SHOULD be sorted alphabetically.

5.4 Usage of Markdown

The use of GitHub-flavored Markdown is permitted in the following fields:

```
$.document.acknowledgments[*].summary
$.document.distribution.text
$.document.involvement.actions[*].summary
$.document.involvement.entities[*].contact.details
$.document.involvement.entity_groups[*].summary
$.document.notes[*].text
$.document.publisher.contact.details
$.document.publisher.issuing_authority
$.document.references[*].summary
$.document.tracking.revision_history[*].summary
$.product_tree.product_groups[*].summary
$.vulnerabilities[*].acknowledgments[*].summary
$.vulnerabilities[*].notes[*].text
$.vulnerabilities[*].references[*].summary
```

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```
$.vulnerabilities[*].remediations[*].details
$.vulnerabilities[*].remediations[*].entitlements[*]
$.vulnerabilities[*].remediations[*].restart_required.details
$.vulnerabilities[*].threats[*].details
```

Other fields SHALL NOT contain Markdown.

5.5 Branch Recursion

The $.product_tree uses a nested structure for branches. Along a single path to a leaf, the recursion of branches is limited to 30 repetitions. Therefore, the longest path to a leaf is:

```
$.product_tree.branches[*].branches[*].branches[*].branches[*].branches[*].branches[*].branches[*].branches[*].branches[*].branches[*].branches[*].branches[*].branches[*].branches[*].branches[*].branche
```

5.6 Hardware and Software within the Product Tree

If a product consists of hardware and software, the hardware part SHALL be presented as one product in the product tree and the software part as another one. To form the overall product, both parts SHALL be combined through a product path.

Example 1:

```
"product_tree": {
  "branches": [
    {
      "branches": [
        {
          "branches": [
            {
              "category": "product_version",
              "name": "1.0",
              "product": {
                "name": "Example Company Controller A 1.0",
                "product_id": "CSAFPID-908070601",
                "product_identification_helper": {
                  "serial_numbers": [
                    "143-D-354"
                  ]
                }
              }
            }
          ],
          "category": "product_name",
          "name": "Controller A"
        },
        {
          "branches": [
            {
              "category": "product_version",
              "name": "4.1",
              "product": {
                "name": "Example Company Controller A Firmware 4.1",
                "product_id": "CSAFPID-908070602",
                "product_identification_helper": {
                  "hashes": [
                    {
                      "file_hashes": [
                        {
                          "algorithm": "sha256",
                          "value": "3fb9d502d096b1dfbcdfe60eed80ddecd98c8771bf21a82bbe1752735c4dc9e2"
                        }
                      ],
                      "filename": "a_4-1.bin"
                    }
                  ]
                }
              }
            },
            {
              "category": "product_version",
              "name": "4.2",
              "product": {
                "name": "Example Company Controller A Firmware 4.2",
                "product_id": "CSAFPID-908070603",
                "product_identification_helper": {
                  "hashes": [
```

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```
                {
                  "file_hashes": [
                    {
                      "algorithm": "sha256",
                      "value": "0a853ce2337f0608489ac596a308dc5b7b19d35a52b10bf31261586ac368b175"
                    }
                  ],
                  "filename": "a_4-2.bin"
                }
              ]
            }
          }
        }
      ],
      "category": "product_name",
      "name": "Controller A Firmware"
    }
  ],
  "category": "vendor",
  "name": "Example Company"
 }
],
"product_paths": [
 {
  "beginning_product_reference": "CSAFPID-908070602",
  "full_product_name": {
   "name": "Example Company Controller A Firmware 4.1 installed on Example Company Controller A 1.0",
   "product_id": "CSAFPID-908070604"
  },
  "subpaths": [
   {
    "category": "installed_on",
    "next_product_reference": "CSAFPID-908070601"
   }
  ]
 },
 {
  "beginning_product_reference": "CSAFPID-908070603",
  "full_product_name": {
   "name": "Example Company Controller A Firmware 4.2 installed on Example Company Controller A 1.0",
   "product_id": "CSAFPID-908070605"
  },
  "subpaths": [
   {
    "category": "installed_on",
    "next_product_reference": "CSAFPID-908070601"
   }
  ]
 }
]
}
```


This requirement is important to allow for correct matching. The serial number 143-D-354 identifies the Example Company Controller A 1.0 which is in this example the hardware in its version 1.0. The hash 3fb9d502d096b1dfbcdfe60eed80ddecd98c8771bf21a82bbe1752735c4dc9e2 identifies the software in the version 4.1; the hash 0a853ce2337f0608489ac596a308dc5b7b19d35a52b10bf31261586ac368b175 identifies the software in the version 4.2. The product paths combine the software and hardware part and form new products. These are used e.g. to assign the product status in the vulnerability section.

A matching tool searches in a first step for the product identification helper, e.g the serial number in an asset database to identify the asset that has this specific hardware and matches the software separately in a second step.

Representing the software version as a child element under elements representing hardware unsettles the consumer whether the version applies to the software or hardware. Also, this would violate the rule regarding the full identification of a product by the product_identification_helper from section 3.1.10.3.

Based on the CVE statistics up to and including the year 2024, in the majority of cases the vulnerabilities reside in software or are remediated via software. Having multiple products with the same product_identification_helper in different product_status for the same vulnerability would make it undecidable for machines what the product_status actually is.

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6. Tests

The first three subsections list a number of tests which all will have a short description and an excerpt of an example which fails the test. The fourth subsection groups tests into preset.

6.1 Mandatory Tests

Mandatory tests SHALL NOT fail at a valid CSAF document. A program SHALL handle a test failure as an error.

6.1.1 Missing Definition of Product ID

For each element of type $['$defs'].product_id_t which is not inside a Full Product Name (type: full_product_name_t) and therefore reference an element within the product_tree it SHALL be tested that the Full Product Name element with the matching product_id exists. The same applies for all items of elements of type $['$defs'].products_t.

The relevant paths for this test are:

```
$.document.involvement.actions[*].product_ids[*]
$.document.notes[*].product_ids[*]
$.product_tree.product_groups[*].product_ids[*]
$.product_tree.product_paths[*].beginning_product_reference
$.product_tree.product_paths[*].subpaths[*].next_product_reference
$.vulnerabilities[*].first_known_exploitation_dates[*].product_ids[*]
$.vulnerabilities[*].flags[*].product_ids[*]
$.vulnerabilities[*].ids[*].product_ids[*]
$.vulnerabilities[*].metrics[*].products[*]
$.vulnerabilities[*].notes[*].product_ids[*]
$.vulnerabilities[*].product_status.first_affected[*]
$.vulnerabilities[*].product_status.first_fixed[*]
$.vulnerabilities[*].product_status.fixed[*]
$.vulnerabilities[*].product_status.known_affected[*]
$.vulnerabilities[*].product_status.known_not_affected[*]
$.vulnerabilities[*].product_status.last_affected[*]
$.vulnerabilities[*].product_status.recommended[*]
$.vulnerabilities[*].product_status.under_investigation[*]
$.vulnerabilities[*].product_status.unknown[*]
$.vulnerabilities[*].remediations[*].product_ids[*]
$.vulnerabilities[*].threats[*].product_ids[*]
```

Example 1 (which fails the test):

```
"product_tree": {
  "product_groups": [
    {
      "group_id": "CSAFGID-1020300",
      "product_ids": [
        "CSAFPID-9080700",
        "CSAFPID-9080701"
      ]
    }
  ]
}
```

Neither CSAFPID-9080700 nor CSAFPID-9080701 were defined in the product_tree.

6.1.2 Multiple Definition of Product ID

For each Product ID (type $['$defs'].product_id_t) in Full Product Name elements (type: $['$defs'].full_product_name_t) it SHALL be tested that the product_id was not already defined within the same document.

The relevant paths for this test are:

```
$.product_tree..branches[*].product.product_id
$.product_tree.full_product_names[*].product_id
$.product_tree.product_paths[*].full_product_name.product_id
```

Example 1 (which fails the test):

```
"product_tree": {
  "full_product_names": [
```

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```
    {
      "product_id": "CSAFPID-9080700",
      "name": "Product A"
    },
    {
      "product_id": "CSAFPID-9080700",
      "name": "Product B"
    }
  ]
}
```

CSAFPID-9080700 was defined twice.

6.1.3 Circular Definition of Product ID

For each new defined Product ID (type $['$defs'].product_id_t) in items of product paths ($.product_tree.product_paths) it SHALL be tested that the product_id does not end up in a circle.

The relevant path for this test is:

```
$.product_tree.product_paths[*].full_product_name.product_id
```

As this can be quite complex a program for large CSAF documents, a program could check first whether a Product ID defined in a product path item is used as beginning_product_reference or next_product_reference. Only for those which fulfill this condition it is necessary to run the full check following the references.

Example 1 (which fails the test):

```
"product_tree": {
  "full_product_names": [
    {
      "name": "Product A",
      "product_id": "CSAFPID-9080700"
    }
  ],
  "product_paths": [
    {
      "beginning_product_reference": "CSAFPID-9080700",
      "full_product_name": {
        "name": "Product B",
        "product_id": "CSAFPID-9080701"
      },
      "subpaths": [
        {
          "category": "installed_on",
          "next_product_reference": "CSAFPID-9080701"
        }
      ]
    }
  ]
}
```

CSAFPID-9080701 refers to itself - this is a circular definition.

6.1.4 Missing Definition of Product Group ID

For each element of type $['$defs'].product_group_id_t which is not inside a Product Group ($.product_tree.product_groups[*]) and therefore reference an element within the product_tree it SHALL be tested that the Product Group element with the matching group_id exists. The same applies for all items of elements of type $['$defs'].product_groups_t.

The relevant paths for this test are:

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```
$.document.involvement.actions[*].group_ids[*]
$.document.notes[*].group_ids[*]
$.vulnerabilities[*].first_known_exploitation_dates[*].group_ids[*]
$.vulnerabilities[*].flags[*].group_ids[*]
$.vulnerabilities[*].ids[*].group_ids[*]
$.vulnerabilities[*].notes[*].group_ids[*]
$.vulnerabilities[*].remediations[*].group_ids[*]
$.vulnerabilities[*].threats[*].group_ids[*]
```

Example 1 (which fails the test):

```
"product_tree": {
  "full_product_names": [
    {
      "name": "Product A",
      "product_id": "CSAFPID-9080700"
    }
  ]
},
"vulnerabilities": [
  {
    "threats": [
      {
        "category": "exploit_status",
        "details": "Reliable exploits integrated in Metasploit.",
        "group_ids": [
          "CSAFGID-1020301"
        ]
      }
    ]
  }
]
```

CSAFGID-1020301 was not defined in the Product Tree.

6.1.5 Multiple Definition of Product Group ID

For each Product Group ID (type $['$defs'].product_group_id_t) Product Group elements ($.product_tree.product_groups[*]) it SHALL be tested that the group_id was not already defined within the same document.

The relevant path for this test is:

```
$.product_tree.product_groups[*].group_id
```

Example 1 (which fails the test):

```
"product_tree": {
  "full_product_names": [
    {
      "product_id": "CSAFPID-9080700",
      "name": "Product A"
    },
    {
      "product_id": "CSAFPID-9080701",
      "name": "Product B"
    },
    {
      "product_id": "CSAFPID-9080702",
      "name": "Product C"
    }
  ],
  "product_groups": [
    {
      "group_id": "CSAFGID-1020300",
      "product_ids": [
        "CSAFPID-9080700",
        "CSAFPID-9080701"
      ]
    },
    {
      "group_id": "CSAFGID-1020300",
```

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```
      "product_ids": [
        "CSAFPID-9080700",
        "CSAFPID-9080702"
      ]
    }
  ]
}
```

CSAFGID-1020300 was defined twice.

6.1.6 Contradicting Product Status

For each item in $.vulnerabilities it SHALL be tested that the same Product ID is not a member of contradicting product status groups (see section 3.2.4.12). The sets formed by the contradicting groups within one vulnerability item SHALL be pairwise disjoint.

The relevant path for this test is:

```
$.vulnerabilities[*].product_status
```

Example 1 (which fails the test):

```
"product_tree": {
  "full_product_names": [
    {
      "product_id": "CSAFPID-9080700",
      "name": "Product A"
    }
  ]
},
"vulnerabilities": [
  {
    "product_status": {
      "known_affected": [
        "CSAFPID-9080700"
      ],
      "known_not_affected": [
        "CSAFPID-9080700"
      ]
    }
  }
]
```

CSAFPID-9080700 is a member of the two contradicting groups “Affected” and “Not affected”.

6.1.7 Multiple Scores with Same Version per Product

For each item in $.vulnerabilities it SHALL be tested that the same Product ID is not a member of more than one CVSS vector with the same version and same source.

Different sources might assign different scores for the same product.

The test also applies for versions not included in CSAF.

The relevant path for this test is:

```
$.vulnerabilities[*].metrics[*]
```

Example 1 (which fails the test):

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```
"product_tree": {
  "full_product_names": [
    {
      "product_id": "CSAFPID-9080700",
      "name": "Product A"
    }
  ]
},
"vulnerabilities": [
  {
    "metrics": [
      {
        "content": {
          "cvss_v3": {
            "version": "3.1",
            "vectorString": "CVSS:3.1/AV:N/AC:L/PR:N/UI:N/S:C/C:H/I:H/A:H",
            "baseScore": 10,
            "baseSeverity": "CRITICAL"
          }
        },
        "products": [
          "CSAFPID-9080700"
        ]
      },
      {
        "content": {
          "cvss_v3": {
            "version": "3.1",
            "vectorString": "CVSS:3.1/AV:L/AC:L/PR:H/UI:R/S:U/C:H/I:H/A:H",
            "baseScore": 6.5,
            "baseSeverity": "MEDIUM"
          }
        },
        "products": [
          "CSAFPID-9080700"
        ]
      }
    ]
  }
]
```

Two CVSS v3.1 scores are given for CSAFPID-9080700 by the document author.

6.1.8 Invalid CVSS

It SHALL be tested that the given CVSS object is valid according to the referenced schema. The test SHALL fail if the object contains additional properties not defined in the referenced schema.

The relevant paths for this test are:

```
$.vulnerabilities[*].metrics[*].content.cvss_v2
$.vulnerabilities[*].metrics[*].content.cvss_v3
$.vulnerabilities[*].metrics[*].content.cvss_v4
```

Example 1 (which fails the test):

```
"cvss_v3": {
  "version": "3.1",
  "vectorString": "CVSS:3.1/AV:L/AC:L/PR:H/UI:R/S:U/C:H/I:H/A:H",
  "baseScore": 6.5
}
```

The required element baseSeverity is missing.

A tool MAY add one or more of the missing properties version, baseScore and baseSeverity based on the values given in vectorString as a quick fix.

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6.1.9 Invalid CVSS Computation

It SHALL be tested that the given CVSS object has the values computed correctly according to the definition.

> The vectorString SHOULD take precedence.

The relevant paths for this test are:

```
$.vulnerabilities[*].metrics[*].content.cvss_v2.baseScore
$.vulnerabilities[*].metrics[*].content.cvss_v2.temporalScore
$.vulnerabilities[*].metrics[*].content.cvss_v2.environmentalScore
$.vulnerabilities[*].metrics[*].content.cvss_v3.baseScore
$.vulnerabilities[*].metrics[*].content.cvss_v3.baseSeverity
$.vulnerabilities[*].metrics[*].content.cvss_v3.temporalScore
$.vulnerabilities[*].metrics[*].content.cvss_v3.temporalSeverity
$.vulnerabilities[*].metrics[*].content.cvss_v3.environmentalScore
$.vulnerabilities[*].metrics[*].content.cvss_v3.environmentalSeverity
$.vulnerabilities[*].metrics[*].content.cvss_v4.baseScore
$.vulnerabilities[*].metrics[*].content.cvss_v4.baseSeverity
```

> Note: CVSS v4 does not define threatScore, threatSeverity, environmentalScore and environmentalSeverity. The existence of these JSON keys in an older version of the schema was fixed with version 4.0.1.

Example 1 (which fails the test):

```
"cvss_v3": {
  "version": "3.1",
  "vectorString": "CVSS:3.1/AV:L/AC:L/PR:H/UI:R/S:U/C:H/I:H/A:H",
  "baseScore": 10.0,
  "baseSeverity": "LOW"
}
```

> Neither baseScore nor baseSeverity has the correct value according to the specification.

> A tool MAY set the correct values as computed according to the specification as a quick fix.

6.1.10 Inconsistent CVSS

It SHALL be tested that the given CVSS properties do not contradict the CVSS vector.

The relevant paths for this test are:

```
$.vulnerabilities[*].metrics[*].content.cvss_v2
$.vulnerabilities[*].metrics[*].content.cvss_v3
$.vulnerabilities[*].metrics[*].content.cvss_v4
```

Example 1 (which fails the test):

```
"cvss_v3": {
  "version": "3.1",
  "vectorString": "CVSS:3.1/AV:N/AC:L/PR:N/UI:N/S:U/C:H/I:H/A:H",
  "baseScore": 9.8,
  "baseSeverity": "CRITICAL",
  "attackVector": "LOCAL",
  "attackComplexity": "LOW",
  "privilegesRequired": "NONE",
  "userInteraction": "NONE",
  "scope": "CHANGED",
  "confidentialityImpact": "HIGH",
  "integrityImpact": "HIGH",
```

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```
  "availabilityImpact": "LOW"
}
```

The values in the CVSS vector differ from values of the properties attackVector, scope and availabilityImpact.

A tool MAY overwrite contradicting values according to the vectorString as a quick fix.

6.1.11 CWE

For each CWE it SHALL be tested that the given CWE exists and is valid in the version provided. Any id that refers to a CWE Category or View SHALL fail the test.

A list of all CWE release archives is available at [CWE-A].

The relevant path for this test is:

```
$.vulnerabilities[*].cwes[*]
```

Example 1 (which fails the test):

```
"cwes": [
  {
    "id": "CWE-79",
    "name": "Improper Input Validation",
    "version": "4.13"
  }
]
```

The CWE-79 exists. However, its name in version 4.13 is Improper Neutralization of Input During Web Page Generation ('Cross-site Scripting').

6.1.12 Language

For each element of type $['$defs'].lang_t it SHALL be tested that the language code is valid and exists. A CSAF validator SHALL differentiate in the error message between the language code that contain subtags not in the registry and those that contain only subtags from the registry but the compiled language code is invalid.

For example, this happens if a subtag for a region is used as primary language subtag.

The relevant paths for this test are:

```
$.document.lang
$.document.source_lang
```

Example 1 (which fails the test):

```
"lang": "EZ"
```

EZ is not a valid language. It is the subtag for the region “Eurozone”.

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For any deprecated subtag, a tool MAY replace it with its preferred value as a quick fix.

6.1.13 PURL

It SHALL be tested that all given PURLs are valid.

It is not sufficient to just test against the pattern provided in section 3.1.10.3.4. The PURL must be validated against the requirements in the [ECMA-427] specification and the additional constraints given in section 3.1.10.3.4.

The relevant paths for this test are:

```
$.product_tree..branches[*].product.product_identification_helper.purls[*]
$.product_tree.full_product_names[*].product_identification_helper.purls[*]
$.product_tree.product_paths[*].full_product_name.product_identification_helper.purls[*]
```

Example 1 (which fails the test):

```
"product_tree": {
  "full_product_names": [
    {
      "name": "Product A",
      "product_id": "CSAFPID-9080700",
      "product_identification_helper": {
        "purls": [
          "pkg:maven/@1.3.4"
        ]
      }
    }
  ]
}
```

Any valid PURL has a name component.

6.1.14 Sorted Revision History

It SHALL be tested that the value of number of items of the revision history are sorted ascending when the items are sorted ascending by date and as a second level criteria number. As the timestamps might use different timezones, the sorting SHALL take timezones into account.

In order to create consistent results in the case of mixed integer and semantic versioning (failing test 6.1.30), implementations SHALL map items with integer versioning to semantic versioning by appending .0.0 to the corresponding values. Any value that is not a valid integer nor semantic version SHOULD be ignored in this test as the schema test will fail. The aforementioned rules are solely used during the processing to create consistent test results.

These processing rules are not a suggested quick fix for test 6.1.30 but implementation guidance for this edge case.

The relevant path for this test is:

```
$.document.tracking.revision_history
```

Example 1 (which fails the test):

```
"revision_history": [
  {
    "date": "2024-01-22T10:00:00.000Z",
    "number": "2",
    "summary": "Second version."
  },
  {
```

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```
    "date": "2024-01-23T10:00:00.000Z",
    "number": "1",
    "summary": "Initial version."
  }
]
```

The first item when sorted by date has a higher version number than the second.

6.1.15 Translator

It SHALL be tested that $.document.source_lang is present and set if the value translator is used for $.document.publisher.category. A CSAF validator SHALL differentiate in the error message between the key being present but having no or an empty value and not being present at all.

The relevant path for this test is:

```
$.document.source_lang
```

Example 1 (which fails the test):

```
"document": {
  // ...
  "publisher": {
    "category": "translator",
    "name": "CSAF TC Translator",
    "namespace": "https://csaf.io/translator"
  },
  "title": "Mandatory test: Translator (failing example 1)",
  // ...
}
```

The required element source_lang is missing.

A tool MAY add the key as a quick fix. In such case, the value still needs to be set - either manually or by other means from the tool, e.g. through a given configuration.

6.1.16 Latest Document Version

It SHALL be tested that document version has the same value as the number in the last item of the revision history when it is sorted ascending by date and as a second level criteria number. As the timestamps might use different timezones, the sorting SHALL take timezones into account. Build metadata is ignored in the comparison. Any pre-release part is also ignored if the document status is draft.

The relevant path for this test is:

```
$.document.tracking.version
```

Example 1 (which fails the test):

```
"tracking": {
  // ...
  "revision_history": [
    {
      "date": "2024-01-21T09:00:00.000Z",
      "number": "1",
      "summary": "Initial version."
    },
    {
      "date": "2024-01-21T10:00:00.000Z",
      "number": "2",
      "summary": "Second version."
```

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```
    }
  ],
  // ...
  "version": "1"
}
```

The value of number of the last item after sorting is 2. However, the document version is 1.

A tool MAY set the document version to the value of number of the last item in the revision history after sorting as a quick fix.

6.1.17 Document Status Draft

It SHALL be tested that document status is draft if the document version is 0 or 0.y.z or contains the pre-release part.

The relevant path for this test is:

```
$.document.tracking.status
```

Example 1 (which fails the test):

```
"tracking": {
  // ...
  "status": "final",
  "version": "0.9.5"
}
```

The $.document.tracking.version is 0.9.5 but the document status is final.

A tool MAY set the document status to draft as a quick fix.

6.1.18 Released Revision History

It SHALL be tested that no item of the revision history has a number of 0 or 0.y.z when the document status is final or interim.

The relevant path for this test is:

```
$.document.tracking.revision_history[*].number
```

Example 1 (which fails the test):

```
"tracking": {
  // ...
  "revision_history": [
    {
      "date": "2023-09-17T10:00:00.000Z",
      "number": "0",
      "summary": "First draft"
    },
    {
      "date": "2024-01-21T10:00:00.000Z",
      "number": "1",
      "summary": "Initial version."
    }
  ],
  "status": "final",
  "version": "1"
}
```

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The document status is final but the revision history includes an item which has 0 as value for number.

A tool MAY remove the items with a number of 0 or 0.y.z from the revision history as a quick fix.

6.1.19 Revision History Entries for Pre-release Versions

It SHALL be tested that no item of the revision history has a number which includes pre-release information.

The relevant path for this test is:

```
$.document.tracking.revision_history[*].number
```

Example 1 (which fails the test):

```
"revision_history": [
  {
    "date": "2023-08-22T10:00:00.000Z",
    "number": "1.0.0-rc",
    "summary": "Release Candidate for initial version."
  },
  {
    "date": "2023-08-23T10:00:00.000Z",
    "number": "1.0.0",
    "summary": "Initial version."
  }
]
```

The revision history contains an item which has a number that indicates that this is pre-release.

A tool MAY merge each item of the revision history that has a number which includes pre-release information into the corresponding item which contains the same number without pre-release information as a quick fix. If no such item exists, the tool MAY just remove the pre-release information from the number instead of merging the item.

6.1.20 Non-Draft Document Version

It SHALL be tested that document version does not contain a pre-release part if the document status is final or interim.

The relevant path for this test is:

```
$.document.tracking.version
```

Example 1 (which fails the test):

```
"tracking": {
  // ...
  "status": "interim",
  "version": "1.0.0-alpha"
}
```

The document status is interim but the document version contains the pre-release part -alpha.

A tool MAY remove the pre-release part from the document version as a quick fix.

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6.1.21 Missing Item in Revision History

It SHALL be tested that items of the revision history do not omit a version number when the items are sorted ascending by date and as a second level criteria number.

> Dates are used as primary sorting criteria as they correspond to publications. The assigned numbers might be incorrect. Issuing parties are advised to investigate the issue and fix errors by providing the factual revision history with one entry per release. Note that this might lead to a situation where an invalid revision history is corrected by modifying the number in a revision history item to a lower value as required by this test. Nevertheless, the newly added item created to release the corrected version of the document to the intended target group will have a newer timestamp and therefore show up correctly at CSAF consumers.

As the timestamps might use different timezones, the sorting SHALL take timezones into account. The error message SHALL differentiate between a version number not present at all and one that is missing in the sorted list. In the case of semantic versioning in accordance with the rules from section 3.1.19.2, Minor and Patch versions MAY end at any point if the next version has a higher Major respectively Minor version. It SHALL also be tested that the first item in such a sorted list has either the version number 0 or 1 in the case of integer versioning or a Major version of 0 or 1 in the case of semantic versioning.

In order to create consistent results in the case of mixed integer and semantic versioning (failing test 6.1.30), implementations SHALL map items with integer versioning to semantic versioning by appending .0.0 to the corresponding values. Any value that is not a valid integer nor semantic version SHOULD be ignored in this test as the schema test will fail. The aforementioned rules are solely used during the processing to create consistent test results.

> These processing rules are not a suggested quick fix for test 6.1.30 but implementation guidance for this edge case.

The relevant path for this test is:

```
$.document.tracking.revision_history
```

Example 1 (which fails the test):

```
"revision_history": [
  {
    "date": "2023-08-22T10:00:00.000Z",
    "number": "1",
    "summary": "Initial version."
  },
  {
    "date": "2024-01-21T10:00:00.000Z",
    "number": "3",
    "summary": "Some other changes."
  }
]
```

> The item for version 2 is missing.

> A tool MAY add a stub for the missing item to the revision history as a quick fix. The stub might miss the date and summary which have to be provided by the user or other data sources.

6.1.22 Multiple Definition in Revision History

It SHALL be tested that items of the revision history do not contain the same version number.

The relevant path for this test is:

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```
$.document.tracking.revision_history
```

Example 1 (which fails the test):

```
"revision_history": [
  {
    "date": "2024-01-20T10:00:00.000Z",
    "number": "1",
    "summary": "Initial version."
  },
  {
    "date": "2024-01-21T10:00:00.000Z",
    "number": "1",
    "summary": "Some other changes."
  }
]
```

The revision history contains two items with the version number 1.

6.1.23 Multiple Use of Same CVE

It SHALL be tested that a CVE is not used in multiple vulnerability items.

The relevant path for this test is:

```
$.vulnerabilities[*].cve
```

Example 1 (which fails the test):

```
"vulnerabilities": [
  {
    "cve": "CVE-2017-0145"
  },
  {
    "cve": "CVE-2017-0145"
  }
]
```

The vulnerabilities array contains two items with the same CVE identifier CVE-2017-0145.

6.1.24 Multiple Definition in Actions

It SHALL be tested that items of the list of actions do differ in more values than just the action_id.

Ignoring the action_id, two items in the list of actions are the same, if:

1. all of the following fields contain the same value:
   - category
   - date
   - status
   - summary
2. and all of the following sets contain the same values:
   - acting_entity_refs
   - dl_vuln_ids
   - group_ids
   - product_ids
   - receiving_entity_refs
   - referenced_action_ids

A dl_vuln_ids mentioning all vulnerabilities in the document is equivalent to the element missing. All groups have to be expanded to their definitions for the test. As the timestamps might use different timezones, the comparison SHALL take timezones into account.

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The relevant path for this test is:

```
$.document.involvement.actions
```

Example 1 (which fails the test):

```
"actions": [
  {
    "action_id": "ACN-1",
    "acting_entity_refs": [
      "EID-0001"
    ],
    "date": "2024-01-10T09:00:00Z",
    "category": "notification",
    "receiving_entity_refs": [
      "EID-0002"
    ],
    "status": "completed",
    "summary": "Reported vulnerability to vendor."
  },
  {
    "action_id": "ACN-2",
    "acting_entity_refs": [
      "EID-0001"
    ],
    "date": "2024-01-10T09:00:00Z",
    "category": "notification",
    "receiving_entity_refs": [
      "EID-0002"
    ],
    "status": "completed",
    "summary": "Reported vulnerability to vendor."
  }
]
```

The list of actions contains two items with the same content when ignoring their action_id.

6.1.25 Multiple Use of Same Hash Algorithm

It SHALL be tested that the same hash algorithm is not used multiple times in one item of file hashes.

The relevant paths for this test are:

```
$.product_tree..branches[*].product.product_identification_helper.hashes[*].file_hashes
$.product_tree.full_product_names[*].product_identification_helper.hashes[*].file_hashes
$.product_tree.product_paths[*].full_product_name.product_identification_helper.hashes[*].file_hashes
```

Example 1 (which fails the test):

```
"product_tree": {
  "full_product_names": [
    {
      "name": "Product A",
      "product_id": "CSAFPID-9080700",
      "product_identification_helper": {
        "hashes": [
          {
            "file_hashes": [
              {
                "algorithm": "sha256",
                "value": "026a37919b182ef7c63791e82c9645e2f897a3f0b73c7a6028c7febf62e93838"
              },
              {
                "algorithm": "sha256",
                "value": "0a853ce2337f0608489ac596a308dc5b7b19d35a52b10bf31261586ac368b175"
              }
            ],
            "filename": "product_a.so"
          }
        ]
      }
    }
```

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```
  ]
}
```

The hash algorithm sha256 is used two times in one item of file hashes.

6.1.26 Prohibited Document Category Name

It SHALL be tested that the document category is not equal to the (case insensitive) name (without the prefix csaf_) or value of any other profile than "CSAF Base". Any occurrences of dash, hyphen, minus, underscore, white space and invisible characters are removed from the values on both sides before the case insensitive match.

The characters listed above are independent of their graphical variants. In general, all corresponding unicode characters are part of the group. For example, dash, hyphen and underscore characters include, but are not limited to:

- Em dash
- En dash
- Figure dash
- Horizontal bar
- Hyphen
- Hyphen-minus
- Non-breaking hyphen
- Low line
- Combining low line
- Fullwidth low line

This applies for both, the comparison against the name and value. Also the value SHALL NOT start with the reserved prefix csaf_ except if the value is exactly csaf_base.

This test does only apply for CSAF documents with the profile "CSAF Base". Therefore, it SHALL be skipped if the document category matches one of the values defined for the profile other than "CSAF Base".

For CSAF 2.1, the test must be skipped for the following values in $.document.category:

```
csaf_base
csaf_deprecated_security_advisory
csaf_informational_advisory
csaf_security_advisory
csaf_security_incident_response
csaf_superseded
csaf_vex
csaf_withdrawn
csaf_vulnerability_report
```

This is the only mandatory test related to the profile "CSAF Base" as the required fields SHALL be checked by validating the JSON schema.

The relevant path for this test is:

```
$.document.category
```

Examples 1 (for currently prohibited values):

```
Csaf_a
CsaF_VeX
CSafDeprecatedSecurity—Advisory
```

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```
csafvex
Deprecated Security Advisory
Informational Advisory
Security      Advisory
security-incident-response
Superseded
V_eX
veX
withdrawn
Vulnerability__rePORT
```

Example 2 (which fails the test):

```
"category": "Security_Incident_Response"
```

The value Security_Incident_Response is the name of a profile where the space was replaced with underscores.

6.1.27 Profile Tests

This subsubsection structures the mandatory tests for the profiles. Not all tests apply for all profiles. Tests SHOULD be skipped if the document category does not match the one given in the test. Each of the following tests SHOULD be treated as they were listed similar to the other tests.

An application MAY group these tests by profiles when providing the additional function to only run one or more selected tests. This results in one virtual test per profile.

6.1.27.1 Document Notes

It SHALL be tested that at least one item in $.document.notes exists which has a category of description, details, general or summary.

The relevant values for $.document.category are:

```
csaf_informational_advisory
csaf_security_incident_response
```

The relevant path for this test is:

```
$.document.notes
```

Example 1 (which fails the test):

```
"notes": [
  {
    "category": "legal_disclaimer",
    "text": "The CSAF document is provided to You \"AS IS\" and \"AS AVAILABLE\" and with all faults and defects without warranty of any kind.",
    "title": "Terms of Use"
  }
]
```

The document notes do not contain an item which has a category of description, details, general or summary.

6.1.27.2 Document References

It SHALL be tested that at least one item in $.document.references exists that contains a link to an external source. The property category SHALL be present for this item.

Other items MAY exist which do not contain the property category. Issuing parties are advised that tools MAY explicitly filter for or prioritize entries that contain the property category with the value of external.

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The relevant values for $.document.category are:

```
csaf_informational_advisory
csaf_security_incident_response
csaf_superseded
```

The relevant path for this test is:

```
$.document.references
```

Example 1 (which fails the test):

```
"references": [
  {
    "category": "self",
    "summary": "The canonical URL.",
    "url": "https://example.com/security/data/csaf/2024/oasis_csaf_tc-csaf_2_1-2024-6-1-27-02-01.json"
  }
]
```

The document references do not contain any item which has the category external.

6.1.27.3 Vulnerabilities

It SHALL be tested that the element $.vulnerabilities does not exist.

The relevant values for $.document.category are:

```
csaf_informational_advisory
csaf_withdrawn
csaf_superseded
```

The relevant path for this test is:

```
$.vulnerabilities
```

Example 1 (which fails the test):

```
"vulnerabilities": [
  {
    "title": "A vulnerability item that SHALL NOT exist"
  }
]
```

The element $.vulnerabilities exists.

A tool MAY change the $.document.category to csaf_base as a quick fix.

6.1.27.4 Product Tree

It SHALL be tested that the element $.product_tree exists.

The relevant values for $.document.category are:

```
csaf_security_advisory
csaf_vex
csaf_deprecated_security_advisory
csaf_vulnerability_report
```

The relevant path for this test is:

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```
$.product_tree
```

Example 1 (which fails the test):

```
{
  "document": {
    // ...
  },
  "vulnerabilities": [
    // ...
  ]
}
```

The element $.product_tree does not exist.

6.1.27.5 Vulnerability Notes

For each item in $.vulnerabilities it SHALL be tested that the element notes exists.

The relevant values for $.document.category are:

```
csaf_security_advisory
csaf_vex
csaf_deprecated_security_advisory
```

The relevant path for this test is:

```
$.vulnerabilities[*].notes
```

Example 1 (which fails the test):

```
"vulnerabilities": [
  {
    "product_status": {
      "known_affected": [
        "CSAFPID-9080700"
      ]
    },
    "title": "A vulnerability item without a note"
  }
]
```

The vulnerability item has no notes element.

6.1.27.6 Product Status

For each item in $.vulnerabilities it SHALL be tested that the element product_status exists.

The relevant values for $.document.category are:

```
csaf_security_advisory
csaf_deprecated_security_advisory
```

The relevant path for this test is:

```
$.vulnerabilities[*].product_status
```

Example 1 (which fails the test):

```
"vulnerabilities": [
  {
    "title": "A vulnerability item without a product status"
```

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```
  }
]
```

The vulnerability item has no product_status element.

6.1.27.7 VEX Product Status

For each item in $.vulnerabilities it SHALL be tested that at least one of the elements fixed, known_affected, known_not_affected, or under_investigation is present in product_status.

The relevant value for $.document.category is:

```
csaf_vex
```

The relevant paths for this test are:

```
$.vulnerabilities[*].product_status.fixed
$.vulnerabilities[*].product_status.known_affected
$.vulnerabilities[*].product_status.known_not_affected
$.vulnerabilities[*].product_status.under_investigation
```

Example 1 (which fails the test):

```
"product_status": {
  "first_fixed": [
    // ...
  ],
  "recommended": [
    // ...
  ]
}
```

None of the elements fixed, known_affected, known_not_affected, or under_investigation is present in product_status.

6.1.27.8 Vulnerability ID

For each item in $.vulnerabilities it SHALL be tested that at least one of the elements cve or ids is present. If no cve is present and all items in ids contain group_ids or product_ids, it SHALL be tested that each product mentioned in product_status[*][*] is assigned at least one item in ids. This is independent from whether the product is referenced directly or indirectly through a product group.

Without this rule, a product could be mentioned in a CSAF Document that has no clear reference to a vulnerability identifier. If a CVE is present, or at least one item in ids without group_ids and product_ids, the corresponding vulnerability identifier applies to the vulnerability itself and therefore to all products mention in this vulnerability.

The relevant values for $.document.category are:

```
csaf_security_advisory
csaf_vex
csaf_vulnerability_report
```

The relevant paths for this test are:

```
$.vulnerabilities[*].cve
$.vulnerabilities[*].ids
```

Example 1 (which fails the test):

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```
"vulnerabilities": [
  {
    "title": "A vulnerability item without a CVE or ID"
  }
]
```

None of the elements cve or ids is present.

6.1.27.9 Impact Statement

For each item in $.vulnerabilities[*].product_status.known_not_affected it SHALL be tested that a corresponding impact statement exist in $.vulnerabilities[*].flags or $.vulnerabilities[*].threats. For the latter one, the category value for such a statement SHALL be impact.

The relevant value for $.document.category is:

```
csaf_vex
```

The relevant paths for this test are:

```
$.vulnerabilities[*].flags
$.vulnerabilities[*].threats
```

Example 1 (which fails the test):

```
"product_tree": {
  "full_product_names": [
    {
      "product_id": "CSAFPID-9080700",
      "name": "Product A"
    },
    {
      "product_id": "CSAFPID-9080701",
      "name": "Product B"
    },
    {
      "product_id": "CSAFPID-9080702",
      "name": "Product C"
    }
  ],
  "product_groups": [
    {
      "group_id": "CSAFGID-0001",
      "product_ids": [
        "CSAFPID-9080700",
        "CSAFPID-9080701"
      ]
    }
  ]
},
"vulnerabilities": [
  {
    // ...
    "product_status": {
      "known_not_affected": [
        "CSAFPID-9080700",
        "CSAFPID-9080701",
        "CSAFPID-9080702"
      ]
    },
    "threats": [
      {
        "category": "impact",
        "details": "The vulnerable code is not present in these products.",
        "group_ids": [
          "CSAFGID-0001"
        ]
      }
    ]
  }
]
```

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There is no impact statement for CSAFPID-9080702.

Note: The impact statement for CSAFPID-9080700 and CSAFPID-9080701 is given through CSAFGID-0001.

6.1.27.10 Action Statement

For each item in $.vulnerabilities[*].product_status.known_affected it SHALL be tested that a corresponding action statement exist in $.vulnerabilities[*].remediations.

The relevant value for $.document.category is:

```
csaf_vex
```

The relevant path for this test is:

```
$.vulnerabilities[*].remediations
```

Example 1 (which fails the test):

```
"product_tree": {
  "full_product_names": [
    {
      "product_id": "CSAFPID-9080700",
      "name": "Product A"
    },
    {
      "product_id": "CSAFPID-9080701",
      "name": "Product B"
    },
    {
      "product_id": "CSAFPID-9080702",
      "name": "Product C"
    }
  ],
  "product_groups": [
    {
      "group_id": "CSAFGID-0001",
      "product_ids": [
        "CSAFPID-9080700",
        "CSAFPID-9080701"
      ],
      "summary": "EOL products"
    }
  ]
},
"vulnerabilities": [
  {
    // ...
    "product_status": {
      "known_affected": [
        "CSAFPID-9080700",
        "CSAFPID-9080701",
        "CSAFPID-9080702"
      ]
    },
    "remediations": [
      {
        "category": "no_fix_planned",
        "details": "These products are end-of-life. Therefore, no fix will be provided.",
        "group_ids": [
          "CSAFGID-0001"
        ]
      }
    ]
  }
]
```

There is no action statement for CSAFPID-9080702.

Note: The action statement for CSAFPID-9080700 and CSAFPID-9080701 is given through CSAFGID-0001.

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6.1.27.11 Vulnerabilities

It SHALL be tested that the element $.vulnerabilities exists.

The relevant values for $.document.category are:

```
csaf_security_advisory
csaf_vex
csaf_deprecated_security_advisory
csaf_vulnerability_report
```

The relevant path for this test is:

```
$.vulnerabilities
```

Example 1 (which fails the test):

```
{
  "$schema": "https://docs.oasis-open.org/csaf/csaf/v2.1/schema/csaf.json",
  "document": {
    // ...
  },
  "product_tree": [
    // ...
  ]
}
```

The element $.vulnerabilities does not exist.

6.1.27.12 Affected Products

For each item in $.vulnerabilities it SHALL be tested that the element product_status/known_affected exists.

The relevant value for $.document.category is:

```
csaf_security_advisory
```

The relevant path for this test is:

```
$.vulnerabilities[*].product_status.known_affected
```

Example 1 (which fails the test):

```
  "product_status": {
    "under_investigation": [
      "CSAFPID-9080700"
    ]
  }
```

The product status does not contain the known_affected element.

6.1.27.13 Corresponding Affected Products

For each product listed in the product status group fixed in any vulnerability, it SHALL be tested that a corresponding version of the product is listed as affected in the same vulnerability.

For a product path including the installed_with relationship the product path leading to but not including the relationship is a corresponding product. Such product path could also be just the product identified by beginning_product_reference if the first subpath element has the category installed_with.

The relevant value for $.document.category is:

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```
csaf_security_advisory
```

The relevant path for this test is:

```
$.vulnerabilities[*].product_status.known_affected
```

Example 1 (which fails the test):

```
{
  // ...
  "product_tree": {
    "branches": [
      {
        "branches": [
          {
            "branches": [
              {
                "category": "product_version",
                "name": "4.2",
                "product": {
                  "name": "Example Company Product A 4.2",
                  "product_id": "CSAFPID-9080700"
                }
              }
            ],
            "category": "product_name",
            "name": "Product A"
          }
        ],
        "category": "vendor",
        "name": "Example Company"
      }
    ]
  },
  "vulnerabilities": [
    {
      // ...
      "product_status": {
        "fixed": [
          "CSAFPID-9080700"
        ]
      }
    }
  ]
}
```

The vulnerability just contains the fixed product but does not list corresponding affected products.

6.1.27.14 Document Notes

It SHALL be tested that at least one item in $.document.notes exists which has a category of description.

The relevant values for $.document.category are:

```
csaf_withdrawn
csaf_superseded
```

The relevant path for this test is:

```
$.document.notes
```

Example 1 (which fails the test):

```
"notes": [
  {
    "category": "legal_disclaimer",
    "text": "The CSAF document is provided to You \"AS IS\" and \"AS AVAILABLE\" and with all faults and defects without warranty of any kind.",
    "title": "Terms of Use"
```

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```
  }
]
```

The document notes do not contain an item which has a category of description.

6.1.27.15 Product Tree

It SHALL be tested that the element $.product_tree does not exist.

The relevant values for $.document.category are:

```
csaf_withdrawn
csaf_superseded
```

The relevant path for this test is:

```
$.product_tree
```

Example 1 (which fails the test):

```
"product_tree": [
  // ...
]
```

The element $.product_tree exists.

6.1.27.16 Revision History

It SHALL be tested that the revision history contains at least two entries.

The relevant values for $.document.category are:

```
csaf_withdrawn
csaf_superseded
```

The relevant path for this test is:

```
$.document.tracking.revision_history
```

Example 1 (which fails the test):

```
"revision_history": [
  {
    "date": "2024-01-24T10:00:00.000Z",
    "number": "1",
    "summary": "Initial version."
  }
],
```

The revision history contains only one entry.

6.1.27.17 Reasoning for Withdrawal

If the document language is English or unspecified, it SHALL be tested that exactly one item in document notes exists that has the title Reasoning for Withdrawal. The category of this item SHALL be description.

The relevant value for $.document.category is:

```
csaf_withdrawn
```

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The relevant path for this test is:

```
$.document.notes
```

Example 1 (which fails the test):

```
"notes": [
  {
    "category": "summary",
    "text": "This CSAF document contained example data and was withdrawn to create test data.",
    "title": "Reasoning for Withdrawal"
  }
],
```

The note has the correct title. However, it uses the wrong category.

6.1.27.18 Reasoning for Supersession

If the document language is English or unspecified, it SHALL be tested that exactly one item in document notes exists that has the title Reasoning for Supersession. The category of this item SHALL be description.

The relevant value for $.document.category is:

```
csaf_superseded
```

The relevant path for this test is:

```
$.document.notes
```

Example 1 (which fails the test):

```
"notes": [
  {
    "category": "details",
    "text": "This CSAF document contained example data and was withdrawn to create test data.",
    "title": "Reasoning for Supersession"
  }
],
```

The note has the correct title. However, it uses the wrong category.

6.1.27.19 Reference to Superseding Document

If the document language is English or unspecified, it SHALL be tested that at least one item in document references exists that has a summary starting with Superseding Document. For each of these items, the category SHALL be present and have the value external.

The relevant value for $.document.category is:

```
csaf_superseded
```

The relevant path for this test is:

```
$.document.references
```

Example 1 (which fails the test):

```
  "references": [
    {
      "category": "external",
      "summary": "Superseded Document",
      "url": "https://example.com/.well-known/csaf/clear/2024/esa-2024-1234.json"
```

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```
    }
  ],
```

There is no reference where the summary starts with the correct term Superseding Document.

6.1.27.20 Vulnerability Notes

If the document language is English or unspecified, it SHALL be tested that at least one item in vulnerability notes exists that has the title Vulnerability Summary or CVE Description. The category of this item SHALL be consistent with the value required in table 2 of section 3.2.4.11.

The relevant value for $.document.category is:

```
csaf_vulnerability_report
```

The relevant path for this test is:

```
$.vulnerability[*].notes
```

Example 1 (which fails the test):

```
  "notes": [
    {
      "category": "description",
      "text": "Example Company Product A 17.4 performs only checks on file extensions with all lowercase letters. An unauthenticated remote attacker can execute arbitrary binaries if their file extension contains uppercase letters or other characters.",
      "title": "Vulnerability Summary"
    }
  ],
```

The note has the correct title. However, it uses the wrong category.

6.1.28 Translation

It SHALL be tested that the given source language and document language are not the same.

The relevant paths for this test are:

```
$.document.lang
$.document.source_lang
```

Example 1 (which fails the test):

```
"document": {
  // ...
  "lang": "en-US",
  // ...
  "source_lang": "en-US",
  // ...
}
```

The document language and the source language have the same value en-US.

Note: A translation from en-US to en-GB would pass the test.

A tool MAY remove the source language as a quick fix.

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6.1.29 Remediation without Product Reference

For each item in $.vulnerabilities[*].remediations it SHALL be tested that it includes at least one of the elements group_ids or product_ids.

The relevant path for this test is:

```
$.vulnerabilities[*].remediations[*]
```

Example 1 (which fails the test):

```
"remediations": [
  {
    "category": "no_fix_planned",
    "details": "These products are end-of-life. Therefore, no fix will be provided."
  }
]
```

The given remediation does not specify to which products it should be applied.

A tool MAY add all products of the affected group of this vulnerability to the remediation as a quick fix.

6.1.30 Mixed Integer and Semantic Versioning

It SHALL be tested that all elements of type $['$defs'].version_t follow either integer versioning or semantic versioning homogeneously within the same document.

In order to create consistent results in the case of a value that is not a valid integer nor semantic version, implementations SHOULD ignore such a value in this test as the schema test will fail.

This processing rule is solely implementation guidance for this edge case.

The relevant paths for this test are:

```
$.document.tracking.revision_history[*].number
$.document.tracking.version
```

Example 1 (which fails the test):

```
"tracking": {
  // ...
  "revision_history": [
    {
      "date": "2024-01-21T09:00:00.000Z",
      "number": "1.0.0",
      "summary": "Initial version."
    },
    {
      "date": "2024-01-21T10:00:00.000Z",
      "number": "2",
      "summary": "Second version."
    }
  ],
  // ...
  "version": "2"
}
```

The document started with semantic versioning (1.0.0) and switched to integer versioning (2).

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A tool MAY assign all items their corresponding value according to integer versioning as a quick fix. In such case, the old number SHOULD be stored in legacy_version.

6.1.31 Version Range in Product Version

For each element of type $['$defs'].branches_t with category of product_version it SHALL be tested that the value of name does not contain a version range.

To implement this test it is deemed sufficient that, when converted to lowercase, the value of name satisfies the two requirements below:

1. It does not contain any of the following operators:

```
<
<=
>
>=
```

2. If interpreted as a list of individual words separated by white space, the list does not contain any of the following keywords:

```
after
all
before
earlier
later
prior
versions
```

The relevant path for this test is:

```
$.product_tree..branches[*].name
```

Example 1 (which fails the test):

```
"branches": [
  {
    "category": "product_version",
    "name": "prior to 4.2",
    // ...
  }
]
```

The version range prior to 4.2 is given for the branch category product_version.

6.1.32 Flag without Product Reference

For each item in $.vulnerabilities[*].flags it SHALL be tested that it includes at least one of the elements group_ids or product_ids.

The relevant path for this test is:

```
$.vulnerabilities[*].flags[*]
```

Example 1 (which fails the test):

```
"flags": [
  {
    "label": "component_not_present"
```

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```
      }
    ]
```

The given flag does not specify to which products it should be applied.

6.1.33 Multiple Flags with VEX Justification Codes per Product

For each item in $.vulnerabilities[*] it SHALL be tested that a Product is not member of more than one Flag item with a VEX justification code (see section 3.2.4.8). This takes indirect relations through Product Groups into account.

Additional flags with a different purpose might be provided in later versions of CSAF. Through the explicit reference of VEX justification codes the test is specified to be forward-compatible.

The relevant path for this test is:

```
$.vulnerabilities[*].flags
```

Example 1 (which fails the test):

```
"product_tree": {
  "full_product_names": [
    {
      "product_id": "CSAFPID-9080700",
      "name": "Product A"
    },
    {
      "product_id": "CSAFPID-9080701",
      "name": "Product B"
    }
  ],
  "product_groups": [
    {
      "group_id": "CSAFGID-0001",
      "product_ids": [
        "CSAFPID-9080700",
        "CSAFPID-9080701"
      ]
    }
  ]
},
"vulnerabilities": [
  {
    // ...
    "flags": [
      {
        "label": "component_not_present",
        "group_ids": [
          "CSAFGID-0001"
        ]
      },
      {
        "label": "vulnerable_code_cannot_be_controlled_by_adversary",
        "product_ids": [
          "CSAFPID-9080700"
        ]
      }
    ],
    // ...
    "product_status": {
      "known_not_affected": [
        "CSAFPID-9080700",
        "CSAFPID-9080701"
      ]
    }
  }
]
```

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There are two flags given for CSAFPID-9080700 - one indirect through CSAFGID-0001 and one direct.

6.1.34 Branches Recursion Depth

For each product defined under $.product_tree.branches[*] it SHALL be tested that the complete JSON path does not contain more than 30 instances of branches.

The relevant path for this test is:

```
$.product_tree..branches[*].product
```

Example 1 (which fails the test):


```
"product_tree": {
  "branches": [
    {
      "branches": [
        {
          "branches": [
            {
              "branches": [
                {
                  "branches": [
                    {
                      "branches": [
                        {
                          "branches": [
                            {
                              "branches": [
                                {
                                  "branches": [
                                    {
                                      "branches": [
                                        {
                                          "branches": [
                                            {
                                              "branches": [
                                                {
                                                  "branches": [
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                                                      "branches": [
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                                                          "branches": [
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                                                              "branches": [
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                                                                  "branches": [
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                                                                                                                                                                                                                                                                                                                            {
                                                                                                                                                                                                                                                                                                                              "branches": [
                                                                                                                                                                                                                                                                                                                                {
                                                                                                                                                                                                                                                                                                                                  "branches": [
                                                                                                                                                                                                                                                                                                                                    {
                                                                                                                                                                                                                                                                                                                                      "branches": [
```


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```
                                                {
                                                  "category": "product_name",
                                                  "name": "branches",
                                                  "product": {
                                                    "name": "<<generate Product name>>",
                                                    "product_id": "CSAFPID-9080700"
                                                  }
                                                }
                                              ],
                                              "category": "product_family",
                                              "name": "31"
                                            }
                                          ],
                                          "category": "product_family",
                                          "name": "with"
                                        }
                                      ],
                                      "category": "product_family",
                                      "name": "test"
                                    }
                                  ],
                                  "category": "product_family",
                                  "name": "the"
                                }
                              ],
                              "category": "product_family",
                              "name": "fail"
                            }
                          ],
                          "category": "product_family",
                          "name": "and"
                        }
                      ],
                      "category": "product_family",
                      "name": "limits"
                    }
                  ],
                  "category": "product_family",
                  "name": "the"
                }
              ],
              "category": "product_family",
              "name": "testing"
            }
          ],
          "category": "product_family",
          "name": "are"
        }
      ],
      "category": "product_family",
      "name": "they"
    }
  ],
  "category": "product_family",
  "name": "but"
}
],
"category": "product_family",
"name": "unrealistic"
}
],
"category": "product_family",
"name": "less"
}
],
"category": "product_family",
"name": "or"
}
],
"category": "product_family",
"name": "more"
}
],
"category": "product_family",
"name": "and"
}
],
"category": "product_family",
"name": "unnecessary"
```


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```
                                    }
                                   ],
                                   "category": "product_family",
                                   "name": "seem"
                                  }
                                 ],
                                 "category": "product_family",
                                 "name": "which"
                                }
                               ],
                               "category": "product_family",
                               "name": "product"
                              }
                             ],
                             "category": "product_family",
                             "name": "hypothetical"
                            }
                           ],
                           "category": "product_family",
                           "name": "this"
                          }
                         ],
                         "category": "product_family",
                         "name": "for"
                        }
                       ],
                       "category": "product_family",
                       "name": "structure"
                      }
                     ],
                     "category": "product_family",
                     "name": "nested"
                    }
                   ],
                   "category": "product_family",
                   "name": "deeply"
                  }
                 ],
                 "category": "product_family",
                 "name": "a"
                }
               ],
               "category": "product_family",
               "name": "uses"
              }
             ],
             "category": "vendor",
             "name": "Example Company"
            }
           ]
          }
```


The complete JSON path contains 31 times branches.

6.1.35 Contradicting Remediations

For each item in $.vulnerabilities[*].remediations it SHALL be tested that a product is not member of contradicting remediation categories (see table 3). This takes indirect relations through product groups into account.

The relevant path for this test is:

```
$.vulnerabilities[*].remediations[*]
```

Example 1 (which fails the test):


```
"remediations": [
 {
  "category": "no_fix_planned",
  "details": "The product is end-of-life. Therefore, no fix will be provided.",
  "product_ids": [
   "CSAFPID-9080700"
  ]
 },
```


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```
    {
      "category": "vendor_fix",
      "details": "Update to version >=14.3 to fix the vulnerability.",
      "product_ids": [
        "CSAFPID-9080700"
      ]
    }
  ]
```

The two remediations given for the product with product ID CSAFPID-9080700 contradict each other.

A tool MAY apply the conversion rules from the conformance target CSAF 2.0 to CSAF 2.1 converter if applicable or remove the product from the remediation with the lower priority. The priority MAY be defined as follows: vendor_fix > mitigation > workaround > fix_planned > no_fix_planned > optional_patch > none_available

6.1.36 Contradicting Product Status Remediation Combination

For each item in $.vulnerabilities[*].remediations it SHALL be tested that a product is not member of a contradicting product status group (see table 4). This takes indirect relations through product groups into account.

The relevant path for this test is:

```
$.vulnerabilities[*].remediations[*]
```

Example 1 (which fails the test):

```
  "product_status": {
    "known_not_affected": [
      "CSAFPID-9080700"
    ]
  },
  "remediations": [
    {
      "category": "vendor_fix",
      "details": "Update to version >=14.3 to fix the vulnerability.",
      "product_ids": [
        "CSAFPID-9080700"
      ]
    }
  ]
```

For the product with product ID CSAFPID-9080700 a vendor_fix is given but the product was not affected at all.

6.1.37 Date and Time

For each item of type string and format date-time it SHALL be tested that it conforms to the rules given in section 2.3.

The relevant paths for this test are:

```
$.document.involvement.actions[*].date
$.document.tracking.current_release_date
$.document.tracking.generator.date
$.document.tracking.initial_release_date
$.document.tracking.revision_history[*].date
$.vulnerabilities[*].disclosure_date
$.vulnerabilities[*].discovery_date
$.vulnerabilities[*].first_known_exploitation_dates[*].date
$.vulnerabilities[*].first_known_exploitation_dates[*].exploitation_date
$.vulnerabilities[*].flags[*].date
$.vulnerabilities[*].metrics[*].content.epss.timestamp
$.vulnerabilities[*].metrics[*].content.ssvc_v2.timestamp
$.vulnerabilities[*].remediations[*].date
$.vulnerabilities[*].threats[*].date
```

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Example 1 (which fails the test):

```
"current_release_date": "2024-01-24 10:00:00.000Z",
```

The current_release_date uses a white space as separator instead the letter T.

6.1.38 Non-Public Sharing Group with Max UUID

It SHALL be tested that a CSAF document using Max UUID as sharing group ID has the TLP label CLEAR.

The relevant path for this test is:

```
$.document.distribution.tlp.label
```

Example 1 (which fails the test):

```
"distribution": {
  "sharing_group": {
    "id": "ffffffff-ffff-ffff-ffff-ffffffffffff",
    "name": "Public"
  },
  "tlp": {
    "label": "RED"
  }
},
```

The sharing group uses the Max UUID but the CSAF document is labeled as TLP:RED.

A tool MAY remove the property sharing_group as a quick fix.

6.1.39 Public Sharing Group with No Max UUID

It SHALL be tested that a CSAF document with the TLP label CLEAR use the Max UUID as sharing group ID if any. The test SHALL pass if no sharing group is present or the Nil UUID is used and the document status is draft.

The relevant path for this test is:

```
$.document.distribution.sharing_group.id
```

Example 1 (which fails the test):

```
"distribution": {
  "sharing_group": {
    "id": "5868d6be-b28a-404e-a245-0b5093b31b8b"
  },
  "tlp": {
    "label": "CLEAR"
  }
},
```

The sharing group is present for the TLP:CLEAR document but it differs from the Max UUID.

A tool MAY update the sharing group id as a quick fix.

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6.1.40 Invalid Sharing Group Name

It SHALL be tested that the value of sharing group name does not equal the reserved values from section 3.2.2.5.1 if the precondition is not fulfilled.

The relevant path for this test is:

```
$.document.distribution.sharing_group.name
```

Example 1 (which fails the test):

```
"distribution": {
  "sharing_group": {
    "id": "5868d6be-b28a-404e-a245-0b5093b31b8b",
    "name": "Public"
  },
  // ...
},
```

The sharing group name is Public but it does not use the Max UUID.

A tool MAY update the sharing group name as a quick fix.

6.1.41 Missing Sharing Group Name

It SHALL be tested that the sharing group name exists and equals the predefined reserved value from section 3.2.2.5.1 if the precondition is fulfilled.

The relevant path for this test is:

```
$.document.distribution.sharing_group.name
```

Example 1 (which fails the test):

```
"distribution": {
  "sharing_group": {
    "id": "ffffffff-ffff-ffff-ffff-ffffffffffff"
  },
  // ...
},
```

The Max UUID is used but the sharing group name does not exist.

A tool MAY add the corresponding sharing group name as a quick fix.

6.1.42 PURL Qualifiers

For each product_identification_helper object containing multiple PURLs it SHALL be tested that the PURLs only differ in their qualifiers.

The relevant paths for this test are:

```
$.product_tree..branches[*].product.product_identification_helper.purls[*]
$.product_tree.full_product_names[*].product_identification_helper.purls[*]
$.product_tree.product_paths[*].full_product_name.product_identification_helper.purls[*]
```

Example 1 (which fails the test):

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```
"product_tree": {
  "full_product_names": [
    {
      "name": "Product A",
      "product_id": "CSAFPID-9080700",
      "product_identification_helper": {
        "purls": [
          "pkg:maven/com.example/logging@1.3.4",
          "pkg:maven/com.example/audit@1.3.4"
        ]
      }
    }
  ]
}
```

The two PURLs differ in the name component.

6.1.43 Use of Multiple Stars in Model Number

For each model number it SHALL be tested that it does not contain multiple unescaped stars.

Multiple * that match zero or multiple characters within a model number introduce ambiguity and are therefore prohibited.

The relevant paths for this test are:

```
$.product_tree..branches[*].product.product_identification_helper.model_numbers[*]
$.product_tree.full_product_names[*].product_identification_helper.model_numbers[*]
$.product_tree.product_paths[*].full_product_name.product_identification_helper.model_numbers[*]
```

Example 1 (which fails the test):

```
"model_numbers": [
  "P*A*"
]
```

The model number contains two unescaped stars.

6.1.44 Use of Multiple Stars in Serial Number

For each serial number it SHALL be tested that it does not contain multiple unescaped stars.

Multiple * that match zero or multiple characters within a serial number introduce ambiguity and are therefore prohibited.

The relevant paths for this test are:

```
$.product_tree..branches[*].product.product_identification_helper.serial_numbers[*]
$.product_tree.full_product_names[*].product_identification_helper.serial_numbers[*]
$.product_tree.product_paths[*].full_product_name.product_identification_helper.serial_numbers[*]
```

Example 1 (which fails the test):

```
"serial_numbers": [
  "P*A*"
]
```

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The serial number contains two unescaped stars.

6.1.45 Inconsistent Disclosure Date

For each vulnerability, it SHALL be tested that the disclosure_date is earlier than or equal to the date of the newest item of the revision_history if the document is labeled TLP:CLEAR and the document status is final or interim. As the timestamps might use different timezones, the sorting and comparison SHALL take timezones into account.

The relevant path for this test is:

```
$.vulnerabilities[*].disclosure_date
```

Example 1 (which fails the test):

```
"document": {
  // ...
  "distribution": {
    "tlp": {
      "label": "CLEAR"
    }
  },
  // ...
  "tracking": {
    // ...
    "revision_history": [
      {
        "date": "2024-01-24T10:00:00.000Z",
        "number": "1",
        "summary": "Initial version."
      }
    ],
    "status": "final",
    // ...
  }
},
"vulnerabilities": [
  {
    "disclosure_date": "2024-02-24T10:00:00.000Z"
  }
]
```

The document is labeled TLP:CLEAR and in status final but the disclosure_date is newer than the date of newest item in the revision_history.

6.1.46 Invalid SSVC

It SHALL be tested that the given SSVC object is valid according to the referenced schema. If the object contains a language code, for example as a part of a namespace, it SHALL also be tested that the language code is valid and exists. A CSAF validator SHALL differentiate in the error message between the language code that contain subtags not in the registry and those that contain only subtags from the registry but the compiled language code is invalid.

The relevant path for this test is:

```
$.vulnerabilities[*].metrics[*].content.ssvc_v2
```

Example 1 (which fails the test):

```
"ssvc_v2": {
  "schemaVersion": "2.0.0",
  "timestamp": "2024-01-24T10:00:00.000Z"
}
```

The required element selections is missing.

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6.1.47 Inconsistent SSVC Target IDs

For each ssvc_v2 object it SHALL be tested that each item in target_ids is either the CVE of the vulnerability given in cve or the text of an item in the ids array of the vulnerability. The test SHALL fail, if the target ID equals the $.document.tracking.id and the CSAF document contains more than one vulnerability.

The relevant path for this test is:

```
$.vulnerabilities[*].metrics[*].content.ssvc_v2.target_ids[*]
```

Example 1 (which fails the test):

```
"vulnerabilities": [
  {
    "cve": "CVE-1900-0001",
    "metrics": [
      {
        "content": {
          "ssvc_v2": {
            // ...
            "target_ids": [
              "CVE-1900-0002"
            ],
            // ...
          }
        },
        // ...
      }
    ]
  }
]
```

The SSVC Target ID does not match the CVE ID.

A tool MAY remove any inconsistent SSVC Target ID as a quick fix.

If the tool removes the last item of the array, it SHALL remove the element target_ids as well.

6.1.48 SSVC Decision Points

For each SSVC decision point given under selections within a registered base namespace, it SHALL be tested that given decision point exists, is valid and the items in values are ordered correctly. The test SHALL pass, if a unregistered namespace is used. Namespaces reserved for special purpose SHALL be treated as per their definition.

A list of all currently registered namespaces is a available in the SSVC documentation at [SSVC-RNS].

A list of all valid decision points of registered namespaces including their values is available at the SSVC repository (see [SSVC-DP]). The items in values need to have the same order as in their definition.

This test also covers decision points which use a registered base namespace with any number of extensions. The rules for namespace extensions as specified by SSVC must be applied in such case. Note that:

- Extensions can limit the number of available values, but not add new ones or change the order.
- Extensions can translate or refine the name and definition, but not change the key.

The relevant path for this test is:

```
$.vulnerabilities[*].metrics[*].content.ssvc_v2.selections[*]
```

Example 1 (which fails the test):

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```
"ssvc_v2": {
  // ...
  "selections": [
    {
      "key": "MI",
      "namespace": "ssvc",
      "values": [
        {
          "key": "N",
          "name": "None"
        },
        {
          "key": "D",
          "name": "Degraded"
        }
      ],
      "version": "1.0.0"
    }
  ],
  // ...
}
```

The SSVC decision point Mission Impact doesn't have the value Degraded in version 1.0.0.

If applicable, a tool MAY sort the items in values according to the order of their definition as a quick fix.

6.1.49 Inconsistent SSVC Timestamp

For each vulnerability, it SHALL be tested that each SSVC timestamp is earlier than or equal to the date of the newest item of the revision_history if the document status is final or interim. As the timestamps might use different timezones, the sorting and comparison SHALL take timezones into account.

The relevant path for this test is:

```
$.vulnerabilities[*].metrics[*].content.ssvc_v2.timestamp
```

Example 1 (which fails the test):

```
"document": {
  // ...
  "tracking": {
    // ...
    "revision_history": [
      {
        "date": "2024-01-24T10:00:00.000Z",
        "number": "1",
        "summary": "Initial version."
      }
    ],
    "status": "final",
    // ...
  }
},
// ...
"vulnerabilities": [
  {
    "metrics": [
      {
        "content": {
          "ssvc_v2": {
            // ...
            "timestamp": "2024-07-13T10:00:00.000Z"
          }
        },
        // ...
      }
    ]
  }
]
```

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The document is in status final but the SSVC timestamp is newer than the date of newest item in the revision_history.

6.1.50 Product Version Range Rules

For each element of type $['$defs'].branches_t with category of product_version_range, it SHALL be tested that the value of name complies with the rules given in section 3.1.3.3.2. VERS types not supported by the implementation SHALL result in a warning which SHALL include the VERS type name used. Nevertheless, all other rules SHALL be checked to the extent possible.

The relevant path for this test is:

```
$.product_tree..branches[*].name
```

Example 1 (which fails the test):

```
{
  "category": "product_version_range",
  "name": "all versions < 4.2.0",
  // ...
}
```

The version range given does not comply with the rules given in section 3.1.3.3.2.

6.1.51 Inconsistent EPSS Timestamp

For each vulnerability, it SHALL be tested that each EPSS timestamp is earlier than or equal to the date of the newest item of the revision_history if the document status is final or interim. As the timestamps might use different timezones, the sorting and comparison SHALL take timezones into account.

The relevant path for this test is:

```
$.vulnerabilities[*].metrics[*].content.epss.timestamp
```

Example 1 (which fails the test):

```
"document": {
  // ...
  "distribution": {
    "tlp": {
      "label": "CLEAR"
    }
  },
  // ...
  "tracking": {
    // ...
    "revision_history": [
      {
        "date": "2024-01-24T10:00:00.000Z",
        "number": "1",
        "summary": "Initial version."
      }
    ],
    "status": "final",
    // ...
  }
},
"vulnerabilities": [
  {
    "cve": "CVE-1900-0001",
    "metrics": [
      {
        "content": {
          "epss": {
            "percentile": "0.999990000",
            "probability": "0.975570000",
            "timestamp": "2024-07-13T10:00:00.000Z"
          }
        },
        "products": [
```

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```
          "CSAFPID-9080700"
        ]
      }
    ]
  }
]
```

The document is in status final but the EPSS timestamp is newer than the date of newest item in the revision_history.

6.1.52 Inconsistent First Known Exploitation Dates

For each First Known Exploitation Dates item, it SHALL be tested that the values of its date and exploitation_date properties are both earlier than or equal to the date of the newest item of the revision_history if the document status is final or interim. As the timestamps might use different timezones, the sorting and comparison SHALL take timezones into account.

The relevant paths for this test are:

```
$.vulnerabilities[*].first_known_exploitation_dates[*].date
$.vulnerabilities[*].first_known_exploitation_dates[*].exploitation_date
```

Example 1 (which fails the test):

```
"document": {
  // ...
  "tracking": {
    // ...
    "revision_history": [
      {
        "date": "2024-01-24T10:00:00.000Z",
        "number": "1",
        "summary": "Initial version."
      }
    ],
    "status": "final",
    // ...
  }
},
// ...
"vulnerabilities": [
  {
    "first_known_exploitation_dates": [
      {
        "date": "2024-01-24T13:00:00.000Z",
        "exploitation_date": "2024-01-24T12:34:56.789Z",
        // ...
      }
    ]
  }
]
```

The document is in status final but the values of the properties date and exploitation_date of the First Known Exploitation Dates item are newer than the date of newest item in the revision_history.

6.1.53 Inconsistent Exploitation Date

For each First Known Exploitation Dates item, it SHALL be tested that the value of exploitation_date is earlier than or equal to value of the sibling element date. As the timestamps might use different timezones, the comparison SHALL take timezones into account.

The relevant path for this test is:

```
$.vulnerabilities[*].first_known_exploitation_dates[*].exploitation_date
```

Example 1 (which fails the test):

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```
"first_known_exploitation_dates": [
  {
    "date": "2024-01-24T10:00:00.000Z",
    "exploitation_date": "2024-02-25T11:00:22.987Z",
    // ...
  }
]
```

The value of exploitation_date is newer than the value of date in the same item.

6.1.54 License Expression

It SHALL be tested that the license expression is valid.

To implement this test, it is deemed sufficient to check for the ABNF defined in annex B of [SPDX31] and the restriction on the DocumentRef part and case sensitivity given in 3.2.2.8. Although a match against the SPDX License List (license-id) or a specific LicenseRef or AdditionalRef list is not in scope for this test, it checks all other constraints given in the ABNF.

The relevant path for this test is:

```
$.document.license_expression
```

Example 1 (which fails the test):

```
  "license_expression": "This is a license text that should not be here.",
```

The license expression contains a license text instead of a SPDX license expression.

A tool MAY provide a function to fix case errors as a quick fix, if applicable.

6.1.55 License Text

If the document language is English or unspecified, and the license_expression contains license identifiers or exceptions that are not listed in the SPDX license list or AboutCode's "ScanCode LicenseDB", it SHALL be tested that exactly one item in document notes exists that has the title License. The category of this item SHALL be legal_disclaimer.

The relevant path for this test is:

```
$.document.notes
```

Example 1 (which fails the test):

```
"document": {
  // ...
  "license_expression": "LicenseRef-www.example.com-no-work-pd",
  "notes": [
    {
      "category": "other",
      "text": "This is not a work and therefore it can't be licensed. Use it as public domain.",
      "title": "License"
    }
  ],
  // ...
}
```

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The note has the correct title. However, it uses the wrong category.

6.1.56 Use of CVSS and Qualitative Severity Rating

For each item in $.vulnerabilities it SHALL be tested that no Qualitative Severity Rating and CVSS values are listed for the tuple of Product ID and source.

Different source might assign different metrics for the same product.

The relevant path for this test is:

```
$.vulnerabilities[*].metrics[*]
```

Example 1 (which fails the test):

```
"product_tree": {
  "full_product_names": [
    {
      "product_id": "CSAFPID-9080700",
      "name": "Product A"
    }
  ]
},
"vulnerabilities": [
  {
    "metrics": [
      {
        "content": {
          "cvss_v3": {
            // ...
          }
        },
        "products": [
          "CSAFPID-9080700"
        ]
      },
      {
        "content": {
          "qualitative_severity_rating": "critical"
        },
        "products": [
          "CSAFPID-9080700"
        ]
      }
    ]
  }
]
```

A CVSS v3.1 score and a Qualitative Severity Rating is given for CSAFPID-9080700 by the document author.

A tool MAY remove the qualitative_severity_rating as a quick fix.

6.1.57 Stacked Branch Categories

For each full_product_name_t element under $.product_tree.branches, it SHALL be tested that branch categories used along the path leading to the full_product_name_t element appear only once in the path. The sole exception is product_family which can occur multiple times. Therefore, product_family SHALL be ignored in the test.

The relevant path for this test is:

```
$.product_tree..branches[*].category
```

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Example 1 (which fails the test):

```
"product_tree": {
  "branches": [
    {
      "branches": [
        {
          // ...
          "category": "vendor",
          "name": "ISDuBA"
        }
      ],
      "category": "vendor",
      "name": "Open Source"
    }
  ]
}
```

The category vendor was used twice along the path.

A tool MAY collapse the stacked branch categories as a quick fix, if applicable.

6.1.58 Use of product_version in one Path with product_version_range

For each full_product_name_t element under $.product_tree.branches, it SHALL be tested that only one of the branch categories product_version and product_version_range is used along the path leading to the full_product_name_t element.

The relevant path for this test is:

```
$.product_tree..branches[*].category
```

Example 1 (which fails the test):

```
{
  "branches": [
    {
      "category": "product_version_range",
      "name": "vers:intdot/<1.1.1",
      "product": {
        // ...
      }
    }
  ],
  "category": "product_version",
  "name": "1.1"
},
```

Both categories product_version and product_version_range were used along the path.

A tool MAY convert the product_version into the product_name element which contains the value of the product_version appended to the original product name as a quick fix, if applicable. In such conversion, a tool MAY convert a previously existing product_name element into a product_family, if applicable.

6.1.59 Single Version as Product Version Range

For each element of type $['$defs'].branches_t with category of product_version_range, it SHALL be tested that the value of name does not identify only a single version.

The relevant path for this test is:

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```
$.product_tree..branches[*].name
```

Example 1 (which fails the test):

```
{
  "category": "product_version_range",
  "name": "vers:intdot/4.2.0",
  // ...
}
```

The version range given identifies just a single version.

A tool MAY change the branch category to product_version and replace the value for name with the version as a quick fix.

6.1.60 Extension Tests

This subsubsection structures the mandatory tests for extensions. Each of the following tests SHOULD be treated as they were listed similar to the other tests.

An application MAY group these tests when providing the additional function to only run one or more selected tests.

6.1.60.1 Content Schema

For each item in an element of type $['$defs'].extensions_t it SHALL be tested that the item is valid against the Extension Content Schema.

The relevant paths for this test are:

```
$.document.x_extensions[*]
$.product_tree..branches[*].product.x_extensions[*]
$.product_tree.full_product_names[*].x_extensions[*]
$.product_tree.product_paths[*].full_product_name.x_extensions[*]
$.vulnerabilities[*].metrics[*].content.x_extensions[*]
$.vulnerabilities[*].x_extensions[*]
$.x_extensions[*]
```

Example 1 (which fails the test):

```
"x_extensions": [
  {
    "$schema": "https://raw.githubusercontent.com/oasis-tcs/csaf/refs/heads/master/csaf_2.1/extension/data/invalid/documentation-01/documentation-01-content_1.0.0.
json",
    "category": "significant",
    "content": {
      "documentation": "This extension is for documentation and test purposed only. It is invalid as it misses the `critical` property. It is not allowed to be used in a
production CSAF."
    }
  }
]
```

The extension is missing the required property critical.

6.1.60.2 Extension Schema

For each item in an element of type $['$defs'].extensions_t it SHALL be tested that the item is valid against the declared CSAF Extension Schema. CSAF Extensions not supported by the implementation SHALL result in a warning which SHALL include the value of $schema of the extension. Such warning SHALL differentiate between the different classes of extensions.

The relevant paths for this test are:

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```
$.document.x_extensions[*]
$.product_tree..branches[*].product.x_extensions[*]
$.product_tree.full_product_names[*].x_extensions[*]
$.product_tree.product_paths[*].full_product_name.x_extensions[*]
$.vulnerabilities[*].metrics[*].content.x_extensions[*]
$.vulnerabilities[*].x_extensions[*]
$.x_extensions[*]
```

Example 1 (which fails the test):

```
"x_extensions": [
  {
    "$schema": "https://raw.githubusercontent.com/oasis-tcs/csaf/refs/heads/master/csaf_2.1/extension/data/valid/documentation-12/documentation-12-content_1.0.0.json",
    // ...
    "content": {
      "notes": [
        "This instance is invalid even though it validates against the CSAF Content Schema.",
        "It misses the property `documentation` as required by the declared schema."
      ]
    },
    // ...
  }
]
```

The extension is missing the property documentation which is required by the declared CSAF Extension Schema.

6.1.60.3 Extension Metadata

For each element of type $['$defs'].extensions_t it SHALL be tested that the requirements provided through its metadata are fulfilled for each CSAF Extension used. CSAF Extensions not supported by the implementation SHALL result in a warning which SHALL include the value of $schema of the extension. Such warning SHALL differentiate between the different classes of extensions.

This includes, but is not limited to:

- The extension is allowed at this path at all.
- The extension is not used more often than allowed.
- The extension is not used with other incompatible extensions.

The relevant paths for this test are:

```
$.document.x_extensions
$.product_tree..branches[*].product.x_extensions
$.product_tree.full_product_names[*].x_extensions
$.product_tree.product_paths[*].full_product_name.x_extensions
$.vulnerabilities[*].metrics[*].content.x_extensions
$.vulnerabilities[*].x_extensions
$.x_extensions
```

Example 1 (which fails the test):

```
{
  // ...
  "x_extensions": [
    {
      "$schema": "https://raw.githubusercontent.com/oasis-tcs/csaf/refs/heads/master/csaf_2.1/extension/data/valid/documentation-11/documentation-11-content_1.0.0.
json",
      "category": "significant",
      // ...
    },
    {
      "$schema": "https://raw.githubusercontent.com/oasis-tcs/csaf/refs/heads/master/csaf_2.1/extension/data/valid/documentation-11/documentation-11-content_1.0.0.
json",
      "category": "supplementary",
```

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```
      // ...
    }
  ]
}
```

The extension is only allowed once in the path $.x_extensions.

6.1.61 Use of Multiple Stars in SKU

For each stock keeping unit it SHALL be tested that it does not contain multiple unescaped stars.

Multiple * that match zero or multiple characters within a model number introduce ambiguity and are therefore prohibited.

The relevant paths for this test are:

```
$.product_tree..branches[*].product.product_identification_helper.skus[*]
$.product_tree.full_product_names[*].product_identification_helper.skus[*]
$.product_tree.product_paths[*].full_product_name.product_identification_helper.skus[*]
```

Example 1 (which fails the test):

```
        "skus": [
          "NL*12*"
        ]
```

The stock keeping unit contains two unescaped stars.

6.1.62 Inconsistent Discovery Date

For each vulnerability, it SHALL be tested that the discovery_date is earlier than or equal to the date of the newest item of the revision_history if the document status is final or interim. Also, the discovery_date SHALL be earlier than or equal to the disclosure_date of the same vulnerability.

As the timestamps might use different timezones, the sorting and comparison SHALL take timezones into account.

The relevant path for this test is:

```
$.vulnerabilities[*].discovery_date
```

Example 1 (which fails the test):

```
"document": {
  // ...
  "tracking": {
    // ...
    "revision_history": [
      {
        "date": "2024-01-24T10:00:00.000Z",
        "number": "1",
        "summary": "Initial version."
      }
    ],
    "status": "final",
    // ...
  }
},
"vulnerabilities": [
  {
    "discovery_date": "2024-02-24T10:00:00.000Z"
  }
]
```

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The document is in status final but the discovery_date is newer than the date of newest item in the revision_history.

6.1.63 Inconsistent Hash Value Length

For each file hash object, it SHALL be tested that the value length aligns with the algorithm. The test SHALL be skipped for algorithms with variable length output. Algorithms not supported by the implementation SHALL result in a warning which SHALL include the value of algorithm. The warning SHALL differentiate between the values mentioned in section 3.1.10.3.2 and those not mentioned there.

The relevant paths for this test are:

```
$.product_tree..branches[*].product.product_identification_helper.hashes[*].file_hashes[*]
$.product_tree.full_product_names[*].product_identification_helper.hashes[*].file_hashes[*]
$.product_tree.product_paths[*].full_product_name.product_identification_helper.hashes[*].file_hashes[*]
```

Example 1 (which fails the test):

```
"product_tree": {
  "full_product_names": [
    {
      "name": "Product A",
      "product_id": "CSAFPID-9080700",
      "product_identification_helper": {
        "hashes": [
          {
            "file_hashes": [
              {
                "algorithm": "sha256",
                "value": "026a37919b182ef7c63791e82c9645e2f897a3f0b73c7a6028c7febf62e93838d0143"
              }
            ],
            "filename": "product_a.so"
          }
        ]
      }
    }
  ]
}
```

The hash claims to be an MD4 but its length (69 characters) is longer than the expected length (64 characters).

6.1.64 Involvement Tests

This subsubsection structures the mandatory tests for $.document.involvement. Each of the following tests SHOULD be treated as they were listed similar to the other tests.

An application MAY group these tests when providing the additional function to only run one or more selected tests.

6.1.64.1 Missing Receiving Entity Refs

For each item in $.document.involvement.actions which has one of the categories confirmation, coordination, dispute or notification it SHALL be tested that the element receiving_entity_refs exists.

The relevant path for this test is:

```
$.document.involvement.actions[*]
```

Example 1 (which fails the test):

```
{
  "acting_entity_refs": [
    "EID-0001"
  ],
  "action_id": "ACN-1",
```

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```
  "category": "notification",
  "date": "2024-01-10T09:00:00Z",
  "status": "completed",
  "summary": "Reported vulnerability to vendor."
},
```

The action does not contain the element receiving_entity_refs.

6.1.64.2 Contradicting Definition in Actions

It SHALL be tested that items of the list of actions do not contradict each other.

Ignoring the action_id, two items in the list of actions contradict each other, if:

1. all of the following fields contain the same value:
   - category
   - date
2. and all of the following sets contain the same values:
   - acting_entity_refs
   - dl_vuln_ids
   - group_ids
   - product_ids
   - receiving_entity_refs
   - referenced_action_ids
3. but have different values for status.

A dl_vuln_ids mentioning all vulnerabilities in the document is equivalent to the element missing. All groups have to be expanded to their definitions for the test. As the timestamps might use different timezones, the comparison SHALL take timezones into account.

If the two dates are different, the test does not fail because status changes along the timeline are expected.

The relevant path for this test is:

```
$.document.involvement.actions
```

Example 1 (which fails the test):

```
"actions": [
  {
    "acting_entity_refs": [
      "EID-0001"
    ],
    "action_id": "ACN-1",
    "category": "notification",
    "date": "2024-01-10T09:00:00Z",
    "receiving_entity_refs": [
      "EID-0002"
    ],
    "status": "discontinued",
    "summary": "Did not report vulnerability to vendor."
  },
  {
    "acting_entity_refs": [
      "EID-0001"
    ],
    "action_id": "ACN-2",
    "category": "notification",
    "date": "2024-01-10T09:00:00Z",
    "receiving_entity_refs": [
      "EID-0002"
    ],
    "status": "completed",
```

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```
    "summary": "Reported vulnerability to vendor."
  }
],
```

The list of actions contains two items with contradicting content.

6.1.64.3 Missing Recipient on Attempt

For each item in $.document.involvement.actions with the status value of attempted it SHALL be tested that the element receiving_entity_refs exists.

The relevant path for this test is:

```
$.document.involvement.actions[*]
```

Example 1 (which fails the test):

```
{
  "acting_entity_refs": [
    "EID-0001"
  ],
  "action_id": "ACN-1",
  "category": "notification",
  "date": "2024-01-10T09:00:00Z",
  "status": "attempted",
  "summary": "Tried to report vulnerability to vendor."
}
```

The action does not contain the element receiving_entity_refs.

6.1.64.4 Actor and Recipient

For each item in $.document.involvement.actions it SHALL be tested that an entity is not actor and recipient. This takes indirect relations through entity groups into account.

The relevant path for this test is:

```
$.document.involvement.actions[*]
```

Example 1 (which fails the test):

```
{
  "acting_entity_refs": [
    "EID-0001"
  ],
  "action_id": "ACN-1",
  "category": "notification",
  "date": "2024-01-10T09:00:00Z",
  "receiving_entity_refs": [
    "EID-0001"
  ],
  "status": "completed",
  "summary": "Reported vulnerability to vendor."
},
```

The action contains EID-0001 as acting and receiving entity.

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CSAF focuses on the exchange of information. Consequently, for vulnerabilities found internal, i.e. a vendor finds a vulnerability in its own product, either the discoverer part is listed independent of the vendor part which results in two entities or action is omitted as no information exchange happened, i.e. both parts are the same.

6.1.64.5 Missing Definition of Entity ID

For each element of type $['$defs'].entity_id_t which is not inside the list of entities ($.document.involvement.entities) and therefore references an item within the list of entities it SHALL be tested that the Entity element with the matching entity_id exists. This test SHALL ignore elements of type $['$defs'].entity_group_id_t in items of elements of type $['$defs'].entity_refs_id_t.

The relevant paths for this test are:

```
$.document.involvement.actions[*].acting_entity_refs[*]
$.document.involvement.actions[*].receiving_entity_refs[*]
$.document.involvement.entity_groups[*].entity_ids[*]
```

Example 1 (which fails the test):

```
"involvement": {
  "actions": [
    {
      "acting_entity_refs": [
        "EID-0001"
      ],
      "action_id": "ACN-1",
      "category": "notification",
      "date": "2024-01-10T09:00:00Z",
      "receiving_entity_refs": [
        "EID-0002"
      ],
      "status": "completed",
      "summary": "Reported vulnerability to vendor."
    },
    // ...
  ],
  "entities": [
    {
      "category": "discoverer",
      "entity_id": "EID-0001",
      "name": "Jane Researcher"
    }
  ]
},
```

The action ACN-1 contains EID-0002 as receiving entity which is not defined in the list of entities.

6.1.64.6 Multiple Definition of Entity ID

For each Entity ID (type: $['$defs'].entity_id_t) in Entity elements ($.document.involvement.entities[*]) it SHALL be tested that the entity_id was not already defined within the same document.

The relevant path for this test is:

```
$.document.involvement.entities[*].entity_id
```

Example 1 (which fails the test):

```
"entities": [
  {
    "category": "discoverer",
    "entity_id": "EID-0001",
    "name": "Jane Researcher"
  },
  // ...
  {
    "category": "discoverer",
    "entity_id": "EID-0001",
```

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```
    "name": "Joe Researcher"
  }
]
```

EID-0001 was defined twice.

6.1.64.7 Missing Definition of Entity Group ID

For each element of type $['$defs'].entity_group_id_t which is not inside the list of entity groups ($.document.involvement.entity_groups) and therefore references an item within the list of entity groups it SHALL be tested that the Entity Group element with the matching entity_group_id exists. This test SHALL ignore elements of type $['$defs'].entity_id_t in items of elements of type $['$defs'].entity_refs_id_t.

The relevant paths for this test are:

```
$.document.involvement.actions[*].acting_entity_refs[*]
$.document.involvement.actions[*].receiving_entity_refs[*]
```

Example 1 (which fails the test):

```
"involvement": {
  "actions": [
    {
      "acting_entity_refs": [
        "EGID-9011"
      ],
      "action_id": "ACN-1",
      "category": "notification",
      "date": "2024-01-10T09:00:00Z",
      "receiving_entity_refs": [
        "EID-0002"
      ],
      "status": "completed",
      "summary": "Reported vulnerability to vendor."
    },
    // ...
  ],
  // ...
  "entity_groups": [
    {
      "entity_group_id": "EGID-9001",
      "entity_ids": [
        "EID-0001",
        "EID-0003"
      ],
      "name": "JJ Research"
    }
  ]
},
```

The action ACN-1 contains EGID-9011 as acting entity which is not defined in the list of entity groups.

6.1.64.8 Multiple Definition of Entity Group ID

For each Entity Group ID (type: $['$defs'].entity_group_id_t) in Entity Group elements ($.document.involvement.entity_groups[*]) it SHALL be tested that the entity_group_id was not already defined within the same document.

The relevant path for this test is:

```
$.document.involvement.entity_groups[*].entity_group_id
```

Example 1 (which fails the test):

```
"entity_groups": [
  {
```

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```
    "entity_group_id": "EGID-9001",
    "entity_ids": [
      "EID-0001",
      "EID-0003"
    ],
    "name": "JJ Research"
  },
  {
    "entity_group_id": "EGID-9001",
    "entity_ids": [
      "EID-0004",
      "EID-0005"
    ],
    "name": "J+J Research"
  }
]
```

EGID-9001 was defined twice.

6.1.64.9 Missing Definition of Action ID

For each element of type $['$defs'].action_id_t which is not defining an action ($.document.involvement.actions[*].action_id) and therefore references an action it SHALL be tested that the Action element with the matching action_id exists.

The relevant path for this test is:

```
$.document.involvement.actions[*].referenced_action_ids[*]
```

Example 1 (which fails the test):

```
"actions": [
  {
    "acting_entity_refs": [
      "EID-0002"
    ],
    "action_id": "ACN-2",
    "category": "confirmation",
    "date": "2024-01-10T15:00:00Z",
    "receiving_entity_refs": [
      "EID-0001"
    ],
    "referenced_action_ids": [
      "ACN-1"
    ],
    "status": "completed",
    "summary": "Confirmed receipt of the report."
  }
],
```

The action ACN-2 references action ACN-1 which is not defined in the list of actions.

6.1.64.10 Multiple Definition of Action ID

For each element of type $['$defs'].action_id_t which is defining an action ($.document.involvement.actions[*].action_id) it SHALL be tested that the action_id was not already defined within the same document.

The relevant path for this test is:

```
$.document.involvement.actions[*].action_id
```

Example 1 (which fails the test):

```
"actions": [
  {
    // ...
    "action_id": "ACN-1",
```

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```
    // ...
  },
  {
    // ...
    "action_id": "ACN-1",
    // ...
  },
  // ...
],
```

ACN-1 was defined twice.

6.1.64.11 Circular Reference of Action ID

For each action containing at least one referenced action id (type $['$defs'].action_id_t) it SHALL be tested that the action does not end up in a circle. This also includes actions containing their own action_id in the referenced_action_ids.

The relevant path for this test is:

```
$.document.involvement.actions[*]
```

Example 1 (which fails the test):

```
{
  // ...
  "action_id": "ACN-1",
  // ...
  "referenced_action_ids": [
    "ACN-1"
  ],
  // ...
}
```

ACN-1 refers to itself - this is a circular reference.

6.1.64.12 Confirmation from Unmentioned Entity

For each item in $.document.involvement.actions with the category value of confirmation it SHALL be tested that all acting entities are mentioned as receiving entities in each action referenced through referenced_action_ids. This takes indirect relations through entity groups into account.

The relevant path for this test is:

```
$.document.involvement.actions[*]
```

Example 1 (which fails the test):

```
"actions": [
  {
    // ...
    "action_id": "ACN-1",
    // ...
    "receiving_entity_refs": [
      "EID-0002"
    ],
    // ...
  },
  // ...
  {
    "acting_entity_refs": [
      "EID-0003"
    ],
    "action_id": "ACN-3",
    "category": "confirmation",
    // ...
    "referenced_action_ids": [
```

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```
      "ACN-1"
    ],
    // ...
  }
],
```

EID-0003 confirms in ACN-3 receiving action ACN-1. However, ACN-1 does not list EID-0003 as receiving entity of this action.

If an entity receives an unsolicited confirmation, it is likely that the acting entity has received the corresponding information through a third party, e.g. a notification was forwarded. However, such a representation could also result from mistake while authoring the CSAF document. To maintain consistency, a confirmation from unmentioned entity is not allowed to be denoted as confirmation but SHALL be represented as notification.

6.1.64.13 Missing Referenced Action IDs

For each item in $.document.involvement.actions which has one of the categories confirmation or dispute it SHALL be tested that the element referenced_action_ids exists.

The relevant path for this test is:

```
$.document.involvement.actions[*]
```

Example 1 (which fails the test):

```
{
  "acting_entity_refs": [
    "EID-0002"
  ],
  "action_id": "ACN-2",
  "category": "confirmation",
  "date": "2024-01-10T15:00:00Z",
  "receiving_entity_refs": [
    "EID-0001"
  ],
  "status": "completed",
  "summary": "Confirmed receipt of the report."
}
```

The action does not contain the element referenced_action_ids.

6.1.65 Document-local Vuln ID Tests

This subsubsection structures the mandatory tests for Document-local Vuln ID. Each of the following tests SHOULD be treated as they were listed similar to the other tests.

An application MAY group these tests when providing the additional function to only run one or more selected tests.

6.1.65.1 Missing Definition of Document-local Vuln ID

For each element of type $['$defs'].dl_vuln_id_t which is not inside a vulnerability ($.vulnerabilities[*].dl_vuln_id) and therefore references a vulnerability it SHALL be tested that the vulnerability element with the matching dl_vuln_id exists.

The relevant path for this test is:

```
$.document.involvement.actions[*].dl_vuln_ids[*]
```

Example 1 (which fails the test):

```
{
  // ...
```

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```
"document": {
  // ...
  "involvement": {
    "actions": [
      {
        // ...
        "dl_vuln_ids": [
          "VULN-1"
        ],
        // ...
      },
      // ...
    ],
    // ...
  },
  // ...
},
"vulnerabilities": [
  {
    "title": "A vulnerability without a document-local vuln ID."
  }
]
}
```

VULN-1 is not defined in the list of vulnerabilities.

6.1.65.2 Multiple Definition of Document-local Vuln ID

For each element of type $['$defs'].dl_vuln_id_t in vulnerabilities ($.vulnerabilities[*].dl_vuln_id) it SHALL be tested that the dl_vuln_id was not already defined within the same document.

The relevant path for this test is:

```
$.vulnerabilities[*].dl_vuln_id
```

Example 1 (which fails the test):

```
"vulnerabilities": [
  {
    "dl_vuln_id": "VULN-1"
  },
  {
    "dl_vuln_id": "VULN-1"
  }
]
```

VULN-1 was defined twice.

6.1.66 Non-Draft Initial Release Date

It SHALL be tested that the value of $.document.tracking.initial_release_date is equal to the value of date of the revision history item that has a number of 1 or 1.0.0 when the document status is final or interim. As the timestamps might use different timezones, the comparison SHALL take timezones into account. The test SHALL fail with a different error message if the date of the revision history item that has a number of 1 or 1.0.0 cannot be determined.

The relevant path for this test is:

```
$.document.tracking.initial_release_date
```

Example 1 (which fails the test):

```
"tracking": {
  // ...
  "initial_release_date": "2024-01-22T10:00:00.000Z",
  "revision_history": [
```

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```
    // ...
    {
      "date": "2024-01-21T10:00:00.000Z",
      "number": "1",
      "summary": "Initial version."
    }
  ],
  "status": "final",
  // ...
}
```

The document status is final but the revision history item which has 1 as value for number has a different date than given through the initial_release_date.

A tool MAY set the initial_release_date from the date of the revision history item with a number of 1 or 1.0.0 as a quick fix. If the year of the initial_release_date changed during the quick fix this can lead to a situation where a CSAF document moves from one year folder to another (see 7.1.11). Issuing parties are strongly advised to double-check whether this reflects the facts. If that is not the case, the issuing party should correct the mistake by updating the revision history.

6.2 Recommended Tests

Recommended tests SHOULD NOT fail at a valid CSAF document without a good reason. Failing such a test does not make the CSAF document invalid. These tests may include information about features which are still supported but expected to be deprecated in a future version of CSAF. A program SHALL handle a test failure as a warning.

6.2.1 Unused Definition of Product ID

For each Product ID (type $['$defs'].product_id_t) in Full Product Name elements (type: $['$defs'].full_product_name_t) it SHALL be tested that the product_id is referenced somewhere within the same document.

This test SHALL be skipped for CSAF documents conforming the profile “Informational Advisory”.

The relevant paths for this test are:

```
$.product_tree..branches[*].product.product_id
$.product_tree.full_product_names[*].product_id
$.product_tree.product_paths[*].full_product_name.product_id
```

Example 1 (which fails the test):

```
"product_tree": {
  "full_product_names": [
    {
      "product_id": "CSAFPID-9080700",
      "name": "Product A"
    }
  ]
}
```

CSAFPID-9080700 was defined but never used.

A tool MAY remove the unused definition as a quick fix. However, such quick fix shall not be applied if the test was skipped.

6.2.2 Missing Remediation

For each Product ID (type $['$defs'].product_id_t) in the Product Status groups Affected and Under investigation it SHALL be tested that a remediation exists.

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The remediation might be of the category none_available or no_fix_planned.

The relevant paths for this test are:

```
$.vulnerabilities[*].product_status.first_affected[*]
$.vulnerabilities[*].product_status.known_affected[*]
$.vulnerabilities[*].product_status.last_affected[*]
$.vulnerabilities[*].product_status.under_investigation[*]
```

Example 1 (which fails the test):

```
"product_tree": {
  "full_product_names": [
    {
      "product_id": "CSAFPID-9080700",
      "name": "Product A"
    }
  ]
},
"vulnerabilities": [
  {
    "product_status": {
      "last_affected": [
        "CSAFPID-9080700"
      ]
    }
  }
]
```

CSAFPID-9080700 has in Product Status last_affected but there is no remediation object for this Product ID.

6.2.3 Missing Metric

For each Product ID (type $['$defs'].product_id_t) in the Product Status groups Affected it SHALL be tested that a metric object exists which covers this product.

The relevant paths for this test are:

```
$.vulnerabilities[*].product_status.first_affected[*]
$.vulnerabilities[*].product_status.known_affected[*]
$.vulnerabilities[*].product_status.last_affected[*]
```

Example 1 (which fails the test):

```
"product_tree": {
  "full_product_names": [
    {
      "product_id": "CSAFPID-9080700",
      "name": "Product A"
    }
  ]
},
"vulnerabilities": [
  {
    "product_status": {
      "first_affected": [
        "CSAFPID-9080700"
      ]
    }
  }
]
```

CSAFPID-9080700 has in Product Status first_affected but there is no metric object which covers this Product ID.

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6.2.4 Build Metadata in Revision History

For each item in revision history it SHALL be tested that number does not include build metadata.

The relevant path for this test is:

```
$.document.tracking.revision_history[*].number
```

Example 1 (which fails the test):

```
"revision_history": [
  {
    "date": "2023-08-23T10:00:00.000Z",
    "number": "1.0.0+exp.sha.ac00785",
    "summary": "Initial version."
  }
]
```

The revision history contains an item which has a number that includes the build metadata +exp.sha.ac00785.

6.2.5 Older Initial Release Date than Revision History

It SHALL be tested that the Initial Release Date is not older than the date of the oldest item in Revision History. As the timestamps might use different timezones, the sorting and comparison SHALL take timezones into account.

The relevant path for this test is:

```
$.document.tracking.initial_release_date
```

Example 1 (which fails the test):

```
"tracking": {
  // ...
  "initial_release_date": "2023-08-22T10:00:00.000Z",
  "revision_history": [
    {
      "date": "2023-09-06T10:00:00.000Z",
      "number": "1",
      "summary": "Initial version."
    },
    {
      "date": "2024-01-21T11:00:00.000Z",
      "number": "2",
      "summary": "Second version."
    }
  ],
  // ...
}
```

The initial release date 2023-08-22T10:00:00.000Z is older than 2023-09-06T10:00:00.000Z which is the date of the oldest item in Revision History.

6.2.6 Older Current Release Date than Revision History

It SHALL be tested that the Current Release Date is not older than the date of the newest item in Revision History. As the timestamps might use different timezones, the sorting and comparison SHALL take timezones into account.

The relevant path for this test is:

```
$.document.tracking.current_release_date
```

Example 1 (which fails the test):

```
"tracking": {
  "current_release_date": "2023-09-06T10:00:00.000Z",
```

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```
    // ...
    "revision_history": [
      {
        "date": "2023-09-06T10:00:00.000Z",
        "number": "1",
        "summary": "Initial version."
      },
      {
        "date": "2024-01-21T11:00:00.000Z",
        "number": "2",
        "summary": "Second version."
      }
    ],
    // ...
  }
```

The current release date 2023-09-06T10:00:00.000Z is older than 2023-09-23T11:00:00.000Z which is the date of the newest item in Revision History.

6.2.7 Missing Date of the Action (Obsolete)

The date stating when an involvement action occurred (“Date of the action”: `$.document.involvement.actions[*].date) is now required by the schema. Therefore, the recommended test is obsolete. This section is kept to document that change and keep the numbering of the remaining sections stable. The test is excluded from any preset and requirement to be executed.

6.2.8 Use of MD5 As the Only Hash Algorithm

It SHALL be tested that the hash algorithm md5 is not the only one present.

Since collision attacks exist for MD5 such value should be accompanied by a second cryptographically stronger hash. This will allow users to double-check the results.

The relevant paths for this test are:

```
$.product_tree..branches[*].product.product_identification_helper.hashes[*].file_hashes
$.product_tree.full_product_names[*].product_identification_helper.hashes[*].file_hashes
$.product_tree.product_paths[*].full_product_name.product_identification_helper.hashes[*].file_hashes
```

Example 1 (which fails the test):

```
"product_tree": {
  "full_product_names": [
    {
      "name": "Product A",
      "product_id": "CSAFPID-9080700",
      "product_identification_helper": {
        "hashes": [
          {
            "file_hashes": [
              {
                "algorithm": "md5",
                "value": "6ae24620ea9656230f49234efd078935"
              }
            ],
            "filename": "product_a.so"
          }
        ]
      }
    }
  ]
}
```

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The hash algorithm md5 is used in one item of hashes without being accompanied by a second hash algorithm.

6.2.9 Use of SHA-1 As the Only Hash Algorithm

It SHALL be tested that the hash algorithm sha1 is not the only one present.

Since collision attacks exist for SHA-1 such value should be accompanied by a second cryptographically stronger hash. This will allow users to double-check the results.

The relevant paths for this test are:

```
$.product_tree..branches[*].product.product_identification_helper.hashes[*].file_hashes
$.product_tree.full_product_names[*].product_identification_helper.hashes[*].file_hashes
$.product_tree.product_paths[*].full_product_name.product_identification_helper.hashes[*].file_hashes
```

Example 1 (which fails the test):

```
"product_tree": {
  "full_product_names": [
    {
      "name": "Product A",
      "product_id": "CSAFPID-9080700",
      "product_identification_helper": {
        "hashes": [
          {
            "file_hashes": [
              {
                "algorithm": "sha1",
                "value": "e067035314dd8673fe1c9fc6b01414fe0950fdc4"
              }
            ],
            "filename": "product_a.so"
          }
        ]
      }
    }
  ]
}
```

The hash algorithm sha1 is used in one item of hashes without being accompanied by a second hash algorithm.

6.2.10 Missing TLP Label (Obsolete)

The TLP label is now required by the schema. Therefore, the recommended test is obsolete. This section is kept to document that change and keep the numbering of the remaining sections stable. The test is excluded from any preset and requirement to be executed.

6.2.11 Missing Canonical URL

It SHALL be tested that the CSAF document has a canonical URL.

To implement this test it is deemed sufficient that one item in $.document.references fulfills all of the following:

- It has the category self.
- The url starts with https://.
- The url ends with the valid filename for the CSAF document according to the rules in section 5.1 which is preceded by a /.

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- The hostname is not empty.

The relevant path for this test is:

```
$.document.references
```

Example 1 (which fails the test):

```
"document": {
  // ...
  "references": [
    {
      "category": "self",
      "summary": "A non-canonical URL.",
      "url": "https://example.com/security/data/csaf/2024/oasis_csaf_tc-csaf_2.1-2024-6-2-11-01_1.json"
    }
  ],
  // ...
  "tracking": {
    // ...
    "id": "OASIS_CSAF_TC-CSAF_2.1-2024-6-2-11-01",
    // ...
    "version": "1"
  },
  // ...
}
```

The only element where the category is self has a URL that does not fulfill the requirement of a valid filename for a CSAF document.

6.2.12 Missing Document Language

It SHALL be tested that the document language member is present and set. A CSAF validator SHALL differentiate in the error message between the key being present but having no or an empty value and not being present at all.

The relevant path for this test is:

```
$.document.lang
```

Example 1 (which fails the test):

```
"document": {
  "category": "csaf_base",
  "csaf_version": "2.1",
  "distribution": {
    "tlp": {
      "label": "CLEAR"
    }
  },
  "publisher": {
    // ...
  },
  // ...
}
```

The document language is not defined.

A tool MAY add the key as a quick fix. In such case, the value still needs to be set - either manually or by other means from the tool, e.g. through a given configuration.

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6.2.13 Sorting

It SHALL be tested that all keys in a CSAF document are sorted alphabetically.

The relevant path for this test is:

```
$
```

Example 1 (which fails the test):

```
"document": {
  "csaf_version": "2.1",
  "category": "csaf_base",
  // ...
}
```

The key csaf_version is not at the right place.

A tool MAY sort the keys as a quick fix.

6.2.14 Use of Private Language

For each element of type $['$defs'].lang_t it SHALL be tested that the language code does not contain subtags reserved for private use.

The relevant paths for this test are:

```
$.document.lang
$.document.source_lang
```

Example 1 (which fails the test):

```
"lang": "qtx"
```

The language code qtx is reserved for private use.

A tool MAY remove such subtag as a quick fix.

6.2.15 Use of Default Language

For each element of type $['$defs'].lang_t it SHALL be tested that the language code is not i-default.

The relevant paths for this test are:

```
$.document.lang
$.document.source_lang
```

Example 1 (which fails the test):

```
"lang": "i-default"
```

The language code i-default is used.

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A tool MAY remove such element as a quick fix.

6.2.16 Missing Product Identification Helper

For each element of type $['$defs'].full_product_name_t it SHALL be tested that it includes the property product_identification_helper.

The relevant paths for this test are:

```
$.product_tree..branches[*].product
$.product_tree.full_product_names[*]
$.product_tree.product_paths[*].full_product_name
```

Example 1 (which fails the test):

```
"full_product_names": [
  {
    "product_id": "CSAFPID-9080700",
    "name": "Product A"
  }
]
```

The product CSAFPID-9080700 does not provide any Product Identification Helper at all.

6.2.17 CVE in Field IDs

For each item in $.vulnerabilities[*].ids it SHALL be tested that it is not a CVE ID.

To implement this test it is deemed sufficient to check after invisible and white space characters have been removed from the value of the property text whether it matches the regex ^CVE-[0-9]{4}-[0-9]{4,}$.

The relevant path for this test is:

```
$.vulnerabilities[*].ids[*]
```

Example 1 (which fails the test):

```
"ids": [
  {
    "system_name": "CVE Project",
    "text": "CVE-2021-44228"
  }
]
```

The CVE-2021-44228 is listed in an item of the ids array instead under cve.

A tool MAY set such element as value for the cve property as a quick fix, if that didn't exist before. Alternatively, it MAY remove such element as a quick fix.

6.2.18 Product Version Range without VERS

For each element of type $['$defs'].branches_t with category of product_version_range it SHALL be tested that the value of name indicates that the product version range is using VERS.

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Compliance with the VERS specification itself is enforced via test 6.1.50. Unknown VERS types are detected via test 6.2.51.

To implement this test it is deemed sufficient that the value of name matches the following regex:

```
^vers:[a-z\\.\\-\\+][a-z0-9\\.\\-\\+]*/.+
```

The warning SHALL clearly advise to carefully check whether VERS can be used or the versions can be enumerated as the use of vls is only a fallback option. For more details, see sections 3.1.3.2 and 3.1.3.3.2.

It is planned to deprecate and finally remove the support for vls in future versions of this standard.

The relevant path for this test is:

```
$.product_tree..branches[*].name
```

Example 1 (which fails the test):

```
"branches": [
  {
    "category": "product_version_range",
    "name": ">4.2",
    // ...
  }
]
```

The version range >4.2 is a valid vls but not valid according to the VERS specification.

6.2.19 CVSS for Fixed Products

For each item the fixed products group (first_fixed and fixed) it SHALL be tested that a CVSS applying to this product has an overall score of 0. The test SHALL pass if none of the Product IDs listed within product status fixed or first_fixed is found in products of any item of the metrics element.

The relevant paths for this test are:

```
$.vulnerabilities[*].product_status.first_fixed[*]
$.vulnerabilities[*].product_status.fixed[*]
```

Example 1 (which fails the test):

```
"product_tree": {
  "full_product_names": [
    {
      "product_id": "CSAFPID-9080700",
      "name": "Product A"
    }
  ]
},
"vulnerabilities": [
  {
    "product_status": {
      "fixed": [
        "CSAFPID-9080700"
      ]
    },
    "metrics": [
      {
        "content": {
          "cvss_v3": {
            "baseScore": 6.5,
            "baseSeverity": "MEDIUM",
```

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```
          "vectorString": "CVSS:3.1/AV:L/AC:L/PR:H/UI:R/S:U/C:H/I:H/A:H",
          "version": "3.1"
        }
      }          ,
      "products": [
        "CSAFPID-9080700"
      ]
    }
  ]
 }
]
```

Neither the environmentalScore nor the properties modifiedAvailabilityImpact, modifiedConfidentialityImpact, modifiedIntegrityImpact nor the corresponding attributes in the vectorString have been set.

A tool MAY remove any Product IDs listed within product status fixed or first_fixed from products of all items of the metrics element.

Alternatively, a tool MAY set those environmental properties according to the CVSS version used that reduce the overall score to 0 and compute the respective score and severity as a quick fix. The following environmental properties have been identified:

- CVSS v2: targetDistribution to NONE
- CVSS v3: all of modifiedAvailabilityImpact, modifiedConfidentialityImpact, and modifiedIntegrityImpact to NONE
- CVSS v4: all of
  - modifiedVulnAvailabilityImpact, modifiedVulnConfidentialityImpact, and modifiedVulnIntegrityImpact to NONE and
  - modifiedSubAvailabilityImpact, modifiedSubConfidentialityImpact, and modifiedSubIntegrityImpact to NEGLIGIBLE

6.2.20 Additional Properties

It SHALL be tested that there is no additional property in the CSAF document that was not defined in the CSAF JSON schema. This also applies for referenced schemas. For CSAF Extensions, the declared schemas SHALL be checked additionally to the CSAF Extension Content Schema. Additional and unevaluated properties allowed by the CSAF Extension Content Schema SHALL not be reported. CSAF Extensions not supported by the implementation SHALL result in a warning which SHALL include the value of $schema of the extension. Such warning SHALL differentiate between the different classes of extensions.

Some of the errors could be reported by combining the JSON schema check with several mandatory tests, this test provides an one stop solution and can be used as quality gate in CSAF publication pipelines or contracts. Therefore, a potential double reporting is not consider an issue.

The relevant path for this test is:

```
$
```

To implement this test it is deemed sufficient to validate the CSAF document against a “strict” version schema that has all references integrated and sets additionalProperties respectively unevaluatedProperties to false at all appropriate places to detect additional properties.

Example 1 (which fails the test):

```
        "cvss_v3": {
          "baseScore": 6.4,
```

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```
        "baseSeverity": "MEDIUM",
        "custom_property": "any",
        "vectorString": "CVSS:3.1/AV:L/AC:H/PR:L/UI:N/S:C/C:N/I:H/A:L",
        "version": "3.1"
      }
```

The key `custom_property` is not defined in the JSON schema.

A tool MAY remove such keys as a quick fix.

6.2.21 Same Timestamps in Revision History

It SHALL be tested that the timestamps of all items in the revision history are pairwise disjoint. As the timestamps might use different timezones, the comparison SHALL take timezones into account.

The relevant path for this test is:

```
$.document.tracking.revision_history[*].date
```

Example 1 (which fails the test):

```
"revision_history": [
  {
    "date": "2024-01-21T10:00:00.000Z",
    "number": "2.0.0",
    "summary": "Second version."
  },
  {
    "date": "2024-01-21T10:00:00.000Z",
    "number": "1.0.0",
    "summary": "Initial version."
  }
]
```

The first and second revision have the same timestamp.

6.2.22 Document Tracking ID in Title

It SHALL be tested that the `$.document.title` does not contain the `$.document.tracking.id`.

The relevant path for this test is:

```
$.document.title
```

Example 1 (which fails the test):

```
  "title": "OASIS_CSAF_TC-CSAF_2.1-2024-6-2-22-01: Recommended test: Document Tracking ID in Title (failing example 1)",
  "tracking": {
    // ...
    "id": "OASIS_CSAF_TC-CSAF_2.1-2024-6-2-22-01",
    // ...
  }
```

The document title contains the document tracking id.

A tool MAY remove the document tracking id from the document title. It SHOULD also remove any separating characters including white space, colon, dash and brackets.

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6.2.23 Usage of Deprecated CWE

For each item in the CWE array it SHALL be tested that the CWE is not deprecated in the given version.

The relevant path for this test is:

```
$.vulnerabilities[*].cwes[*]
```

Example 1 (which fails the test):

```
"cwes": [
  {
    "id": "CWE-596",
    "name": "DEPRECATED: Incorrect Semantic Object Comparison",
    "version": "4.13"
  }
]
```

The CWE-596 is deprecated in version 4.13.

A tool MAY suggest to replace the deprecated CWE with its replacement or closest equivalent.

6.2.24 Usage of Non-Latest CWE Version

For each item in the CWE array it SHALL be tested that the latest CWE version available at the time of the last revision was used. The test SHALL fail if a later CWE version was used.

The relevant path for this test is:

```
$.vulnerabilities[*].cwes[*]
```

Example 1 (which fails the test):

```
"document": {
  // ...
  "tracking": {
    "current_release_date": "2024-01-21T10:00:00.000Z",
    // ...
  }
},
"vulnerabilities": [
  {
    "cwes": [
      {
        "id": "CWE-256",
        "name": "Plaintext Storage of a Password",
        "version": "4.12"
      }
    ]
  }
]
```

The CWE version listed is 4.12. However, version 4.13 was most recent version when the document was released on 2024-01-21T10:00:00.000Z.

A tool MAY suggest to use the latest version available at the time of the current_release_date. This is most likely also the overall latest CWE version as modifications to a CSAF document lead to a new current_release_date.

6.2.25 Usage of CWE Not Allowed for Vulnerability Mapping

For each item in the CWE array it SHALL be tested that the vulnerability mapping is allowed.

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Currently, this includes the two usage state Allowed and Allowed-with-Review.

Note: The property Usage within the MappingNotesType was introduced in version 7.0 of the CWE schema definition. As a consequence, this information might not be available before CWE version 4.12.

The relevant path for this test is:

```
$.vulnerabilities[*].cwes[*]
```

Example 1 (which fails the test):

```
"cwes": [
  {
    "id": "CWE-20",
    "name": "Improper Input Validation",
    "version": "4.13"
  }
]
```

The usage of CWE-20 is discouraged as "is commonly misused in low-information vulnerability reports when lower-level CWEs could be used instead, or when more details about the vulnerability are available" [CWE-20].

6.2.26 Usage of CWE Allowed with Review for Vulnerability Mapping

For each item in the CWE array it SHALL be tested that the vulnerability mapping is allowed without review.

Reasoning: CWEs marked with a vulnerability mapping state of Allowed-with-Review should only be used if a thorough review was done. This test helps to flag such mappings which can be used to trigger processes that ensure the extra review, e.g. by a senior analyst.

Note: The property Usage within the MappingNotesType was introduced in version 7.0 of the CWE schema definition. As a consequence, this information might not be available before CWE version 4.12.

The test SHALL NOT fail on items that are not allowed at all.

Those are covert through tests 6.1.11 and 6.2.25

The relevant path for this test is:

```
$.vulnerabilities[*].cwes[*]
```

Example 1 (which fails the test):

```
"cwes": [
  {
    "id": "CWE-1023",
    "name": "Incomplete Comparison with Missing Factors",
    "version": "4.13"
  }
]
```

The usage of CWE-1023 is allowed with review as the "CWE entry is a Class and might have Base-level children that would be more appropriate". [CWE-1023]

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6.2.27 Discouraged Product Status Remediation Combination

For each item in $.vulnerabilities[*].remediations, it SHALL be tested that a Product is not member of a discouraged product status group remediation category combination (see table 4). This takes indirect relations through Product Groups into account.

The relevant path for this test is:

```
$.vulnerabilities[*].remediations[*]
```

Example 1 (which fails the test):

```
"product_status": {
  "known_not_affected": [
    "CSAFPID-9080700"
  ]
},
"remediations": [
  {
    "category": "fix_planned",
    "details": "The fix should be available in Q4 2024.",
    "product_ids": [
      "CSAFPID-9080700"
    ]
  }
]
```

For the product with product ID CSAFPID-9080700 a fix is planned but the product was not affected at all.

6.2.28 Usage of Max UUID

It SHALL be tested that the Max UUID is not used as sharing group id.

The relevant path for this test is:

```
$.document.distribution.sharing_group.id
```

Example 1 (which fails the test):

```
"distribution": {
  "sharing_group": {
    "id": "ffffffff-ffff-ffff-ffff-ffffffffffff",
    "name": "Public"
  },
  // ...
},
```

The sharing group id uses the Max UUID.

A tool MAY remove the property sharing_group as a quick fix.

6.2.29 Usage of Nil UUID

It SHALL be tested that the Nil UUID is not used as sharing group id.

The relevant path for this test is:

```
$.document.distribution.sharing_group.id
```

Example 1 (which fails the test):

```
"distribution": {
  "sharing_group": {
```

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```
      "id": "00000000-0000-0000-0000-000000000000",
      "name": "No sharing allowed"
    },
    // ...
  },
```

The sharing group id uses the Nil UUID.

A tool MAY remove the property sharing_group as a quick fix.

6.2.30 Usage of Sharing Group on TLP:CLEAR

It SHALL be tested that no sharing group is used if the document is TLP:CLEAR.

The relevant path for this test is:

```
$.document.distribution.sharing_group
```

Example 1 (which fails the test):

```
  "distribution": {
    "sharing_group": {
      "id": "ffffffff-ffff-ffff-ffff-ffffffffffff",
      "name": "Public"
    },
    "tlp": {
      "label": "CLEAR"
    }
  },
```

The CSAF document is TLP:CLEAR but a sharing group is given.

A tool MAY remove the property sharing_group as a quick fix.

6.2.31 Hardware and Software

For each product containing at least one of the Product Identification Helpers serial_numbers or model_numbers it SHALL be tested that a product path exists referencing this product.

This tests detects a potential situation where hardware and software have been mixed in the product_tree. Note: This test will fail if the CSAF document contains in its product_tree only hardware. However, this is expected and considered a good reason for the test to fail. This does not make the CSAF document invalid.

The relevant paths for this test are:

```
$.product_tree..branches[*].product
$.product_tree.full_product_names[*]
$.product_tree.product_paths[*].full_product_name
```

Example 1 (which fails the test):

```
"product_tree": {
  "branches": [
    {
      "branches": [
        {
```

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```
        "branches": [
          {
            "category": "product_version",
            "name": "4.1",
            "product": {
              "name": "Example Company Controller A Firmware 4.1",
              "product_id": "CSAFPID-908070601",
              "product_identification_helper": {
                "serial_numbers": [
                  "143-D-354"
                ]
              }
            }
          }
        ],
        "category": "product_name",
        "name": "Controller A"
      }
    ],
    "category": "vendor",
    "name": "Example Company"
  }
 ]
}
```

The product_tree mentions the hardware product Example Company Controller A and combines it with the Firmware version 4.1.

A correct representation for hardware and software in the product_tree is given in example [1 (of section 5.6)].

6.2.32 Use of Same Product Identification Helper for Different Products

For each Product Identification Helper category it SHALL be tested that the same value is not used for multiple products in this category.

This test detects a potentially incorrect constructed product tree. Note: This test will fail if the CSAF document contains in its product_tree the old and new name of a product that was renamed. However, this is expected and considered a good reason for the test to fail. This does not make the CSAF document invalid.

For the comparison, arrays need to be treated as unordered sets which need to be pairwise disjoint. When comparing two helper-objects' array properties whose items are themselves JSON objects (e.g. hashes or file_hashes), the array needs to be treated as an unordered set, and each object is compared by deep-equality of its key/value pairs (ignoring key-order).

The relevant paths for this test are:

```
$.product_tree..branches[*].product.product_identification_helper
$.product_tree.full_product_names[*].product_id.product_identification_helper
$.product_tree.product_paths[*].full_product_name.product_id.product_identification_helper
```

Example 1 (which fails the test):

```
"product_tree": {
 "branches": [
  {
   "branches": [
    {
     "branches": [
      {
       "category": "product_version",
       "name": "1.0",
       "product": {
        "name": "Example Company Product A 1.0",
```

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```
            "product_id": "CSAFPID-908070601",
            "product_identification_helper": {
              "serial_numbers": [
                "143-D-354"
              ]
            }
          }
        },
        {
          "category": "product_version",
          "name": "2.0",
          "product": {
            "name": "Example Company Product A 2.0",
            "product_id": "CSAFPID-908070602",
            "product_identification_helper": {
              "serial_numbers": [
                "143-D-354"
              ]
            }
          }
        }
      ],
      "category": "product_name",
      "name": "Product A"
    }
  ],
  "category": "vendor",
  "name": "Example Company"
 }
]
}
```

Both products are identified by the same serial number 143-D-354.

6.2.33 Disclosure Date Newer than Revision History

For each vulnerability, it SHALL be tested that the disclosure_date is earlier than or equal to the date of the newest item of the revision_history if the disclosure_date is in the past at the time of the test execution. As the timestamps might use different timezones, the sorting and comparison SHALL take timezones into account.

The result of the test is dependent upon the time of the execution of the test - it might change for a given CSAF document over time. However, the latest version of a CSAF document should always pass the test.

The relevant path for this test is:

```
$.vulnerabilities[*].disclosure_date
```

Example 1 (which fails the test):

```
"document": {
 // ...
 "distribution": {
  "tlp": {
   "label": "GREEN"
  }
 },
 // ...
 "tracking": {
  "current_release_date": "2024-01-24T10:00:00.000Z",
  // ...
  "initial_release_date": "2024-01-24T10:00:00.000Z",
  "revision_history": [
   {
    "date": "2024-01-24T10:00:00.000Z",
    "number": "1",
    "summary": "Initial version."
   }
  ],
```

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```
    "status": "final",
    "version": "1"
  }
},
"vulnerabilities": [
  {
    "disclosure_date": "2024-02-24T10:00:00.000Z"
  }
]
```

The disclosure_date is in the past but newer than the date of newest item in the revision_history.

6.2.34 Usage of Unknown SSVC Decision Point Base Namespace

For each SSVC decision point given under selections, it SHALL be tested that the base namespace is a registered one. Namespaces reserved for special purpose SHALL be treated as per their definition.

This test fails on unregistered namespaces as well as registered ones not yet supported by the implementation.

To implement this determine the base namespace in this test, it is deemed sufficient to remove any characters including and after the first /.

The relevant path for this test is:

```
$.vulnerabilities[*].metrics[*].content.ssvc_v2.selections[*].namespace
```

Example 1 (which fails the test):

```
"ssvc_v2": {
  "schemaVersion": "2.0.0",
  "selections": [
    {
      // ...
      "namespace": "x_example.unregistered#some-yet-unknown-or-maybe-private-namespace",
      // ...
    }
  ],
  // ...
}
```

The namespace x_example.unregistered#some-yet-unknown-or-maybe-private-namespace is not a registered namespace. Its decision point definitions might therefore not be known to the reader of the document.

6.2.35 Usage of Unregistered SSVC Decision Point Base Namespace in TLP:CLEAR Document

For each SSVC decision point given under selections, it SHALL be tested that the base namespace is not an unregistered one if the document is labeled TLP:CLEAR. Namespaces reserved for special purpose SHALL be treated as per their definition.

To implement this determine the base namespace in this test, it is deemed sufficient to remove any characters including and after the first /.

The relevant path for this test is:

```
$.vulnerabilities[*].metrics[*].content.ssvc_v2.selections[*].namespace
```

Example 1 (which fails the test):

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```
{
  "document": {
    // ...
    "distribution": {
      "tlp": {
        "label": "CLEAR"
      }
    },
    // ...
  }
  "vulnerabilities": [
    {
      "metrics": [
        {
          "content": {
            "ssvc_v2": {
              // ...
              "selections": [
                {
                  // ...
                  "namespace": "x_example.unregistered#namespace",
                  // ...
                }
              ],
              // ...
            }
          },
          // ...
        }
      ]
    }
  ]
}
```

The namespace x_example.unregistered#namespace is an unregistered base namespace. Its decision point definitions might therefore not be known to the reader of the document.

6.2.36 Usage of SSVC Decision Point Namespace with Extension in TLP:CLEAR Document

For each SSVC decision point given under selections, it SHALL be tested that the namespace does not use an extension if the document is labeled TLP:CLEAR. Namespaces reserved for special purpose SHALL be treated as per their definition.

To implement the check for an extensions part in the namespace, it is deemed sufficient to check whether the namespace contains a /.

The relevant path for this test is:

```
$.vulnerabilities[*].metrics[*].content.ssvc_v2.selections[*].namespace
```

Example 1 (which fails the test):

```
{
  "document": {
    // ...
    "distribution": {
      "tlp": {
        "label": "CLEAR"
      }
    },
    // ...
  }
  "vulnerabilities": [
    {
      "metrics": [
        {
          "content": {
          "ssvc_v2": {
            // ...
```

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```
          "selections": [
            {
              // ...
              "namespace": "ssvc//.example.test#refined-technical-impacts",
              // ...
            }
            ],
            // ...
          }
        },
        // ...
      }
    ]
  }
 ]
}
```

The namespace contains the extension .example.test#refined-technical-impacts. Its decision point definitions might therefore not be known to the reader of the document. However, as per SSVC specification, extensions cannot extend an existing decision point with new values. Thus, they can be treated as decision points from the base namespace, if the extension is unknown.

6.2.37 Usage of Unknown SSVC Decision Point Namespace without Resource

For each SSVC object containing a decision point with a full namespace that is not registered, it SHALL be tested that a Decision Point Resource exists for each one that provides additional context about the decision points from this namespace. Namespaces reserved for special purpose SHALL be treated as per their definition.

A full namespace includes any extension. To implement this test it is deemed sufficient to check whether the summary contains the full namespace.

Namespaces reserved for special purpose SHALL be treated as per their definition.

The relevant path for this test is:

```
$.vulnerabilities[*].metrics[*].content.ssvc_v2.decision_point_resources
```

Example 1 (which fails the test):

```
"content": {
  "ssvc_v2": {
    "decision_point_resources": [
      {
        "summary": "Specification for Decision Points within 'x_example.test#some-other-collection'",
        "uri": "https://test.example/ssvc-collection/test-some-other-collection"
      }
    ],
    // ...
    "selections": [
      {
        // ...
        "namespace": "x_example.test#without-resource/de-DE",
        // ...
      }
    ],
    // ...
  }
},
```

The namespace x_example.test#without-resource/de-DE is not a registered one, however no decision_point_resources mentions the namespace in its summary.

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6.2.38 Usage of Deprecated Profile

It SHALL be tested that the $.document.category does not start with csaf_deprecated_.

> To implement this test, it is deemed sufficient to do a "starts with" check. In contrast to test 6.1.26, this test detects the use of a profile defined by CSAF that is deprecated.

The relevant path for this test is:

```
$.document.category
```

Example 1 (which fails the test):

```
"category": "csaf_deprecated_security_advisory",
```

> The document category starts with csaf_deprecated_.

6.2.39 Profile Tests

This subsubsection structures the recommended tests for the profiles. Not all tests apply for all profiles. Tests SHOULD be skipped if the document category does not match the one given in the test. Each of the following tests SHOULD be treated as they were listed similar to the other tests.

6.2.39.1 Missing Fixed Product

For each product listed in the product status group affected in any vulnerability, it SHALL be tested that a corresponding version of the product is listed as fixed in the same vulnerability. The test SHALL be skipped if there is a clear indication, that such a version of the product does not exist. Indicators include a remediation item with one of the categories fix_planned, no_fix_planned or none_available referring to the affected product. The test SHALL NOT be skipped, if there is an indication, that such a version of the product might exist. Indicators include an affected product version range with the comparator < in the last version constraint and a remediation item with the categories vendor_fix referring to the affected product.

The relevant value for $.document.category is:

```
csaf_security_advisory
```

The relevant path for this test is:

```
$.vulnerabilities[*].product_status
```

Example 1 (which fails the test):

```
"vulnerabilities": [
  {
    // ...
    "product_status": {
      "known_affected": [
        "CSAFPID-9080700"
      ]
    },
    "remediations": [
      {
        "category": "vendor_fix",
        "details": "Update to the latest version, at least version 4.2.",
        "product_ids": [
          "CSAFPID-9080700"
        ]
      }
    ]
  }
]
```

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The fixed product is not listed in the advisory but there is a clear indication that such product exists as there is a remediation with category vendor_fix.

A tool MAY create the missing fixed product based on the data available in the advisory as a quick fix.

6.2.39.2 Language Specific Reasoning for Withdrawal

If the document language is specified but not English, it SHALL be tested that exactly one item in document notes exists that has the language specific translation of the term Reasoning for Withdrawal as title. The category of this item SHALL be description. If no language specific translation has been recorded, the test SHALL be skipped and output an information to the user that no such translation is known.

A list of the language specific translations is kept at the OASIS CSAF TC.

The relevant value for $.document.category is:

```
csaf_withdrawn
```

The relevant path for this test is:

```
$.document.notes
```

Example 1 (which fails the test):

```
"notes": [
  {
    "category": "summary",
    "text": "Das CSAF Document enthielt Beispieldaten und wurde zurückgezogen, um Testdaten zu erzeugen.",
    "title": "Begründung für die Zurückziehung"
  }
],
```

The note has the correct title. However, it uses the wrong category.

6.2.39.3 Language Specific Reasoning for Supersession

If the document language is specified but not English, it SHALL be tested that exactly one item in document notes exists that has the language specific translation of the term Reasoning for Supersession as title. The category of this item SHALL be description. If no language specific translation has been recorded, the test SHALL be skipped and output an information to the user that no such translation is known.

A list of the language specific translations is kept at the OASIS CSAF TC.

The relevant value for $.document.category is:

```
csaf_superseded
```

The relevant path for this test is:

```
$.document.notes
```

Example 1 (which fails the test):

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```
  "notes": [
    {
      "category": "summary",
      "text": "Das CSAF Dokument enthielt Beispieldaten und wurde ersetzt, um Testdaten zu erzeugen.",
      "title": "Begründung für die Ersetzung"
    }
  ],
```

The note has the correct title. However, it uses the wrong category.

6.2.39.4 Language Specific Superseding Document

If the document language is specified but not English, it SHALL be tested that at least one item in document references exists that starts with the language specific translation of the term Superseding Document as summary. For each of these items, the category SHALL be present and have the value external. If no language specific translation has been recorded, the test SHALL be skipped and output an information to the user that no such translation is known.

A list of the language specific translations is kept at the OASIS CSAF TC.

The relevant value for $.document.category is:

```
csaf_superseded
```

The relevant path for this test is:

```
$.document.references
```

Example 1 (which fails the test):

```
  "references": [
    {
      "category": "external",
      "summary": "Ersetztes Dokument",
      "url": "https://example.com/.well-known/csaf/clear/2024/esa-2024-1234.json"
    }
  ],
```

There is no reference where the summary starts with the correct term Ersetzendes Dokument.

6.2.39.5 Extension in Superseded or Withdrawn Document

It SHALL be tested that the document does not contain an extension.

The relevant values for $.document.category are:

```
csaf_withdrawn
csaf_superseded
```

The relevant paths for this test are:

```
$.document.x_extensions
$.x_extensions
```

Example 1 (which fails the test):

```
{
  // ...
  "x_extensions": [
    {
```

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```
      "$schema": "https://raw.githubusercontent.com/oasis-tcs/csaf/refs/heads/master/csaf_2.1/extension/data/valid/documentation-11/documentation-11-content_1.0.0.
json",
      // ...
    }
  ]
}
```

The document contains a CSAF Extension.

6.2.39.6 Language Specific Vulnerability Notes

If the document language is specified but not English, it SHALL be tested that at least one item in vulnerability notes exist that has the language specific translation of the term Vulnerability Summary or CVE Description as title. The category of this item SHALL be consistent with the value required in table 2 of section 3.2.4.11. If no language specific translation has been recorded, the test SHALL be skipped and output an information to the user that no such translation is known.

A list of the language specific translations is kept at the OASIS CSAF TC.

The relevant value for $.document.category is:

```
csaf_vulnerability_report
```

The relevant path for this test is:

```
$.vulnerabilities[*].notes
```

Example 1 (which fails the test):

```
  "notes": [
    {
      "category": "summary",
      // ...
      "title": "CVE Beschreibung"
    }
  ],
```

The note has the correct title. However, it uses the wrong category.

6.2.39.7 Document Acknowledgments

It SHALL be tested that the element $.document.acknowledgments exists.

Public acknowledgment helps ensure that discoverers and contributing organizations receive appropriate credit after coordinated disclosure. Omitting the acknowledgments element indicates that the discoverer does not wish to receive public acknowledgment following disclosure. In such case, the test fails by design. Discoverers that do not wish to receive public acknowledgment MAY safely ignore the test.

The relevant value for $.document.category is:

```
csaf_vulnerability_report
```

The relevant path for this test is:

```
$.document.acknowledgments
```

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Example 1 (which fails the test):

```
"document": {
  "category": "csaf_vulnerability_report"
  // ...
}
```

The element $.document.acknowledgments does not exist.

Recommendation:

It is recommended to add $.document.acknowledgments so the vulnerability report identifies who should receive public acknowledgment and provides a suggested phrasing for that. For an example, see [1 (of section 3.1.1.5)] in section 3.1.1.5.

6.2.39.8 Aggregate Severity

It SHALL be tested that the element $.document.aggregate_severity exists.

An aggregate severity helps recipients quickly understand the overall urgency of the vulnerability report during coordination.

The relevant value for $.document.category is:

```
csaf_vulnerability_report
```

The relevant path for this test is:

```
$.document.aggregate_severity
```

Example 1 (which fails the test):

```
"document": {
  "category": "csaf_vulnerability_report",
  // ...
}
```

The element $.document.aggregate_severity does not exist.

A tool MAY add the highest value of all CVSS Severity values of all vulnerabilities as quick fix.

6.2.39.9 TLP-Label for Vulnerability Report

It SHALL be tested that the TLP label is set to AMBER.

The vulnerability report profile recommends TLP:AMBER as reasonable default so the report can be shared on a need-to-know basis during private coordination. Note that there are cases, where the document distribution needs to be limited more (e.g. TLP:AMBER+STRICT). Also, there are cases, where there is no added value in limiting the distribution to TLP:AMBER. For example, if the report contains only already public information as a full disclosure happened before.

The relevant value for $.document.category is:

```
csaf_vulnerability_report
```

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The relevant path for this test is:

```
$.document.distribution.tlp.label
```

Example 1 (which fails the test):

```
"distribution": {
  "tlp": {
    "label": "RED"
  }
}
```

The vulnerability report uses a TLP label other than the recommended value AMBER.

A tool MAY set the TLP label to AMBER as a quick fix.

6.2.39.10 Vulnerability Metrics

It SHALL be tested that $.vulnerabilities[*].metrics exists.

Metrics such as CVSS, EPSS, or SSVC help recipients evaluate technical severity and prioritize their response.

The relevant value for $.document.category is:

```
csaf_vulnerability_report
```

The relevant path for this test is:

```
$.vulnerabilities[*].metrics
```

Example 1 (which fails the test):

```
"vulnerabilities": [
  {
    "cve": "CVE-1900-0001"
  }
]
```

The vulnerability does not have a metrics element.

6.2.39.11 Vulnerability References

It SHALL be tested that $.vulnerabilities[*].references exists.

Public references help vendors and coordinators review supporting evidence, reproduction material, and exploitation claims.

Additional information SHOULD NOT be published if the solely purpose of the publication is to be reference in the vulnerability report. The issuing party SHOULD just reference material already published. Any unpublished material MAY be included into the vulnerability report, if applicable.

The relevant value for $.document.category is:

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```
csaf_vulnerability_report
```

The relevant path for this test is:

```
$.vulnerabilities[*].references
```

Example 1 (which fails the test):

```
"vulnerabilities": [
  {
    "cve": "CVE-1900-0001"
  }
]
```

> The vulnerability does not have a references element.

6.2.39.12 Vulnerability Threats

It SHALL be tested that $.vulnerabilities[*].threats exists.

> Threat statements capture reported impact or exploitation status and help the receiving party understand the security consequences.

The relevant value for $.document.category is:

```
csaf_vulnerability_report
```

The relevant path for this test is:

```
$.vulnerabilities[*].threats
```

Example 1 (which fails the test):

```
"vulnerabilities": [
  {
    "cve": "CVE-1900-0001"
  }
]
```

> The vulnerability does not have a threats element.

6.2.39.13 Vulnerability Title

It SHALL be tested that $.vulnerabilities[*].title exists.

> A concise vulnerability title helps the participating parties discuss, triage, and track the report consistently.

The relevant value for $.document.category is:

```
csaf_vulnerability_report
```

The relevant path for this test is:

```
$.vulnerabilities[*].title
```

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Example 1 (which fails the test):

```
"vulnerabilities": [
  {
    "cve": "CVE-1900-0001"
  }
]
```

The vulnerability does not have a title element.

6.2.40 Product Description without Product Reference

For each product description it SHALL be tested that it includes at least one of the elements group_ids or product_ids.

If the document language is English or unspecified, the product description can be identified by checking for a note containing the corresponding category and title combination from 3.2.2.9. For other languages, the language specific translation is used.

If no language specific translation has been recorded, the test SHALL be skipped and output an information to the user that no translation for product description is known.

The relevant path for this test is:

```
$.document.notes[*]
```

Example 1 (which fails the test):

```
  "notes": [
    {
      "category": "description",
      "text": "Product A is a local time tracking tool. It is mainly used by software developers and can be connected with most modern time-tracking systems.",
      "title": "Product Description"
    }
  ],
```

The given note item does not specify to which products it applies to.

6.2.41 Old EPSS Timestamp

For each vulnerability, it SHALL be tested that the youngest EPSS timestamp is not more than 15 days older than to the date of the newest item of the revision_history if the document status is final or interim. As the timestamps might use different timezones, the sorting and comparison SHALL take timezones into account.

The relevant path for this test is:

```
$.vulnerabilities[*].metrics[*].content.epss.timestamp
```

Example 1 (which fails the test):

```
"document": {
  // ...
  "distribution": {
    "tlp": {
      "label": "CLEAR"
    }
  },
  // ...
  "tracking": {
    // ...
    "revision_history": [
      {
```

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```
        "date": "2024-01-24T10:00:00.000Z",
        "number": "1",
        "summary": "Initial version."
       }
     ],
     "status": "final",
     // ...
   }
 },
 "vulnerabilities": [
   {
     "cve": "CVE-1900-0001",
     "metrics": [
       {
         "content": {
           "epss": {
             "percentile": "0.999990000",
             "probability": "0.975570000",
             "timestamp": "2024-07-13T10:00:00.000Z"
           }
         },
         "products": [
           "CSAFPID-9080700"
         ]
       }
     ]
   }
 ]
```

The document is in status final but the EPSS timestamp is more than 15 days older than the date of newest item in the revision_history.

6.2.42 Inconsistent Product Identification Helper

For each product identification helper which resides in branches, it SHALL be tested that the product identification helper contain at least the same information as the categorized strings. Information that cannot be represented in the specific product identification helper SHALL be omitted from the comparison.

To implement this test it is deemed sufficient to follow the process below. It is based on a static mapping of branch categories to the corresponding part of the product identification helpers. The latter one are referred to as counterparts.

1. Determine the counterparts in the specific product identification helper for all categories used along the path up to the product.
2. Check whether the counterparts in the product identification helper are set.
3. Check whether each counterpart set does not contain a wildcard unless the value of the categorized string would indicate that. Only exception is the version part in a CPE, which can be set to * for a product version range due to the matching exception in section 3.1.10.3.
4. Check whether the version part algins between the categorized strings and the product identification helper. Omit this check for a version part of * in a CPE.
5. For CPE only: Check whether extra information is included in the CPE that is not in the categorized string. This might be a counterpart that is set but the corresponding categorized string is missing along th path.

The relevant paths for this test are:

```
$.product_tree..branches[*].product.product_identification_helper.cpe
$.product_tree..branches[*].product.product_identification_helper.purls[*]
```

Example 1 (which fails the test):

```
"product_tree": {
  "branches": [
    {
```

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```
      "category": "vendor",
      "name": "Example Company",
      "branches": [
       {
        "category": "product_name",
        "name": "Product A",
        "branches": [
         {
          "category": "product_version",
          "name": "2.2.0",
          "product": {
           "product_id": "CSAFPID-9080700",
           "name": "Example Company Product A 2.2.0",
           "product_identification_helper": {
            "cpe": "cpe:2.3:h:example:product_a:2.2.0:*:*:de:*:*:*:*"
           }
          }
         },
         {
          "category": "product_version",
          "name": "2.3.0",
          "product": {
           "product_id": "CSAFPID-9080701",
           "name": "Example Company Product A 2.3.0",
           "product_identification_helper": {
            "cpe": "cpe:2.3:h:example:product_a:2.3.0:PL17:*:*:*:*:*:*"
           }
          }
         }
        ]
       }
      ]
     }
    ]
}
```

The CPE of the first product contains a language part that is not given through the categorized strings. The CPE of the second product contains an update part that is not given through the categorized strings.

6.2.43 Missing License Expression

It SHALL be tested that the license expression is present and set. A CSAF validator SHALL differentiate in the error message between the key being present but having no or an empty value and not being present at all.

The relevant path for this test is:

```
$.document.license_expression
```

Example 1 (which fails the test):

```
"document": {
  // ...
  "lang": "en-US",
  "publisher": {
    // ...
  },
  // ...
}
```

The license expression is not defined.

A tool MAY add the key as a quick fix. In such case, the value still needs to be set - either manually or by other means from the tool, e.g. through a given configuration.

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6.2.44 Deprecated License Identifier

It SHALL be tested that all license identifier and exceptions used are not deprecated. This SHALL be tested for the SPDX license list and AboutCode's "ScanCode LicenseDB". The test MAY be skipped for other license inventoring entities.

The relevant path for this test is:

```
$.document.license_expression
```

Example 1 (which fails the test):

```
"license_expression": "GFDL-1.1",
```

The license identifier GFDL-1.1 was deprecated as of version 3.0.

6.2.45 Non-Existing License Identifier

It SHALL be tested that all license identifier and exceptions used exist. This SHALL be tested for the SPDX license list and AboutCode's "ScanCode LicenseDB". The test MAY be skipped for other license inventoring entities.

The relevant path for this test is:

```
$.document.license_expression
```

Example 1 (which fails the test):

```
"license_expression": "A-License-Identifier-That-Does-Not-Exist",
```

The license identifier does not exist in the SPDX license list.

6.2.46 Language Specific License Text

If the document language is specified but not English, and the license_expression contains license identifiers or exceptions that are not listed in the SPDX license list or AboutCode's "ScanCode LicenseDB", it SHALL be tested that exactly one item in document notes exists that has the language specific translation of the term License as title. The category of this item SHALL be legal_disclaimer. If no language specific translation has been recorded, the test SHALL be skipped and output an information to the user that no such translation is known.

The relevant path for this test is:

```
$.document.notes
```

Example 1 (which fails the test):

```
"document": {
  // ...
  "lang": "de-DE",
  "license_expression": "LicenseRef-www.example.com-no-work",
  "notes": [
    {
      "category": "summary",
      "text": "Es handelt sich nicht um ein urheberrechtliches Werk, da die Schöpfungshöhe nicht erreicht ist.",
      "title": "Lizenz"
    }
  ],
  // ...
}
```

The note has the correct title. However, it uses the wrong category.

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6.2.47 Use of Qualitative Severity Rating by Issuing Party

For each item in metrics provided by the issuing party it SHALL be tested that it does not use the qualitative severity rating.

> This covers all items in metrics that do not have a source property and those where the source is equal to the canonical URL. It does not cover assessments made by third parties.

The relevant path for this test is:

```
$.vulnerabilities[*].metrics[*]
```

Example 1 (which fails the test):

```
"product_tree": {
  "full_product_names": [
    {
      "product_id": "CSAFPID-9080700",
      "name": "Product A"
    }
  ]
},
"vulnerabilities": [
  {
    "metrics": [
      {
        "content": {
          "qualitative_severity_rating": "low"
        },
        "products": [
          "CSAFPID-9080700"
        ]
      }
    ]
  }
]
```

> The document issuer provided metric for CSAFPID-9080700 uses a qualitative_severity_rating.

> A tool MAY recommend other metrics or guide a CVSS assessment.

6.2.48 Misuse at Vendor Name

For each item in branches with category vendor it SHALL be tested that the name is not Open Source. Any occurrences of dash, hyphen, minus, underscore, white space and invisible characters are removed from the value on both sides before the case insensitive match.

> Some issuing parties use Open Source as a vendor name for all open source products. This usage hinders efficient matching and is most likely not true. However, if there is a vendor whose official name is exactly Open Source it would be expected and acceptable to fail this test. The same applies for any other name that results in opensource after the removal step from sentence two. This does not apply, if the official name differs, e.g. Open Source Ltd.. Users are advised to carefully review the results of this test.

The relevant path for this test is:

```
$.product_tree..branches[*].name
```

Example 1 (which fails the test):

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```
"product_tree": {
  "branches": [
    {
      "branches": [
        {
          // ...
          "category": "product_name",
          "name": "ISDuBA"
        }
      ],
      "category": "vendor",
      "name": "Open Source"
    }
  ]
}
```

The issuing party used Open Source as vendor name for the open source product ISDuBA.

A tool MAY replace the vendor name with the correct one as a quick fix.

6.2.49 Upper Open Ended Product Version Range

For each element of type $['$defs'].branches_t with category of product_version_range, it SHALL be tested that the value of name is not an upper open ended product version range.

Usually, the last including version constraint of an upper open ended product version range contains as comparator either > or >=.

The relevant path for this test is:

```
$.product_tree..branches[*].name
```

Example 1 (which fails the test):

```
{
  "category": "product_version_range",
  "name": "vers:intdot/>=4.2.0",
  // ...
}
```

The version range has no specific upper end point and states a right-open interval.

If the upper open ended product version range consists only of one version constraint, a tool MAY convert it into the product version it starts with as a quick fix, if applicable. If the upper open ended product version range consists of more than one version constraint and the upper open range is the last version constraint, a tool MAY convert it into the product version range ending with the version of the upper open ended product version as a quick fix, if applicable. Note that a change of inclusion boundaries is not allowed.

6.2.50 Overlapping Product Version Range

This subsubsection structures the recommended tests for overlapping product version ranges. The following definitions apply to all tests:

- Two product version ranges a and b are overlapping if the intersection of their sets of product versions a' and b' does not result in an empty set.

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As the intersect operation is commutative it is sufficient to test one order.

- A product version c overlaps with a product version range a if c is included in a.
- An element of type $['$defs'].full_product_name_t which has a branch category of product_version_range in the path leading to it, is called "element with product version range" (EPVR).
- The corresponding Product ID identifying such an EPVR is called "EPVR Product ID" (EPVRPID).
- For each element of type $['$defs'].branches_t with category of product_version_range which is called "currently tested product version range" (CTPVR), all sibling elements on the same level form the "group of sibling elements of the product version range" (GSEPVR).
- The group GSEPVR is divided in three pairwise disjoint sets:
  1. Elements that have the branch category product_version_range form the "product version range sibling set" (PVRSS).
  2. Elements that have the branch category product_version form the "product version sibling set" (PVSS).
  3. Elements that do not fall into 1 or 2 form the "other branch category sibling set" (OBCSS).
- PVRSS is divided in three pairwise disjoint sets:
  1. Elements that use valid VERS form the "product version range sibling set - vers" (PVRSS-vers).
  2. Elements that use valid vls form the "product version range sibling set - vls" (PVRSS-vls).
  3. Elements that do not fall into 1 or 2 form the "product version range sibling set - invalid" (PVRSS-invalid).
- PVRSS elements can also be grouped into three pairwise disjoint sets:
  1. Elements that contain the branches as only optional property form the "product version range sibling set with branches" (PVRSS+b).
  2. Elements that contain the product as only optional property form the "product version range sibling set with leaf" (PVRSS+l).
  3. Elements that do not fall into 1 or 2 form the "product version range sibling set - invalid branch" (PVRSS+i).
- Similar, PVSS elements can be grouped into three pairwise disjoint sets:
  1. Elements that contain the branches as only optional property form the "product version sibling set with branches" (PVSS+b).
  2. Elements that contain the product as only optional property form the "product version sibling set with leaf" (PVSS+l).
  3. Elements that do not fall into 1 or 2 form the "product version sibling set - invalid branch" (PVSS+i).
- Attributes of PVRSS elements can be combined:
  1. Product version range sibling set with branches - VERS (PVRSS+b-vers): Intersection of PVRSS+b and PVRSS-vers
  2. Product version range sibling set with branches - vls (PVRSS+b-vls): Intersection of PVRSS+b and PVRSS-vls
  3. Product version range sibling set with leaf - VERS (PVRSS+l-vers): Intersection of PVRSS+l and PVRSS-vers
  4. Product version range sibling set with leaf - vls (PVRSS+l-vls): Intersection of PVRSS+l and PVRSS-vls
  5. Product version range sibling set with branches - invalid (PVRSS+b-invalid): Intersection of PVRSS+b and PVRSS-invalid
  6. Product version range sibling set with leaf - invalid (PVRSS+l-invalid): Intersection of PVRSS+l and PVRSS-invalid
  7. Product version range sibling set - invalid branch - VERS (PVRSS+i-vers): Intersection of PVRSS+i and PVRSS-vers
  8. Product version range sibling set - invalid branch - vls (PVRSS+i-vls): Intersection of PVRSS+i and PVRSS-vls
  9. Product version range sibling set - invalid branch - invalid (PVRSS+i-invalid): Intersection of PVRSS+i and PVRSS-invalid

Elements in the sets PVRSS+i and PVSS+i are detectable via the validation of the JSON schema and therefore ignored in these tests. The exact characteristic of the product version range is of minor importance as the tests cannot produce any meaningful result in such an invalid product tree. Elements in PVRSS-invalid should be detected by test 6.1.50.

6.2.50.1 Overlapping Product Version Range with VERS in Contradicting Product Status Group

For each item in $.vulnerabilities all EPVRPID in the product status groups SHALL be identified. For each EPVR (as CTPVR), it SHALL be tested that the Product IDs of all elements in PVRSS+l-vers that overlap with CTPVR are not member of a contradicting product status groups (see section 3.2.4.12).

The relevant path for this test is:

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```
$.vulnerabilities[*].product_status
```

Example 1 (which fails the test):

```
"product_tree": {
 "branches": [
  {
   "branches": [
    {
     "branches": [
      {
       "category": "product_version_range",
       "name": "vers:intdot/>2.1.0|<4.2.0",
       "product": {
        "name": "Example Company Product A >2.1.0|<4.2.0",
        "product_id": "CSAFPID-9080700"
       }
      },
      {
       "category": "product_version_range",
       "name": "vers:intdot/>1.2.0|<2.4.0",
       "product": {
        "name": "Example Company Product A >1.2.0|<2.4.0",
        "product_id": "CSAFPID-9080701"
       }
      }
     ],
     "category": "product_name",
     "name": "Product A"
    }
   ],
   "category": "vendor",
   "name": "Example Company"
  }
 ]
},
"vulnerabilities": [
 {
  "product_status": {
   "known_affected": [
    "CSAFPID-9080700"
   ],
   "known_not_affected": [
    "CSAFPID-9080701"
   ]
  }
 }
]
```

The product version ranges of CSAFPID-9080700 and CSAFPID-9080701 overlap but they are members of the contradicting groups “Affected” and “Not affected”. For the version range vers:intdot/>2.1.0|<2.4.0 it cannot be determined which product status is applicable.

A tool MAY split overlapping product version ranges into non-overlapping ones to simplify the resolution as a quick fix.

6.2.50.2 Overlapping Product Version Range with vls in Contradicting Product Status Group

For each item in $.vulnerabilities all EPVRPID in the product status groups SHALL be identified. For each EPVR (as CTPVR), it SHALL be tested that the Product IDs of all elements in PVRSS+l-vls that overlap with CTPVR are not member of a contradicting product status groups (see section 3.2.4.12).

The relevant path for this test is:

```
$.vulnerabilities[*].product_status
```

Example 1 (which fails the test):

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```
"product_tree": {
  "branches": [
    {
      "branches": [
        {
          "branches": [
            {
              "category": "product_version_range",
              "name": ">2.1.0|<4.2.0",
              "product": {
                "name": "Example Company Product A >2.1.0|<4.2.0",
                "product_id": "CSAFPID-9080700"
              }
            },
            {
              "category": "product_version_range",
              "name": ">1.2.0|<2.4.0",
              "product": {
                "name": "Example Company Product A >1.2.0|<2.4.0",
                "product_id": "CSAFPID-9080701"
              }
            }
          ],
          "category": "product_name",
          "name": "Product A"
        }
      ],
      "category": "vendor",
      "name": "Example Company"
    }
  ]
},
"vulnerabilities": [
  {
    "product_status": {
      "known_affected": [
        "CSAFPID-9080700"
      ],
      "known_not_affected": [
        "CSAFPID-9080701"
      ]
    }
  }
]
```

The product version ranges of CSAFPID-9080700 and CSAFPID-9080701 overlap but they are members of the contradicting groups “Affected” and “Not affected”. For the version range >2.1.0|<2.4.0 it cannot be determined which product status is applicable.

A tool MAY split overlapping product version ranges into non-overlapping ones to simplify the resolution as a quick fix.

6.2.50.3 Overlapping Product Version Range with Product Version in Contradicting Product Status Group

For each item in $.vulnerabilities all EPVRPID in the product status groups SHALL be identified. For each EPVR (as CTPVR), it SHALL be tested that the Product IDs of all elements in PVSS+I that overlap with CTPVR are not member of a contradicting product status groups (see section 3.2.4.12).

The relevant path for this test is:

```
$.vulnerabilities[*].product_status
```

Example 1 (which fails the test):

```
"product_tree": {
  "branches": [
    {
      "branches": [
        {
```

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```
        "branches": [
          {
            "category": "product_version_range",
            "name": "vers:intdot/>2.1.0|<4.2.0",
            "product": {
              "name": "Example Company Product A >2.1.0|<4.2.0",
              "product_id": "CSAFPID-9080700"
            }
          },
          {
            "category": "product_version",
            "name": "2.4.0",
            "product": {
              "name": "Example Company Product A 2.4.0",
              "product_id": "CSAFPID-9080701"
            }
          }
        ],
        "category": "product_name",
        "name": "Product A"
      }
    ],
    "category": "vendor",
    "name": "Example Company"
  }
 ]
},
"vulnerabilities": [
 {
  "product_status": {
   "known_affected": [
    "CSAFPID-9080700"
   ],
   "known_not_affected": [
    "CSAFPID-9080701"
   ]
  }
 }
]
```

The product version ranges of CSAFPID-9080700 and CSAFPID-9080701 overlap but they are members of the contradicting groups "Affected" and "Not affected". For the product version 2.4.0 it cannot be determined which product status is applicable.

A tool MAY exclude the overlapping product version from the product version range as a quick fix.

6.2.51 Unknown VERS Type

For each element of type $['$defs'].branches_t with category of product_version_range which indicates that it is using vers, it SHALL be tested that the VERS type is officially registered and supported by the implementation. The warning SHALL differentiate between officially registered VERS types and those that are not in this state.

Different implementations might support different VERS types. Usually, unknown VERS types hinder the automated evaluation of VERS. However, it is expected that the test is able to recognize all officially registered VERS types. The differentiation will help users analyzing the result of the test and addressing the issue appropriately.

The relevant path for this test is:

```
$.product_tree..branches[*].name
```

Example 1 (which fails the test):

```
{
 "category": "product_version_range",
 "name": "vers:someweirdunknownverstype/<4.2.0|>3.91.1|!=3.2.0|<1.2",
```

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```
  // ...
}
```

The VERS type someweirdunknownverstype is a not an officially registered one. Note: An implementation would also have to state whether it supports this VERS type.

6.2.52 Unknown Hash Algorithm

For each element of type $['$defs'].full_product_name_t..product_identification_helper..hashes[*]..file_hashes[*]..algorithm, it SHALL be tested that the hash algorithm is supported by the implementation. The warning SHALL differentiate between the values mentioned in section 3.1.10.3.2 and those not mentioned there.

Different implementations might support different hash algorithms. Usually, unknown hash algorithms might hinder the automated matching of hashes. However, it is expected that the test is able to recognize all algorithms mentioned in the standard. The differentiation will help users analyzing the result of the test and addressing the issue appropriately.

The relevant paths for this test are:

```
$.product_tree..branches[*].product.product_identification_helper.hashes[*].file_hashes[*].algorithm
$.product_tree.full_product_names[*].product_identification_helper.hashes[*].file_hashes[*].algorithm
$.product_tree.product_paths[*].full_product_name.product_identification_helper.hashes[*].file_hashes[*].algorithm
```

Example 1 (which fails the test):

```
{
  "algorithm": "unknown-algorithm",
  "value": "026a37919b182ef7c63791e82c9645e2f897a3f0b73c7a6028c7febf62e93838"
}
```

The hash algorithm unknown-algorithm is not listed in section 3.1.10.3.2. Note: An implementation would also have to state whether it supports this algorithm.

6.2.53 Matching Text for Registered ID System

For each item in $.vulnerabilities[*].ids that has the value of a registered vulnerability ID system as system_name, it SHALL be tested that the text in the CSAF document matches the text_pattern given by the "Registry for Vulnerability ID Systems for CSAF" (RVISC).

The RVISC is available at [RVISC].

The relevant path for this test is:

```
$.vulnerabilities[*].ids[*].text
```

Example 1 (which fails the test):

```
  "ids": [
    {
      "system_name": "https://github.com/oasis-tcs/csaf",
      "text": "Issue 1217"
    }
  ]
```

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The system_name contains a registered vulnerability ID system but the value of text does not match the text_pattern stated for "OASIS Open CSAF TC GitHub Issues" in the RVISC.

6.2.54 Extension Tests

This subsubsection structures the recommended tests for extensions. Each of the following tests SHOULD be treated as they were listed similar to the other tests.

An application MAY group these tests when providing the additional function to only run one or more selected tests.

6.2.54.1 Registered Extension

For each item in an element of type $['$defs'].extensions_t it SHALL be tested that the item is a registered CSAF Extension. The test SHALL be skipped for official CSAF Extensions.

The relevant paths for this test are:

```
$.document.x_extensions[*]
$.product_tree..branches[*].product.x_extensions[*]
$.product_tree.full_product_names[*].x_extensions[*]
$.product_tree.product_paths[*].full_product_name.x_extensions[*]
$.vulnerabilities[*].metrics[*].content.x_extensions[*]
$.vulnerabilities[*].x_extensions[*]
$.x_extensions[*]
```

Example 1 (which fails the test):

```
"x_extensions": [
  {
    "$schema": "https://raw.githubusercontent.com/oasis-tcs/csaf/refs/heads/master/csaf_2.1/extension/data/valid/documentation-11/documentation-11-content_1.0.0.json",
    // ...
  }
]
```

The extension is neither an official nor a registered CSAF Extension.

6.2.54.2 Official Extension

For each item in an element of type $['$defs'].extensions_t it SHALL be tested that the item is an official CSAF Extension.

The relevant paths for this test are:

```
$.document.x_extensions[*]
$.product_tree..branches[*].product.x_extensions[*]
$.product_tree.full_product_names[*].x_extensions[*]
$.product_tree.product_paths[*].full_product_name.x_extensions[*]
$.vulnerabilities[*].metrics[*].content.x_extensions[*]
$.vulnerabilities[*].x_extensions[*]
$.x_extensions[*]
```

Example 1 (which fails the test):

```
"x_extensions": [
  {
    "$schema": "https://raw.githubusercontent.com/oasis-tcs/csaf/refs/heads/master/csaf_2.1/extension/data/valid/documentation-11/documentation-11-content_1.0.0.json",
    // ...
  }
]
```

The extension is not an official CSAF Extension.

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6.2.54.3 Critical Extension

For each item in an element of type $['$defs'].extensions_t it SHALL be tested that the item is not a critical extension.

> It is sufficient to check whether the value of critical is false.

The relevant paths for this test are:

```
$.document.x_extensions[*].critical
$.product_tree..branches[*].product.x_extensions[*].critical
$.product_tree.full_product_names[*].x_extensions[*].critical
$.product_tree.product_paths[*].full_product_name.x_extensions[*].critical
$.vulnerabilities[*].metrics[*].content.x_extensions[*].critical
$.vulnerabilities[*].x_extensions[*].critical
$.x_extensions[*].critical
```

Example 1 (which fails the test):

```
"x_extensions": [
  {
    // ...
    "critical": true
  }
]
```

> The extension is critical.

6.2.54.4 Usage of Experimental Extension in TLP:CLEAR Document

For each item in an element of type $['$defs'].extensions_t it SHALL be tested that no experimental extension is used if the document is labeled TLP:CLEAR.

The relevant paths for this test are:

```
$.document.x_extensions[*]
$.product_tree..branches[*].product.x_extensions[*]
$.product_tree.full_product_names[*].x_extensions[*]
$.product_tree.product_paths[*].full_product_name.x_extensions[*]
$.vulnerabilities[*].metrics[*].content.x_extensions[*]
$.vulnerabilities[*].x_extensions[*]
$.x_extensions[*]
```

Example 1 (which fails the test):

```
{
  // ...
  "document": {
    // ...
    "distribution": {
      "tlp": {
        "label": "CLEAR"
      }
    },
    // ...
  }
  "x_extensions": [
    {
      "$schema": "https://raw.githubusercontent.com/oasis-tcs/csaf/refs/heads/master/csaf_2.1/extension/data/valid/documentation-11/documentation-11-content_1.0.0.
json",
      // ...
    }
  ]
}
```

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The extension is an experimental CSAF Extension. Its properties and meaning might therefore not be known to the reader of the document.

6.2.55 Public OpenPGP Key URL

It SHALL be tested that the URL given as value of public_openpgp_key_url in the CSAF document delivers a valid public OpenPGP key allowing encryption as ASCII armored file with the matching content type. The test SHALL be skipped if the URL results in a client or server error.

As these might be temporary errors, they are reported through test 6.3.6.

The relevant path for this test is:

```
$.document.publisher.contact.public_openpgp_key_url
```

Example 1 (which fails the test):

```
"contact": {
  // ...
  "public_openpgp_key_url": "https://data.test.csaf.dev/csaf_2.1/test/validator/auxiliary/openpgp/expired.asc"
},
```

The public_openpgp_key_url points to an expired OpenPGP key.

6.2.56 Involvement Tests

This subsubsection structures the recommended tests for $.document.involvement. Each of the following tests SHOULD be treated as they were listed similar to the other tests.

An application MAY group these tests when providing the additional function to only run one or more selected tests.

6.2.56.1 Confirmation before Action

For each item in $.document.involvement.actions with category confirmation it SHALL be tested that the date of the action is later than or equal to the newest item of actions it refers to through referenced_action_ids. As the timestamps might use different timezones, the sorting and comparison SHALL take timezones into account.

The relevant path for this test is:

```
$.document.involvement.actions[*]
```

Example 1 (which fails the test):

```
"actions": [
  {
    "acting_entity_refs": [
      "EID-0001"
    ],
    "action_id": "ACN-1",
    "category": "notification",
    "date": "2024-01-11T09:00:00Z",
    "receiving_entity_refs": [
      "EID-0002"
    ],
    "status": "completed",
    "summary": "Reported vulnerability to vendor."
  },
  {
    "acting_entity_refs": [
```

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```
      "EID-0002"
    ],
    "action_id": "ACN-2",
    "category": "confirmation",
    "date": "2024-01-02T15:00:00Z",
    "receiving_entity_refs": [
      "EID-0001"
    ],
    "referenced_action_ids": [
      "ACN-1"
    ],
    "status": "completed",
    "summary": "Confirmed receipt of the report."
  }
],
```

The date of ACN-2 is older than the date of the newest item it refers to.

6.2.56.2 Unused Definition of Entity Group ID without Summary

For each Entity Group ID (type: $['$defs'].entity_group_id_t) in Entity Group elements ($.document.involvement.entity_groups[*]) which does not contain a summary element it SHALL be tested that the entity_group_id is referenced somewhere within the same document.

Without being referenced or containing a summary, a reader has no clue how the group is related to the document and why it was added.

The relevant path for this test is:

```
$.document.involvement.entity_groups[*].entity_group_id
```

Example 1 (which fails the test):

```
"involvement": {
  "actions": [
    {
      "acting_entity_refs": [
        "EID-0001"
      ],
      // ...
      "receiving_entity_refs": [
        "EID-0002"
      ],
      // ...
    },
    {
      "acting_entity_refs": [
        "EID-0002"
      ],
      // ...
      "receiving_entity_refs": [
        "EID-0001"
      ],
      // ...
    }
  ],
  // ...
  "entity_groups": [
    {
      "entity_group_id": "EGID-9001",
      "entity_ids": [
        "EID-0001",
        "EID-0003"
      ],
      "name": "JJ Research"
    }
  ]
},
```

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The unreferenced entity group EGID-9001 does not contain a summary.

A tool MAY suggest to add a summary to such definition as a quick fix.

6.2.56.3 Unused Definition of Entity ID

For each Entity ID (type: $['$defs'].entity_id_t) in Entity elements ($.document.involvement.entities[*]) it SHALL be tested that the entity_id is referenced somewhere within the same document.

The relevant path for this test is:

```
$.document.involvement.entities[*].entity_id
```

Example 1 (which fails the test):

```
"involvement": {
  "actions": [
    {
      "acting_entity_refs": [
        "EID-0001"
      ],
      // ...
      "receiving_entity_refs": [
        "EID-0002"
      ],
      // ...
    },
    {
      "acting_entity_refs": [
        "EID-0002"
      ],
      // ...
      "receiving_entity_refs": [
        "EID-0001"
      ],
      // ...
    }
  ],
  "entities": [
    // ...
    {
      "category": "discoverer",
      "entity_id": "EID-0003",
      "name": "Joe Researcher"
    }
  ]
},
```

EID-0003 was defined but never used.

A tool MAY remove the unused definition as a quick fix.

6.2.56.4 Explicit Reference to all Vulnerabilities

For each item in $.document.involvement.actions it SHALL be tested that the action does not mention all vulnerability items of the document explicitly.

If all vulnerabilities are referenced through the dl_vuln_ids element, it is semantically identical to an action without the dl_vuln_ids element. As the latter is shorter, it is the preferred way in CSAF to represent this scenario.

Note to implementers: This test cannot fail, if

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1. at least one vulnerability item exists that does not define a dl_vuln_id element (as such item cannot be mention in an action) or
2. the length of the dl_vuln_ids array in the action is smaller that the length of the array $.vulnerabilities.

The relevant path for this test is:

```
$.document.involvement.actions[*].dl_vuln_ids[*]
```

Example 1 (which fails the test):

```
{
  // ...
  "document": {
    // ...
    "involvement": {
      "actions": [
        {
          // ...
          "action_id": "ACN-1",
          // ...
          "dl_vuln_ids": [
            "VULN-1"
          ],
          // ...
        },
        // ...
      ],
      // ...
    },
    // ...
  },
  "vulnerabilities": [
    {
      "dl_vuln_id": "VULN-1",
      "title": "A vulnerability explicitly referenced."
    }
  ]
}
```

ACN-1 mentions all vulnerabilites from the CSAF document explicitly.

A tool MAY remove such dl_vuln_ids element as a quick fix.

6.2.56.5 Inconsistent Future Action Date

For each item in $.document.involvement.actions with one the status values attempted, completed or discontinued it SHALL be tested that the date of the action is earlier than or equal to the newest item of the revision_history. As the timestamps might use different timezones, the sorting and comparison SHALL take timezones into account.

The aforementioned status indicate that the action occurred already which would have happened after the release of the document.

The relevant path for this test is:

```
$.document.involvement.actions[*].date
```

Example 1 (which fails the test):

```
"document": {
  // ...
```

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```
  "involvement": {
   "actions": [
    // ...
    {
     "acting_entity_refs": [
      "EID-0002"
     ],
     "action_id": "ACN-3",
     "category": "coordination",
     "date": "2024-03-02T15:00:00Z",
     "receiving_entity_refs": [
      "EID-0001"
     ],
     "referenced_action_ids": [
      "ACN-1"
     ],
     "status": "completed",
     "summary": "Closed case after release of CSAF document."
    }
   ],
   // ...
  },
  // ...
  "tracking": {
   // ...
   "revision_history": [
    {
     "date": "2024-01-24T10:00:00.000Z",
     "number": "1",
     "summary": "Initial version."
    }
   ],
   // ...
  }
 }
```

The date of ACN-3 is newer than the date of the newest item in the revision_history.

A tool MAY remove such element as a quick fix.

6.2.57 Unused Definition of Document-local Vuln ID

For each element of type $['$defs'].dl_vuln_id_t in vulnerabilities ($.vulnerabilities[*].dl_vuln_id) it SHALL be tested that the dl_vuln_id is referenced somewhere within the same document.

The relevant path for this test is:

```
$.vulnerabilities[*].dl_vuln_id
```

Example 1 (which fails the test):

```
{
 // ...
 "document": {
  // ...
  "involvement": {
   "actions": [
    {
     "acting_entity_refs": [
      "EID-0001"
     ],
     "action_id": "ACN-1",
     "category": "notification",
     "date": "2024-01-01T09:00:00Z",
     "receiving_entity_refs": [
      "EID-0002"
     ],
     "status": "completed",
     "summary": "Reported vulnerability to vendor."
```

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```
        },
        {
          "acting_entity_refs": [
            "EID-0002"
          ],
          "action_id": "ACN-2",
          "category": "confirmation",
          "date": "2024-01-02T15:00:00Z",
          "receiving_entity_refs": [
            "EID-0001"
          ],
          "referenced_action_ids": [
            "ACN-1"
          ],
          "status": "completed",
          "summary": "Confirmed receipt of the report."
        }
      ],
      // ...
    },
    // ...
  },
  "vulnerabilities": [
    {
      "dl_vuln_id": "VULN-1",
      "title": "A vulnerability with an unreferenced Document-local Vuln ID."
    }
  ]
}
```

VULN-1 was defined but never used.

A tool MAY remove the unused definition as a quick fix. However, such quick fix shall not be applied if the test was skipped.

6.3 Informative Tests

Informative tests provide insights in common mistakes and bad practices. They MAY fail at a valid CSAF document. It is up to the issuing party to decide whether this was an intended behavior and can be ignore or should be treated. These tests MAY include information about recommended usage. A program SHALL handle a test failure as a information.

6.3.1 Use of CVSS v2 As the Only Scoring System

For each item in the list of metrics which contains the cvss_v2 object under content it SHALL be tested that is not the only scoring item present. The test SHALL pass if a second scoring object is available regarding the specific product.

One source might just provide CVSS v2. As long as at least one different source provides a different scoring system for the same products, the test passes.

The relevant path for this test is:

```
$.vulnerabilities[*].metrics
```

Example 1 (which fails the test):

```
"product_tree": {
  "full_product_names": [
    {
      "product_id": "CSAFPID-9080700",
      "name": "Product A"
    }
  ]
},
"vulnerabilities": [
```

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```
  {
    "metrics": [
      {
        "content": {
          "cvss_v2": {
            "version": "2.0",
            "vectorString": "AV:N/AC:L/Au:N/C:C/I:C/A:C",
            "baseScore": 10
          }
        },
        "products": [
          "CSAFPID-9080700"
        ]
      }
    ]
  }
]
```

There is only a CVSS v2 score given for CSAFPID-9080700.

Recommendation:

It is recommended to (also) use the CVSS v4.0.

6.3.2 Use of CVSS v3.0

For each item in the list of metrics which contains the cvss_v3 object under content it SHALL be tested that CVSS v3.0 is not used.

The relevant paths for this test are:

```
$.vulnerabilities[*].metrics[*].content.cvss_v3.version
$.vulnerabilities[*].metrics[*].content.cvss_v3.vectorString
```

Example 1 (which fails the test):

```
"cvss_v3": {
  "version": "3.0",
  "vectorString": "CVSS:3.0/AV:L/AC:L/PR:H/UI:R/S:U/C:H/I:H/A:H",
  "baseScore": 6.5,
  "baseSeverity": "MEDIUM"
}
```

The CVSS v3.0 is used.

Recommendation:

It is recommended to upgrade to CVSS v3.1.

A tool MAY upgrade to CVSS v3.1 as a quick fix. However, if such quick fix is supported the tool SHALL also recompute the baseScore and baseSeverity. The same applies for temporalScore and temporalSeverity respectively environmentalScore and environmentalSeverity if the necessary fields for computing their value are present and set.

6.3.3 Missing CVE

It SHALL be tested that the element cve is present and set. A CSAF validator SHALL differentiate in the error message between the key being present but having no or an empty value and not being present at all.

The relevant path for this test is:

```
$.vulnerabilities[*].cve
```

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Example 1 (which fails the test):

```
"vulnerabilities": [
  {
    "title": "BlueKeep"
  }
]
```

The element cve is not present.

Recommendation:

It is recommended to provide a CVE number to support the users efforts to find more details about a vulnerability and potentially track it through multiple advisories. If no CVE exists for that vulnerability, it is recommended to get one assigned.

6.3.4 Missing CWE

It SHALL be tested that at least one CWE is given.

The relevant path for this test is:

```
$.vulnerabilities[*].cwes
```

Example 1 (which fails the test):

```
"vulnerabilities": [
  {
    "cve": "CVE-2019-0708",
    "title": "BlueKeep"
  }
]
```

No CWE number is given.

6.3.5 Use of Short Hash

It SHALL be tested that the length of the hash value is not shorter than 64 characters.

The relevant paths for this test are:

```
$.product_tree..branches[*].product.product_identification_helper.hashes[*].file_hashes[*].value
$.product_tree.full_product_names[*].product_identification_helper.hashes[*].file_hashes[*].value
$.product_tree.product_paths[*].full_product_name.product_identification_helper.hashes[*].file_hashes[*].value
```

Example 1 (which fails the test):

```
"product_tree": {
  "full_product_names": [
    {
      "name": "Product A",
      "product_id": "CSAFPID-9080700",
      "product_identification_helper": {
        "hashes": [
          {
            "file_hashes": [
              {
                "algorithm": "md4",
                "value": "3202b50e2e5b2fcd75e284c3d9d5f8d6"
              }
            ],
            "filename": "product_a.so"
          }
        ]
      }
    }
```

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```
  ]
}
```

The length of the hash value is only 32 characters long.

6.3.6 Use of Non-Self Referencing URLs Failing to Resolve

For each URL which is not in the category self it SHALL be tested that it resolves with a HTTP status code from the 2xx (Successful) or 3xx (Redirection) class.

This test does not apply for any item in an array of type references_t with the category self. For details about the HTTP status code classes see [RFC7231].

The relevant paths for this test are:

```
$.document.acknowledgments[*].urls[*]
$.document.aggregate_severity.namespace
$.document.distribution.tlp.url
$.document.references[*].url
$.document.publisher.contact.public_openpgp_key_url
$.document.publisher.namespace
$.product_tree.branches[*].product.product_identification_helper.sbom_urls[*]
$.product_tree.branches[*].product.product_identification_helper.x_generic_uris[*].namespace
$.product_tree.branches[*].product.product_identification_helper.x_generic_uris[*].uri
$.product_tree..branches[*].product.product_identification_helper.sbom_urls[*]
$.product_tree..branches[*].product.product_identification_helper.x_generic_uris[*].namespace
$.product_tree..branches[*].product.product_identification_helper.x_generic_uris[*].uri
$.product_tree.full_product_names[*].product_identification_helper.sbom_urls[*]
$.product_tree.full_product_names[*].product_identification_helper.x_generic_uris[*].namespace
$.product_tree.full_product_names[*].product_identification_helper.x_generic_uris[*].uri
$.product_tree.product_paths[*].full_product_name.product_identification_helper.sbom_urls[*]
$.product_tree.product_paths[*].full_product_name.product_identification_helper.x_generic_uris[*].namespace
$.product_tree.product_paths[*].full_product_name.product_identification_helper.x_generic_uris[*].uri
$.vulnerabilities[*].acknowledgments[*].urls[*]
$.vulnerabilities[*].references[*].url
$.vulnerabilities[*].remediations[*].url
```

Example 1 (which fails the test):

```
"references": [
  {
    "summary": "A URL that does not resolve with HTTP status code in the interval between (including) 200 and (excluding) 400.",
    "url": "https://example.invalid"
  }
]
```

The category is not set and therefore treated as its default value external. A request to that URL does not resolve with a status code from the 2xx (Successful) or 3xx (Redirection) class.

6.3.7 Use of Self Referencing URLs Failing to Resolve

For each item in an array of type references_t with the category self it SHALL be tested that the URL referenced resolves with a HTTP status code less than 400.

This test will most likely fail if the CSAF document is in a status before the initial release. For details about the HTTP status code classes see [RFC7231].

The relevant paths for this test are:

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```
$.document.references[*].url
$.vulnerabilities[*].references[*].url
```

Example 1 (which fails the test):

```
"references": [
  {
    "category": "self",
    "summary": "A URL that does not resolve with HTTP status code in the interval between (including) 200 and (excluding) 400.",
    "url": "https://example.invalid"
  }
]
```

The category is self and a request to that URL does not resolve with a status code from the 2xx (Successful) or 3xx (Redirection) class.

6.3.8 Spell Check

If the document language is given it SHALL be tested that a spell check for the given language does not find any mistakes. The test SHALL be skipped if the document language is not set. It SHALL fail if the given language is not supported. The value of $.document.category SHOULD NOT be tested if the CSAF document does not use the profile “CSAF Base”.

The relevant paths for this test are:

```
$.document.acknowledgments[*].names[*]
$.document.acknowledgments[*].organization
$.document.acknowledgments[*].summary
$.document.aggregate_severity.text
$.document.category
$.document.distribution.text
$.document.involvement.actions[*].summary
$.document.involvement.entities[*].contact.details
$.document.involvement.entities[*].name
$.document.involvement.entity_groups[*].name
$.document.involvement.entity_groups[*].summary
$.document.notes[*].audience
$.document.notes[*].text
$.document.notes[*].title
$.document.publisher.contact.details
$.document.publisher.issuing_authority
$.document.publisher.name
$.document.references[*].summary
$.document.title
$.document.tracking.aliases[*]
$.document.tracking.generator.engine.name
$.document.tracking.revision_history[*].summary
$.product_tree..branches[*].name
$.product_tree..branches[*].product.name
$.product_tree.full_product_names[*].name
$.product_tree.product_groups[*].summary
$.product_tree.product_paths[*].full_product_name.name
$.vulnerabilities[*].acknowledgments[*].names[*]
$.vulnerabilities[*].acknowledgments[*].organization
$.vulnerabilities[*].acknowledgments[*].summary
$.vulnerabilities[*].notes[*].audience
$.vulnerabilities[*].notes[*].text
$.vulnerabilities[*].notes[*].title
$.vulnerabilities[*].references[*].summary
$.vulnerabilities[*].remediations[*].details
$.vulnerabilities[*].remediations[*].entitlements[*]
$.vulnerabilities[*].remediations[*].restart_required.details
$.vulnerabilities[*].threats[*].details
$.vulnerabilities[*].title
```

Example 1 (which fails the test):

```
"document": {
  // ...
  "lang": "en",
  "notes": [
```

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```
    {
      "category": "summary",
      "text": "Secruity researchers found multiple vulnerabilities in XYZ."
    }
  ],
  // ...
}
```

There is a spelling mistake in Secruity.

6.3.9 Branch Categories

For each element of type $['$defs'].full_product_name_t in $.product_tree.branches it SHALL be tested that ancestor nodes along the path exist which use the following branch categories vendor -> product_name -> product_version in that order starting with the Product tree node.

Other branch categories can be used before, after or between the aforementioned branch categories without making the test invalid.

The relevant path for this test is:

```
$.product_tree.branches
```

Example 1 (which fails the test):

```
"branches": [
  {
    "category": "vendor",
    "name": "Example Company",
    "branches": [
      {
        "category": "product_name",
        "name": "Product A",
        "branches": [
          {
            "category": "patch_level",
            "name": "91",
            "product": {
              "product_id": "CSAFPID-9080700",
              "name": "Example Company Product A Update 91"
            }
          }
        ]
      }
    ]
  }
]
```

The product CSAFPID-9080700 does not have any ancestor with the branch category product_version.

6.3.10 Usage of Product Version Range

For each element of type $['$defs'].branches_t it SHALL be tested that the category is not product_version_range.

It is usually hard decide for machines whether a product version matches a product version ranges. Therefore, it is recommended to avoid version ranges and enumerate versions wherever possible.

The relevant path for this test is:

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```
$.product_tree..branches[*].category
```

Example 1 (which fails the test):

```
            "category": "product_version_range",
```

The category product_version_range was used.

6.3.11 Usage of V as Version Indicator

For each element of type $['$defs'].branches_t with category of product_version it SHALL be tested that the value of name does not start with v or V before the version.

To implement this test it is deemed sufficient that the value of name does not match the following regex:

```
^[vV][0-9]
```

The relevant path for this test is:

```
$.product_tree..branches[*].name
```

Example 1 (which fails the test):

```
        "branches": [
          {
            "category": "product_version",
            "name": "v4.2",
            // ...
          }
        ]
```

The product version starts with a v.

6.3.12 Missing CVSS v4.0

For each item in the list of metrics that contains any CVSS object it SHALL be tested that a cvss_v4 object is present. The test SHALL fail, if any Product ID (type $['$defs'].product_id_t) in the product status group Affected (see section 3.2.4.12) is not covered by any CVSS object.

The relevant path for this test is:

```
$.vulnerabilities[*].metrics[*].content
```

Example 1 (which fails the test):

```
"product_tree": {
  "full_product_names": [
    {
      "product_id": "CSAFPID-9080700",
      "name": "Product A"
    }
  ]
},
"vulnerabilities": [
  {
    "metrics": [
      {
        "content": {
          "cvss_v3": {
```

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```
            "version": "3.1",
            "vectorString": "CVSS:3.1/AV:N/AC:L/PR:N/UI:N/S:C/C:H/I:H/A:H",
            "baseScore": 10,
            "baseSeverity": "CRITICAL"
          }
        },
        "products": [
          "CSAFPID-9080700"
        ]
      }
    ]
  }
]
```

There is no CVSS v4.0 score given for CSAFPID-9080700.

6.3.13 Usage of Non-Latest SSVC Decision Point Version

For each SSVC decision point given under selections with a registered namespace, it SHALL be tested the latest decision point version available at the time of the timestamp was used. The test SHALL fail if a later version was used. Namespaces reserved for special purpose SHALL be treated as per their definition.

A list of all valid decision points of registered namespaces including their values is available at the SSVC repository (see [SSVC-DP]).

The relevant path for this test is:

```
$.vulnerabilities[*].metrics[*].content.ssvc_v2.selections[*]
```

Example 1 (which fails the test):

```
"ssvc_v2": {
  // ...
  "selections": [
    {
      "key": "MI",
      "name": "Mission Impact",
      "namespace": "ssvc",
      // ...
      "version": "1.0.0"
    }
  ],
  "timestamp": "2024-01-24T10:00:00.000Z"
}
```

At the timestamp 2024-01-24T10:00:00.000Z version 2.0.0 of the SSVC decision point Mission Impact was already available.

6.3.14 Usage of Unregistered SSVC Decision Point Base Namespace in Non-TLP:CLEAR Document

For each SSVC decision point given under selections, it SHALL be tested that the base namespace is not an unregistered one if the document is not labeled TLP:CLEAR. Namespaces reserved for special purpose SHALL be treated as per their definition.

To implement this determine the base namespace in this test, it is deemed sufficient to remove any characters including and after the first /.

The relevant path for this test is:

```
$.vulnerabilities[*].metrics[*].content.ssvc_v2.selections[*].namespace
```

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Example 1 (which fails the test):

```
{
 "document": {
  // ...
  "distribution": {
   "tlp": {
    "label": "GREEN"
   }
  },
  // ...
 }
 "vulnerabilities": [
  {
   "metrics": [
    {
     "content": {
      "ssvc_v2": {
       // ...
       "selections": [
        {
         // ...
         "namespace": "x_example.unregistered#namespace",
         // ...
        }
       ],
       // ...
      }
     },
     // ...
    }
   ]
  }
 ]
}
```

The namespace x_example.unregistered#namespace is an unregistered base namespace. Its decision point definitions might therefore not be known to the reader of the document.

6.3.15 Usage of SSVC Decision Point Namespace with Extension in Non-TLP:CLEAR Document

For each SSVC decision point given under selections, it SHALL be tested that the namespace does not use an extension if the document is not labeled TLP:CLEAR. Namespaces reserved for special purpose SHALL be treated as per their definition.

To implement the check for an extensions part in the namespace, it is deemed sufficient to check whether the namespace contains a /.

The relevant path for this test is:

```
$.vulnerabilities[*].metrics[*].content.ssvc_v2.selections[*].namespace
```

Example 1 (which fails the test):

```
{
 "document": {
  // ...
  "distribution": {
   "tlp": {
    "label": "GREEN"
   }
  },
  // ...
 }
 "vulnerabilities": [
  {
   "metrics": [
    {
     "content": {
```

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```
          "ssvc_v2": {
            // ...
            "selections": [
              {
                // ...
                "namespace": "ssvc//.example.test#refined-technical-impacts",
                // ...
              }
            ],
            // ...
          }
        },
        // ...
      }
    ]
  }
 ]
}
```

The namespace contains the extension .example.test#refined-technical-impacts. Its decision point definitions might therefore not be known to the reader of the document.

6.3.16 Grammar Check

If the document language is given it SHALL be tested that a grammar check for the given language does not find any mistakes. The test SHALL be skipped if the document language is not set. It SHALL fail if the given language is not supported.

The relevant paths for this test are:

```
$.document.acknowledgments[*].summary
$.document.aggregate_severity.text
$.document.distribution.text
$.document.involvement.actions[*].summary
$.document.involvement.entities[*].contact.details
$.document.involvement.entity_groups[*].summary
$.document.notes[*].audience
$.document.notes[*].text
$.document.notes[*].title
$.document.publisher.contact.details
$.document.publisher.issuing_authority
$.document.references[*].summary
$.document.title
$.document.tracking.revision_history[*].summary
$.product_tree.product_groups[*].summary
$.vulnerabilities[*].acknowledgments[*].summary
$.vulnerabilities[*].notes[*].audience
$.vulnerabilities[*].notes[*].text
$.vulnerabilities[*].notes[*].title
$.vulnerabilities[*].references[*].summary
$.vulnerabilities[*].remediations[*].details
$.vulnerabilities[*].remediations[*].entitlements[*]
$.vulnerabilities[*].remediations[*].restart_required.details
$.vulnerabilities[*].threats[*].details
$.vulnerabilities[*].title
```

Example 1 (which fails the test):

```
"document": {
  // ...
  "lang": "en",
  "notes": [
    {
      "category": "summary",
      "text": "The security hardening guide must followed for ensure secure operations of a products."
    }
  ],
  // ...
}
```

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Multiple grammar mistakes exist:

- a be is missing between must and followed
- the for needs to be a to
- a products is also incorrect as a indicates singular.

6.3.17 Use of Unregistered License

It SHALL be tested that the all license identifiers and exceptions are listed either in the official SPDX license identifier list or AboutCode's "ScanCode LicenseDB".

The relevant path for this test is:

```
$.document.license_expression
```

Example 1 (which fails the test):

```
"license_expression": "LicenseRef-www.example.com-no-work-pd",
```

The license_expression contains a license identifier that is neither listed in the official SPDX license identifier list nor AboutCode's "ScanCode LicenseDB".

6.3.18 Use of Qualitative Severity Rating

For each item in metrics it SHALL be tested that it does not use the qualitative severity rating.

This covers all items in metrics regardless of their origin. Even though the Qualitative Severity Rating is a specified property, its usage is discouraged as it provides no information about the assessment inputs. Nevertheless, it can be useful to provide the such limited assessment result, especially for supply chain vulnerabilities where the upstream product just provides a qualitative severity rating.

The relevant path for this test is:

```
$.vulnerabilities[*].metrics[*]
```

Example 1 (which fails the test):

```
"product_tree": {
  "full_product_names": [
    {
      "product_id": "CSAFPID-9080700",
      "name": "Product A"
    }
  ]
},
"vulnerabilities": [
  {
    "metrics": [
      {
        "content": {
          "qualitative_severity_rating": "low"
        },
        "products": [
          "CSAFPID-9080700"
        ],
        "source": "https://upstream.example/advisories/42/an-advisory-with-just-a-severity-without-other-metrics"
      }
    ]
  }
]
```

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The upstream provided metric for CSAFPID-9080700 uses a qualitative_severity_rating.

6.3.19 Overlapping Product Version Range

This subsubsection structures the informative tests for overlapping product version ranges. These tests provide weak indicators of potential errors in the construction of the product tree. The abbreviations for groups are defined in section 6.2.50.

6.3.19.1 Overlapping Product Version Range with VERS in Same Product Status Group

For each item in $.vulnerabilities all EPVRPID in the product status groups SHALL be identified. For each EPVR (as CTPVR), it SHALL be tested that the Product IDs of all elements in PVRSS+l-vers that overlap with CTPVR are not member of the same product status group (see section 3.2.4.12).

The relevant path for this test is:

```
$.vulnerabilities[*].product_status
```

Example 1 (which fails the test):

```
"product_tree": {
  "branches": [
    {
      "branches": [
        {
          "branches": [
            {
              "category": "product_version_range",
              "name": "vers:intdot/>2.1.0|<4.2.0",
              "product": {
                "name": "Example Company Product A >2.1.0|<4.2.0",
                "product_id": "CSAFPID-9080700"
              }
            },
            {
              "category": "product_version_range",
              "name": "vers:intdot/>1.2.0|<2.4.0",
              "product": {
                "name": "Example Company Product A >1.2.0|<2.4.0",
                "product_id": "CSAFPID-9080701"
              }
            }
          ],
          "category": "product_name",
          "name": "Product A"
        }
      ],
      "category": "vendor",
      "name": "Example Company"
    }
  ]
},
"vulnerabilities": [
  {
    "product_status": {
      "first_fixed": [
        "CSAFPID-9080701"
      ],
      "fixed": [
        "CSAFPID-9080700"
      ]
    }
  }
]
```

The product version ranges of CSAFPID-9080700 and CSAFPID-9080701 overlap and they are members of the same group “Fixed”.

A tool MAY split overlapping product version ranges into non-overlapping ones to simplify the resolution as a quick fix.

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6.3.19.2 Overlapping Product Version Range with vls in Same Product Status Group

For each item in $.vulnerabilities all EPVRPID in the product status groups SHALL be identified. For each EPVR (as CTPVR), it SHALL be tested that the Product IDs of all elements in PVRSS+l-vls that overlap with CTPVR are not member of the same product status group (see section 3.2.4.12).

The relevant path for this test is:

```
$.vulnerabilities[*].product_status
```

Example 1 (which fails the test):

```
"product_tree": {
 "branches": [
  {
   "branches": [
    {
     "branches": [
      {
       "category": "product_version_range",
       "name": ">2.1.0|<4.2.0",
       "product": {
        "name": "Example Company Product A >2.1.0|<4.2.0",
        "product_id": "CSAFPID-9080700"
       }
      },
      {
       "category": "product_version_range",
       "name": ">1.2.0|<2.4.0",
       "product": {
        "name": "Example Company Product A >1.2.0|<2.4.0",
        "product_id": "CSAFPID-9080701"
       }
      }
     ],
     "category": "product_name",
     "name": "Product A"
    }
   ],
   "category": "vendor",
   "name": "Example Company"
  }
 ]
},
"vulnerabilities": [
 {
  "product_status": {
   "first_fixed": [
    "CSAFPID-9080701"
   ],
   "fixed": [
    "CSAFPID-9080700"
   ]
  }
 }
]
```

The product version ranges of CSAFPID-9080700 and CSAFPID-9080701 overlap and they are members of the same group “Fixed”.

A tool MAY split overlapping product version ranges into non-overlapping ones to simplify the resolution as a quick fix.

6.3.19.3 Overlapping Product Version Range with Product Version in Same Product Status Group

For each item in $.vulnerabilities all EPVRPID in the product status groups SHALL be identified. For each EPVR (as CTPVR), it SHALL be tested that the Product IDs of all elements in PVSS that overlap with CTPVR are not member of the same product status group (see section 3.2.4.12).

The relevant path for this test is:

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```
$.vulnerabilities[*].product_status
```

Example 1 (which fails the test):

```
"product_tree": {
 "branches": [
  {
   "branches": [
    {
     "branches": [
      {
       "category": "product_version_range",
       "name": "vers:intdot/>2.1.0|<4.2.0",
       "product": {
        "name": "Example Company Product A >2.1.0|<4.2.0",
        "product_id": "CSAFPID-9080700"
       }
      },
      {
       "category": "product_version",
       "name": "2.4.0",
       "product": {
        "name": "Example Company Product A 2.4.0",
        "product_id": "CSAFPID-9080701"
       }
      }
     ],
     "category": "product_name",
     "name": "Product A"
    }
   ],
   "category": "vendor",
   "name": "Example Company"
  }
 ]
},
"vulnerabilities": [
 {
  "product_status": {
   "first_fixed": [
    "CSAFPID-9080701"
   ],
   "fixed": [
    "CSAFPID-9080700"
   ]
  }
 }
]
```

The product version ranges of CSAFPID-9080700 and CSAFPID-9080701 overlap and they are members of the same group “Fixed”.

A tool MAY exclude the overlapping product version from the product version range as a quick fix.

6.3.19.4 Overlapping Product Version Range with Product Version Range in Branch

For each item in $.vulnerabilities all EPVRPID in the product status groups SHALL be identified. For each EPVR (as CTPVR), it SHALL be tested that all product version ranges of elements in PVRSS+b do not overlap with CTPVR.

The relevant path for this test is:

```
$.product_tree..branches[*].name
```

Example 1 (which fails the test):

```
{
 "branches": [
  {
   "branches": [
```

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```
        // ...
      ],
      "category": "product_version_range",
      "name": "vers:intdot/>2.1.0|<4.2.0"
    },
    {
      "branches": [
        // ...
      ],
      "category": "product_version_range",
      "name": "vers:intdot/>1.2.0|<2.4.0"
    }
  ],
  // ...
}
```

The product version ranges overlap.

A tool MAY split overlapping product version ranges into non-overlapping ones to simplify the resolution as a quick fix.

6.3.19.5 Overlapping Product Version Range with Product Version in Branch

For each item in $.vulnerabilities all EPVRPID in the product status groups SHALL be identified. For each EPVR (as CTPVR), it SHALL be tested that all product versions of elements in PVSS+b do not overlap with CTPVR.

The relevant path for this test is:

```
$.product_tree..branches[*].name
```

Example 1 (which fails the test):

```
{
  "branches": [
    {
      "branches": [
        // ...
      ],
      "category": "product_version_range",
      "name": "vers:intdot/>2.1.0|<4.2.0"
    },
    {
      "branches": [
        // ...
      ],
      "category": "product_version",
      "name": "3.0.0"
    }
  ],
  // ...
}
```

The product version is part of the product version range.

6.3.20 Use of Unregistered ID System

For each item in $.vulnerabilities[*].ids it SHALL be tested that the value of system_name belongs to a registered vulnerability ID system in RVISC.

The RVISC is available at [RVISC].

The relevant path for this test is:

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```
$.vulnerabilities[*].ids[*].system_name
```

Example 1 (which fails the test):

```
"ids": [
  {
    "system_name": "OASIS Open CSAF TC GitHub Issues",
    "text": "Issue 1217"
  }
]
```

The system_name is not one that is registered in RVISC.

6.3.21 Extension Tests

This subsubsection structures the informative tests for extensions. Each of the following tests SHOULD be treated as they were listed similar to the other tests.

An application MAY group these tests when providing the additional function to only run one or more selected tests.

6.3.21.1 Extension Category Essential

For each item in an element of type $['$defs'].extensions_t it SHALL be tested that the extension category of the item is not essential.

It is sufficient to check whether the value of category is not essential.

The relevant paths for this test are:

```
$.document.x_extensions[*].category
$.product_tree..branches[*].product.x_extensions[*].category
$.product_tree.full_product_names[*].x_extensions[*].category
$.product_tree.product_paths[*].full_product_name.x_extensions[*].category
$.vulnerabilities[*].metrics[*].content.x_extensions[*].category
$.vulnerabilities[*].x_extensions[*].category
$.x_extensions[*].category
```

Example 1 (which fails the test):

```
"x_extensions": [
  {
    // ...
    "category": "essential",
    // ...
  }
]
```

The extension category is essential. A reader that does not understand this extension might miss information that is essential to understand the content of the CSAF document.

6.3.21.2 Usage of Experimental Extension in Non-TLP:CLEAR Document

For each item in an element of type $['$defs'].extensions_t it SHALL be tested that no experimental extension is used if the document is not labeled TLP:CLEAR.

The relevant paths for this test are:

```
$.document.x_extensions[*]
$.product_tree..branches[*].product.x_extensions[*]
```

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```
$.product_tree.full_product_names[*].x_extensions[*]
$.product_tree.product_paths[*].full_product_name.x_extensions[*]
$.vulnerabilities[*].metrics[*].content.x_extensions[*]
$.vulnerabilities[*].x_extensions[*]
$.x_extensions[*]
```

Example 1 (which fails the test):

```
{
  // ...
  "document": {
    // ...
    "distribution": {
      "tlp": {
        "label": "GREEN"
      }
    },
    // ...
  }
  "x_extensions": [
    {
      "$schema": "https://raw.githubusercontent.com/oasis-tcs/csaf/refs/heads/master/csaf_2.1/extension/data/valid/documentation-11/documentation-11-content_1.0.0.
json",
      // ...
    }
  ]
}
```

The extension is an experimental CSAF Extension. Its properties and meaning might therefore not be known to the reader of the document.

6.3.21.3 Usage of Extension at Document Level

It SHALL be tested that the element $.document.x_extensions does not exist.

The relevant path for this test is:

```
$.document.x_extensions
```

Example 1 (which fails the test):

```
{
  // ...
  "document": {
    // ...
    "x_extensions": [
      // ...
    ]
  }
}
```

The element $.document.x_extensions exists.

6.3.21.4 Usage of Extension in Product Tree Branch Path

It SHALL be tested that the element x_extensions does not exist in any path that starts with $.product_tree.branches.

The relevant path for this test is:

```
$.product_tree..branches[*].product.x_extensions
```

Example 1 (which fails the test):

```
"product_tree": {
  "branches": [
```

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```
    {
      "branches": [
        {
          "branches": [
            {
              // ...
              "product": {
                // ...
                "x_extensions": [
                  // ...
                ]
              }
            }
          ],
          // ...
        }
      ],
      // ...
    }
  ]
}
```

The element x_extensions exists in a path that starts with $.product_tree.branches.

6.3.21.5 Usage of Extension in Product Tree Full Product Names Path

It SHALL be tested that the element x_extensions does not exist in any path that starts with $.product_tree.full_product_names.

The relevant path for this test is:

```
$.product_tree.full_product_names[*].x_extensions
```

Example 1 (which fails the test):

```
"product_tree": {
  "full_product_names": [
    {
      // ...
      "x_extensions": [
        // ...
      ]
    }
  ]
}
```

The element x_extensions exists in a path that starts with $.product_tree.full_product_names.

6.3.21.6 Usage of Extension in Product Tree Product Paths Path

It SHALL be tested that the element x_extensions does not exist in any path that starts with $.product_tree.product_paths.

The relevant path for this test is:

```
$.product_tree.product_paths[*].full_product_name.x_extensions
```

Example 1 (which fails the test):

```
"product_tree": {
  // ...
  "product_paths": [
    {
      // ...
      "full_product_name": {
        // ...
        "x_extensions": [
          // ...
        ]
```

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```
      },
      // ...
    }
  ]
}
```

The element x_extensions exists in a path that starts with $.product_tree.product_paths.

6.3.21.7 Usage of Extension in Vulnerabilities Metrics Path

It SHALL be tested that the element x_extensions does not exist in any path that starts with $.vulnerabilities[*].metrics.

The relevant path for this test is:

```
$.vulnerabilities[*].metrics[*].content.x_extensions
```

Example 1 (which fails the test):

```
"vulnerabilities": [
 {
  "metrics": [
   {
    "content": {
     "x_extensions": [
      // ...
     ]
    },
    // ...
   }
  ]
 }
]
```

The element x_extensions exists in a path that starts with $.vulnerabilities[*].metrics.

6.3.21.8 Usage of Extension at Vulnerabilities Level

For each item $.vulnerabilities it SHALL be tested that the element x_extensions does not exist.

The relevant path for this test is:

```
$.vulnerabilities[*].x_extensions
```

Example 1 (which fails the test):

```
{
 // ...
 "vulnerabilities": [
  {
   // ...
   "x_extensions": [
    // ...
   ]
  }
 ]
}
```

The element x_extensions exists inside a vulnerabilities item.

6.3.21.9 Usage of Extension at Root Level

It SHALL be tested that the element $.x_extensions does not exist.

The relevant path for this test is:

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```
$.x_extensions
```

Example 1 (which fails the test):

```
{
  // ...
  "x_extensions": [
    // ...
  ]
}
```

The element $.x_extensions exists.

6.3.22 Nested Product Path

For each item in $.product_tree.product_paths[*] it SHALL be tested that all referenced Product IDs are not defined under a full_product_name_t element within items of $.product_tree.product_paths.

A product defined by a product path that contains another product path is usually harder to match. Nevertheless, there are scenarios that require such a nesting to correctly describe the reality or provide additional matching information.

The relevant paths for this test are:

```
$.product_tree.product_paths[*].beginning_product_reference
$.product_tree.product_paths[*].subpaths[*].next_product_reference
```

Example 1 (which fails the test):

```
"product_tree": {
  "full_product_names": [
    {
      "name": "Product A",
      "product_id": "CSAFPID-908070601"
    },
    {
      "name": "Product B",
      "product_id": "CSAFPID-908070602"
    },
    {
      "name": "Product C",
      "product_id": "CSAFPID-908070603"
    }
  ],
  "product_paths": [
    {
      "beginning_product_reference": "CSAFPID-908070601",
      "full_product_name": {
        "name": "Product A installed on Product B",
        "product_id": "CSAFPID-908070604"
      },
      "subpaths": [
        {
          "category": "installed_on",
          "next_product_reference": "CSAFPID-908070602"
        }
      ]
    }
    {
      "beginning_product_reference": "CSAFPID-908070604",
      "full_product_name": {
        "name": "Product A installed on Product B installed on Product C",
        "product_id": "CSAFPID-908070605"
      },
      "subpaths": [
        {
          "category": "installed_on",
          "next_product_reference": "CSAFPID-908070603"
```

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```
      }
     ]
    }
   ]
}
```

The Product with the ID CSAFPID-908070605 is defined through a product reference which belongs to a product formed by a product path.

6.3.23 Profile Tests

This subsubsection structures the informative tests for the profiles. Not all tests apply for all profiles. Tests SHOULD be skipped if the document category does not match the one given in the test. Each of the following tests SHOULD be treated as they were listed similar to the other tests.

6.3.23.1 Involvement

It SHALL be tested that $.document.involvement exists.

Recording involvement actions helps the participating parties understand the disclosure timeline and who has engaged in the process.

The relevant value for $.document.category is:

```
csaf_vulnerability_report
```

The relevant path for this test is:

```
$.document.involvement
```

Example 1 (which fails the test):

```
document: {
  // ...
  "distribution": {
    "tlp": {
      "label": "AMBER"
    }
  },
  "publisher": {
    "category": "other",
    "name": "OASIS CSAF TC",
    "namespace": "https://csaf.io"
  },
  // ...
}
```

The vulnerability does not record any coordination activity in $.document.involvement.

Recommendation:

It is recommended that issuing parties use $.document.involvement to record coordination milestones during the CVD process.

6.3.23.2 Disclosure Date

It SHALL be tested that $.vulnerabilities[*].disclosure_date exists.

The relevant value for $.document.category is:

```
csaf_vulnerability_report
```

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The relevant path for this test is:

```
$.vulnerabilities[*].disclosure_date
```

Example 1 (which fails the test):

```
"vulnerabilities": [
  {
    "cve": "CVE-1900-0001"
  }
]
```

The vulnerability does not have a disclosure_date element.

Recommendation:

It is recommended that issuing parties use the disclosure date.

6.3.24 Public OpenPGP Key URL User ID

It SHALL be tested that the URL given as value of public_openpgp_key_url in the CSAF document delivers a public OpenPGP key with an email in the user ID matching the sibling property email. The test SHALL be skipped if the URL results in a client or server error or the OpenPGP key retrieved is not in ASCII-armored format. The test SHALL report if the user ID is not set or empty.

The cases excluded are handled through tests 6.3.6 and 6.2.55.

The relevant path for this test is:

```
$.document.publisher.contact.public_openpgp_key_url
```

Example 1 (which fails the test):

```
  "contact": {
    "email": "test-6.3.24@csaf.example",
    "public_openpgp_key_url": "https://data.test.csaf.dev/csaf_2.1/test/validator/auxiliary/openpgp/5-years-valid.asc"
  },
```

The public_openpgp_key_url points to an OpenPGP key where the user ID does not match the email given in the sibling property email.

Note: A difference between the email address in the user ID and the value of the property email does not necessarily mean that the receiver of the email is unable to decrypt the messages.

Recommendation:

It is recommended that issuing parties conduct an analysis to make an informed decision based on pros and cons regarding the inclusion of a matching email address into the user ID of the public OpenPGP key. Sections 3.2.2.10.2 and 8 contain advise to take into consideration.

6.3.25 Non-Latest SSVC Decision Point Version at Validation Time

For each SSVC decision point given under selections with a registered namespace, it SHALL be tested the latest decision point version available at the time the test is performed was used. Namespaces reserved for special purpose SHALL be treated as per their definition.

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The result of the test is dependent upon the time of the execution of the test - it might change for a given CSAF document over time. This test is intended to be used during the editorial process of CSAF documents. The significance of the test result regarding released CSAF documents is limited. It is not expected to update a released CSAF document if this test fails and no other new information is available. Observatories and other reporting entities are requested to not include this test in their general score.

Usage of a later version of a SSVC Decision Point will cause test 6.3.13 to fail. The reason for this is most likely an outdated data set of SSVC decision points. A list of all valid decision points of registered namespaces including their values is available at the SSVC registry (see [SSVC-OR]).

The relevant path for this test is:

```
$.vulnerabilities[*].metrics[*].content.ssvc_v2.selections[*]
```

Example 1 (which fails the test):

```
"ssvc_v2": {
  // ...
  "selections": [
    {
      "key": "MI",
      "name": "Mission Impact",
      "namespace": "ssvc",
      // ...
      "version": "1.0.0"
    }
  ],
  "timestamp": "2024-01-24T10:00:00.000Z"
}
```

At the time of the test execution 2026-10-07T10:00:00.000Z version 2.0.0 of the SSVC decision point Mission Impact was already available.

6.3.26 Involvement Tests

This subsubsection structures the informative tests for $.document.involvement. Each of the following tests SHOULD be treated as they were listed similar to the other tests.

An application MAY group these tests when providing the additional function to only run one or more selected tests.

6.3.26.1 Vulnerability Not Covered by Involvement

For each vulnerability item it SHALL be tested that the item is covered by at least one action in $.document.involvement. This takes implicit coverage through omitting dl_vuln_ids into account.

The relevant path for this test is:

```
$.vulnerabilities[*]
```

Example 1 (which fails the test):

```
{
  // ...
  "document": {
    // ...
    "involvement": {
      "actions": [
        {
          // ...
          "dl_vuln_ids": [
```

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```
          "VULN-1"
        ],
        // ...
      },
      {
        // ...
        "dl_vuln_ids": [
          "VULN-1"
        ],
        // ...
      }
    ],
    // ...
  },
  // ...
},
"vulnerabilities": [
  {
    "cve": "CVE-1900-0001",
    "dl_vuln_id": "VULN-1",
    "title": "A vulnerability with a document-local vuln ID and CVE."
  },
  {
    "cve": "CVE-1900-0002",
    "title": "A vulnerability discovered independently."
  }
]
}
```

The vulnerability CVE-1900-0002 was not referenced in the involvement.

6.3.26.2 Unused Definition of Entity Group ID

For each Entity Group ID (type: $['$defs'].entity_group_id_t) in Entity Group elements ($.document.involvement.entity_groups[*]) it SHALL be tested that the entity_group_id is referenced somewhere within the same document.

The relevant path for this test is:

```
$.document.involvement.entity_groups[*].entity_group_id
```

Example 1 (which fails the test):

```
"involvement": {
  "actions": [
    {
      "acting_entity_refs": [
        "EID-0001"
      ],
      // ...
      "receiving_entity_refs": [
        "EID-0002"
      ],
      // ...
    },
    {
      "acting_entity_refs": [
        "EID-0002"
      ],
      // ...
      "receiving_entity_refs": [
        "EID-0001"
      ],
      // ...
    }
  ],
  // ...
  "entity_groups": [
    {
      "entity_group_id": "EGID-9001",
      "entity_ids": [
        "EID-0001",
        "EID-0003"
      ],
```

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```
      "name": "JJ Research"
    }
  ]
},
```

EGID-9001 was not referenced elsewhere in the document.

A tool MAY remove the unused definition as a quick fix.

6.3.26.3 Multiple Use of Same Entity Group Name

For each name in $.document.involvement.entity_groups[*] it SHALL be tested that the name was not already used for another entity group within the same document.

For the justified scenarios, the test should be ignored.

The relevant path for this test is:

```
$.document.involvement.entity_groups[*].name
```

Example 1 (which fails the test):

```
"entity_groups": [
  {
    "entity_group_id": "EGID-9001",
    "entity_ids": [
      "EID-0001",
      "EID-0003"
    ],
    "name": "JJ Research"
  },
  {
    "entity_group_id": "EGID-9002",
    "entity_ids": [
      "EID-0001",
      "EID-0003"
    ],
    "name": "JJ Research"
  }
]
```

JJ Research was used twice.

6.3.26.4 Multiple Use of Same Entity Name

For each name in $.document.involvement.entities[*] it SHALL be tested that the name was not already used for another entity within the same document.

Such a construct is expected to happen, e.g. if a entity is reached out to under multiple addresses or the entity has different roles. For the justified scenarios, the test should be ignored.

The relevant path for this test is:

```
$.document.involvement.entities[*].name
```

Example 1 (which fails the test):

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```
"entities": [
  {
    "category": "discoverer",
    "entity_id": "EID-0001",
    "name": "Jane Researcher"
  },
  // ...
  {
    "category": "discoverer",
    "entity_id": "EID-0003",
    "name": "Jane Researcher"
  }
]
```

Jane Researcher was used twice.

6.4 Test Presets

A test preset is a predefined set of tests that was given a name. It MAY contain any number of tests. Two presets MAY overlap. The content of a preset MAY vary in different CSAF versions. A CSAF validator SHALL support every official preset that solely include tests that are implemented by the CSAF validator. A CSAF validator MAY provide or support additional presets. A CSAF validator SHALL implement all tests for any supported preset.

Names of presets not defined in this CSAF standard SHALL have the following prefix before their name:

- x_: for any CSAF validator specific preset.

> Multiple CSAF validators might use the same preset name for different sets of tests. Users are advised to carefully check the documentation of the tools to avoid incorrect assumptions.

- org_ followed by an organization identifier and an underscore (_): for any preset specified by an organization as a part of a public definition that can be implemented by different CSAF validators. The organization identifier SHALL only use the characters identified by the pattern [0-9a-zA-Z-.]. The organization identifier SHALL be registered with the OASIS CSAF TC prior to the publication of the definition.
- csaf_: for any preset defined later on by the OASIS CSAF TC.

To avoid collisions especially in the x_ namespace, the reverse DNS notation for a domain under the author's own control MAY be used.

Namespaces starting with the following prefixes as derived from [RFC2606] are reserved:

- x_test and org_test: for testing purposes
- x_example and org_example: for documentation and testing purposes
- x_invalid and org_invalid: as always invalid

Namespaces starting with the following prefixes are reserved by the OASIS CSAF TC:

- x_csaf
- org_csaf

Those namespaces are solely assignable by the OASIS CSAF TC.

Official presets are defined in different parts of the standard.

6.4.1 Presets Defined through Test Subsections

The following presets are defined through subsections of the test section:

- mandatory: all tests given in section 6.1
- recommended: all tests given in section 6.2
- informative: all tests given in section 6.3

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6.4.2 Presets Defined through Conformance Targets

The following presets are defined through conformance targets:

- schema: Check against the JSON schema (see section 9.1.14)
- basic: schema + mandatory (see section 9.1.14)
- extended: basic + recommended (see section 9.1.15)
- full: extended + informative (see section 9.1.16)
- additional: all implemented tests that satisfy the CSAF Additional Test conformance profile (see section 9.1.27)

> Note: The last preset MAY vary between different implementations as it is implementation specific.

As presets are sets, the operator + SHALL be interpreted as the union operation.

6.4.3 Additional Presets

Additional presets are defined as follows:

- non-external-request-free:
  - Description: Any test that cannot be executed without a request into the Internet or a different network.
  - Set:
    - 6.2.55
    - 6.3.6
    - 6.3.7
    - 6.3.24
- external-request-free:
  - Description: Any test that can be executed without a request into the Internet or a different network.

    > This is intended to be used for browser-based tools as external requests may result in CORS issues. Request over network to a tool that is delivered with or an install requirement for a CSAF validator are not considered external.

  - Set: full excluding tests from preset non-external-request-free
- consistent-revision-history:
  - Description: Any test that is related to the revision history and ensures consistence within it.
  - Set:
    - 6.1.14
    - 6.1.18
    - 6.1.19
    - 6.1.21
    - 6.1.22
    - 6.1.37
    - 6.1.66
    - 6.2.4
    - 6.2.5
    - 6.2.6
    - 6.2.21
    - 6.2.33
- consistent-date-times:
  - Description: Any test that is related to timestamps in CSAF and avoids invalid situations.
  - Set:
    - 6.1.37
    - 6.1.45
    - 6.1.49
    - 6.1.51

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  - 6.1.52
  - 6.1.53
  - 6.1.62
  - 6.2.56.1
  - 6.2.56.5
- ssvc:
  - Description: Any test that is related to SSVC in CSAF.
  - Set:
    - 6.1.46
    - 6.1.47
    - 6.1.48
    - 6.1.49
    - 6.2.3
    - 6.2.34
    - 6.2.35
    - 6.2.36
    - 6.2.37
    - 6.3.13
    - 6.3.14
    - 6.3.15
- extensions:
  - Description: Any test that is related to CSAF Extensions.
  - Set:
    - 6.1.60.1
    - 6.1.60.2
    - 6.1.60.3
    - 6.2.39.5
    - 6.2.54.1
    - 6.2.54.2
    - 6.2.54.3
    - 6.2.54.4
    - 6.3.21.1
    - 6.3.21.2
    - 6.3.21.3
    - 6.3.21.4
    - 6.3.21.5
    - 6.3.21.6
    - 6.3.21.7
    - 6.3.21.8
    - 6.3.21.9
- extensions-exist:
  - Description: Detects CSAF Extensions in all places.
  - Set:
    - 6.3.21.3
    - 6.3.21.4
    - 6.3.21.5
    - 6.3.21.6
    - 6.3.21.7
    - 6.3.21.8
    - 6.3.21.9
- validation-time:
  - Description: Any test whose result depends on the time it is run.
  - Set:
    - 6.2.33
    - 6.3.25

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6.5 Test Result Elevation

Test result elevation allows to modify the test result of one or more test or preset. The following options are defined:

- all-as-error: updates warnings to errors and information to errors for all test results it is applied to
- as-level-up: updates warnings to errors and information to warnings (in this order) for all test results it is applied to
- info-as-error: updates information to errors for all test results it is applied to
- info-as-warning: updates information to warnings for all test results it is applied to
- warning-as-error: updates warnings to errors for all test results it is applied to

CSAF validators SHOULD implement the options in a way that if no additional parameters are given they apply to all executed tests. If parameters are provided, the test result elevation applies only to the tests specified in the parameters. CSAF validators SHALL determine the overall validation status after the test result elevation.

For example, a CSAF document failing test 6.2.1 would be valid without test result elevation. If the CSAF validator is instructed to use the test result elevation warning-as-error, the CSAF validator determines that the same CSAF document is invalid as the warning resulting from the failing test 6.2.1 is reported as error and therefore changes the overall result. These options help issuing parties to ensure data quality within their CSAF documents as they can be used in automated checks, e.g. CI/CD pipelines.

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7. Distributing CSAF Documents

This section lists requirements and roles defined for distributing CSAF documents. The first subsection provides all requirements - the second one the roles. It is mandatory to fulfill the basic role “CSAF Publisher”. The last section provides specific rules for the process of retrieving CSAF documents.

7.1 Requirements

The requirements in this subsection are consecutively numbered to be able to refer to them directly. The order does not give any hint about the importance. Not all requirements have to be fulfilled to conform to this specification - the sets of requirements per conformance clause are defined in section 7.2.

7.1.1 Requirement 1: Valid CSAF Document

The document is a valid CSAF document (cf. Conformance clause 1).

7.1.2 Requirement 2: Filename

The CSAF document has a filename according to the rules in section 5.1.

7.1.3 Requirement 3: TLS

The CSAF document is per default retrievable from a website which uses TLS for encryption and server authenticity. The CSAF document SHALL NOT be downloadable from a location which does not encrypt the transport when crossing organizational boundaries to maintain the chain of custody.

7.1.4 Requirement 4: TLP:CLEAR

If the CSAF document is labeled TLP:CLEAR, it SHALL be freely accessible.

This does not exclude that such a document is also available in an access protected customer portal. However, there SHALL be one copy of the document available for people without access to the portal.

> Reasoning: If an advisory is already in the media, an end user should not be forced to collect the pieces of information from a press release but be able to retrieve the CSAF document.

7.1.5 Requirement 5: TLP:AMBER, TLP:AMBER+STRICT and TLP:RED

CSAF documents labeled TLP:AMBER, TLP:AMBER+STRICT or TLP:RED SHALL be access protected. If they are provided via a web server this SHALL be done under a different path than for TLP:CLEAR, TLP:GREEN and unlabeled CSAF documents. TLS client authentication, access tokens or any other automatable authentication method SHALL be used.

An issuing party MAY agree with the recipients to use any kind of secured drop at the recipients’ side to avoid putting them on their own website. However, it SHALL be ensured that the documents are still access protected.

7.1.6 Requirement 6: No Redirects

Redirects SHOULD NOT be used. If they are inevitable only HTTP Header redirects SHALL be used.

> Reasoning: Clients should not parse the payload for navigation and some, as e.g. curl, do not follow any other kind of redirects.

If any redirects are used, there SHOULD NOT be more than 10, and SHALL NOT be more than 20 consecutive redirects.

> This aligns with “HTTP-redirect fetch” section of the [FETCH] specification.

7.1.7 Requirement 7: provider-metadata.json

The party SHALL provide a valid provider-metadata.json according to the schema CSAF provider metadata for its own metadata. The publisher object SHOULD match the one used in the CSAF documents of the issuing party but can be set to whatever value a CSAF aggregator SHOULD display over any individual publisher values in the CSAF documents themselves.

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This information is used to collect the data for CSAF aggregators, listers and end users. The CSAF provider metadata schema ensures the consistency of the metadata for a CSAF provider across the ecosystem. Other approaches, like extracting the publisher object from CSAF documents, are likely to fail if the object differs between CSAF documents.

It is suggested to put the file provider-metadata.json adjacent to the ROLIE feed documents (requirement 15) or in the main directory adjacent to the year folders (requirement 14), changes.csv (requirement 13) and the index.txt (requirement 12). Suggested locations to store the provider-metadata.json are:

- https://www.example.com/.well-known/csaf/provider-metadata.json
- https://domain.example/security/data/csaf/provider-metadata.json
- https://psirt.domain.example/advisories/csaf/provider-metadata.json
- https://domain.example/security/csaf/provider-metadata.json

Example 1 (minimal with ROLIE document):

```
{
  "$schema": "https://docs.oasis-open.org/csaf/csaf/v2.1/schema/provider.json",
  "canonical_url": "https://www.example.com/.well-known/csaf/provider-metadata.json",
  "distributions": [
    {
      "rolie": {
        "feeds": [
          {
            "last_updated": "2024-01-24T20:20:56.169Z",
            "summary": "All TLP:CLEAR advisories of Example Company.",
            "tlp_label": "CLEAR",
            "url": "https://www.example.com/.well-known/csaf/feed-tlp-clear.json"
          }
        ]
      }
    }
  ],
  "last_updated": "2024-01-24T20:20:56.169Z",
  "list_on_CSAF_aggregators": true,
  "metadata_version": "2.1",
  "mirror_on_CSAF_aggregators": true,
  "public_openpgp_keys": [
    {
      "fingerprint": "8F5F267907B2C4559DB360DB2294BA7D2B2298B1",
      "url": "https://keys.example.net/vks/v1/by-fingerprint/8F5F267907B2C4559DB360DB2294BA7D2B2298B1"
    }
  ],
  "publisher": {
    "category": "vendor",
    "name": "Example Company ProductCERT",
    "namespace": "https://psirt.example.com"
  },
  "role": "csaf_trusted_provider"
}
```

The maintained_until and maintained_from properties can be used to indicate that the distributions contained in provider-metadata.json at the given canonical_url are only guaranteed to be maintained until or after the specified date and time. This SHOULD be used to support a transition period between CSAF 2.0 and CSAF 2.1 (cf. section 7.4). CSAF downloaders (cf. section 9.1.23) and programs retrieving or providing a provider-metadata.json SHOULD evaluate this property and emit a warning if the current date is less than 90 days away from maintained_until and an error if the current date exceeds maintained_until. The programs MAY provide a non-default option to use the provider-metadata.json anyway. Furthermore, such programs SHOULD evaluate the maintained_from property and output a warning if the current date is still before the maintained_from timestamp.

If a CSAF publisher (cf. section 7.2.1) does not provide the provider-metadata.json, an aggregator SHOULD contact the CSAF publisher in question to determine the values for list_on_CSAF_aggregators and mirror_on_CSAF_aggregators. If that is impossible or if the CSAF publisher is unresponsive the following values SHALL be used:

```
"list_on_CSAF_aggregators": true,
"mirror_on_CSAF_aggregators": false
```

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This prevents that CSAF documents of a CSAF publisher which have been collected by one CSAF aggregator A are mirrored again on a second CSAF aggregator B. Such cascades are prone to outdated information. If the first aggregator A collects the CSAF documents on best effort and B copies the files from A and announces that this is done weekly, one might assume that B's CSAF documents are more recent. However, that is not the case as B's information depends on A.

7.1.8 Requirement 8: security.txt

In the security.txt there SHALL be at least one field CSAF which points to the provider-metadata.json (requirement 7). If this field indicates a web URI, then it SHALL begin with "https://" (as per section 2.7.2 of [RFC7230]). See [SECURITY-TXT] for more details.

The security.txt was published as [RFC9116] in April 2022. The CSAF field was officially added through the IANA registry.

Examples 1:

```
CSAF: https://www.example.com/.well-known/csaf/provider-metadata.json
CSAF: https://domain.example/security/data/csaf/provider-metadata.json
CSAF: https://psirt.domain.example/advisories/csaf/provider-metadata.json
CSAF: https://domain.example/security/csaf/provider-metadata.json
```

It is possible to advertise more than one provider-metadata.json by adding multiple CSAF fields, e.g. in case of changes to the organizational structure through mergers and acquisitions. However, this SHOULD NOT be done and removed as soon as possible. A valid use case for temporarily including multiple entries would be a transition phase between different CSAF versions, in which documents and provider metadata of both versions are served simultaneously (cf. section 7.4). If one of the URLs fulfills requirement 9, it SHALL be set as the first CSAF entry in the security.txt.

7.1.9 Requirement 9: Well-Known URL for provider-metadata.json

The URL path /.well-known/csaf/provider-metadata.json under the main domain of the issuing authority serves directly the provider-metadata.json according to requirement 7. That implies that redirects SHALL NOT be used. The use of the scheme "HTTPS" is required. See [RFC8615] for more details.

Example 1:

```
https://www.example.com/.well-known/csaf/provider-metadata.json
```

As specified in 7.4, the value of canonical_url MAY differ from the URL that was requested as a part of this requirement. Such state is intended and SHALL NOT be reported as error.

7.1.10 Requirement 10: DNS Path

Assuming that the organization's main domain is domain.example, the DNS record csaf.data.security.domain.example SHALL resolve to the IP address of a web server which serves directly the provider-metadata.json according to requirement 7.

The domain.example is just a placeholder for the organization's main domain. For the organization with the main domain being example.com, the necessary DNS record is csaf.data.security.example.com.

That implies that redirects SHALL NOT be used. The use of the scheme "HTTPS" is required.

7.1.11 Requirement 11: One Folder per Year

The CSAF documents SHALL be located within folders named <YYYY> where <YYYY> is the year given in the value of $.document.tracking.initial_release_date.

Examples 1:

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```
2024
2023
```

7.1.12 Requirement 12: index.txt

The file index.txt SHALL contain the list of all filenames of CSAF documents which are located in the sub-directories with their filenames. Each entry SHALL be terminated by a newline sequence. The last entry MAY skip the newline sequence.

Example 1:

```
2025/esa-2025-421324.json
2024/esa-2024-620109.json
2024/esa-2024-470520.json
2024/esa-2024-430524.json
2023/esa-2023-450116.json
```

> This can be used to download all CSAF documents.

The file index.txt SHALL be located in the folder given as directory URL under $.distributions[*].directory.url in the provider-metadata.json.

> If different TLP labels are used, multiple index.txt exist. However, they are located in the corresponding folders and contain only the filenames of files for that TLP label.

Example 2:

```
.well-known/
├─ csaf/
│  ├─ amber/
│  │  ├─ 2023/
│  │  │  ├─ esa-2023-540205.json
│  │  ├─ 2025/
│  │  │  ├─ esa-2025-440412.json
│  │  ├─ changes.csv
│  │  ├─ index.txt
│  ├─ amber+strict/
│  │  ├─ 2024/
│  │  │  ├─ esa-2024-451013.json
│  │  ├─ 2025/
│  │  │  ├─ esa-2025-540204.json
│  │  ├─ changes.csv
│  │  ├─ index.txt
│  ├─ clear/
│  │  ├─ 2023/
│  │  │  ├─ esa-2023-450116.json
│  │  ├─ 2024/
│  │  │  ├─ esa-2024-430524.json
│  │  │  ├─ esa-2024-470520.json
│  │  │  ├─ esa-2024-620109.json
│  │  ├─ 2025/
│  │  │  ├─ esa-2025-421324.json
│  │  ├─ changes.csv
│  │  ├─ index.txt
│  ├─ green/
│  │  ├─ 2024/
│  │  │  ├─ esa-2024-430316.json
│  │  ├─ 2025/
│  │  │  ├─ esa-2025-431009.json
│  │  ├─ changes.csv
│  │  ├─ index.txt
│  ├─ red/
│  │  ├─ 2022/
│  │  │  ├─ esa-2022-431406.json
│  │  │  ├─ esa-2022-431407.json
│  │  ├─ changes.csv
│  │  ├─ index.txt
```

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The example [2] uses five index.txt files - one for each TLP label. The corresponding provider-metadata.json excerpt is given in example [3]. Example [1] depicts the content of the index.txt within the folder clear located at https://www.example.com/.well-known/csaf/clear/index.txt.

Example 3:

```
"distributions": [
  {
    "directory": {
      "tlp_label": "CLEAR",
      "url": "https://www.example.com/.well-known/csaf/clear/",
    }
  },
  {
    "directory": {
      "tlp_label": "GREEN",
      "url": "https://www.example.com/.well-known/csaf/green/",
    }
  },
  {
    "directory": {
      "tlp_label": "AMBER",
      "url": "https://www.example.com/.well-known/csaf/amber/",
    }
  },
  {
    "directory": {
      "tlp_label": "AMBER+STRICT",
      "url": "https://www.example.com/.well-known/csaf/amber+strict/",
    }
  },
  {
    "directory": {
      "tlp_label": "RED",
      "url": "https://www.example.com/.well-known/csaf/red/",
    }
  }
],
```

7.1.13 Requirement 13: changes.csv

The file changes.csv contains a list of CSAF documents in the current TLP level that were changed recently. Therefore, it SHALL contain the filename as well as the value of $.document.tracking.current_release_date for each CSAF document in the sub-directories without a heading; lines SHALL be sorted by the current_release_date timestamp with the latest one first. The changes.csv SHALL be a valid comma separated values format as defined by [RFC4180] without double quotes.

Note: As a consequence of section 7.1.2 Requirement 2 for filenames and section 7.1.11 Requirement for directory names, there must not be any characters within the changes.csv that would require quoting.

Example 1:

```
2024/esa-2024-430524.json,2025-07-21T11:14:37Z
2025/esa-2025-421324.json,2025-01-10T00:51:10Z
2024/esa-2024-470520.json,2025-01-01T15:34:54Z
2023/esa-2023-450116.json,2024-12-01T17:09:02Z
2024/esa-2024-620109.json,2024-07-30T23:59:29Z
```

Note: As CSAF 2.0 requires quotes, an [RFC4180] parser can read both format revisions.

The file changes.csv SHALL be located in the folder given as directory URL under $.distributions[*].directory.url in the provider-metadata.json.

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The example [2 (of section 7.1.12)] uses five changes.csv files - one for each TLP label. The corresponding provider-metadata.json excerpt is given in example [3 (of section 7.1.12)]. Example [1] depicts the content of the changes.csv within the folder clear located at https://www.example.com/.well-known/csaf/clear/changes.csv.

7.1.14 Requirement 14: Directory Listings

Server-side generated directory listing SHALL be enabled to support manual navigation.

As the content of the directory listing is more or less static, there is little to no benefit in using of client-side scripts. Moreover, client-side scripts, like JavaScript, are usually not evaluated in text-based browsers and are also hard to check programmatically.

7.1.15 Requirement 15: ROLIE Feed

Resource-Oriented Lightweight Information Exchange (ROLIE) is a standard to ease discovery of security content. ROLIE is built on top of the Atom Publishing Format and Protocol, with specific requirements that support publishing security content. All CSAF documents with the same TLP level SHALL be listed in a single ROLIE feed (summary feed). Additional ROLIE feeds might exist that contain only a subset of the CSAF documents. The selection criteria SHOULD be described through the summary. At least one of the feeds

- TLP:CLEAR
- TLP:GREEN

SHALL exist. Each ROLIE feed document SHALL be a JSON file that conforms with [RFC8322].

The ROLIE feed document SHALL contain a feed category with the registered ROLIE information type csaf. The scheme for this category SHALL be urn:ietf:params:rolie:category:information-type.

Example 1:

```
{
  "feed": {
    "id": "example-csaf-feed-tlp-clear",
    "title": "Example CSAF feed (TLP:CLEAR)",
    "link": [
      {
        "rel": "self",
        "href": "https://psirt.domain.example/advisories/csaf/feed-tlp-clear.json"
      }
    ],
    "category": [
      {
        "scheme": "urn:ietf:params:rolie:category:information-type",
        "term": "csaf"
      }
    ],
    "updated": "2024-01-01T12:00:00.000Z",
    "entry": [
      {
        "id": "ESA-2024-001",
        "title": "Multiple vulnerabilities in ABC 0.0.2",
        "link": [
          {
            "rel": "self",
            "href": "https://psirt.domain.example/advisories/csaf/2024/esa-2024-001.json"
          },
          {
            "rel": "hash",
            "href": "https://psirt.domain.example/advisories/csaf/2024/esa-2024-001.json.sha512"
          },
          {
            "rel": "signature",
            "href": "https://psirt.domain.example/advisories/csaf/2024/esa-2024-001.json.asc"
          }
        ],
        "published": "2024-01-01T11:00:00.000Z",
        "updated": "2024-01-01T12:00:00.000Z",
        "summary": {
```

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```
        "content": "Multiple vulnerabilities were fixed in ABC 0.0.3"
       },
       "content": {
        "type": "application/json",
        "src": "https://psirt.domain.example/advisories/csaf/2024/esa-2024-001.json"
       },
       "format": {
        "schema": "https://docs.oasis-open.org/csaf/csaf/v2.1/schema/csaf.json",
        "version": "2.1"
       }
      }
     ]
    }
}
```

Any existing hash file (requirement 18) SHALL be listed in the corresponding entry of the ROLIE feed as an item of the array link having the rel value of hash. Any existing signature file (requirement 19) SHALL be listed in the corresponding entry of the ROLIE feed as an item of the array link having the rel value of signature.

7.1.16 Requirement 16: ROLIE Service Document

The use and therefore the existence of ROLIE service document is optional. If it is used, each ROLIE service document SHALL be a JSON file that conforms with [RFC8322] and lists the ROLIE feed documents. Additionally, it can also list the corresponding ROLIE category documents. The ROLIE service document SHOULD use the filename service.json and reside next to the provider-metadata.json.

Example 1:

```
{
 "service": {
  "workspace": [
   {
    "title": "Public CSAF feed",
    "collection": [
     {
      "title": "Example CSAF feed (TLP:CLEAR)",
      "href": "https://psirt.domain.example/advisories/csaf/feed-tlp-clear.json",
      "categories": {
       "category": [
        {
         "scheme": "urn:ietf:params:rolie:category:information-type",
         "term": "csaf"
        }
       ]
      }
     }
    ]
   }
  ]
 }
}
```

7.1.17 Requirement 17: ROLIE Category Document

The use and therefore the existence of ROLIE category document is optional. If it is used, each ROLIE category document SHALL be a JSON file that conforms with [RFC8322]. A ROLIE category document SHOULD reside next to the corresponding ROLIE feed. ROLIE categories SHOULD be used for to further dissect CSAF documents by one or more of the following criteria:

- document category
- document language
- values of the branch category within the Product Tree including but not limited to
  - vendor
  - product_family
  - product_name
  - product_version
- type of product

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Examples 1:

```
CPU
Firewall
Monitor
PLC
Printer
Router
Sensor
Server
```

- areas or sectors, the products are used in

  Examples 2:

```
Chemical
Commercial
Communication
Critical Manufacturing
Dams
Energy
Healthcare
Water
```

- any other categorization useful to the consumers

Example 3:

```
{
  "categories": {
    "category": [
      {
         "term": "Example Company Product A"
      },
      {
         "term": "Example Company Product B"
      }
    ]
  }
}
```

7.1.18 Requirement 18: Integrity

All CSAF documents SHALL have at least one hash file computed with a secure cryptographic hash algorithm (e.g. SHA-512 or SHA-3) to ensure their integrity. The filename is constructed by appending the file extension which is given by the algorithm.

MD5 and SHA1 SHALL NOT be used.

Example 1:

```
File name of CSAF document: esa-2022-02723.json
File name of SHA-256 hash file: esa-2022-02723.json.sha256
File name of SHA-512 hash file: esa-2022-02723.json.sha512
```

The file content SHALL start with the first byte of the hexadecimal hash value. The hash value SHALL be represented in lowercase. Any subsequent data (like a filename) which is optional SHALL be separated by at least one space.

Example 2:

```
ea6a209dba30a958a78d82309d6cdcc6929fcb81673b3dc4d6b16fac18b6ff38  esa-2022-02723.json
```

If a ROLIE feed exists, each hash file SHALL be listed in it as described in requirement 15.

7.1.19 Requirement 19: Signatures

All CSAF documents SHALL have at least one OpenPGP signature file which is provided under the same filename which is extended by the appropriate extension. This signature SHALL be presented as an ASCII armored file. See [RFC4880] for more details.

Example 1:

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```
File name of CSAF document: esa-2022-02723.json
File name of signature file: esa-2022-02723.json.asc
```

If a ROLIE feed exists, each signature file SHALL be listed in it as described in requirement 15.

At all times, signatures SHALL remain valid for a minimum of 30 days and ideally for at least 90 days. When executing CSAF document signatures, the signing party SHOULD adhere to or surpass the prevailing best practices and recommendations regarding key length. Tools SHOULD treat the violation of the rules given in the first sentence as:

- warning if the signature is only valid for 90 days or less at the time of the verification,
- error, which MAY be ignored by the user per option, if the signature is only valid for 30 days or less at the time of the verification and
- error if the signature is expired at the time of the verification.

Signatures are a technical construct. A current signature does not guarantee that newer information is not also available. The date representing the state of knowledge on which a CSAF document is based is described by the current_release_date in the CSAF document. For this reason and because the signatures are located in separate files, signatures can be replaced without modifying the document itself. Therefore, issuing parties SHALL replace signatures without updating the corresponding CSAF documents.

7.1.20 Requirement 20: Public OpenPGP Key

The public part of the OpenPGP key used to sign the CSAF documents SHALL be available. This key file SHALL be presented as an ASCII armored file. It SHOULD also be available at a public key server.

> For example, the public part of the OpenPGP key could be placed in a directory openpgp adjacent to the provider-metadata.json.

The OpenPGP key SHOULD have a strength that is considered secure.

> Guidance on OpenPGP key strength can be retrieved from technical guidelines of competent authorities.

7.1.21 Requirement 21: List of CSAF Providers

The file aggregator.json SHALL be present and valid according to the JSON schema CSAF aggregator. It SHALL NOT be stored adjacent to a provider-metadata.json.

The file aggregator.json SHOULD be accessible at the registered path in the .well-known directory: /.well-known/csaf-aggregator/aggregator.json.

Examples 1:

```
https://aggregator.example/.well-known/csaf-aggregator/aggregator.json
https://aggregator.example/.well-known/csaf-aggregator/v2.1/aggregator.json
```

The file aggregator.json SHOULD only list the latest version of the metadata of a CSAF provider.

Example 2:

```
{
  "$schema": "https://docs.oasis-open.org/csaf/csaf/v2.1/schema/aggregator.json",
  "aggregator": {
    "category": "lister",
    "contact_details": "Example CSAF Lister can be reached at contact_us@lister.example, or via our website at https://lister.example/security/csaf/aggregator/contact.",
    "issuing_authority": "This service is provided as it is. It is free for everybody.",
    "name": "Example CSAF Lister",
    "namespace": "https://lister.example"
  },
  "aggregator_version": "2.1",
  "canonical_url": "https://aggregator.example/.well-known/csaf-aggregator/aggregator.json",
  "csaf_providers": [
    {
      "metadata": {
        "last_updated": "2024-01-12T20:20:56.169Z",
        "publisher": {
          "category": "vendor",
```

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```
          "name": "Example Company ProductCERT",
          "namespace": "https://psirt.example.com"
        },
        "role": "csaf_provider",
        "url": "https://www.example.com/.well-known/csaf/provider-metadata.json"
      }
    },
    {
      "metadata": {
        "last_updated": "2024-01-12T21:35:38.000Z",
        "publisher": {
          "category": "coordinator",
          "name": "Example Coordinator CERT",
          "namespace": "https://cert.example"
        },
        "role": "csaf_trusted_provider",
        "url": "https://cert.example/advisories/csaf/provider-metadata.json"
      }
    }
  ],
  "last_updated": "2024-01-24T22:35:38.978Z"
}
```

7.1.22 Requirement 22: Two Disjoint Issuing Parties

The file aggregator.json (requirement 21) lists at least two disjoint CSAF providers (including CSAF trusted providers) or one CSAF publisher and one CSAF provider (including CSAF trusted provider).

7.1.23 Requirement 23: Mirror

The CSAF documents for each issuing party that is mirrored SHALL be in a different folder. The folder name SHOULD be retrieved from the name of the issuing authority. This folders SHALL be adjacent to the aggregator.json (requirement 21). Each such folder SHALL at least:

- provide a provider-metadata.json for the current issuing party.
- provide the ROLIE feed document according to requirement 15 which links to the local copy of the CSAF document.

Example 1:

```
{
  "$schema": "https://docs.oasis-open.org/csaf/csaf/v2.1/schema/aggregator.json",
  "aggregator": {
    "category": "aggregator",
    "contact_details": "Example Aggregator can be reached at contact_us@aggregator.example, or via our website at https://aggregator.example/security/csaf/aggregator/
contact.",
    "issuing_authority": "This service is provided as it is. It is free for everybody.",
    "name": "Example Aggregator",
    "namespace": "https://aggregator.example"
  },
  "aggregator_version": "2.1",
  "canonical_url": "https://aggregator.example/.well-known/csaf-aggregator/aggregator.json",
  "csaf_providers": [
    {
      "metadata": {
        "last_updated": "2024-01-12T20:20:56.169Z",
        "publisher": {
          "category": "vendor",
          "name": "Example Company ProductCERT",
          "namespace": "https://psirt.example.com"
        },
        "role": "csaf_provider",
        "url": "https://www.example.com/.well-known/csaf/provider-metadata.json"
      },
      "mirrors": [
        "https://aggregator.example/.well-known/csaf-aggregator/Example_Company_ProductCERT/provider-metadata.json"
      ]
    },
    {
      "metadata": {
        "last_updated": "2024-01-12T21:35:38.000Z",
        "publisher": {
          "category": "coordinator",
          "name": "Example Coordinator CERT",
          "namespace": "https://cert.example"
        },
        "role": "csaf_trusted_provider",
```

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```
        "url": "https://cert.example/advisories/csaf/provider-metadata.json"
      },
      "mirrors": [
        "https://aggregator.example/.well-known/csaf-aggregator/Example_Coordinator_CERT/provider-metadata.json"
      ]
    }
  ],
  "last_updated": "2024-01-24T22:35:38.978Z"
}
```

7.1.24 Requirement 24: HTTP User-Agent

Access to the CSAF related files and directories provided, for both metadata and documents, SHALL be allowed independent of the value of HTTP User-Agent.

> Limit the value of HTTP User-Agents to a certain set would hinder adoption of tools retrieving the files.

The only exception is that the temporary blocking of certain HTTP User-Agents is allowed to mitigate an ongoing security incident (e.g. a DoS attack on the web server serving the CSAF files). However, a less severe measure with a similar effect SHOULD be used. CSAF related files and directories SHOULD be exempted from temporary blocking. The temporary blocking SHOULD be removed as soon as possible, at latest two weeks after the security incident process was completed.

> Also confer to the TC's guidance on content delivery networks and caching.

7.1.25 Requirement 25: Access-Control-Allow-Origin

For any CSAF documents and related metadata, the web server SHOULD set the HTTP header `Access-Control-Allow-Origin: *`.

> The HTTP header enables users to access the CSAF data with web browser based clients.

The value of the HTTP header MAY be altered to allow just specified domains. In such case, the response SHOULD follow the recommendations of [FETCH] including but not limited to those about the `Vary` header.

> Such restriction may allow the allow-listed domains to send credentials.

7.2 Roles

This subsection groups the requirements from the previous subsection into named sets which target the roles with the same name. This allows end users to request their suppliers to fulfill a certain set of requirements. A supplier can use roles for advertising and marketing.

The roles "CSAF Publisher", "CSAF Provider", and "CSAF Trusted Provider" are intended directly for issuing parties and form the first group. The second group consists of the roles "CSAF Lister" and "CSAF Aggregator". They collect data from the aforementioned issuing parties of the first group and provide them in a single place to aid in automation. Parties of the second group can also issue their own advisories. However, they SHALL follow the rules for the first group for that.

Both, a CSAF lister and a CSAF aggregator, decide based on their own rules which issuing parties to list respectively to mirror. However, an issuing party MAY apply to be listed or mirrored.

Issuing parties SHALL indicate through the value `false` in `list_on_CSAF_aggregators` if they do not want to be listed. Issuing parties SHALL indicate through the value `false` in `mirror_on_CSAF_aggregators` if they do not want to be mirrored.

The values are independent. The combination of the value `false` in `list_on_CSAF_aggregators` and `true` in `mirror_on_CSAF_aggregators` implies that the issuing party does not want to be listed without having the CSAF documents mirrored. Therefore, a CSAF aggregator can list that issuing party if it mirrors the files.

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7.2.1 Role: CSAF Publisher

A distributing party satisfies the "CSAF Publisher" role if the party:

- satisfies the requirements 1 to 4 in section 7.1.
- distributes only CSAF documents on behalf of its own.

7.2.2 Role: CSAF Provider

A CSAF publisher satisfies the "CSAF Provider" role if the party fulfills the following three groups of requirements:

Firstly, the party:

- satisfies the "CSAF Publisher" role profile.
- additionally satisfies the requirements 5 to 7, 24 and 25 in section 7.1.

Secondly, the party:

- satisfies at least one of the requirements 8 to 10 in section 7.1.

Thirdly, the party:

- satisfies the requirements 11 to 14 in section 7.1 or requirements 15 to 17 in section 7.1.

> If the party uses the ROLIE-based distribution, it must also satisfy requirements 15 to 17. If it uses the directory-based distribution, it must also satisfy requirements 11 to 14.

7.2.3 Role: CSAF Trusted Provider

A CSAF provider satisfies the "CSAF Trusted Provider" role if the party:

- satisfies the "CSAF Provider" role profile.
- additionally satisfies the requirements 18 to 20 in section 7.1.

7.2.4 Role: CSAF Lister

A distributing party satisfies the "CSAF Lister" role if the party:

- satisfies the requirements 6, 21, 22, 24 and 25 in section 7.1.
- uses the value lister for $.aggregator.category.
- does not list any mirror pointing to a domain under its own control.

> The purpose of this role is to provide a list of URLs where to find CSAF documents. It is not assumed that the list will be complete.

7.2.5 Role: CSAF Aggregator

A distributing party satisfies the "CSAF Aggregator" role if the party:

- satisfies the requirements 1 to 6 and 21 to 25 in section 7.1.
- uses the value aggregator for $.aggregator.category.
- lists a mirror for at least two disjoint issuing parties pointing to a domain under its own control.
- links the public part of the OpenPGP key used to sign CSAF documents for each mirrored issuing party in the corresponding provider-metadata.json.
- provides for each CSAF document that is mirrored a signature (requirement 19) and a hash (requirement 18). Both SHALL be listed in the ROLIE feed. If the issuing party provides those files for a CSAF document, they SHOULD be copied as well. If the issuing party does not provide those files, they SHALL be created by the CSAF aggregator. Such a signature does not imply any liability of CSAF aggregator for the content of the corresponding CSAF document. It just confirms that the CSAF document provided has not been modified after being downloaded from the issuing party. A CSAF aggregator MAY add additional signatures and hashes for a CSAF document.

Additionally, a CSAF aggregator MAY list one or more issuing parties that it does not mirror.

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The purpose of this role is to provide a single point where CSAF documents can be retrieved. Multiple CSAF aggregators are expected to exist around the world. None of them is required to mirror all CSAF documents of all issuing parties. CSAF aggregators can be provided for free or as a paid service.

To aid in automation, CSAF aggregators MAY mirror CSAF documents from CSAF publishers. Regarding the terms of use they SHOULD consult with the issuing party. The purpose of this option is that a consumer can retrieve CSAF documents from a CSAF publisher as if this issuing party would be a CSAF trusted provider. To reach that goal, a CSAF aggregator collects the CSAF documents from the CSAF publisher and mirrors it. The collection process MAY be automated or manual. CSAF aggregators announce the collection interval through the field update_interval in the corresponding item of the CSAF publishers list (csaf_publishers) in their aggregator.json. To minimize the implementation efforts and process overhead, a CSAF aggregator MAY upload the CSAF documents of a CSAF publisher into an internal instance of a CSAF provider software. Such construct is called "CSAF Proxy Provider" as it can be mirrored by the CSAF aggregator software. However, such a CSAF proxy provider SHALL NOT be accessible from anyone else than the CSAF aggregator itself. Otherwise, that would violate the second rule of section 7.2.1. Therefore, it is recommended to expose the CSAF proxy provider only on localhost and allow the access only from the CSAF aggregator software.

7.3 Retrieving Rules

The retrieving process executes in two phases: Finding the provider-metadata.json (requirement 7 in section 7.1) and retrieving CSAF documents.

A retrieving party SHOULD do the first phase every time. Based on the setup and use case of the retrieving party it MAY choose to do it less often, e.g. only when adding new or updating distributing parties. In that case, it SHOULD to check regularly whether new information is available.

7.3.1 Finding provider-metadata.json

Direct locating: The following process SHOULD be used to determine the location of a provider-metadata.json (requirement 7 in section 7.1) based on the main domain of the issuing party.

First, an ordered list of possible provider-metadata.json candidates SHOULD be generated in the following way:

1. Checking the Well-known URL (requirement 9 in section 7.1)
2. Checking the security.txt (requirement 8 in section 7.1)
3. If the above steps fail to produce any candidates: Checking the DNS path (requirement 10 in section 7.1)

Second, select one or more provider-metadata.json to use from the list of valid candidates. If the retrieving party is only able to process one provider-metadata.json, the first one in the list SHOULD be chosen.

The term "checking" used in the listing above SHOULD be understood as follows: Try to access the resource and test whether the response provides an expected result as defined in the respective requirement of that step as defined in section 7.1. If that is the case, the response is added to the list of candidates - otherwise not. If the resource yields more than one response, the responses are added to the list in the order they are returned from the resource.

Indirect locating: A retrieving party MAY choose to determine the location of a provider-metadata.json by retrieving its location from an aggregator.json (requirement 21 in section 7.1) of a CSAF lister or CSAF aggregator.

7.3.2 Retrieving CSAF Documents

Given a provider-metadata.json, the following process SHOULD be used to retrieve CSAF documents:

1. Parse the provider-metadata.json to determine whether the directory-based (requirements 11 to 14 in section 7.1) or ROLIE-based distribution (requirements 15 to 17 in section 7.1) is used. If both are present, the ROLIE information SHOULD be preferred.
2. For any CSAF trusted provider, the hash and signature files (requirements 18 to 19 in section 7.1) SHOULD be retrieved together with the CSAF document. They SHALL be checked before further processing the CSAF document.
3. Test the CSAF document against the schema.

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4. Execute mandatory tests on the CSAF document.

7.3.3 Finding aggregator.json

Direct locating: The file aggregator.json SHOULD be located at the registered .well-known path based on the main domain of the aggregator as specified in requirement 21 in section 7.1.

Indirect locating: A retrieving party MAY choose to determine the location of an aggregator.json through an out-of-band channel, e.g. an email list, news articles or a public website.

7.4 Transition between CSAF 2.0 and CSAF 2.1

This subsection details the process that SHOULD be followed when transitioning the distribution of documents from CSAF 2.0 to CSAF 2.1. Different scenarios can be encountered:

- Providers will continue to use CSAF 2.0. This SHOULD be avoided, as the CSAF ecosystem greatly profits from adoption of the new standard version.
- Providers will immediately upgrade all documents as well as the provider-metadata.json to CSAF 2.1. While this benefits the adoption of CSAF 2.1, consumers that still rely on CSAF 2.0 are cut off.
- Providers will begin a transition period, in which they continue to serve existing documents in CSAF 2.0, gradually updating the existing document base (e.g. by using a CSAF 2.0 to CSAF 2.1 converter as described in 9.1.18) and publishing new documents using CSAF 2.1.

7.4.1 Announcing the Transition

In the last scenario, a temporary parallel distribution of CSAF 2.0 and CSAF 2.1 documents and provider metadata is RECOMMENDED. The provider SHOULD announce a transition period containing three points in time:

- The beginning of the transition period, where the provider is starting to serve CSAF 2.1 documents, while CSAF 2.0 being authoritative.

> It is expected that the CSAF 2.1 files can be used in production from this point onward.

- The roll-over-date at which CSAF 2.1 becomes authoritative but CSAF 2.0 is still supported.
- The end of the transition period, after which CSAF 2.0 is not supported, i.e. maintained, any more.

The announcement MAY contain also the following information:

- The first date, when CSAF 2.1 documents are available and automatically retrievable from the server through a CSAF 2.1 provider-metadata.json.

> This might be a public beta prior to the begin of the transition period. However, those documents SHOULD NOT be used in production as they may be unstable due to their beta status.

- The last date, when CSAF 2.0 documents are available and automatically retrievable from the server through a CSAF 2.0 provider-metadata.json.

> This date is usually at the end of the transition period.

7.4.2 Transition Process for a CSAF Provider

The following process SHOULD be followed:

- A provider-metadata.json in conformance to CSAF 2.0 SHOULD be placed at /.well-known/csaf/v2.0/provider-metadata.json.
  - Its canonical_url SHALL be set to an URL corresponding to the /.well-known/csaf/v2.0/provider-metadata.json path.

> The property maintained_until was not defined in CSAF 2.0 and therefore cannot be used - otherwise it would be set to the end of the transition period.

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  - The URL SHALL also be added as an entry in the security.txt, if used.
- A provider-metadata.json in conformance to CSAF 2.1 SHOULD be placed at /.well-known/csaf/v2.1/provider-metadata.json.
  - Its canonical_url SHALL be set to an URL corresponding to the /.well-known/csaf/v2.1/provider-metadata.json path.
  - Optionally, the property maintained_from can be set to an appropriate date in the future, if the service is not considered stable yet.

> The transition officially starts at the date and time noted in the property maintained_from as this is the point in time when both CSAF 2.0 and CSAF 2.1 documents are retrievable in production-grade quality.

  - The URL SHALL also be added to the security.txt, if used.
- At the begin of the transition period, a provider-metadata.json in conformance to CSAF 2.0 SHOULD be placed at /.well-known/csaf/provider-metadata.json.
  - The content of the resource SHALL be equal to the resource accessible at /.well-known/csaf/v2.0/provider-metadata.json.
  - For file-based distribution servers, this MAY be achieved by using a symlink. Redirects SHALL NOT be used (cf. to requirement 7.1.9)
- Sometime before the roll-over-date, all existing CSAF 2.0 documents SHOULD be converted to CSAF 2.1.
- A the roll-over-date, a provider-metadata.json in conformance to CSAF 2.1 SHOULD be placed at /.well-known/csaf/provider-metadata.json.
  - The content of the resource SHALL be equal to the resource accessible at /.well-known/csaf/v2.1/provider-metadata.json.
  - For file-based distribution servers, this MAY be achieved by using a symlink. Redirects SHALL NOT be used (cf. to requirement 7.1.9)
- At the end of the transition period, the URL of the CSAF 2.0 provider-metadata.json SHOULD be removed from the security.txt.
  - The unmaintained CSAF 2.0 directory structure and files SHOULD be removed or made inaccessible.
  - The CSAF 2.0 documents MAY be archived.

If a DNS path (cf. section 7.1.10) is used instead of the well-known URL, the same process SHOULD be followed taking the rules below into account:

- The leading and authoritative provider-metadata.json is always served according to the DNS path requirement (see section 7.1.10).

> This implies that a CSAF 2.0 provider-metadata.json is served at the beginning and replaced by the CSAF 2.1 provider-metadata.json at the roll-over-date.

- It is recommended to use the folders v2.0 and v2.1 to differentiate between the CSAF versions.

7.4.3 Archive of CSAF Document from Previous Version

The following rules apply for the archival of CSAF document from a previous version:

- This archive SHOULD be located in /.well-known/csaf/archive/ and use the file name v2.0.tar.zst, v2.0.tar.bz2 or v2.0.tar.xz.
- The CSAF documents within the archive SHALL be sorted into folders according to requirement 11 in section 7.1.11 and be accompanied by a hash according to requirement 18 7.1.18.
- The archive SHALL be accompanied by a hash of the same algorithm.
- Existing signatures MAY also be included into the archive. It is NOT RECOMMENDED to renew the signatures in the archive unless the archive is updated with new content.

7.4.4 Transition Process for a CSAF Aggregator

Similarly, to the transition process for a CSAF provider, the same process SHOULD be used for a CSAF aggregator. It is RECOMMENDED to use the following URLs during the process:

- /.well-known/csaf-aggregator/aggregator.json for the currently valid aggregator metadata.
- /.well-known/csaf-aggregator/v2.0/aggregator.json for a valid CSAF 2.0 aggregator.json
- /.well-known/csaf-aggregator/v2.1/aggregator.json for a valid CSAF 2.1 aggregator.json

A CSAF 2.1 aggregator SHALL only sync and list CSAF 2.1 publishers and providers.

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8. Safety, Security, and Data Protection Considerations

All safety, security, and data protection requirements relevant to the context in which CSAF documents are used SHALL be translated into, and consistently enforced through, CSAF implementations and processes.

CSAF documents are based on JSON, thus the security considerations of [RFC8259] apply and are repeated here as service for the reader:

> Generally, there are security issues with scripting languages. JSON is a subset of JavaScript but excludes assignment and invocation.
>
> Since JSON's syntax is borrowed from JavaScript, it is possible to use that language's eval() function to parse most JSON texts (but not all; certain characters such as U+2028 LINE SEPARATOR and U+2029 PARAGRAPH SEPARATOR are legal in JSON but not JavaScript). This generally constitutes an unacceptable security risk, since the text could contain executable code along with data declarations. The same consideration applies to the use of eval()-like functions in any other programming language in which JSON texts conform to that language's syntax.

[RFC8259] also states that JSON keys SHOULD be unique. However, this is not enforced. As some CSAF documents are created with human interaction, there is a risk of duplicate JSON keys. CSAF documents with duplicate JSON keys can lead situations where one JSON key is displayed but the other one is processed which can lead to unexpected outcomes - especially if automated actions are triggered. CSAF tools SHOULD handle this situation appropriately. Appropriate actions can differ depending on the intended use and environment.

In addition, CSAF documents may be rendered by consumers in various human-readable formats like HTML or PDF. Thus, for security reasons, CSAF producers and consumers SHALL adhere to the following:

- CSAF producers SHOULD NOT emit messages that contain HTML, even though GitHub-flavored Markdown is permitted. To include HTML, source code, or any other content that may be interpreted or executed by a CSAF consumer, e.g. to provide a proof-of-concept, the issuing party SHALL use Markdown's fenced code blocks or inline code option.
- Deeply nested markup can cause a stack overflow in the Markdown processor [GFMENG]. To reduce this risk, CSAF consumers SHALL use a Markdown processor that is hardened against such attacks. **Note**: One example is the GitHub fork of the cmark Markdown processor [GFMCMARK].
- To reduce the risk posed by possibly malicious CSAF files that do contain arbitrary HTML (including, for example, data:image/svg+xml), CSAF consumers SHALL either disable HTML processing (for example, by using the --safe option in the cmark Markdown processor) or run the resulting HTML through an HTML sanitizer.
- To reduce the risk posed by possibly malicious links within a CSAF document (including, for example, javascript: links), CSAF consumers SHALL either remove all actions from links (for example, by displaying them as standard text) or render only those actionable that are known to be safe (for example, determining that via the media type). CSAF consumers that are not prepared to deal with the security implications of formatted messages SHALL NOT attempt to render them and SHALL instead fall back to the corresponding plain text messages. As also any other programming code can be contained within a CSAF document, CSAF consumers SHALL ensure that none of the values of a CSAF document is run as code. Moreover, it SHALL be treated as unsafe (user) input.

> Additional, supporting mitigation measures like retrieving only CSAF documents from trusted sources and check their integrity and signature before parsing the document SHOULD be in place to reduce the risk further.

The distribution requirements of CSAF data allow to specify domains as the value of the HTTP header Access-Control-Allow-Origin. While a wildcard (*) as header value usually prevents implementing browsers from sending credentials during the CORS request, the restriction to specified domains often enables sending credentials. Allowing several specified domains results in using dynamics on the server, which can widen the attack surface by using more code and configuration. Furthermore, this might reveal information about internal structures, e.g. which domains are allowed to send credentials, or which tools are used. Given that credentials from a browser are a potent tool in the event of an attack, restricting the origins seems to imply a higher risk and therefore less secure than allowing all domains without credentials.

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As setting the Access-Control-Allow-Origin header potentially allows for cross site request forgery, it SHOULD only be served on files and directories containing CSAF data. For any restricted feeds, standard authentication methods SHOULD be used that are not send by web browsers if the wildcard is used as header value.

Implementors are reminded that the evaluation of JSONPath can lead to multiple elements within the result array. Especially CSAF validators SHALL NOT stop the evaluation of the result array after the first instance unless instructed explicitly by a non-default option even though the specification might just expect one instance. Otherwise, test results could be incomplete or wrong. To avoid ambiguity, it is RECOMMENDED to use JSON pointer (see [RFC6901]) when identifying or referring to a specific key or instance within a CSAF document.

CSAF tools SHALL have carefully chosen default options when processing CSAF Documents. Especially CSAF validators have a risk of falling for DoS attacks as they could process data from untrusted sources. Therefore, it is RECOMMENDED to limit the amount of messages that can be issued by a test or during a run to a reasonable default. In such case, the CSAF validator SHALL indicate that there might have been more messages but processing stopped because of the limit.

CSAF producers, CSAF consumers and CSAF validators SHOULD NOT automatically retrieve JSON schemas from a URL declared in CSAF documents as this poses a security risk. Loading files from an untrusted source can result in information leakage or remotely triggered automated exploitation. If CSAF producers, CSAF consumers or CSAF validators provide an option to interactively or automatically load missing schemes, they SHALL point out the risks of setting the option and actively warn the user about it. Such option SHALL NOT be set by default. CSAF producers, CSAF consumers and CSAF validators SHOULD keep a local copy of all schemas necessary to fulfill their tasks. Tool developers SHOULD ensure to regularly check that those copies still reflect the current version of those schemas as they could have been altered, e.g. by an Errata to fix a bug. Such checks and potential updates SHOULD be done at least before a release. The same applies for necessary data from registries and other sources.

> Such checks can be automated, e.g. in the CI/CD pipeline.

CSAF documents can contain personally identifiable information. This includes but is not limited to names, e.g. in acknowledgments, as well as email addresses, e.g. in involvement. Issuing parties SHALL ensure that they are allowed to publish such information.

> Acknowledging entities or individuals is always possible with their consent. Functional email addresses are usually not consider to be personally identifiable information. The same applies for names of organizations.

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9. Conformance

In the only subsection of this section, the conformance targets and clauses are listed. The clauses, matching the targets one to one, are listed in separate sub-subsections of the targets listing subsection.

The order in which targets, and their corresponding clauses appear is somewhat arbitrary as there is no natural order on such diverse roles participating in the document exchanging ecosystem.

Except for the target **CSAF Document**, all other 35 targets span a taxonomy of the complex CSAF ecosystems existing in and between diverse security advisory generating, sharing, and consuming communities.

The capabilities are not necessarily organized in increasing quality levels for targets because the security advisory sharing communities follow the chain link model. Instead, a single minimum capability level for every target is given to maintain important goals of providing a common framework for security advisories:

- Fast production, sharing, and actionable consumption of security advisories
- Consistent end to end automation through collaborating actors
- Clear baseline across the communities per this specification
- Additional per-community cooperative extensions which may flow back into future updates of this specification

9.1 Conformance Targets

This document defines requirements for the file format and for certain software components that interact with it. The entities ("conformance targets") for which this document defines requirements are:

- **CSAF Document**: A security advisory text document in the format defined by this document.
- **CSAF Producer**: A program which emits output in the CSAF format.
- **CSAF Direct Producer**: An analysis tool which acts as a CSAF Producer.
- **CSAF Converter**: A CSAF Producer that transforms the output of an analysis tool from its native output format into the CSAF format.
- **CVRF CSAF Converter**: A CSAF Producer which takes a CVRF document as input and converts it into a valid CSAF Document.
- **CSAF Content Management System**: A program that is able to create, review and manage CSAF Documents and is able to preview their details as required by CSAF Viewer.
- **CSAF Post-Processor**: A CSAF Producer that transforms an existing CSAF Document into a new CSAF Document, for example, by removing or redacting elements according to sharing policies.
- **CSAF Modifier**: A CSAF Post-Processor which takes a CSAF Document as input and modifies the structure or values of properties. The output is a valid CSAF Document.
- **CSAF Translator**: A CSAF Post-Processor which takes a CSAF Document as input and translates values of properties into another language. The output is a valid CSAF Document.
- **CSAF Consumer**: A program that reads and interprets a CSAF Document.
- **CSAF Viewer**: A CSAF Consumer that reads a CSAF Document, displays a list of the results it contains, and allows an end user to view each result in the context of the artifact in which it occurs.
- **CSAF Management System**: A program that is able to manage CSAF Documents and is able to display their details as required by CSAF Viewer.
- **CSAF Asset Matching System**: A program that connects to or is an asset database and is able to manage CSAF Documents as required by CSAF Management System as well as matching them to assets of the asset database.
- **CSAF Basic Validator**: A program that reads a document and checks it against the JSON schema and performs mandatory tests.
- **CSAF Extended Validator**: A CSAF Basic Validator that additionally performs recommended tests.
- **CSAF Full Validator**: A CSAF Extended Validator that additionally performs informative tests.
- **CSAF SBOM Matching System**: A program that connects to or is an SBOM database and is able to manage CSAF Documents as required by CSAF Management System as well as matching them to SBOM components of the SBOM database.
- **CSAF 2.0 to CSAF 2.1 Converter**: A CSAF Producer which takes a CSAF 2.0 Document as input and converts it into a valid CSAF 2.1 Document.

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- **CSAF Library**: A library that implements CSAF data capabilities.
- **CSAF Library with Basic Validation**: A CSAF Library that also satisfies the conformance target "CSAF Basic Validator".
- **CSAF Library with Extended Validation**: A CSAF Library that also satisfies the conformance target "CSAF Extended Validator".
- **CSAF Library with Full Validation**: A CSAF Library that also satisfies the conformance target "CSAF Full Validator".
- **CSAF Downloader**: A program that retrieves CSAF Documents in an automated fashion.
- **CSAF Withdrawer**: A CSAF Post-Processor that transforms a given CSAF into a Withdrawn one.
- **CSAF Superseder**: A CSAF Post-Processor that transforms a given CSAF into a Superseded one.
- **CSAF RVISC ID Updater**: A CSAF Post-Processor that updates vulnerability IDs in a given CSAF based on the entries in [RVISC].
- **CSAF Additional Test**: A test that is not yet defined in section 6.
- **CSAF Extension**: A specified JSON object conveying additional information different from the content that can be conveyed with the CSAF Core elements.
- **CSAF Extension Schema**: A JSON schema specifying the content and properties of a CSAF Extension.
- **CSAF Extension Overlay Test**: A test whose execution depends on the presence of the specifying CSAF Extension that extends or replaces a test standardized in this specification or a CSAF Additional Test.
- **CSAF Extension Additional Test**: A test whose execution depends on the presence of the specifying CSAF Extension that provides additional checks in the context of the CSAF Extension or the CSAF Document the extension is embedded in.
- **CSAF Extension Test**: A test that is either a CSAF Extension Overlay Test or a CSAF Extension Additional Test.
- **CSAF Extension Specification**: The specification of a single CSAF Extension and related material.
- **CSAF Extension Bundle**: A of compilation of machine-readable artifacts related to a single CSAF Extension.
- **CSAF Extension Package**: A of compilation of all artifacts related to a single CSAF Extension.
- **CSAF Extension Collection**: A set of multiple CSAF Extension Package.

9.1.1 Conformance Clause 1: CSAF Document

A text file or data stream satisfies the "CSAF Document" conformance profile if it:

- is valid JSON encoded using UTF-8.
- conforms to the syntax and semantics defined in section 2.2.
- conforms to the syntax and semantics defined in section 2.3.
- conforms to the syntax and semantics defined in section 2.4.
- conforms to the syntax and semantics defined in section 3.
- satisfies at least one profile defined in section 4.
- conforms to the syntax and semantics defined in section 5.
- does not fail any mandatory test defined in section 6.1 respectively its corresponding CSAF Extension Test.
- contains no extension that does not fulfill the conformance profile "CSAF Extension".
- contains no extension at any other path than the specified ones in sections 3.1.10.4, 3.2.2.15, 3.2.4.10.1, 3.2.4.17 and 3.2.5.

9.1.2 Conformance Clause 2: CSAF Producer

A program satisfies the "CSAF Producer" conformance profile if the program:

- produces output in the CSAF format, according to the conformance profile "CSAF Document".
- satisfies those normative requirements in sections 3 and 8 that are designated as applying to CSAF Producers.
- satisfies those normative requirements in sections 2.4 and 8 that are designated as applying to CSAF Tools.

9.1.3 Conformance Clause 3: CSAF Direct Producer

An analysis tool satisfies the "CSAF Direct Producer" conformance profile if the analysis tool:

- satisfies the "CSAF Producer" conformance profile.
- additionally satisfies those normative requirements in section 3 that are designated as applying to "direct producers" or to "analysis tools".
- does not emit any objects, properties, or values which, according to section 3, are intended to be produced only by converters.

9.1.4 Conformance Clause 4: CSAF Converter

A converter satisfies the "CSAF Converter" conformance profile if the converter:

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- satisfies the “CSAF Producer” conformance profile.
- additionally satisfies those normative requirements in section 3 that are designated as applying to converters.
- does not emit any objects, properties, or values which, according to section 3, are intended to be produced only by direct producers.

9.1.5 Conformance Clause 5: CVRF CSAF Converter

A program satisfies the “CVRF CSAF Converter” conformance profile if the program fulfills the following two groups of requirements:

Firstly, the program:

- satisfies the “CSAF Producer” conformance profile.
- takes only CVRF documents as input.
- issues a warning that an additional property was detected and not converted if it detects an additional property in the input. Such a warning SHALL include the additional property and its path. The CVRF CSAF Converter SHALL ignore that additional property during the conversion.
- includes in every error and warning the relevant paths and values from the original file that triggered the alert, unless otherwise specified in this standard.
- additionally satisfies the normative requirements given below (see also table 5).

Secondly, the program fulfills the following for all items of:

- value type string with format date-time: If the value contains a 60 in the seconds place, the CVRF CSAF Converter SHALL replace the seconds and their fractions with 59.999999. In addition, the converter issues a warning that leap seconds are now prohibited in CSAF and the value has been replaced. The CVRF CSAF Converter SHOULD indicate in such warning message whether the value was a valid leap second or not.
- type $['$defs'].branches_t:
  - If any prod:Branch instance has the type Legacy, Realm, or Resource, the CVRF CSAF Converter SHALL replace those with the category product_name. In addition, the converter issues a warning that those types do not exist in CSAF 2.1 and have been replaced with the category product_name.

    > There is a chance, that this replacement is incorrect or another category is a better fit. Users of the converter are advised to check the content of such documents to make sure the conversion is correct or at least not misleading.

  - If any Branch Type appears multiple times along a path under /prod:ProductTree/prod:Branch and the Branch Type does not map to an excepted category according to test 6.1.57, the CVRF CSAF Converter SHALL try to convert the data into a valid product tree by applying the following steps to the path:

    1. If the stacked Branch Type is Vendor, the vendor items named Open Source, NOASSERTION undefined and unknown (white space, dash, hyphen, minus, underscore, invisible character and case insensitive on both sides) SHALL be removed.
    2. If the stacked Branch Type is Product Version and the item directly before the first Product Version is a Product Name:
       - the category of the original Product Name item SHALL be changed to product_family and
       - the category of the first Product Version item SHALL be changed to product_name and
       - the value of the newly created product_family item SHALL be prepended at the value of the newly created product_name item.

    If the CVRF CSAF Converter is able to create a valid product tree, it SHALL issue a warning that an invalid product tree with stacked branch types was detected and resolved. Such a warning SHALL include the invalid path as well as the branch types that were present multiple times.

    > A tool MAY provide a non-default option to suppress this conversion step.

    If the CVRF CSAF Converter is unable to create a valid product tree, it SHALL issue an error that an invalid product tree with stacked branch types was detected and could not be resolved. Such a error SHALL include the invalid path as well as the branch types that were present multiple times.

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> A tool MAY provide a non-default option to output the invalid document.

- type $['$defs'].full_product_name_t..product_identification_helper.cpe: If a CPE is invalid, the CVRF CSAF Converter SHOULD remove the invalid value and issue a warning that an invalid CPE was detected and removed.
- type $['$defs'].version_t: If any element doesn't match the semantic versioning, replace the all elements of type $['$defs'].version_t with the corresponding integer version. For that, CVRF CSAF Converter sorts the items of $.document.tracking.revision_history by number ascending according to the rules of CVRF. Then, it replaces the value of number with the index number in the array (starting with 1). The value of $.document.tracking.version is replaced by value of number of the corresponding revision item. The match SHALL be calculated by the original values used in the CVRF document. If this conversion was applied, for each Revision the original value of cvrf:Number SHALL be set as legacy_version in the converted document.
- $.document.acknowledgments[*].organization and $.vulnerabilities[*].acknowledgments[*].organization: If more than one cvrf:Organization instance is given, the CVRF CSAF Converter converts the first one into the organization. In addition, the converter issues a warning that information might be lost during conversion of document or vulnerability acknowledgment.
- $.document.category:
  - If the cvrf:DocumentType is Security Advisory (white space and case-insensitive), the CVRF CSAF Converter SHALL try to convert the data into a valid CSAF Document in this profile according to CSAF 2.1.

    > A tool MAY offer rules to create the missing fixed products from version ranges, if applicable. A tool MAY offer rules to assign missing CVEs or vulnerability IDs. Nevertheless, issuing parties are responsible to keep track of assigned IDs to prevent the assignment of the same ID for different issues.

    If the CVRF CSAF Converter is unable to create a valid CSAF 2.1 Document according to the profile, it SHALL set the category value to csaf_deprecated_security_advisory.
  - If one or more CVRF elements containing an xml:lang attribute exist and their value is English or the document language of the CVRF document is unspecified, the following rules apply:
    - If the cvrf:DocumentTitle starts with the string Superseded or the cvrf:DocumentType starts with Superseded (case-insensitive), the CVRF CSAF Converter SHALL try to convert all data into a valid CSAF Document in the profile "Superseded" according to CSAF 2.1.
    - If the cvrf:DocumentTitle starts with the string Withdrawn or the cvrf:DocumentType starts with Withdrawn (case-insensitive), the CVRF CSAF Converter SHALL try to convert all data into a valid CSAF Document in the profile "Withdrawn" according to CSAF 2.1.

    > A tool MAY provide a non-default option to remove or transform certain or all elements that hinder the creation of a valid CSAF Document according to the profile.

    > A tool MAY support this detection for other languages.

    If the CVRF CSAF Converter is unable to create a valid CSAF 2.1 Document according to the profile, it SHALL set the category according to the conversion rules and issue a warning a potentially withdrawn CSAF Document was created which would result in an invalid CSAF.
- $.document.involvement:
  - If the user passes an involvement element to the CVRF CSAF Converter, the tool SHALL insert that element into $.document.involvement and omit the conversion of all vuln:Involvements elements.
  - If any vulnerability item contains an involvements element and the user provided an involvement element, the CVRF CSAF Converter SHOULD issue a warning that all vuln:Involvements elements omitted as the user provided involvement takes precedence.

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- If any vulnerability item contains an involvements element and the user did not provide an involvement element, the CVRF CSAF Converter SHALL try to convert the data into a valid CSAF Document by apply the following process:

1. The CVRF CSAF Converter assigns each vulnerability containing an vuln:Involvements element a document-local vuln ID with the prefix VULN-, followed by the current value of counter for the vulnerabilities containing an vuln:Involvements element. The counter starts with 1 and is zero-padded to four digits before appended to the prefix.

   If the counter is at 1000 no additional padding is needed anymore.

2. The CVRF CSAF Converter SHALL create for each vuln:Involvement item a conversion ID which is constructed as follows:
   - The filename is computed from the $.document.tracking.id according to the rules of 5.1. Remaining + in that string are replaced by _. The file extension .json is removed from the string.
   - This string is extended by appending the dl_vuln_id of the vulnerability the vuln:Involvement item is found in followed by the item number inside the current vuln:Involvements. The counter for the item number starts with 1 and is zero-padded to four digits. This conversion ID identifies an involvement item with in a CSAF document locally and provides collision resistance if the filenames of the processed files are sufficiently different.
3. The CVRF CSAF Converter SHALL create for each Party mentioned in the vuln:Involvement a new entry in entities and assign a entity_id by prepending the mandatory prefix EID- to the conversion ID of that vuln:Involvement item.

   This create a new entry in $.document.involvement.entities per vuln:Involvement item.

   The CVRF CSAF Converter SHALL convert the value of Party into the value of category. If the value of Party is Other, the CVRF CSAF Converter SHOULD issue a warning that some parties have been regrouped into new category values. An option to suppress this warning SHALL exist. In addition, an option SHOULD be provided to set one of the new values directly.

   Entities can be corrected through a mapping in a later step.

4. The CVRF CSAF Converter SHALL create an entity for the publisher of the CVRF document with the same name and category as stated in the cvrf:DocumentPublisher element. The entity_id SHALL be EID-Publisher.
5. If the user provides a mapping which contains entities to CVRF CSAF Converter, the tool SHALL replace entries that contain a matching entity_id with the corresponding entry from the mapping. The CVRF CSAF Converter SHALL skip this step, if not mapping was provided by the user.
6. Each item of the vuln:Involvements inside the current vulnerability is converted using the following subprocess:
   - The CVRF CSAF Converter SHALL create a new action and add the document-local vuln ID of the current vulnerability into dl_vuln_ids of the action.
   - The action_id is constructed by prepending the prefix ACN- to the conversion ID of that vuln:Involvement item.
   - If the vuln:Description was given, the CVRF CSAF Converter SHALL convert its value as value into the summary element.
   - If the date was given in the vuln:Description, the CVRF CSAF Converter SHALL convert its value as value into the date element. If the date is missing, the CVRF CSAF Converter SHOULD stop the subprocess and discard the action unless the user instructed the tool to add the element with an empty value. In the first case, the CVRF CSAF Converter SHALL issue a warning that an involvement without date was detected and removed. If the CVRF CSAF Converter was instructed to add an empty date, it SHALL issue a warning that an involvement without date was detected and converted into an invalid action with empty date.

     In such case, the user is expected to add missing dates from other sources.

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- The CVRF CSAF Converter SHALL use the table 5 to set the category and status.
- The CVRF CSAF Converter SHOULD have an option to extract additional details about the entity from the vuln:Description, if applicable. If that option is given, the CVRF CSAF Converter SHALL insert the additionally extracted details in the corresponding fields and issue a warning that additional details were found and used to enrich the entity. Such warning SHOULD provide the full enriched entity element and the vuln:Description that the content was extracted from.
- The CVRF CSAF Converter SHALL set the acting_entity_refs and receiving_entity_refs according to the content of the current vuln:Involvement item using the corresponding entry from the list of entities. As default, the entity that corresponds to the Party is referenced in the acting_entity_refs. If the vuln:Description shows that the Party is the receiver, the entity that corresponds to the Party is referenced in the receiving_entity_refs.
- To determine the entity that is not given in the CVRF document, the CVRF CSAF Converter SHALL apply the following steps in this order:
  1. If the missing entity can be derived from the context, e.g. through the content of the vuln:Description, the respective entity SHALL be used. In most cases, this will lead to inserting EID-Publisher into the receiving_entity_refs respectively acting_entity_refs. The CVRF CSAF Converter SHALL NOT make up an entity not given in the vuln:Description.
  2. If the context suggests that the vuln:Involvement item matches one of the categories that do not need a recipient, the respective category SHALL be used. The CVRF CSAF Converter SHALL reference the corresponding action referenced_action_ids if required by the category value. If the value cannot be determined with certainty, this step is unsuccessful.
  3. If the steps mentioned above are unsuccessful or inconclusive, the CVRF CSAF Converter SHALL stop the subprocess and discard the action unless the user instructed the tool to add the action with partial data. In the first case, the CVRF CSAF Converter SHALL issue a warning that an involvement without sufficient data was detected, could not be converted automatically and therefore was removed. If the CVRF CSAF Converter was instructed to add the action with partial data, it SHALL issue a warning that an involvement without sufficient data was detected and only partially converted.

7. The CVRF CSAF Converter SHALL provide an option to collapse entities. If that option is selected, the CVRF CSAF Converter SHALL map all items from entities into groups where each group contains only members that differ only in their entity_id. The first one of each group which becomes the representative of that group. References to other elements of the group are replaced with a reference to the representative of the group. Then, the CVRF CSAF Converter SHALL remove all items from entities except for the representative of the groups.
8. The CVRF CSAF Converter SHALL provide an option to collapse actions. If that option is selected, the CVRF CSAF Converter SHALL map all items from actions which are not referenced in any referenced_action_ids into groups where each group contains only members that differ only in their dl_vuln_ids. The first one of each group which becomes the representative of that group. The CVRF CSAF Converter SHALL collect all values from dl_vuln_ids within one group and insert the values into dl_vuln_ids of the representative of that group. Then, the CVRF CSAF Converter SHALL remove all items from actions except for the representative of the groups. During the conversion, the CVRF CSAF Converter SHALL NOT invent data or set values in any other way than described above. If the CVRF CSAF Converter is unable to create a valid CSAF 2.1 Document, it SHALL remove the involvement from the CSAF Document and issue a warning that vuln:Involvements could not be converted automatically and therefore have been removed.

- $.document.lang: If one or more CVRF elements containing an xml:lang attribute exist and contain the exact same value, the CVRF CSAF Converter converts this value into lang. If the values of xml:lang attributes are not equal, the CVRF CSAF Converter issues a warning that the language could not be determined and possibly a document with multiple languages was produced. In addition, it SHOULD also present all values of xml:lang attributes as a set in the warning.
- $.document.license_expression: If any cvrf:Note item with Type Legal Disclaimer contains valid SPDX license expressions, the CVRF CSAF Converter SHALL apply the following rules to the list of candidates:
  - If the list contains only one element, the CVRF CSAF Converter SHALL set this value as value of license_expression. In addition, the converter issues a warning that a license expression was found and set as the document license expression.
  - If multiple SPDX license expressions are found, the CVRF CSAF Converter issues a warning that multiple SPDX license expressions were found and therefore no document license expression could be determined. This warning SHALL include the list of all SPDX license expressions and their associated paths.

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A tool MAY implement an option to suppress this conversion.

- $.document.notes: If any cvrf:Note item contains one of the category and title combinations specified in table 1 of section 3.2.2.9, where the title is extended, the CVRF CSAF Converter SHALL try to identify whether that extension is a specific product name, version or family. In such case, the CVRF CSAF Converter SHALL try to add the corresponding products to the note item and issue a warning that a potential product specific note has been discovered and products have been assigned to it. Such warning SHALL also include the note and the assigned products. If the CVRF CSAF Converter is unable to create a valid object, it SHALL remove the reference to the products and issue a warning that a potential product specific note has been discovered and no products could be assigned to it.
- $.document.publisher.contact:
  - If a cvrf:ContactDetails element is present, the CVRF CSAF Converter SHALL convert its content into the value of the details property.
  - If a URL to a Public OpenPGP Key was provided in the cvrf:ContactDetails element and the URL fulfills the requirements of the property public_openpgp_key_url, the CVRF CSAF Converter SHALL set it as the content of that property.
  - The CVRF CSAF Converter SHOULD extract the matching email address from the content of the cvrf:ContactDetails element, if applicable, or use the first email address and set it as value of email.
  - The CVRF CSAF Converter SHALL provide configuration options and corresponding arguments to set these values if they are not present. If values from both sources are present, the program SHOULD prefer the latter one. It SHOULD provide an option to overwrite the values extracted from the document. The program SHALL NOT use hard-coded values.
- $.document.publisher.name and $.document.publisher.namespace: Sets the value as given in the configuration of the program or the corresponding argument the program was invoked with. If values from both sources are present, the program SHOULD prefer the latter one. The program SHALL NOT use hard-coded values.
- $.document.tracking.id: If the element cvrf:ID contains any newline sequence or leading or trailing white space, the CVRF CSAF Converter removes those characters. In addition, the converter issues a warning that the ID was changed.
- $.document.tracking.initial_release_date: If the value would fail test 6.1.66, the CVRF CSAF Converter SHALL set the date of the revision history item with a number of 1 or 1.0.0 as value for the initial_release_date and issue a warning that an inconsistent initial release date was detected and replaced with the date from the revision history item with a number of 1 or 1.0.0.

If the year of the initial_release_date changed during this conversion step it can lead to a situation where a CSAF document moves from one year folder to another (see 7.1.11). Issuing parties are strongly advised to double-check whether that conversion reflects the facts. If that is not the case, the issuing party should correct the mistake in the input file.

- $.product_tree.product_paths[*]: For each element prod:Relationship, the CVRF CSAF Converter SHALL apply the following rules:
  - The value of the attribute ProductReference is set as the value of beginning_product_reference.
  - The first prod:FullProductName instance is converted into the full_product_name. If more than one prod:FullProductName instance is given, the CVRF CSAF Converter SHALL issue a warning that information might be lost during conversion of product paths. Such warning SHOULD contain the values of all prod:FullProductName instances skipped.
  - The CVRF CSAF Converter constructs the first item of subpaths by converting the value of RelationType into category and the value of the attribute RelatesToProductReference into next_product_reference.

A tool MAY provide an option to collapse chained product path elements into the appropriate number of product path elements, if this is possible without the loss of correct information. Usually, collapsing chained product paths is not possible if
- a product (type: full_product_name_t) defined by a product path element in the chain is referenced in other parts of the document, or
- a product (type: full_product_name_t) defined by a product path element in the chain contains a product_identification_helper element. For examples, see appendix Appendix D.

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- $.vulnerabilities[*].cwes[*]:
  - The CVRF CSAF Converter SHALL remove all preceding and trailing white space from the name.
  - The CVRF CSAF Converter SHALL determine the CWE specification version the given CWE was selected from by using the latest version that matches the id and name exactly and was published prior to the value of $.document.tracking.current_release_date of the source document. If no such version exists, the first matching version published after the value of $.document.tracking.current_release_date of the source document SHOULD be used.

    This is done to create a deterministic conversion.

    If the CWE does not match at all, the CVRF CSAF Converter SHALL omit this CWE and issue a warning that an invalid CWE was found and has been removed.
  - If a vuln:CWE instance refers to a CWE category or view, the CVRF CSAF Converter SHALL omit this instance and issues a warning that this CWE has been removed as its usage is not allowed in vulnerability mappings.
- $.vulnerabilities[*].disclosure_date: If a vuln:ReleaseDate was given, the CVRF CSAF Converter SHALL convert its value into the disclosure_date element.
- $.vulnerabilities[*].ids: If a vuln:ID element is given, the CVRF CSAF Converter converts it into the first item of the ids array. Then, the CVRF CSAF Converter SHALL execute the following steps at the converted object:
  - If a system_name was given, the CVRF CSAF Converter SHALL test whether it belongs to a registered vulnerability ID system in RVISC.
  - If the system_name belongs to an RVISC entry, the CVRF CSAF Converter SHALL execute test 6.2.53.
    - If the test passes, no further action is needed.
    - If the test fails, the CVRF CSAF Converter SHALL try to convert the entry based on the mapping given in [RVISC-M].
      - If at least one matching mapping is ambiguous, the CVRF CSAF Converter SHALL issue a warning that an ID from a registered vulnerability system was detected and not converted as the mapping was ambiguous. Such warning SHALL contain all matching mappings known from [RVISC-M]. The program SHOULD provide an option to force a user-defined conversion.
      - If the mapping succeeds and passes test 6.2.53, the CVRF CSAF Converter SHALL issue a warning that an ID from a registered vulnerability system was detected and converted.
      - If the mapping succeeds but does not pass test 6.2.53 or the mapping fails, the CVRF CSAF Converter SHALL issue a warning that an ID from a registered vulnerability system was detected and but could not be converted automatically. Such warning SHALL state the reason for failure. This includes also if the mapping is not implemented.

    A tool MAY provide a non-default option to suppress this conversion step.

    The output SHALL include the original values and, if applicable, the converted ones.
  - If the system_name does not belong to an RVISC entry, the CVRF CSAF Converter SHALL try to convert the entry based on the mapping given in [RVISC-M].
    - If no matching mapping exists, the CVRF CSAF Converter SHALL issue a warning that an ID from a potentially unregistered vulnerability system was detected and no change occurred.
    - If the mapping succeeds and passes test 6.2.53, the CVRF CSAF Converter SHALL issue a warning that an ID from a vulnerability system with a known mapping was detected and converted.
    - If the mapping succeeds but does not pass test 6.2.53 or the mapping fails otherwise, the CVRF CSAF Converter SHALL issue a warning that an ID from a vulnerability system with a known mapping was detected and but could not be converted automatically. Such warning SHALL state the reason for failure. This includes also if the mapping is not implemented.

    A tool MAY provide a non-default option to suppress this conversion step.

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The output SHALL include the original values and, if applicable, the converted ones.

- $.vulnerabilities[*].metrics[*]:
  - For any CVSS v4 element, the CVRF CSAF Converter SHALL compute the baseSeverity from the baseScore according to the rules of the applicable CVSS standard. (CSAF CVRF v1.2 predates CVSS v4.0.)
  - For any CVSS v3 element, the CVRF CSAF Converter SHALL compute the baseSeverity from the baseScore according to the rules of the applicable CVSS standard.
  - If no product_id is given, the CVRF CSAF Converter appends all Product IDs which are listed under ../product_status in the arrays known_affected, first_affected and last_affected. If none of these arrays exist, the CVRF CSAF Converter issues an error that no matching Product ID was found for this score element.
  - If a vectorString is missing, the CVRF CSAF Converter issues an error that the CVSS element could not be converted as the CVSS vector was missing. A CVRF CSAF Converter MAY offer a configuration option to delete such elements.
  - If there are CVSS v3.0 and CVSS v3.1 Vectors available for the same product, the CVRF CSAF Converter discards the CVSS v3.0 information and provide in CSAF only the CVSS v3.1 information.
  - To determine, which minor version of CVSS v3 is used and to evaluate a CVSS v4 that was wrongly inserted in a CVSS v3 element, the CVRF CSAF Converter uses the following steps:

    1. Retrieve the CVSS version from the CVSS vector, if present.

       Example 1:

       ```
       CVSS:3.1/AV:N/AC:L/PR:N/UI:N/S:U/C:N/I:N/A:H => 3.1
       ```

    2. Retrieve the CVSS version from the CVSS element's namespace, if present. The CVRF CSAF Converter issues a warning that this value was guessed from the element's namespace.

       Example 2:

       ```
       xmlns:cvssv31="https://www.first.org/cvss/cvss-v3.1.xsd"
       <!-- -->
       <cvssv31:ScoreSetV3>
       ```

       is handled the same as

       Example 3:

       ```
       <ScoreSetV3 xmlns="https://www.first.org/cvss/cvss-v3.1.xsd">
       ```

    3. Retrieve the CVSS version from the CVSS namespace given in the root element, if present. The CVRF CSAF Converter issues a warning that this value was guessed from the global namespace. If more than one CVSS namespace is present and the element is not clearly defined via the namespace, this step SHALL be skipped without a decision.

       Example 4:

       ```
       xmlns:cvssv3="https://www.first.org/cvss/cvss-v3.0.xsd" => 3.0
       ```

    4. Retrieve the CVSS version from a config value, which defaults to 3.0. (As CSAF CVRF v1.2 predates CVSS v3.1.) The CVRF CSAF Converter issues a warning that this value was taken from the config.

- $.vulnerabilities[*].metrics[*].content.cvss_v4: If an external reference in the vulnerability linking to the official FIRST.org CVSS v4.0 calculator exists, the CVRF CSAF Converter SHALL convert the vector given in the fragment into a cvss_v4 object linked to all affected products of the vulnerability.

  A tool MAY implement an option to suppress this conversion.

  If the CVRF CSAF Converter is unable to construct a valid object with the information given, the CVRF CSAF Converter SHALL remove the invalid cvss_v4 object and issue a warning that the automatic conversion of the CVSS v4.0 reference failed. Such warning SHOULD include the specific error that occurred.

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- $.vulnerabilities[*].notes: If any vuln:Note item contains one of the category and title combinations specified in table 2 of section 3.2.4.11, where the title is extended, the CVRF CSAF Converter SHALL try to identify whether that extension is a specific product name, version or family. In such case, the CVRF CSAF Converter SHALL try to add the corresponding products to the note item and issue a warning that a potential product specific note has been discovered and products have been assigned to it. Such warning SHALL also include the note and the assigned products. If the CVRF CSAF Converter is unable to create a valid object, it SHALL remove the reference to the products and issue a warning that a potential product specific note has been discovered and no products could be assigned to it.
- $.vulnerabilities[*].remediations[*]:
  - If neither product_ids nor group_ids are given, the CVRF CSAF Converter appends all Product IDs which are listed under ../product_status in the arrays known_affected, first_affected and last_affected into product_ids. If none of these arrays exist, the CVRF CSAF Converter issues an error that no matching Product ID was found for this remediation element.
  - The CVRF CSAF Converter SHALL convert any remediation with the type Vendor Fix into the category optional_patch if the product in question is in one of the product status groups "Not Affected" or "Fixed" for this vulnerability. Otherwise, the category vendor_fix SHALL be set. If multiple products are associated with the remediation - either directly or through a product group - and the products belong to different product status groups, the CVRF CSAF Converter SHALL duplicate the remediation, change the category in one instance to optional_patch and distribute the products accordingly as stated by the conversion rule.
  - The CVRF CSAF Converter SHALL convert any remediation with the type None Available into the category fix_planned if the product in question is also listed in a remediation of the type Vendor Fix with a Date in the future or no Date at all. Consequently, the product SHALL be removed from the remediation of the category vendor_fix. If it was the last product in that remediation, the remediation SHALL be removed.
  - The CVRF CSAF Converter SHALL remove any product from a remediation with the type None Available if the product in question is also listed in a remediation of the type Vendor Fix with a Date in the past or to the exact same time. If it was the last product in that remediation, the remediation SHALL be removed.
  - In any other case, the CVRF CSAF Converter SHALL preserve the product in the remediation of the category none_available.
  - The CVRF CSAF Converter SHALL issue a warning if a remediation was added, deleted or the value of the category was changed, including the products it was changed for.

Mapping of involvements status:

Table 5: Mapping of values into $.document.involvement.actions[*] from CSAF CVRF 1.2 to CSAF 2.1.

CSAF CVRF 1.2 Status	CSAF 2.1 status	CSAF 2.1 Default category
Completed	completed	coordination
Contact Attempted	attempted	notification
Disputed	completed	dispute
In Progress	in_progress	coordination
Not Contacted	deferred	notification
Open	outstanding	coordination

9.1.6 Conformance Clause 6: CSAF Content Management System

A CSAF Content Management System satisfies the "CSAF Content Management System" conformance profile if the content management system:

- satisfies the "CSAF Producer" conformance profile.
- satisfies the "CSAF Viewer" conformance profile.
- provides at least the following management functions:
  - create new CSAF Documents
  - prefill CSAF Documents based on values given in the configuration (see below)
  - create a new version of an existing CSAF Document
  - checkout old versions of a CSAF Document
  - show all differences between versions of a CSAF Document

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- list all CSAF Documents within the system
- delete CSAF Documents from the system
- review CSAF Documents in the system
- approve CSAF Documents
- search for CSAF Documents by values of required fields at document-level or their children within the system
- search for CSAF Documents by values of cve within the system
- search for CSAF Documents based on properties of product_tree
- filter on all properties which it is required to search for
- export of CSAF Documents
- show an audit log for each CSAF Document
- identify the latest version of CSAF Documents with the same $.document.tracking.id
- suggest a $.document.tracking.id based on the given configuration.
- option to suggest an ID for a vulnerability based on the given configuration.
- initialize a vulnerability element from a given CVE by loading the corresponding CVE JSON from a location provided in the configuration. Local instances implementing the respective CVE API SHALL be supported.
- support the creation of $.document.involvement by tracking the relationship between vulnerabilities and actions in order to make sure the information in $.document.involvement stays correct when vulnerability items are added.
- option to automatically assign a dl_vuln_id for each vulnerability added.
- track of the version of CSAF Documents automatically and increment according to the versioning scheme (see also subsections of 3.1.19) selected in the configuration.
- check that the document version is set correctly based on the changes in comparison to the previous version (see also subsections of 3.1.19).
- suggest to use the document status interim if a CSAF Document is updated more frequent than the given threshold in the configuration (default: 3 weeks)
- suggest to publish a new version of the CSAF Document with the document status final if the document status was interim and no new release has be done during the given threshold in the configuration (default: 6 weeks)

> Note that the terms “publish”, “publication” and their derived forms are used in this conformance profile independent of whether the specified target group is the public or a closed group.

- support the following workflows:
  - “New Advisory”: create a new advisory, request a review, provide review comments or approve it, resolve review comments; if the review approved it, the approval for publication can be requested; if granted the document status changes to final (or interim based on the selection in approval or configuration) and the advisory is provided for publication (manual or time-based)
  - “Update Advisory”: open an existing advisory, create new revision & change content, request a review, provide review comments or approve it, resolve review comments; if the review approved it, the approval for publication can be requested; if granted the document status changes to final (or interim based on the selection in approval or configuration) and the advisory is provided for publication (manual or time-based)

- offers both: publication immediately or at a given date/time.
- automates handling of date/time and version.
- provides an API to retrieve all CSAF Documents which are currently in the status published.
- optionally provides an API to import or create new advisories from outside systems (e.g. bug tracker, CVD platform,…).

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- optionally provides an API to update advisories from outside systems (e.g. add $.document.involvement.actions from ticket system).
- provides a user management and support at least the following roles:
  - *Registered*: Able to see all published CSAF Documents (but only in the published version).
  - *Author*: inherits *Registered* permissions and also can Create and Edit Own (mostly used for automated creation, see above)
  - *Editor*: inherits *Author* permissions and can Edit (mostly used in PSIRT)
  - *Publisher*: inherits *Editor* permissions and can Change state and Review any (mostly used as HEAD of PSIRT or team lead)
  - *Reviewer*: inherits *Registered* permissions and can Review advisories assigned to him (might be a subject matter expert or management)
  - *Manager*: inherits *Publisher* permissions and can Delete; User management up to *Publisher*
  - *Administrator*: inherits *Manager* permissions and can Change the configuration
- may use groups to support client separation (multitenancy) and therefore restrict the roles to actions within their group. In this case, there SHALL be a *Group configurator* which is able to change the values which are used to prefill fields in new advisories for that group. He might also do the user management for the group up to a configured level.
- prefills the following fields in new CSAF Documents with the values given below or based on the templates from configuration:
  - $['$schema'] with the value prescribed by the schema
  - $.document.category (based on the templates from configuration)
  - $.document.csaf_version with the value prescribed by the schema
  - $.document.distribution
    - sharing_group (based on the templates from configuration; per default a sharing group is omitted if not configured explicitly)
    - tlp (based on the templates from configuration; default label for drafts: TLP:AMBER+STRICT; default label for interim and final: TLP:CLEAR)
  - $.document.lang
  - $.document.license_expression (based on the templates from configuration)
  - $.document.notes
    - legal_disclaimer (Terms of use from the configuration)
    - general (General Security recommendations from the configuration)
  - $.document.publisher and children
  - $.document.tracking.current_release_date with the current date
  - $.document.tracking.generator and children
  - $.document.tracking.initial_release_date with the current date
  - $.document.tracking.revision_history
    - date with the current date
    - number (based on the templates according to the versioning scheme configured)
    - summary (based on the templates from configuration; default: “Initial version.”)
  - $.document.tracking.status with draft
  - $.document.tracking.version with the value of number the latest $.document.tracking.revision_history[*] element
- When updating an existing CSAF Document:
  - prefills all fields which have be present in the existing CSAF Document
  - adds a new item in $.document.tracking.revision_history[*]
  - updates the following fields with the values given below or based on the templates from configuration:
    - $['$schema'] with the value prescribed by the schema
    - $.document.csaf_version with the value prescribed by the schema
    - $.document.lang
    - $.document.notes
      - legal_disclaimer (Terms of use from the configuration)
      - general (General Security recommendations from the configuration)
    - $.document.publisher and children
    - $.document.tracking.current_release_date with the current date

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  - $.document.tracking.generator and children
  - the new item in $.document.tracking.revision_history[*]
    - date with the current date
    - number (based on the templates according to the versioning scheme configured)
  - $.document.tracking.status with draft
  - $.document.tracking.version with the value of number the latest $.document.tracking.revision_history[*] element

- before allowing $.document.tracking.status to transition from draft to interim or final:
  - enforces that the current document contains all required fields according to the profile specified in $.document.category
  - removes all $.vulnerabilities[*].dl_vuln_ids that are not referenced
  - enforces that the current document is a valid CSAF Document according to this specification
  - offers to set $.document.distribution and children to one of the values from the configuration for released documents (default label for released documents is TLP:CLEAR with no sharing_group present)

9.1.7 Conformance Clause 7: CSAF Post-Processor

A CSAF Post-Processor satisfies the "CSAF Post-Processor" conformance profile if the post-processor:

- satisfies the "CSAF Consumer" conformance profile.
- satisfies the "CSAF Producer" conformance profile.
- additionally satisfies those normative requirements in section 3 that are designated as applying to post-processors.

9.1.8 Conformance Clause 8: CSAF Modifier

A program satisfies the "CSAF Modifier" conformance profile if the program fulfills the two following groups of requirements:

The program:

- satisfies the "CSAF Post-Processor" conformance profile.
- adds, deletes or modifies at least one property, array, object or value of a property or item of an array.
- does not emit any objects, properties, or values which, according to section 9, are intended to be produced only by CSAF Translators.
- satisfies the normative requirements given below.

The resulting modified document:

- does not have the same $.document.tracking.id as the original document. The modified document can use a completely new $.document.tracking.id or compute one by appending the original $.document.tracking.id as a suffix after an ID from the naming scheme of the issuer of the modified version. It SHOULD NOT use the original $.document.tracking.id as a prefix.
- includes a reference to the original advisory as first element of the array $.document.references[*].

9.1.9 Conformance Clause 9: CSAF Translator

A program satisfies the "CSAF Translator" conformance profile if the program fulfills the two following groups of requirements:

The program:

- satisfies the "CSAF Post-Processor" conformance profile.
- translates at least one value.
- preserves the same semantics and form across translations.
- satisfies the normative requirements given below and does not add or remove other elements than required below.

The resulting translated document:

- does not use the same $.document.tracking.id as the original document. The translated document can use a completely new $.document.tracking.id or compute one by using the original $.document.tracking.id as a prefix and adding an ID from the naming scheme of the issuer of the translated version. It SHOULD NOT use the original $.document.tracking.id as a suffix. If an issuer uses a CSAF Translator to publish his advisories in multiple languages they MAY use the combination of the original $.document.tracking.id and translated $.document.lang as a $.document.tracking.id for the translated document.

> Note that the term "publish" is used in this conformance profile independent of whether the specified target group is the public or a closed group.

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- provides the $.document.lang property with a value matching the language of the translation.
- provides the $.document.source_lang to contain the language of the original document (and SHOULD only be set by CSAF Translators).
- has the value translator set in $.document.publisher.category
- includes a reference to the original advisory as first element of the array $.document.references[*].
- MAY contain translations for elements in arrays of references_t after the first element. However, it SHALL keep the original URLs as references at the end.

9.1.10 Conformance Clause 10: CSAF Consumer

A processor satisfies the "CSAF Consumer" conformance profile if the processor:

- reads CSAF Documents and interprets them according to the semantics defined in section 3 and 5.
- satisfies those normative requirements in section 2.4, 3, 5 and 8 that are designated as applying to CSAF Consumers.
- satisfies those normative requirements in sections 2.4 and 8 that are designated as applying to CSAF Tools.

9.1.11 Conformance Clause 11: CSAF Viewer

A viewer satisfies the "CSAF Viewer" conformance profile if the viewer fulfills the three following groups of requirements:

The viewer:

- satisfies the "CSAF Consumer" conformance profile.
- satisfies those normative requirements in section 3, 5 and 8 that are designated as applying to CSAF Viewers.
- satisfies the normative requirements given below.

For each CVSS score in $.vulnerabilities[*].metrics[*] the viewer:

- preferably shows the vector if there is an inconsistency between the vector and any other sibling attribute.
- SHOULD prefer the item of metrics[*] for each product_id which originates from the document author (and therefore has no property source) and has the highest CVSS Base Score and newest CVSS version (in that order) if a product_id is listed in more than one item of metrics[*].

The viewer SHOULD output the data of extensions that it does not have implemented as text.

9.1.12 Conformance Clause 12: CSAF Management System

A CSAF Management System satisfies the "CSAF Management System" conformance profile if the management system:

- satisfies the "CSAF Viewer" conformance profile.
- provides at least the following management functions:
  - add new CSAF Documents (e.g. from file system or URL) to the system
  - list all CSAF Documents within the system
  - delete CSAF Documents from the system
  - comment on CSAF Documents in the system
  - mark CSAF Documents as read in the system
  - search for CSAF Documents by values of required fields at document-level or their children within the system
  - search for CSAF Documents by values of cve within the system
  - search for CSAF Documents based on properties of $.product_tree
  - filter on all properties which it is required to search for
  - sort on all properties which it is required to search for
  - sort on CVSS scores and $.document.aggregate_severity.text
- identifies the latest version of CSAF Documents with the same $.document.tracking.id.
- is able to show the difference between 2 versions of a CSAF Document with the same $.document.tracking.id.

9.1.13 Conformance Clause 13: CSAF Asset Matching System

A CSAF Asset Matching System satisfies the "CSAF Asset Matching System" conformance profile if the asset matching system:

- satisfies the "CSAF Management System" conformance profile.
- is an asset database or connects to one.

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- matches the CSAF Documents within the system to the respective assets. This might be done with a probability which gives the end user the chance to broaden or narrow the results. The process of matching is also referred to as "run of the asset matching module".
- provides for each product of the asset database a list of matched advisories.
- provides for each asset of the asset database a list of matched advisories.
- provides for each CSAF Document a list of matched product of the asset database.
- provides for each CSAF Document a list of matched asset of the asset database.
- provides for each vulnerability within a CSAF Document the option to mark a matched asset in the asset database as "not remediated", "remediation in progress", or "remediation done". A switch to mark all assets at once MAY be implemented.
- does not bring up a newer revision of a CSAF Document as a new match if the remediation for the matched product or asset has not changed.
- detects the usage semantic version (as described in section 3.1.19.2).
- is able to trigger a run of the asset matching module:
  - manually:
    - per CSAF Document
    - per list of CSAF Documents
    - per asset
    - per list of assets
  - automatically:
    - when a new CSAF Document is inserted (for this CSAF Document)
    - when a new asset is inserted (for this asset)
    - when the Major version in a CSAF Document with semantic versioning changes (for this CSAF Document)

    These also apply if more than one CSAF Document or asset was added. To reduce the computational efforts the runs can be pooled into one run which fulfills all the tasks at once (batch mode).
  - Manually and automatically triggered runs SHOULD NOT be pooled.
- provides at least the following statistics for the count of assets:
  - matching that CSAF Document at all
  - marked with a given status

9.1.14 Conformance Clause 14: CSAF Basic Validator

A program satisfies the "CSAF Basic Validator" conformance profile if the program:

- reads documents and performs a check against the JSON schema, including also the format validation (cf. section 2.2).
- performs all tests of the preset mandatory as given in section 6.4.1.
- does not change the CSAF Documents unless explicitly invoked to do so (e.g. for applying quick fixes).
- satisfies those normative requirements in sections 2.4, 3, 6.1, 6.4, and 8 that are designated as applying to CSAF validators.
- satisfies those normative requirements in sections 2.4 and 8 that are designated as applying to CSAF Tools.
- issues a warning if an "not implemented warning" occurs as the validation status might not be correct.

A CSAF Basic Validator MAY provide one or more additional functions:

- Only run one or more selected mandatory tests.
- Apply quick fixes as specified in the standard.
- Apply additional quick fixes as implemented by the vendor.
- satisfies those normative requirements in section 6.5 that are designated as applying to CSAF validators.
- Provide an option to fail after the first error.
- Provide an option to generate all error messages. Such option SHOULD NOT be the default as it could lead into a DoS situation.

A CSAF Basic Validator MAY implement CSAF Additional Tests. In that case, it SHALL make through its documentation available which tests are implemented.

9.1.15 Conformance Clause 15: CSAF Extended Validator

A CSAF Basic Validator satisfies the "CSAF Extended Validator" conformance profile if the CSAF Basic Validator:

- satisfies the "CSAF Basic Validator" conformance profile.

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- additionally performs all tests of the preset recommended as given in section 6.4.1.
- additionally satisfies those normative requirements in sections 6.2 and 6.5 that are designated as applying to CSAF validators.

A CSAF Extended Validator MAY provide an additional function to only run one or more selected recommended tests.

9.1.16 Conformance Clause 16: CSAF Full Validator

A CSAF Extended Validator satisfies the "CSAF Full Validator" conformance profile if the CSAF Extended Validator:

- satisfies the "CSAF Extended Validator" conformance profile.
- additionally performs all tests of the preset informative as given in section 6.4.1.
- provides an option to additionally use a custom dictionary for test 6.3.8.
- additionally satisfies those normative requirements in section 6.3 that are designated as applying to CSAF validators.

A CSAF Full Validator MAY provide an additional function to only run one or more selected informative tests.

9.1.17 Conformance Clause 17: CSAF SBOM Matching System

A CSAF SBOM Matching System satisfies the "CSAF SBOM Matching System" conformance profile if the SBOM matching system:

- satisfies the "CSAF Management System" conformance profile.
- is an SBOM database or connects to one.

> A repository or any other location that can be queried for SBOMs and their content is also considered an SBOM database.

- matches the CSAF Documents within the system to the respective SBOM components. This might be done with a probability which gives the user the chance to broaden or narrow the results. The process of matching is also referred to as "run of the SBOM matching module".
- provides for each SBOM of the SBOM database a list of matched advisories.
- provides for each SBOM component of the SBOM database a list of matched advisories.
- provides for each CSAF Document a list of matched SBOMs of the SBOM database.
- provides for each CSAF Document a list of matched SBOM components of the SBOM database.
- provides for each vulnerability within a CSAF Document the option to mark a matched SBOM component in the SBOM database as "not remediated", "remediation in progress", or "remediation done". A switch to mark all SBOM component at once MAY be implemented.
- does not bring up a newer revision of a CSAF Document as a new match if the remediation for the matched SBOM or SBOM component has not changed.
- detects the usage semantic version (as described in section 3.1.19.2).
- is able to trigger a run of the SBOM matching module:
  - manually:
    - per CSAF Document
    - per list of CSAF Documents
    - per SBOM component
    - per list of SBOM components
  - automatically:
    - when a new CSAF Document is inserted (for this CSAF Document)
    - when a new SBOM component is inserted (for this SBOM component)
    - when the Major version in a CSAF Document with semantic versioning changes (for this CSAF Document)

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> These also apply if more than one CSAF Document or SBOM component was added. To reduce the computational efforts the runs can be pooled into one run which fulfills all the tasks at once (batch mode).

> Manually and automatically triggered runs should not be pooled.

- provides at least the following statistics for the count of SBOM component:
  - matching that CSAF Document at all
  - marked with a given status

9.1.18 Conformance Clause 18: CSAF 2.0 to CSAF 2.1 Converter

A program satisfies the "CSAF 2.0 to CSAF 2.1 Converter" conformance profile if the program fulfills the following two groups of requirements:

Firstly, the program:

- satisfies the "CSAF Producer" conformance profile.
- takes CSAF 2.0 Documents as input.
- rejects invalid CSAF 2.0 Documents unless forced by the user to proceed with a certain option. The existence of such option is OPTIONAL.

> If a program has no option to force the processing of invalid CSAF 2.0 documents, this does not necessarily mean that the program cannot be a compliant CSAF 2.0 to CSAF 2.1 Converter.

- issues a warning that an additional property was detected and not converted if it detects an additional property in the input. Such a warning SHALL include the additional property and its path. The CSAF 2.0 to CSAF 2.1 Converter SHALL ignore that additional property during the conversion.
- includes in every error and warning the relevant paths and values from the original file that triggered the alert, unless otherwise specified in this standard.
- additionally satisfies the normative requirements given below (see also table 6 and table 7).

Secondly, the program fulfills the following for all items of:

- value type `string` with format `date-time`: If the value contains a `60` in the seconds place, the CSAF 2.0 to CSAF 2.1 Converter SHALL replace the seconds and their fractions with `59.999999`. In addition, the converter issues a warning that leap seconds are now prohibited in CSAF and the value has been replaced. The CSAF 2.0 to CSAF 2.1 Converter SHOULD indicate in such warning message whether the value was a valid leap second or not.
- type `$['$defs'].branches_t`:
  - If a branch item uses the category `legacy`, the CSAF 2.0 to CSAF 2.1 Converter SHALL replace it with the category `product_name`. In addition, the converter issues a warning that this type does not exist in CSAF 2.1 and has been replaced with the category `product_name`.

> There is a chance, that this replacement is incorrect or another category is a better fit. Users of the converter are advised to check the content of such documents to make sure the conversion is correct or at least not misleading.

  - If any branch category appears multiple times along a path under `$.product_tree.branches` and the category is not an excepted one according to test 6.1.57, the CSAF 2.0 to CSAF 2.1 Converter SHALL try to convert the data into a valid product tree by applying the following steps to the path:
    1. If the stacked branch category is `vendor`, the vendor items named `Open Source`, `NOASSERTION`, `undefined` and `unknown` (white space, dash, hyphen, minus, underscore, invisible character and case insensitive on both sides) SHALL be removed.
    2. If the stacked branch category is `product_version` and the item directly before the first `product_version` is a `product_name`:

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- the category of the original product_name item SHALL be changed to product_family and
- the category of the first product_version item SHALL be changed to product_name and
- the value of the newly created product_family item SHALL be prepended at the value of the newly created product_name item.

If the CSAF 2.0 to CSAF 2.1 Converter is able to create a valid product tree, it SHALL issue a warning that an invalid product tree with stacked branch categories was detected and resolved. Such a warning SHALL include the invalid path as well as the branch categories that were present multiple times.

A tool MAY provide a non-default option to suppress this conversion step.

If the CSAF 2.0 to CSAF 2.1 Converter is unable to create a valid product tree, it SHALL issue an error that an invalid product tree with stacked branch categories was detected and could not be resolved. Such a error SHALL include the invalid path as well as the branch categories that were present multiple times.

A tool MAY provide a non-default option to output the invalid document.

- If the branch categories product_version and product_version_range appear along a path under $.product_tree.branches, the CSAF 2.0 to CSAF 2.1 Converter SHALL try to convert the data into a valid product tree by applying the following steps to the path:

1. If the branch category product_version occurs before the product_version_range and the item directly before the first product_version is a product_name:
   - the category of the original product_name item SHALL be changed to product_family and
   - the category of the first product_version item SHALL be changed to product_name and
   - the value of the newly created product_family item SHALL be prepended at the value of the newly created product_name item.
2. If the branch category product_version occurs before the product_version_range and the item directly before the first product_version is a product_family:
   - the category of the first product_version item SHALL be changed to product_name and
   - the value of the direct ancestor product_family item SHALL be prepended at the value of the newly created product_name item.

If the CSAF 2.0 to CSAF 2.1 Converter is able to create a valid product tree, it SHALL issue a warning that an invalid product tree with branch categories product_version and product_version_range in one path was detected and resolved. Such a warning SHALL include the invalid path as well as the branch category items changed.

A tool MAY provide a non-default option to suppress this conversion step.

If the CSAF 2.0 to CSAF 2.1 Converter is unable to create a valid product tree, it SHALL issue an error that an invalid product tree with branch categories product_version and product_version_range in one path was detected and could not be resolved. Such a error SHALL include the invalid path as well as the branch category items.

A tool MAY provide a non-default option to output the invalid document.

- If the value of name of an item categorized as product_version_range and contains an upper open ended product version range, the CSAF 2.0 to CSAF 2.1 Converter SHALL try to convert the data into a valid product tree by applying the following steps to the path:

1. If value of name consists only of one version constraint:
   - the category of the original product_version_range item SHALL be changed to product_version and
   - the version SHALL be extracted from the original value and set as new value of name.

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> The product version range vers:intdot/>=1.0 is changed to the product version 1.0. Note that a conversion of vers:intdot/>1.0 to the product version 1.0 would change the inclusion boundary and is not allowed.

2. If value of name consists of more than one version constraint and the upper open range is the last version constraint:
   - the category of the original product_version_range item SHALL be kept and
   - the value name SHALL be converted into the product version range ending with the version of the upper open ended product version if that does not change the inclusion boundaries.

> The product version range vers:intdot/>0.9|<=1.0|>=2.4 is changed to the product version range vers:intdot/>0.9|<=1.0|2.4. Note that a conversion of vers:intdot/<0.3|>1.0 to the product version range vers:intdot/<0.3|1.0 would change the inclusion boundary and is not allowed.

If the CSAF 2.0 to CSAF 2.1 Converter is able to create a valid product tree, it SHALL issue a warning that an invalid product tree with an upper open ended product version range in one path was detected and resolved. Such a warning SHALL include the invalid path as well as the original and new value.

> A tool MAY provide a non-default option to suppress this conversion step.

If the CSAF 2.0 to CSAF 2.1 Converter is unable to create a valid product tree, it SHALL issue an error that an invalid product tree with an upper open ended product version range in one path was detected and could not be resolved. Such a error SHALL include the invalid path as well as the original and new value.

> A tool MAY provide a non-default option to output the invalid document.

- If the value of name of an item categorized as product_version_range contains just a single version, the CSAF 2.0 to CSAF 2.1 Converter SHALL try to convert the data into a valid product tree by applying the following steps to the path:
  1. If value of name is in the VERS format:
     - the category of the original product_version_range item SHALL be changed to product_version and
     - the version SHALL be extracted from the version constraint and set as new value of name.
  2. If value of name is in the vls format:
     - the category of the original product_version_range item SHALL be changed to product_version and
     - the value name SHALL be kept unchanged.

  If the CSAF 2.0 to CSAF 2.1 Converter is able to create a valid product tree, it SHALL issue a warning that an invalid product tree with a product_version declared as product_version_range in one path was detected and resolved. Such a warning SHALL include the invalid path as well as value of the product version.

  > A tool MAY provide a non-default option to suppress this conversion step.

  If the CSAF 2.0 to CSAF 2.1 Converter is unable to create a valid product tree, it SHALL issue an error that an invalid product tree with a product_version declared as product_version_range in one path was detected and could not be resolved. Such a error SHALL include the invalid path as well as value of the product version.

  > A tool MAY provide a non-default option to output the invalid document.

- type $['$defs'].full_product_name_t..product_identification_helper..cpe: If a CPE is invalid, the CSAF 2.0 to CSAF 2.1 Converter SHOULD remove the invalid value and issue a warning that an invalid CPE was detected and removed.

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- type $['$defs'].full_product_name_t..product_identification_helper..hashes: The CSAF 2.0 to CSAF 2.1 Converter SHALL keep the order of the items during the conversion. If a value occurs multiple times, the CSAF 2.0 to CSAF 2.1 Converter SHALL convert only the first one and therefore making sure that all items are unique.
- type $['$defs'].full_product_name_t..product_identification_helper..hashes[*]..file_hashes: The CSAF 2.0 to CSAF 2.1 Converter SHALL keep the order of the items during the conversion. If a value occurs multiple times, the CSAF 2.0 to CSAF 2.1 Converter SHALL convert only the first one and therefore making sure that all items are unique.
- type $['$defs'].full_product_name_t..product_identification_helper..hashes[*]..file_hashes[*]..algorithm: If the algorithm is known to the implementation or mentioned in this standard, the CSAF 2.0 to CSAF 2.1 Converter SHALL ensure its spelling is exactly as prescribed by this standard. If the algorithm is unknown to the implementation, the CSAF 2.0 to CSAF 2.1 Converter SHALL convert it to lowercase and issue a warning that an unknown hash algorithm was detected and converted.

  A tool MAY provide a non-default option to suppress this conversion step.

- type $['$defs'].full_product_name_t..product_identification_helper..hashes[*]..file_hashes[*]..value: The CSAF 2.0 to CSAF 2.1 Converter SHALL convert the value into a lowercase string.
- type $['$defs'].full_product_name_t..product_identification_helper..model_numbers[*]:

  The values were implicitly open ended in CSAF 2.0 which resulted in ambiguity. A fixed-length matching was not possible. CSAF 2.1 requires that values intended to be interpreted as "starts with" be explicitly marked with an asterisk (*) at the end.

  - If a model number is given that does not end on a star, the CSAF 2.0 to CSAF 2.1 Converter SHOULD add a * to the end and issue a warning that a partial model number was detected and a star has been added.
  - If the model number contains a \, the CSAF 2.0 to CSAF 2.1 Converter SHALL escape it by inserting an additional \ before the character.
  - If the model number contains multiple unescaped * after the conversion, the CSAF 2.0 to CSAF 2.1 Converter SHALL remove the entry and issue a warning that a model number with multiple stars was detected and removed.

  A tool MAY provide a non-default option to interpret all model numbers as complete and therefore does not add any stars.

  A tool MAY provide a non-default option to interpret the ? in all model numbers as part of the model number itself and therefore escape it.

  A tool MAY provide a non-default option to interpret the * in all model numbers as part of the model number itself and therefore escape it.

- type $['$defs'].full_product_name_t..product_identification_helper..purls: If a $['$defs'].full_product_name_t..product_identification_helper..purl is given, the CSAF 2.0 to CSAF 2.1 Converter SHALL convert it into the first item of the corresponding purls array.
- type $['$defs'].full_product_name_t..product_identification_helper..sbom_urls: The CSAF 2.0 to CSAF 2.1 Converter SHALL keep the order of the items during the conversion. If a value occurs multiple times, the CSAF 2.0 to CSAF 2.1 Converter SHALL convert only the first one and therefore making sure that all items are unique.
- type $['$defs'].full_product_name_t..product_identification_helper..serial_numbers[*]:

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> The values were implicitly open ended in CSAF 2.0 which resulted in ambiguity. A fixed-length matching was not possible. CSAF 2.1 requires that values intended to be interpreted as “starts with” be explicitly marked with an asterisk (*) at the end.

- If a serial number is given that does not end on a star, the CSAF 2.0 to CSAF 2.1 Converter SHOULD add a * to the end and issue a warning that a partial serial number was detected and a star has been added.
- If the serial number contains a \, the CSAF 2.0 to CSAF 2.1 Converter SHALL escape it by inserting an additional \ before the character.
- If the serial number contains multiple unescaped * after the conversion, the CSAF 2.0 to CSAF 2.1 Converter SHALL remove the entry and issue a warning that a serial number with multiple stars was detected and removed.

> A tool MAY provide a non-default option to interpret all serial numbers as complete and therefore does not add any stars.

> A tool MAY provide a non-default option to interpret the ? in all serial numbers as part of the serial number itself and therefore escape it.

> A tool MAY provide a non-default option to interpret the * in all serial numbers as part of the serial number itself and therefore escape it.

- type $['$defs'].full_product_name_t..product_identification_helper..skus: The CSAF 2.0 to CSAF 2.1 Converter SHALL keep the order of the items during the conversion. If a value occurs multiple times, the CSAF 2.0 to CSAF 2.1 Converter SHALL convert only the first one and therefore making sure that all items are unique.
- type $['$defs'].full_product_name_t..product_identification_helper..skus[*]:

> The values were implicitly open ended in CSAF 2.0 which resulted in ambiguity. A fixed-length matching was not possible. CSAF 2.1 requires that values intended to be interpreted as “starts with” be explicitly marked with an asterisk (*) at the end.

- If a stock keeping unit is given that does not end on a star, the CSAF 2.0 to CSAF 2.1 Converter SHOULD add a * to the end and issue a warning that a partial stock keeping unit was detected and a star has been added.
- If the stock keeping unit contains a \, the CSAF 2.0 to CSAF 2.1 Converter SHALL escape it by inserting an additional \ before the character.
- If the stock keeping unit contains multiple unescaped * after the conversion, the CSAF 2.0 to CSAF 2.1 Converter SHALL remove the entry and issue a warning that a stock keeping unit with multiple stars was detected and removed.

> A tool MAY provide a non-default option to interpret all stock keeping units as complete and therefore does not add any stars.

> A tool MAY provide a non-default option to interpret the ? in all stock keeping units as part of the stock keeping unit itself and therefore escape it.

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A tool MAY provide a non-default option to interpret the * in all stock keeping units as part of the stock keeping unit itself and therefore escape it.

- type $['$defs'].full_product_name_t..product_identification_helper..x_generic_uris: The CSAF 2.0 to CSAF 2.1 Converter SHALL keep the order of the items during the conversion. If a value occurs multiple times, the CSAF 2.0 to CSAF 2.1 Converter SHALL convert only the first one and therefore making sure that all items are unique.
- $['$schema']: The CSAF 2.0 to CSAF 2.1 Converter SHALL set property with the value prescribed by the schema.
- $.document.category:
  - If the category equals csaf_security_advisory, the CSAF 2.0 to CSAF 2.1 Converter SHALL try to convert the data into a valid CSAF Document in this profile according to CSAF 2.1. For any version range of affected products that uses the strict <, i.e. not <=, as comparator of the last version constraint, the CSAF 2.0 to CSAF 2.1 Converter SHOULD add a new product with the version of the last constraint and add that in the appropriate places as fixed. The CSAF 2.0 to CSAF 2.1 Converter SHALL issue a warning that a product was added to the product_tree and the corresponding $.vulnerabilities[*]. Such warning SHALL contain the full product name and its path as well as the paths of the $.vulnerabilities[*] it was added to. The CSAF 2.0 to CSAF 2.1 Converter MAY have a non-default option to assign organization-wide IDs to any vulnerability not containing a CVE nor any other vulnerability ID and therefore failing test 6.1.27.8. Issuing parties are responsible to keep track of assigned IDs to prevent the assignment of the same ID for different issues. If the CSAF 2.0 to CSAF 2.1 Converter is unable to create a valid CSAF 2.1 Document according to the profile, it SHALL set the category value to csaf_deprecated_security_advisory.
  - If the $.document.lang is English or unspecified, the following rules apply:
    - If the $.document.title starts with the string Superseded or the $.document.category has the value Superseded (case-insensitive), the CSAF 2.0 to CSAF 2.1 Converter SHALL try to convert all data into a valid CSAF Document in the profile "Superseded" according to CSAF 2.1.
    - If the $.document.title starts with the string Withdrawn or the $.document.category has the value Withdrawn (case-insensitive), the CSAF 2.0 to CSAF 2.1 Converter SHALL try to convert all data into a valid CSAF Document in the profile "Withdrawn" according to CSAF 2.1.

    A tool MAY provide a non-default option to remove or transform certain or all elements that hinder the creation of a valid CSAF Document according to the profile.

    A tool MAY support this detection for other languages.

    If the CSAF 2.0 to CSAF 2.1 Converter is unable to create a valid CSAF 2.1 Document according to the profile, it SHALL set the category of the original CSAF Document and issue a warning a potentially withdrawn CSAF Document was created which would result in an invalid CSAF.
- $.document.csaf_version: The CSAF 2.0 to CSAF 2.1 Converter SHALL update the value to 2.1.
- $.document.distribution.tlp.label: If a TLP label is given, the CSAF 2.0 to CSAF 2.1 Converter SHALL convert it according to table 6.

  If $.document.distribution.text contains the string TLP v2.0: TLP: followed by a valid TLP v2.0 label as defined in the CSAF 2.1 JSON schema, the CSAF 2.0 to CSAF 2.1 Converter SHOULD provide an option to use this label instead.

  If the TLP label changes during such a conversion in a way not listed in table 6, the CSAF 2.0 to CSAF 2.1 Converter SHALL issue a warning that the TLP label was taken from the distribution text. This warning SHALL include both values: the value converted using the table and the value from the distribution text.

  This is a common case for CSAF 2.0 Documents labeled as TLP:RED but actually intended to be TLP:AMBER+STRICT.

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If no TLP label was given, the CSAF 2.0 to CSAF 2.1 Converter SHOULD assign TLP:CLEAR and issue a warning that the default TLP has been set.

- $.document.involvement:
  - If the user passes an involvement element to the CSAF 2.0 to CSAF 2.1 Converter, the tool SHALL insert that element into $.document.involvement and omit the conversion of all elements in $.vulnerabilities[*].involvements.
  - If any vulnerability item contains an involvements element and the user provided an involvement element, the CSAF 2.0 to CSAF 2.1 Converter SHOULD issue a warning that all elements in $.vulnerabilities[*].involvements omitted as the user provided involvement takes precedence.
  - If any vulnerability item contains an involvements element and the user did not provide an involvement element, the CSAF 2.0 to CSAF 2.1 Converter SHALL try to convert the data into a valid CSAF Document by apply the following process:

    1. The CSAF 2.0 to CSAF 2.1 Converter assigns each vulnerability containing an involvements element a document-local vuln ID with the prefix VULN-, followed by the current value of counter for the vulnerabilities containing an involvements element. The counter starts with 1 and is zero-padded to four digits before appended to the prefix.

       > If the counter is at 1000 no additional padding is needed anymore.

    2. The CSAF 2.0 to CSAF 2.1 Converter SHALL create for each involvements item a conversion ID which is constructed as follows:
       - The filename is computed from the $.document.tracking.id according to the rules of 5.1. Remaining + in that string are replaced by _. The file extension .json is removed from the string.
       - This string is extended by appending the dl_vuln_id of the vulnerability the involvements item is found in followed by the item number inside the current involvements. The counter for the item number starts with 1 and is zero-padded to four digits. This conversion ID identifies an involvement item with in a CSAF document locally and provides collision resistance if the filenames of the processed files are sufficiently different.
    3. The CSAF 2.0 to CSAF 2.1 Converter SHALL create for each party mentioned in the involvements a new entry in entities and assign a entity_id by prepending the mandatory prefix EID- to the conversion ID of that involvements item.

       > This create a new entry in $.document.involvement.entities per involvements item.

       The CSAF 2.0 to CSAF 2.1 Converter SHALL convert the value of party into the value of category. If the value of party is other, the CSAF 2.0 to CSAF 2.1 Converter SHOULD issue a warning that some parties have been regrouped into new category values. An option to suppress this warning SHALL exist. In addition, an option SHOULD be provided to set one of the new values directly.

       > Entities can be corrected through a mapping in a later step.

    4. The CSAF 2.0 to CSAF 2.1 Converter SHALL create an entity for the publisher of the CSAF 2.0 document with the same name and category as stated in the $.document.publisher object. The entity_id SHALL be EID-Publisher.
    5. If the user provides a mapping which contains entities to CSAF 2.0 to CSAF 2.1 Converter, the tool SHALL replace entries that contain a matching entity_id with the corresponding entry from the mapping. The CSAF 2.0 to CSAF 2.1 Converter SHALL skip this step, if not mapping was provided by the user.
    6. Each item of the involvements array inside the current vulnerability is converted using the following subprocess:
       - The CSAF 2.0 to CSAF 2.1 Converter SHALL create a new action and add the document-local vuln ID of the current vulnerability into dl_vuln_ids of the action.
       - The action_id is constructed by prepending the prefix ACN- to the conversion ID of that involvements item.
       - If the summary was given, the CSAF 2.0 to CSAF 2.1 Converter SHALL convert its value as value into the summary element.

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- If the date was given, the CSAF 2.0 to CSAF 2.1 Converter SHALL convert its value as value into the date element. If the date is missing, the CSAF 2.0 to CSAF 2.1 Converter SHOULD stop the subprocess and discard the action unless the user instructed the tool to add the element with an empty value. In the first case, the CSAF 2.0 to CSAF 2.1 Converter SHALL issue a warning that an involvement without date was detected and removed. If the CSAF 2.0 to CSAF 2.1 Converter was instructed to add an empty date, it SHALL issue a warning that an involvement without date was detected and converted into an invalid action with empty date.

  In such case, the user is expected to add missing dates from other sources.

- The CSAF 2.0 to CSAF 2.1 Converter SHALL use the table 7 to set the category and status.
- The CSAF 2.0 to CSAF 2.1 Converter SHOULD have an option to extract additional details about the entity from the summary, if applicable. If that option is given, the CSAF 2.0 to CSAF 2.1 Converter SHALL insert the additionally extracted details in the corresponding fields and issue a warning that additional details were found and used to enrich the entity. Such warning SHOULD provide the full enriched entity element and the summary that the content was extracted from.
- The CSAF 2.0 to CSAF 2.1 Converter SHALL set the acting_entity_refs and receiving_entity_refs according to the content of the current involvements item using the corresponding entry from the list of entities. As default, the entity that corresponds to the party is referenced in the acting_entity_refs. If the summary shows that the party is the receiver, the entity that corresponds to the party is referenced in the receiving_entity_refs.
- To determine the entity that is not given in the CSAF 2.0 document, the CSAF 2.0 to CSAF 2.1 Converter SHALL apply the following steps in this order:
  1. If the missing entity can be derived from the context, e.g. through the content of the summary, the respective entity SHALL be used. In most cases, this will lead to inserting EID-Publisher into the receiving_entity_refs respectively acting_entity_refs. The CSAF 2.0 to CSAF 2.1 Converter SHALL NOT make up an entity not given in the summary.
  2. If the context suggests that the involvements item matches one of the categories that do not need a recipient, the respective category SHALL be used. The CSAF 2.0 to CSAF 2.1 Converter SHALL reference the corresponding action referenced_action_ids if required by the category value. If the value cannot be determined with certainty, this step is unsuccessful.
  3. If the steps mentioned above are unsuccessful or inconclusive, the CSAF 2.0 to CSAF 2.1 Converter SHALL stop the subprocess and discard the action unless the user instructed the tool to add the action with partial data. In the first case, the CSAF 2.0 to CSAF 2.1 Converter SHALL issue a warning that an involvement without sufficient data was detected, could not be converted automatically and therefore was removed. If the CSAF 2.0 to CSAF 2.1 Converter was instructed to add the action with partial data, it SHALL issue a warning that an involvement without sufficient data was detected and only partially converted.

7. The CSAF 2.0 to CSAF 2.1 Converter SHALL provide an option to collapse entities. If that option is selected, the CSAF 2.0 to CSAF 2.1 Converter SHALL map all items from entities into groups where each group contains only members that differ only in their entity_id. The first one of each group which becomes the representative of that group. References to other elements of the group are replaced with a reference to the representative of the group. Then, the CSAF 2.0 to CSAF 2.1 Converter SHALL remove all items from entities except for the representative of the groups.
8. The CSAF 2.0 to CSAF 2.1 Converter SHALL provide an option to collapse actions. If that option is selected, the CSAF 2.0 to CSAF 2.1 Converter SHALL map all items from actions which are not referenced in any referenced_action_ids into groups where each group contains only members that differ only in their dl_vuln_ids. The first one of each group which becomes the representative of that group. The CSAF 2.0 to CSAF 2.1 Converter SHALL collect all values from dl_vuln_ids within one group and insert the values into dl_vuln_ids of the representative of that group. Then, the CSAF 2.0 to CSAF 2.1 Converter SHALL remove all items from actions except for the representative of the groups. During the conversion, the CSAF 2.0 to CSAF 2.1 Converter SHALL NOT invent data or set values in any other way than described above. If the CSAF 2.0 to CSAF 2.1 Converter is unable to create a valid CSAF 2.1 Document, it SHALL remove the involvement from the CSAF Document and issue a warning that $.vulnerabilities[*].involvements could not be converted automatically and therefore have been removed.

- $.document.license_expression: If any $.document.notes item with category legal_disclaimer contains a valid SPDX license expression, the CSAF 2.0 to CSAF 2.1 Converter SHALL apply the following rules for the list of candidates:

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- If this list contains only one element, the CSAF 2.0 to CSAF 2.1 Converter SHALL set this value as value of license_expression. In addition, the converter issues a warning that license expression was found and set as document license expression.
- If multiple SPDX license expressions are found, the CSAF 2.0 to CSAF 2.1 Converter issues a warning that multiple SPDX license expressions were found and therefore no document license expression could be determined. Such a warning SHALL include the list of all SPDX license expressions and their associated paths.

A tool MAY implement an option to suppress this conversion.

- $.document.notes: If any $.document.notes item contains one of the category and title combinations specified in table 1 of section 3.2.2.9, where the title is extended, the CSAF 2.0 to CSAF 2.1 Converter SHALL try to identify whether that extension is a specific product name, version or family. In such case, the CSAF 2.0 to CSAF 2.1 Converter SHALL try to add the corresponding products to the note item and issue a warning that a potential product specific note has been discovered and products have been assigned to it. Such warning SHALL also include the note and the assigned products. If the CSAF 2.0 to CSAF 2.1 Converter is unable to create a valid object, it SHALL remove the reference to the products and issue a warning that a potential product specific note has been discovered and no products could be assigned to it.
- $.document.publisher.category: If the value is other, the CSAF 2.0 to CSAF 2.1 Converter SHOULD issue a warning that some parties have been regrouped into the new value multiplier. An option to suppress this warning SHALL exist. In addition, an option SHOULD be provided to set the value to multiplier.
- $.document.publisher.contact:
  - If a $.document.publisher.contact_details element is present, the CSAF 2.0 to CSAF 2.1 Converter SHALL convert its content into the value of the details property.
  - If a URL to a Public OpenPGP Key was provided in the $.document.publisher.contact_details element and the URL fulfills the requirements of the property public_openpgp_key_url, the CSAF 2.0 to CSAF 2.1 Converter SHALL set it as the content of that property.
  - The CSAF 2.0 to CSAF 2.1 Converter SHOULD extract the matching email address from the content of the $.document.publisher.contact_details element, if applicable, or use the first email address and set it as value of email.
  - The CSAF 2.0 to CSAF 2.1 Converter SHALL provide configuration options and corresponding arguments to set these values if they are not present. If values from both sources are present, the program SHOULD prefer the latter one. It SHOULD provide an option to overwrite the values extracted from the document. The program SHALL NOT use hard-coded values.
- $.document.tracking.initial_release_date: If the value would fail test 6.1.66, the CSAF 2.0 to CSAF 2.1 Converter SHALL set the date of the revision history item with a number of 1 or 1.0.0 as value for the initial_release_date and issue a warning that an inconsistent initial release date was detected and replaced with the date from the revision history item with a number of 1 or 1.0.0.

If the year of the initial_release_date changed during this conversion step it can lead to a situation where a CSAF document moves from one year folder to another (see 7.1.11). Issuing parties are strongly advised to double-check whether that conversion reflects the facts. If that is not the case, the issuing party should correct the mistake in the input file.

- $.document.title: If the value contains the $.document.tracking.id, the CSAF 2.0 to CSAF 2.1 Converter SHALL remove the $.document.tracking.id from the $.document.title. In addition, separating characters including but not limited to white space, colon, dash and brackets SHALL be removed.
- $.product_tree.product_paths[*]: For each element in $.product_tree.relationships[*], the CSAF 2.0 to CSAF 2.1 Converter SHALL apply the following rules:
  - The value of product_reference is set as the value of beginning_product_reference.
  - The CSAF 2.0 to CSAF 2.1 Converter constructs the first item of subpaths by setting the value of category into category and the value of relates_to_product_reference into next_product_reference.

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A tool MAY provide an option to collapse chained product path elements into the appropriate number of product path elements, if this is possible without the loss of correct information. Usually, collapsing chained product paths is not possible if
- a product (type: full_product_name_t) defined by a product path element in the chain is referenced in other parts of the document, or
- a product (type: full_product_name_t) defined by a product path element in the chain contains a product_identification_helper element. For examples, see appendix Appendix D.

- $.vulnerabilities[*].cwes[*]:
  - The CSAF 2.0 to CSAF 2.1 Converter SHALL remove all preceding and trailing white space from the name.
  - The CSAF 2.0 to CSAF 2.1 Converter SHALL determine the CWE specification version the given CWE was selected from by using the latest version that matches the id and name exactly and was published prior to the value of $.document.tracking.current_release_date of the source document. If no such version exists, the first matching version published after the value of $.document.tracking.current_release_date of the source document SHOULD be used.

    This is done to create a deterministic conversion.

    The tool SHOULD implement an option to use the latest available CWE version at the time of the conversion that still matches.
- $.vulnerabilities[*].disclosure_date: If a release_date was given, the CSAF 2.0 to CSAF 2.1 Converter SHALL convert its value as value into the disclosure_date element.
- $.vulnerabilities[*].ids: If a system_name was given, the CSAF 2.0 to CSAF 2.1 Converter SHALL test whether it belongs to a registered vulnerability ID system in RVISC.
  - If the system_name belongs to an RVISC entry, the CSAF 2.0 to CSAF 2.1 Converter SHALL execute test 6.2.53.
    - If the test passes, no further action is needed.
    - If the test fails, the CSAF 2.0 to CSAF 2.1 Converter SHALL try to convert the entry based on the mapping given in [RVISC-M].
      - If at least one matching mapping is ambiguous, the CSAF 2.0 to CSAF 2.1 Converter SHALL issue a warning that an ID from a registered vulnerability system was detected and not converted as the mapping was ambiguous. Such warning SHALL contain all matching mappings known from [RVISC-M]. The program SHOULD provide an option to force a user-defined conversion.
      - If the mapping succeeds and passes test 6.2.53, the CSAF 2.0 to CSAF 2.1 Converter SHALL issue a warning that an ID from a registered vulnerability system was detected and converted.
      - If the mapping succeeds but does not pass test 6.2.53 or the mapping fails, the CSAF 2.0 to CSAF 2.1 Converter SHALL issue a warning that an ID from a registered vulnerability system was detected and but could not be converted automatically. Such warning SHALL state the reason for failure. This includes also if the mapping is not implemented.

    A tool MAY provide a non-default option to suppress this conversion step.

    The output SHALL include the original values and, if applicable, the converted ones.
  - If the system_name does not belong to an RVISC entry, the CSAF 2.0 to CSAF 2.1 Converter SHALL try to convert the entry based on the mapping given in [RVISC-M].
    - If no matching mapping exists, the CSAF 2.0 to CSAF 2.1 Converter SHALL issue a warning that an ID from a potentially unregistered vulnerability system was detected and no change occurred.
    - If the mapping succeeds and passes test 6.2.53, the CSAF 2.0 to CSAF 2.1 Converter SHALL issue a warning that an ID from a vulnerability system with a known mapping was detected and converted.
    - If the mapping succeeds but does not pass test 6.2.53 or the mapping fails otherwise, the CSAF 2.0 to CSAF 2.1 Converter SHALL issue a warning that an ID from a vulnerability system with a known mapping was detected and

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but could not be converted automatically. Such warning SHALL state the reason for failure. This includes also if the mapping is not implemented.

A tool MAY provide a non-default option to suppress this conversion step.

The output SHALL include the original values and, if applicable, the converted ones.

- $.vulnerabilities[*].metrics[*].content.cvss_v4: If an external reference in the vulnerability linking to the official FIRST.org CVSS v4.0 calculator exists, the CSAF 2.0 to CSAF 2.1 Converter SHALL convert the vector given in the fragment into a cvss_v4 object linked to all affected products of the vulnerability.

A tool MAY implement an option to suppress this conversion.

If the CSAF 2.0 to CSAF 2.1 Converter is unable to construct a valid object with the information given, the CSAF 2.0 to CSAF 2.1 Converter SHALL remove the invalid cvss_v4 object and issue a warning that the automatic conversion of the CVSS v4.0 reference failed. Such warning SHOULD include the specific error that occurred.

- $.vulnerabilities[*].metrics[*].content.ssvc_v2: If a SSVC vector or decision points of an SSVC vector are given in an item of notes of the current vulnerability using the title SSVC and the category other, the CSAF 2.0 to CSAF 2.1 Converter SHALL convert that data into the ssvc_v2 object within the current vulnerability. If the CSAF 2.0 to CSAF 2.1 Converter is able to construct a valid object without losing any information, the corresponding notes item SHALL be removed. If the CSAF 2.0 to CSAF 2.1 Converter is unable to construct a valid object with the information given, the CSAF 2.0 to CSAF 2.1 Converter SHALL remove the invalid ssvc_v2 object, keep the original item of notes and issue a warning that the automatic conversion of the SSVC data failed. If the CSAF 2.0 to CSAF 2.1 Converter would lose information during the conversion, the CSAF 2.0 to CSAF 2.1 Converter SHALL remove the ssvc_v2 object, keep the original item of notes and issue a warning that the automatic conversion of the SSVC data would lead to losing information.
- $.vulnerabilities[*].notes: If any $.vulnerabilities[*].notes item contains one of the category and title combinations specified in table 2 of section 3.2.4.11, where the title is extended, the CSAF 2.0 to CSAF 2.1 Converter SHALL try to identify whether that extension is a specific product name, version or family. In such case, the CSAF 2.0 to CSAF 2.1 Converter SHALL try to add the corresponding products to the note item and issue a warning that a potential product specific note has been discovered and products have been assigned to it. Such warning SHALL also include the note and the assigned products. If the CSAF 2.0 to CSAF 2.1 Converter is unable to create a valid object, it SHALL remove the reference to the products and issue a warning that a potential product specific note has been discovered and no products could be assigned to it.
- $.vulnerabilities[*].remediations[*]:
  - The CSAF 2.0 to CSAF 2.1 Converter SHALL convert any remediation with the category vendor_fix into the category optional_patch if the product in question is in one of the product status groups “Not Affected” or “Fixed” for this vulnerability. Otherwise, the category vendor_fix SHALL stay the same. If multiple products are associated with the remediation - either directly or through a product group - and the products belong to different product status groups, the CSAF 2.0 to CSAF 2.1 Converter SHALL duplicate the remediation, change the category in one instance to optional_patch and distribute the products accordingly as stated by the conversion rule.
  - The CSAF 2.0 to CSAF 2.1 Converter SHALL convert any remediation with the category none_available into the category fix_planned if the product in question is also listed in a remediation of the category vendor_fix with a date in the future or no date at all. Consequently, the product SHALL be removed from the remediation of the category vendor_fix. If it was the last product in that remediation, the remediation SHALL be removed.
  - The CSAF 2.0 to CSAF 2.1 Converter SHALL remove any product from a remediation with the category none_available if the product in question is also listed in a remediation of the category vendor_fix with a date in the past or to the exact same time. If it was the last product in that remediation, the remediation SHALL be removed.
  - In any other case, the CSAF 2.0 to CSAF 2.1 Converter SHALL preserve the product in the remediation of the category none_available.
  - The CSAF 2.0 to CSAF 2.1 Converter SHALL issue a warning if a remediation was added, deleted or the value of the category was changed, including the products it was changed for.

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A tool MAY implement options to convert other Markdown formats to GitHub-flavored Markdown.

A tool MAY implement an additional, non-default option to output an invalid document that can be fixed afterwards. Solely in this case, any of the rules above MAY be ignored to avoid data loss.

Mapping TLP labels per CSAF version:

Table 6: Comparison of values for $.document.distribution.tlp.label across CSAF versions 2.0 and 2.1.

CSAF 2.0 (using TLP v1.0)	CSAF 2.1 (using TLP v2.0)
TLP:WHITE	TLP:CLEAR
TLP:GREEN	TLP:GREEN
TLP:AMBER	TLP:AMBER
TLP:RED	TLP:RED

Mapping of involvements status:

Table 7: Mapping of values into $.document.involvement.actions[*] from CSAF versions 2.0 to 2.1.

CSAF 2.0 status	CSAF 2.1 status	CSAF 2.1 Default category
completed	completed	coordination
contact_attempted	attempted	notification
disputed	completed	dispute
in_progress	in_progress	coordination
not_contacted	deferred	notification
open	outstanding	coordination

9.1.19 Conformance Clause 19: CSAF Library

A library satisfies the "CSAF Library" conformance profile if the library:

- implements all elements as data structures conforming to the syntax and semantics defined in section 2.2, 2.3, 3, 4 and 5.
- checks all elements according to the patterns provided in the JSON schema.
- has a function that checks version ranges.
- has a function that helps to create version ranges.
- provides for each element functions that allow to create, add, modify and delete that element.
- has a function that reads a CSAF Document into the data structure from a
  - file system.
  - URL.
  - data stream.
- provides function for sorting the keys and sorts the keys automatically on output.
- has a function that outputs the data structure as CSAF Document
  - on the file system.
  - as string.
  - into a data stream.
- has a function to determine the filename according to 5.1 and sets the filename per default when saving a CSAF Document.
- generates a new product_id for each new element of type full_product_name_t unless an ID is given during the creation.
- generates a new group_id for each new element of type product_group_id_t unless an ID is given during the creation.
- provides a function to retrieve all elements of type product_id_t with its corresponding full_product_name_t/name and full_product_name_t/product_identification_helper.
- provides a function to retrieve all product_identification_helper and their mapping to elements of type product_id_t.

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- provides a function to retrieve a VEX status mapping for all data, which includes the combination of vulnerability, product, product status and, where necessary according to the profile, the impact statement respectively the action statement.
- provides a function to generate a full_product_name_t/name within branches through concatenating the name values separated by white space of the elements along the path towards this leaf.
- provides a function to generate a combined subpath name for an element of type subpath_t through concatenating separated by white space:
  - the category with _ replaced by white space and
  - the name value of the product identified by the Product ID in next_product_reference.
- provides a function to generate a full_product_name_t/name within product_paths through concatenating separated by white space:
  - the name value of the product identified by the Product ID in beginning_product_reference and
  - the combined subpath name for all elements in subpaths in their order.
- calculates the CVSS scores and severities for existing data for all CVSS versions.
- validates the CVSS scores and severities for existing data for all CVSS versions.
- optionally generates a new dl_vuln_id or sets the one provided during the creation for each new vulnerability.
- provides a function to remove all dl_vuln_id elements that are not referenced.

The library MAY implement an option to retrieve the keys unsorted.

9.1.20 Conformance Clause 20: CSAF Library with Basic Validation

A CSAF Library satisfies the "CSAF Library with Basic Validation" conformance profile if the CSAF Library:

- satisfies the "CSAF Library" conformance profile.
- satisfies the "CSAF Basic Validator" conformance profile.
- validates the CSAF Document before output according to the "CSAF Basic Validator" and presents the validation result accordingly.
- provide a function to validate the data structure in its current state according to the "CSAF Basic Validator" and presents the validation result accordingly.

A CSAF Library does not satisfies the "CSAF Library with Basic Validation" conformance profile if the CSAF Library uses an external library or program for the "CSAF Basic Validator" part and does not enforce its presence.

9.1.21 Conformance Clause 21: CSAF Library with Extended Validation

A CSAF Library satisfies the "CSAF Library with Extended Validation" conformance profile if the CSAF Library:

- satisfies the "CSAF Library" conformance profile.
- satisfies the "CSAF Extended Validator" conformance profile.
- validates the CSAF Document before output according to the "CSAF Extended Validator" and presents the validation result accordingly.
- provide a function to validate the data structure in its current state according to the "CSAF Extended Validator" and presents the validation result accordingly.

A CSAF Library does not satisfies the "CSAF Library with Extended Validation" conformance profile if the CSAF Library uses an external library or program for the "CSAF Extended Validator" part and does not enforce its presence.

9.1.22 Conformance Clause 22: CSAF Library with Full Validation

A CSAF Library satisfies the "CSAF Library with Full Validation" conformance profile if the CSAF Library:

- satisfies the "CSAF Library" conformance profile.
- satisfies the "CSAF Full Validator" conformance profile.
- validates the CSAF Document before output according to the "CSAF Full Validator" and presents the validation result accordingly.
- provide a function to validate the data structure in its current state according to the "CSAF Full Validator" and presents the validation result accordingly.

A CSAF Library does not satisfies the "CSAF Library with Full Validation" conformance profile if the CSAF Library uses an external library or program for the "CSAF Full Validator" part and does not enforce its presence.

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9.1.23 Conformance Clause 23: CSAF Downloader

A program satisfies the "CSAF Downloader" conformance profile if the program:

- conforms to the process defined in section 7.3 by executing all parts that are applicable to the given role.
- supports directory-based and ROLIE-based retrieval.
- is able to execute both steps from section 7.3 separately.
- uses a program-specific HTTP User Agent, e.g. consisting of the name and version of the program.
- satisfies those normative requirements in section 7 that are designated as applying to CSAF Downloaders.

> A tool MAY implement an option to store CSAF Documents that fail any of the steps in section 7.3.2.

9.1.24 Conformance Clause 24: CSAF Withdrawer

A program satisfies the "CSAF Withdrawer" conformance profile if the program:

- satisfies the "CSAF Post-Processor" conformance profile.
- keeps the original $.document.tracking.id.
- adds a new item to the revision history stating the revision metadata of the withdrawal.
- adds the reasoning for withdrawal as specified in section 4.7.
- removes the $.product_tree.
- removes the $.vulnerabilities.

> A tool MAY implement an option to additionally remove any element that would hinder the production of a valid CSAF.

9.1.25 Conformance Clause 25: CSAF Superseder

A program satisfies the "CSAF Superseder" conformance profile if the program:

- satisfies the "CSAF Post-Processor" conformance profile.
- keeps the original $.document.tracking.id.
- adds a new item to the revision history stating the revision metadata of the supersession.
- adds the reasoning for supersession as specified in section 4.8.
- adds the reference to the superseding document as specified in section 4.8.
- removes the $.product_tree.
- removes the $.vulnerabilities.

> A tool MAY implement an option to additionally remove any element that would hinder the production of a valid CSAF.

9.1.26 Conformance Clause 26: CSAF RVISC ID Updater

A program satisfies the "CSAF RVISC ID Updater" conformance profile if the program fulfills the two following groups of requirements:

The program:

- satisfies the "CSAF Post-Processor" conformance profile.
- applies the corresponding assignment from [RVISC-M] to each item in $.vulnerabilities[*].ids[*] whose system_name is not contained in [RVISC-R].
- applies the corresponding assignment from [RVISC-M] to each item in $.vulnerabilities[*].ids[*] whose system_name is contained in [RVISC-R] but the text does not conform the entry.
- issues a warning and skips the conversion for any matching ambiguous assignment from [RVISC-M] unless forced by the user to proceed with a certain option. Such warning SHALL contain all matching ambiguous assignment from [RVISC-M].
- satisfies the normative requirements given below.

The program SHALL provide the following options:

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- an option to insert an automatically generated revision history entry detailing the changes applied and make necessary updates to elements in $.document.tracking (commit mode).
- an option to set selected or all parameters for the commit mode manually which take precedence over the automated generated values.
- an option to do a dry-run which does not apply the changes but just displays them.
- an option to interactively accept or discard changes.
- an option to ignore certain values for system_name.
- an option to output all items in $.vulnerabilities[*].ids[*] whose system_name is not contained in [RVISC-R].
- an option to output all items in $.vulnerabilities[*].ids[*] whose system_name is contained in [RVISC-R] but the text does not conform the entry.
- an option to map an existing system_name to a new value or apply a transformation to a text based on the system_name value and a precondition.
- an option to force the tool to proceed with the conversion applying an ambiguous assignment from [RVISC-M] as selected by the user.
- an option to list all matching ambiguous assignment from [RVISC-M].

9.1.27 Conformance Clause 27: CSAF Additional Test

A test satisfies the "CSAF Additional Test" conformance profile if it:

- covers a requirement of the standard.
- operates on a CSAF Document.
- outputs a result object that aligns with result objects provided for tests specified in section 6.
- has a name starting with AdditionalTest_ followed by a name of the specifying entity conforming to the rules for prefixes of test presets (cf. section 6.4) and an unique name for the test within that entity.
- has its definition specified in the same structure as in section 6.

9.1.28 Conformance Clause 28: CSAF Extension

A JSON object satisfies the "CSAF Extension" conformance profile if it:

- conforms to the syntax and semantics defined in section 2.4.
- validates against the CSAF Extension Content Schema and the schema given through its $schema property.
- uses the value of $id of its CSAF Extension Schema as the value of its $schema property.
- conveys additional information that cannot be conveyed with the CSAF Core elements.
- does not convey any content that can be conveyed with the CSAF Core elements.
- does not contradict the content or purpose of this specification.

9.1.29 Conformance Clause 29: CSAF Extension Schema

A JSON schema satisfies the "CSAF Extension Schema" conformance profile if fulfills the following two groups of requirements:

Firstly, it:

- describes a JSON object satisfying the "CSAF Extension" conformance profile.
- validates against the CSAF Extension Metaschema.
- uses the CSAF Extension Metaschema as the value of its $schema property.
- is versioned according to [SemVer].
- has an $id that conforms to the pattern given for the property $schema in section 2.4.4.1.
- does not allow unevaluated properties. It MAY include patternProperties if they are typed.

> patternProperties are typed if they can't have more than one type and that type is given through the schema.

Secondly, it:

- SHOULD avoid unpredictable JSON key names.

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> Implementations usually ignore any additional properties as they are hard to access, process and may imply security risks.

9.1.30 Conformance Clause 30: CSAF Extension Overlay Test

A test satisfies the "CSAF Extension Overlay Test" conformance profile if it:

- is executed only if the specifying CSAF Extension is present.

> To reuse a test from a different CSAF Extension, it is sufficient to assign it a name according to the rules below and point to the specification of the test it reuses. The referenced specification in such case must be accessible to anyone that can retrieve the referencing CSAF Extension.

- extends or replaces a test specified in this standard in section 6 or a CSAF Additional Test.
- does not adversely affect the purpose of the test in question.
- has a name starting with `Extension_Overlay_Test_` followed by a name of the specifying entity conforming to the rules for prefixes of test presets (cf. section 6.4) and the name of the tests that it replaces.
- has its definition specified in the same structure as in section 6.

The CSAF Extension Overlay Test SHOULD only differ in the paths or the document categories it is applied to.

> This ensures maximum compatibility with other extensions as multiple CSAF Extension can define CSAF Extension Overlay Tests for the same test which are then executed as a union set.

9.1.31 Conformance Clause 31: CSAF Extension Additional Test

A test satisfies the "CSAF Extension Additional Test" conformance profile if it:

- is executed only if the specifying CSAF Extension is present.

> To reuse a test from a different CSAF Extension, it is sufficient to assign it a name according to the rules below and point to the specification of the test it reuses. The referenced specification in such case must be accessible to anyone that can retrieve the referencing CSAF Extension.

- provides additional checks in the context of the CSAF Extension or the CSAF Document the extension is embedded in.
- has a name starting with `Extension_Additional_Test_` followed by a name of the specifying entity conforming to the rules for prefixes of test presets (cf. section 6.4) and an unique number for the test within that entity.
- has its definition specified in the same structure as in section 6.

9.1.32 Conformance Clause 32: CSAF Extension Test

A test satisfies the "CSAF Extension Test" conformance profile if it:

- satisfies the "CSAF Extension Overlay Test" conformance profile or the "CSAF Extension Additional Test" conformance profile.

9.1.33 Conformance Clause 33: CSAF Extension Specification

A specification satisfies the "CSAF Extension Specification" conformance profile if:

- it defines exactly one extension satisfying the "CSAF Extension" conformance profile.
- it defines exactly one JSON schema satisfying the "CSAF Extension Schema" conformance profile.
- all tests defined in it satisfy the "CSAF Extension Test" conformance profile.
- it contains all tests needed for the validation of the extension.

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> This includes tests that check for prerequisites in the CSAF document, if applicable. Referencing an existing test is considered to fulfill the "contains" requirement, if the referenced test is accessible to anyone that can access the referencing specification.

- all requirements above are fulfilled for the same extension.

9.1.34 Conformance Clause 34: CSAF Extension Bundle

A compilation of artifacts satisfies the "CSAF Extension Bundle" conformance profile if:

- it contains exactly one extension satisfying the "CSAF Extension" conformance profile.
- it contains exactly one JSON schema satisfying the "CSAF Extension Schema" conformance profile.
- it contains exactly one JSON object satisfying the "CSAF Extension Metadata Schema".
- all necessary test files to implement the "CSAF Extension Test" for this extension.
- has a name that it can be referred to.

9.1.35 Conformance Clause 35: CSAF Extension Package

A compilation of artifacts satisfies the "CSAF Extension Package" conformance profile if:

- it contains exactly one specification satisfying the "CSAF Extension Specification" conformance profile.
- it contains exactly one compilation of artifacts satisfying the "CSAF Extension Bundle" conformance profile.
- has a name that it can be referred to.

A CSAF Extension Package MAY contain additional documentation and artifacts to guide implementers.

9.1.36 Conformance Clause 36: CSAF Extension Collection

A set of artifacts satisfies the "CSAF Extension Collection" conformance profile if it:

- contains one or more named compilation of artifacts that satisfy the "CSAF Extension Package" conformance profile.
- does not contain any compilation of artifacts that looks like a CSAF Extension Package but does not satisfy the "CSAF Extension Package" conformance profile.

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Appendix A. Acknowledgments

(This appendix does not form an integral part of this specification and is informational.)

The following individuals were members of the OASIS CSAF Technical Committee during the creation of this specification and their contributions are gratefully acknowledged:

CSAF TC Members:

First Name	Last Name	Company
Aashiq	Ramachandran	Cyware Labs
Alexandre	Dulaunoy	CIRCL
Anthony	Berglas	Cryptsoft Pty Ltd.
Avkash	Kathiriya	Cyware Labs
Bernd	Grobauer	Siemens AG
Celia	Savidge	Dell Technologies
Chok	Poh	Oracle
Christian	Banse	Fraunhofer-Gesellschaft e.V. - Fraunhofer AISEC
Christoph	Plutte	Ericsson AB
Dan	West	Microsoft
Dave	Tamasi	Google LLC
David	Waltermire	NIST
Denny	Page	Individual
Dejan	Bosanac	Red Hat
Diane	Morris	Cisco Systems
Dina	Truxius	Federal Office for Information Security (BSI) Germany
Feng	Cao	Oracle
Jane	Ginn	Individual
Jared	Bird	Hewlett Packard Enterprise (HPE)
Jason	Keirstead	Pobal Cyber Ltd
Jean-Didier	Stéfaniak	Dell Technologies
Jeff	Schutt	Cisco Systems
Jennifer	Victor	Dell Technologies
Jessica	Fitzgerald-McKay	National Security Agency
Jessie	Vaught	Red Hat
Johannes	Lorenz	Siemens AG
Justin	Corlett	Cryptsoft Pty Ltd.
Justin	Murphy	US DHS Cybersecurity and Infrastructure Security Agency (CISA)
Martin	Prpic	Red Hat
Michael	Reeder	Dell Technologies
Michael	Schueler	Cisco Systems
Nick	Leali	Cisco Systems
Nicole	Parrish	Mitre Corporation

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First Name	Last Name	Company
Oliver	Chang	Google LLC
Omar	Santos	Cisco Systems
Patrick	Maroney	AT&T
Qin	Long	Alibaba Cloud Computing Ltd.
Richard	Struse	Mitre Corporation
Rushyendra	Velamuri	Dell Technologies
Samuel	Cameron	Cisco Systems
Sergii	Demianchuk	Cisco Systems
Shrey	Bagga	Cisco Systems
Sonny	van Lingen	Huawei Technologies Co., Ltd.
Stefan	Hagen	Individual
Tania	Ward	Dell Technologies
Thomas	Proell	Siemens AG
Thomas	Schaffer	Cisco Systems
Thomas	Schmidt	Federal Office for Information Security (BSI) Germany
Tim	Hudson	Cryptsoft Pty Ltd.
Tobias	Limmer	Siemens AG
Tyler	Townes	BlackBerry Limited
Vic	Chung	SAP SE
Vijay	Sarvepalli	Carnegie Mellon University
Vincent	Danen	Red Hat
Vivek	Nair	Microsoft
William	Ho Thiam Hock	Huawei Technologies Co., Ltd.
Xiaoyu	Ge	Huawei Technologies Co., Ltd.
Zebin	Zhang	Huawei Technologies Co., Ltd.

The following individuals were members of the OASIS CSAF Technical Committee during the creation of the previous version (CSAF 2.0) of this specification and their contributions are gratefully acknowledged:

CSAF TC Members:

First Name	Last Name	Company
Alexandre	Dulaunoy	CIRCL
Anthony	Berglas	Cryptsoft Pty Ltd.
Art	Manion	Carnegie Mellon University
Aukjan	van Belkum	EclecticIQ
Ben	Sooter	Electric Power Research Institute (EPRI)
Bernd	Grobauer	Siemens AG
Bruce	Rich	Cryptsoft Pty Ltd.
Chok	Poh	Oracle

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First Name	Last Name	Company
Dan	West	Microsoft
David	Waltermire	NIST
Denny	Page	TIBCO Software Inc.
Duncan	Sparrell	sFractal Consulting LLC
Eric	Johnson	TIBCO Software Inc.
Ethan	Rahn	Arista Networks
Feng	Cao	Oracle
Greg	Scott	Cryptsoft Pty Ltd.
Harold	Booth	NIST
Jason	Masters	TELUS
Jennifer	Victor	Dell
Jessica	Fitzgerald-McKay	National Security Agency
Jonathan	Bitle	Kaiser Permanente
Justin	Corlett	Cryptsoft Pty Ltd.
Kazuo	Noguchi	Hitachi, Ltd.
Kent	Landfield	McAfee
Langley	Rock	Red Hat
Martin	Prpic	Red Hat
Masato	Terada	Hitachi, Ltd.
Mike	Gorski	Cisco Systems
Nicole	Parrish	Mitre Corporation
Omar	Santos	Cisco Systems
Patrick	Maroney	AT&T
Rhonda	Levy	Cisco Systems
Richard	Struse	Mitre Corporation
Ritwik	Ghoshal	Oracle
Robert	Coderre	Accenture
Robert	Keith	Accenture
Stefan	Hagen	Individual
Tania	Ward	Dell
Ted	Bedwell	Cisco Systems
Thomas	Proell	Siemens AG
Thomas	Schmidt	Federal Office for Information Security (BSI) Germany
Tim	Hudson	Cryptsoft Pty Ltd.
Tobias	Limmer	Siemens AG
Tony	Cox	Cryptsoft Pty Ltd.
Vincent	Danen	Red Hat

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First Name	Last Name	Company
Will	Rideout	Arista Networks
Xiaoyu	Ge	Huawei Technologies Co., Ltd.

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Appendix B. Revision History

(This appendix does not form an integral part of this specification and is informational.)

Revision tracking is publicly available in the version control system at https://github.com/oasis-tcs/csaf/commits/master.

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Appendix C. Guidance on the Size of CSAF Documents

This appendix provides informative guidance on the size of CSAF documents.

The TC carefully considered all known aspects to provide size limits for CSAF documents for this version of the specification with the result that hard limits should not be enforced. However, since there is the need for guidance to ensure interoperability in the ecosystem, the TC provides a set of soft limits. A CSAF document which exceeds those, can still be valid but it might not be processable for some parties.

All *CSAF consumers* should be able to process CSAF documents which comply with the limits below. All *CSAF producers* should not produce CSAF documents which exceed those limits.

> If you come across a case where these limits are exceeded, please provide feedback to the TC.

Boolean values are usually represented as a native data type and therefore omitted here. Structures in the `content` object of items of type `$['$defs'].extensions_t` are not listed in the subsections below as they depend upon on the extension.

C.1 File Size

A CSAF document including any extensions used in the document in the specified JSON format encoded in UTF-8 should conform to known size limits of current technologies parsing JSON content, e.g.: 150 MiB.

> The CSAF documents observed in the wild expose strongly varying sizes as per the use cases they serve. At least one database technology in wide use for storing CSAF documents rejects insert attempts when the transformed BSON size exceeds 16 megabytes. The BSON format optimizes for accessibility and not size. So, small integers and small strings may incur more overhead in the BSON format than in JSON. In addition, the BSON format adds length information for the entries inside the document, which adds to the size when storing CSAF document content in a BSON format.

C.2 Array Length

An array should not have more than:

- 10 000 items for
  - $.document.acknowledgments
  - $.document.acknowledgments[*].names
  - $.document.acknowledgments[*].urls
  - $.document.tracking.aliases
  - $.document.x_extensions
  - $.product_tree..branches[*].product.product_identification_helper.hashes
  - $.product_tree..branches[*].product.product_identification_helper.hashes[*].file_hashes
  - $.product_tree..branches[*].product.product_identification_helper.purls
  - $.product_tree..branches[*].product.product_identification_helper.sbom_urls
  - $.product_tree..branches[*].product.product_identification_helper.x_generic_uris
  - $.product_tree..branches[*].product.x_extensions
  - $.product_tree.full_product_names[*].product_identification_helper.hashes
  - $.product_tree.full_product_names[*].product_identification_helper.hashes[*].file_hashes
  - $.product_tree.full_product_names[*].product_identification_helper.purls
  - $.product_tree.full_product_names[*].product_identification_helper.sbom_urls
  - $.product_tree.full_product_names[*].product_identification_helper.x_generic_uris
  - $.product_tree.full_product_names[*].x_extensions
  - $.product_tree.product_paths[*].full_product_name.product_identification_helper.hashes
  - $.product_tree.product_paths[*].full_product_name.product_identification_helper.hashes[*].file_hashes
  - $.product_tree.product_paths[*].full_product_name.product_identification_helper.purls
  - $.product_tree.product_paths[*].full_product_name.product_identification_helper.sbom_urls
  - $.product_tree.product_paths[*].full_product_name.product_identification_helper.x_generic_uris
  - $.product_tree.product_paths[*].full_product_name.x_extensions
  - $.vulnerabilities[*].acknowledgments

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- $.vulnerabilities[*].acknowledgments[*].names
- $.vulnerabilities[*].acknowledgments[*].urls
- $.vulnerabilities[*].cwes
- $.vulnerabilities[*].ids
- $.vulnerabilities[*].metrics[*].content.ssvc_v2.decision_point_resources
- $.vulnerabilities[*].metrics[*].content.ssvc_v2.selections
- $.vulnerabilities[*].metrics[*].content.ssvc_v2.selections[*].values
- $.vulnerabilities[*].metrics[*].content.ssvc_v2.target_ids
- $.vulnerabilities[*].metrics[*].content.x_extensions
- $.vulnerabilities[*].remediations[*].entitlements
- $.vulnerabilities[*].x_extensions
- $.x_extensions

- 40 000 items for
  - $.document.involvement.actions[*].acting_entity_refs
  - $.document.involvement.actions[*].receiving_entity_refs
  - $.document.involvement.entities
  - $.document.involvement.entity_groups
  - $.document.notes
  - $.document.references
  - $.vulnerabilities[*].metrics[*].content.ssvc_v2.references
  - $.vulnerabilities[*].notes
  - $.vulnerabilities[*].references
- 100 000 for
  - $.document.involvement.actions
  - $.document.involvement.actions[*].dl_vuln_ids
  - $.document.involvement.actions[*].referenced_action_ids
  - $.document.tracking.revision_history
  - $.product_tree.branches
  - $.product_tree..branches
  - $.product_tree..branches[*].product.product_identification_helper.model_numbers
  - $.product_tree..branches[*].product.product_identification_helper.serial_numbers
  - $.product_tree..branches[*].product.product_identification_helper.skus
  - $.product_tree.full_product_names
  - $.product_tree.full_product_names[*].product_identification_helper.model_numbers
  - $.product_tree.full_product_names[*].product_identification_helper.serial_numbers
  - $.product_tree.full_product_names[*].product_identification_helper.skus
  - $.product_tree.product_groups[*].product_ids
  - $.product_tree.product_paths[*].full_product_name.product_identification_helper.model_numbers
  - $.product_tree.product_paths[*].full_product_name.product_identification_helper.serial_numbers
  - $.product_tree.product_paths[*].full_product_name.product_identification_helper.skus
  - $.product_tree.product_paths[*].subpaths
  - $.vulnerabilities
- 10 000 000 for
  - $.product_tree.product_groups
  - $.product_tree.product_paths
  - $.vulnerabilities[*].remediations[*].group_ids
- 100 000 000 for
  - $.document.involvement.actions[*].group_ids
  - $.document.involvement.actions[*].product_ids
  - $.document.notes[*].group_ids
  - $.document.notes[*].product_ids
  - $.vulnerabilities[*].first_known_exploitation_dates
  - $.vulnerabilities[*].first_known_exploitation_dates[*].group_ids
  - $.vulnerabilities[*].first_known_exploitation_dates[*].product_ids
  - $.vulnerabilities[*].flags
  - $.vulnerabilities[*].flags[*].group_ids

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- $.vulnerabilities[*].flags[*].product_ids
- $.vulnerabilities[*].ids[*].group_ids
- $.vulnerabilities[*].ids[*].product_ids
- $.vulnerabilities[*].metrics
- $.vulnerabilities[*].metrics[*].products
- $.vulnerabilities[*].notes[*].group_ids
- $.vulnerabilities[*].notes[*].product_ids
- $.vulnerabilities[*].product_status.first_affected
- $.vulnerabilities[*].product_status.first_fixed
- $.vulnerabilities[*].product_status.fixed
- $.vulnerabilities[*].product_status.known_affected
- $.vulnerabilities[*].product_status.known_not_affected
- $.vulnerabilities[*].product_status.last_affected
- $.vulnerabilities[*].product_status.recommended
- $.vulnerabilities[*].product_status.under_investigation
- $.vulnerabilities[*].product_status.unknown
- $.vulnerabilities[*].remediations
- $.vulnerabilities[*].remediations[*].product_ids
- $.vulnerabilities[*].threats
- $.vulnerabilities[*].threats[*].group_ids
- $.vulnerabilities[*].threats[*].product_ids

C.3 String Length

A string should not have a length greater than:

- 1000 for
  - $.document.acknowledgments[*].names[*]
  - $.document.acknowledgments[*].organization
  - $.document.aggregate_severity.text
  - $.document.category
  - $.document.distribution.sharing_group.name
  - $.document.involvement.actions[*].acting_entity_refs[*]
  - $.document.involvement.actions[*].action_id
  - $.document.involvement.actions[*].group_ids[*]
  - $.document.involvement.actions[*].dl_vuln_ids[*]
  - $.document.involvement.actions[*].product_ids[*]
  - $.document.involvement.actions[*].receiving_entity_refs[*]
  - $.document.involvement.actions[*].referenced_action_ids[*]
  - $.document.involvement.entities[*].contact.email
  - $.document.involvement.entities[*].entity_id
  - $.document.involvement.entities[*].name
  - $.document.involvement.entity_groups[*].entity_group_id
  - $.document.involvement.entity_groups[*].name
  - $.document.lang
  - $.document.license_expression
  - $.document.notes[*].audience
  - $.document.notes[*].group_ids[*]
  - $.document.notes[*].product_ids[*]
  - $.document.notes[*].title
  - $.document.publisher.contact.email
  - $.document.publisher.name
  - $.document.source_lang
  - $.document.title
  - $.document.tracking.aliases[*]
  - $.document.tracking.generator.engine.name
  - $.document.tracking.generator.engine.version

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- $.document.tracking.id
- $.document.tracking.revision_history[*].legacy_version
- $.document.tracking.revision_history[*].number
- $.document.tracking.version
- $.product_tree..branches[*].name
- $.product_tree..branches[*].product.name
- $.product_tree..branches[*].product.product_id
- $.product_tree..branches[*].product.product_identification_helper.hashes[*].file_hashes[*].algorithm
- $.product_tree..branches[*].product.product_identification_helper.hashes[*].file_hashes[*].value
- $.product_tree..branches[*].product.product_identification_helper.hashes[*].filename
- $.product_tree..branches[*].product.product_identification_helper.model_numbers[*]
- $.product_tree..branches[*].product.product_identification_helper.serial_numbers[*]
- $.product_tree..branches[*].product.product_identification_helper.skus[*]
- $.product_tree.full_product_names[*].name
- $.product_tree.full_product_names[*].product_id
- $.product_tree.full_product_names[*].product_identification_helper.hashes[*].file_hashes[*].algorithm
- $.product_tree.full_product_names[*].product_identification_helper.hashes[*].file_hashes[*].value
- $.product_tree.full_product_names[*].product_identification_helper.hashes[*].filename
- $.product_tree.full_product_names[*].product_identification_helper.model_numbers[*]
- $.product_tree.full_product_names[*].product_identification_helper.serial_numbers[*]
- $.product_tree.full_product_names[*].product_identification_helper.skus[*]
- $.product_tree.product_groups[*].group_id
- $.product_tree.product_groups[*].product_ids[*]
- $.product_tree.product_paths[*].full_product_name.name
- $.product_tree.product_paths[*].full_product_name.product_id
- $.product_tree.product_paths[*].full_product_name.product_identification_helper.hashes[*].file_hashes[*].algorithm
- $.product_tree.product_paths[*].full_product_name.product_identification_helper.hashes[*].file_hashes[*].value
- $.product_tree.product_paths[*].full_product_name.product_identification_helper.hashes[*].filename
- $.product_tree.product_paths[*].full_product_name.product_identification_helper.model_numbers[*]
- $.product_tree.product_paths[*].full_product_name.product_identification_helper.serial_numbers[*]
- $.product_tree.product_paths[*].full_product_name.product_identification_helper.skus[*]
- $.product_tree.product_paths[*].beginning_product_reference
- $.product_tree.product_paths[*].subpaths[*].next_product_reference
- $.vulnerabilities[*].acknowledgments[*].names[*]
- $.vulnerabilities[*].acknowledgments[*].organization
- $.vulnerabilities[*].cve
- $.vulnerabilities[*].cwes[*].id
- $.vulnerabilities[*].cwes[*].name
- $.vulnerabilities[*].cwes[*].version
- $.vulnerabilities[*].dl_vuln_id
- $.vulnerabilities[*].flags[*].group_ids[*]
- $.vulnerabilities[*].flags[*].product_ids[*]
- $.vulnerabilities[*].first_known_exploitation_dates[*].group_ids[*]
- $.vulnerabilities[*].ids[*].group_ids[*]
- $.vulnerabilities[*].ids[*].product_ids[*]
- $.vulnerabilities[*].ids[*].system_name
- $.vulnerabilities[*].ids[*].text
- $.vulnerabilities[*].metrics[*].content.cvss_v2.vectorString
- $.vulnerabilities[*].metrics[*].content.cvss_v3.vectorString
- $.vulnerabilities[*].metrics[*].content.cvss_v4.vectorString
- $.vulnerabilities[*].metrics[*].content.epss.percentile
- $.vulnerabilities[*].metrics[*].content.epss.probability
- $.vulnerabilities[*].metrics[*].content.ssvc_v2.selections[*].key
- $.vulnerabilities[*].metrics[*].content.ssvc_v2.selections[*].namespace
- $.vulnerabilities[*].metrics[*].content.ssvc_v2.selections[*].values[*].key
- $.vulnerabilities[*].metrics[*].content.ssvc_v2.selections[*].values[*].name

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- $.vulnerabilities[*].metrics[*].content.ssvc_v2.selections[*].version
- $.vulnerabilities[*].metrics[*].content.ssvc_v2.target_ids[*]
- $.vulnerabilities[*].metrics[*].products[*]
- $.vulnerabilities[*].notes[*].audience
- $.vulnerabilities[*].notes[*].group_ids[*]
- $.vulnerabilities[*].notes[*].product_ids[*]
- $.vulnerabilities[*].notes[*].title
- $.vulnerabilities[*].product_status.first_affected[*]
- $.vulnerabilities[*].product_status.first_fixed[*]
- $.vulnerabilities[*].product_status.fixed[*]
- $.vulnerabilities[*].product_status.known_affected[*]
- $.vulnerabilities[*].product_status.known_not_affected[*]
- $.vulnerabilities[*].product_status.last_affected[*]
- $.vulnerabilities[*].product_status.recommended[*]
- $.vulnerabilities[*].product_status.under_investigation[*]
- $.vulnerabilities[*].product_status.unknown[*]
- $.vulnerabilities[*].remediations[*].group_ids[*]
- $.vulnerabilities[*].remediations[*].product_ids[*]
- $.vulnerabilities[*].threats[*].group_ids[*]
- $.vulnerabilities[*].threats[*].product_ids[*]
- $.vulnerabilities[*].title

- 10 000 for
  - $.document.acknowledgments[*].summary
  - $.document.distribution.text
  - $.document.involvement.actions[*].summary
  - $.document.involvement.entities[*].contact.details
  - $.document.involvement.entity_groups[*].summary
  - $.document.publisher.contact.details
  - $.document.publisher.issuing_authority
  - $.document.references[*].summary
  - $.document.tracking.revision_history[*].summary
  - $.product_tree..branches[*].product.product_identification_helper.cpe
  - $.product_tree..branches[*].product.product_identification_helper.purls[*]
  - $.product_tree.full_product_names[*].product_identification_helper.cpe
  - $.product_tree.full_product_names[*].product_identification_helper.purls[*]
  - $.product_tree.product_groups[*].summary
  - $.product_tree.product_paths[*].full_product_name.product_identification_helper.cpe
  - $.product_tree.product_paths[*].full_product_name.product_identification_helper.purls[*]
  - $.vulnerabilities[*].acknowledgments[*].summary
  - $.vulnerabilities[*].metrics[*].content.ssvc_v2.decision_point_resources[*].summary
  - $.vulnerabilities[*].metrics[*].content.ssvc_v2.references[*].summary
  - $.vulnerabilities[*].references[*].summary
  - $.vulnerabilities[*].remediations[*].entitlements[*]
- 30 000 for
  - $.document.notes[*].text
  - $.vulnerabilities[*].notes[*].text
  - $.vulnerabilities[*].metrics[*].content.ssvc_v2.selections[*].definition
  - $.vulnerabilities[*].metrics[*].content.ssvc_v2.selections[*].values[*].definition
- 250 000 for
  - $.vulnerabilities[*].remediations[*].details
  - $.vulnerabilities[*].remediations[*].restart_required.details
  - $.vulnerabilities[*].threats[*].details

C.4 Date

The maximum length of strings representing a temporal value is given by the format specifier. This applies to:

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- $.document.involvement.actions[*].date
- $.document.tracking.current_release_date
- $.document.tracking.generator.date
- $.document.tracking.initial_release_date
- $.document.tracking.revision_history[*].date
- $.vulnerabilities[*].disclosure_date
- $.vulnerabilities[*].discovery_date
- $.vulnerabilities[*].first_known_exploitation_dates[*].date
- $.vulnerabilities[*].first_known_exploitation_dates[*].exploitation_date
- $.vulnerabilities[*].flags[*].date
- $.vulnerabilities[*].metrics[*].content.epss.timestamp
- $.vulnerabilities[*].metrics[*].content.ssvc_v2.timestamp
- $.vulnerabilities[*].remediations[*].date
- $.vulnerabilities[*].threats[*].date

C.5 Enum

A string which is an enum has a fixed maximum length given by its longest value.

> Later versions of CSAF might add, modify or delete possible values which could change the longest value. Therefore, this sizes should not be implemented as fixed limits if forward compatibility is desired.

anyOf(const) and allOf(const) are treated as enum values in this guidance.

It seems to be safe to assume that the length of each value is not greater than 50. This applies to:

- $.document.distribution.tlp.label (12)
- $.document.notes[*].category (16)
- $.document.involvement.actions[*].category (14)
- $.document.involvement.actions[*].status (12)
- $.document.involvement.entities[*].category (19)
- $.document.publisher.category (11)
- $.document.references[*].category (8)
- $.document.tracking.status (7)
- $.document.x_extensions[*].category (13)
- $.product_tree..branches[*].category (21)
- $.product_tree..branches[*].product.x_extensions[*].category (13)
- $.product_tree.full_product_names[*].x_extensions[*].category (13)
- $.product_tree.product_paths[*].subpaths[*].category (21)
- $.product_tree.product_paths[*].full_product_name.x_extensions[*].category (13)
- $.vulnerabilities[*].flags[*].label (49)
- $.vulnerabilities[*].metrics[*].content.cvss_v2.accessComplexity (6)
- $.vulnerabilities[*].metrics[*].content.cvss_v2.accessVector (16)
- $.vulnerabilities[*].metrics[*].content.cvss_v2.authentication (8)
- $.vulnerabilities[*].metrics[*].content.cvss_v2.availabilityImpact (8)
- $.vulnerabilities[*].metrics[*].content.cvss_v2.availabilityRequirement (11)
- $.vulnerabilities[*].metrics[*].content.cvss_v2.collateralDamagePotential (11)
- $.vulnerabilities[*].metrics[*].content.cvss_v2.confidentialityImpact (8)
- $.vulnerabilities[*].metrics[*].content.cvss_v2.confidentialityRequirement (11)
- $.vulnerabilities[*].metrics[*].content.cvss_v2.exploitability (16)
- $.vulnerabilities[*].metrics[*].content.cvss_v2.integrityImpact (8)
- $.vulnerabilities[*].metrics[*].content.cvss_v2.integrityRequirement (11)
- $.vulnerabilities[*].metrics[*].content.cvss_v2.remediationLevel (13)
- $.vulnerabilities[*].metrics[*].content.cvss_v2.reportConfidence (14)
- $.vulnerabilities[*].metrics[*].content.cvss_v2.targetDistribution (11)
- $.vulnerabilities[*].metrics[*].content.cvss_v2.version (3)

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- $.vulnerabilities[*].metrics[*].content.cvss_v3.attackComplexity (4)
- $.vulnerabilities[*].metrics[*].content.cvss_v3.attackVector (16)
- $.vulnerabilities[*].metrics[*].content.cvss_v3.availabilityImpact (4)
- $.vulnerabilities[*].metrics[*].content.cvss_v3.availabilityRequirement (11)
- $.vulnerabilities[*].metrics[*].content.cvss_v3.baseSeverity (8)
- $.vulnerabilities[*].metrics[*].content.cvss_v3.confidentialityImpact (4)
- $.vulnerabilities[*].metrics[*].content.cvss_v3.confidentialityRequirement (11)
- $.vulnerabilities[*].metrics[*].content.cvss_v3.environmentalSeverity (8)
- $.vulnerabilities[*].metrics[*].content.cvss_v3.exploitCodeMaturity (16)
- $.vulnerabilities[*].metrics[*].content.cvss_v3.integrityImpact (4)
- $.vulnerabilities[*].metrics[*].content.cvss_v3.integrityRequirement (11)
- $.vulnerabilities[*].metrics[*].content.cvss_v3.modifiedAttackComplexity (11)
- $.vulnerabilities[*].metrics[*].content.cvss_v3.modifiedAttackVector (16)
- $.vulnerabilities[*].metrics[*].content.cvss_v3.modifiedAvailabilityImpact (11)
- $.vulnerabilities[*].metrics[*].content.cvss_v3.modifiedConfidentialityImpact (11)
- $.vulnerabilities[*].metrics[*].content.cvss_v3.modifiedIntegrityImpact (11)
- $.vulnerabilities[*].metrics[*].content.cvss_v3.modifiedPrivilegesRequired (11)
- $.vulnerabilities[*].metrics[*].content.cvss_v3.modifiedScope (11)
- $.vulnerabilities[*].metrics[*].content.cvss_v3.modifiedUserInteraction (11)
- $.vulnerabilities[*].metrics[*].content.cvss_v3.privilegesRequired (4)
- $.vulnerabilities[*].metrics[*].content.cvss_v3.remediationLevel (13)
- $.vulnerabilities[*].metrics[*].content.cvss_v3.reportConfidence (11)
- $.vulnerabilities[*].metrics[*].content.cvss_v3.scope (9)
- $.vulnerabilities[*].metrics[*].content.cvss_v3.temporalSeverity (8)
- $.vulnerabilities[*].metrics[*].content.cvss_v3.userInteraction (8)
- $.vulnerabilities[*].metrics[*].content.cvss_v3.version (3)
- $.vulnerabilities[*].metrics[*].content.cvss_v4.attackComplexity (4)
- $.vulnerabilities[*].metrics[*].content.cvss_v4.attackRequirements (7)
- $.vulnerabilities[*].metrics[*].content.cvss_v4.attackVector (8)
- $.vulnerabilities[*].metrics[*].content.cvss_v4.Automatable (11)
- $.vulnerabilities[*].metrics[*].content.cvss_v4.availabilityRequirement (11)
- $.vulnerabilities[*].metrics[*].content.cvss_v4.baseSeverity (8)
- $.vulnerabilities[*].metrics[*].content.cvss_v4.confidentialityRequirement (11)
- $.vulnerabilities[*].metrics[*].content.cvss_v4.exploitMaturity (16)
- $.vulnerabilities[*].metrics[*].content.cvss_v4.integrityRequirement (11)
- $.vulnerabilities[*].metrics[*].content.cvss_v4.modifiedAttackComplexity (11)
- $.vulnerabilities[*].metrics[*].content.cvss_v4.modifiedAttackRequirements (11)
- $.vulnerabilities[*].metrics[*].content.cvss_v4.modifiedAttackVector (11)
- $.vulnerabilities[*].metrics[*].content.cvss_v4.modifiedPrivilegesRequired (11)
- $.vulnerabilities[*].metrics[*].content.cvss_v4.modifiedSubAvailabilityImpact (11)
- $.vulnerabilities[*].metrics[*].content.cvss_v4.modifiedSubConfidentialityImpact (11)
- $.vulnerabilities[*].metrics[*].content.cvss_v4.modifiedSubIntegrityImpact (11)
- $.vulnerabilities[*].metrics[*].content.cvss_v4.modifiedUserInteraction (11)
- $.vulnerabilities[*].metrics[*].content.cvss_v4.modifiedVulnAvailabilityImpact (11)
- $.vulnerabilities[*].metrics[*].content.cvss_v4.modifiedVulnConfidentialityImpact (11)
- $.vulnerabilities[*].metrics[*].content.cvss_v4.modifiedVulnIntegrityImpact (11)
- $.vulnerabilities[*].metrics[*].content.cvss_v4.privilegesRequired (4)
- $.vulnerabilities[*].metrics[*].content.cvss_v4.providerUrgency (11)
- $.vulnerabilities[*].metrics[*].content.cvss_v4.Recovery (13)
- $.vulnerabilities[*].metrics[*].content.cvss_v4.Safety (11)
- $.vulnerabilities[*].metrics[*].content.cvss_v4.subAvailabilityImpact (4)
- $.vulnerabilities[*].metrics[*].content.cvss_v4.subConfidentialityImpact (4)
- $.vulnerabilities[*].metrics[*].content.cvss_v4.subIntegrityImpact (4)
- $.vulnerabilities[*].metrics[*].content.cvss_v4.userInteraction (7)
- $.vulnerabilities[*].metrics[*].content.cvss_v4.valueDensity (12)

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- $.vulnerabilities[*].metrics[*].content.cvss_v4.version (3)
- $.vulnerabilities[*].metrics[*].content.cvss_v4.vulnAvailabilityImpact (4)
- $.vulnerabilities[*].metrics[*].content.cvss_v4.vulnConfidentialityImpact (4)
- $.vulnerabilities[*].metrics[*].content.cvss_v4.vulnerabilityResponseEffort (11)
- $.vulnerabilities[*].metrics[*].content.cvss_v4.vulnIntegrityImpact (4)
- $.vulnerabilities[*].metrics[*].content.qualitative_severity_rating (8)
- $.vulnerabilities[*].metrics[*].content.x_extensions[*].category (13)
- $.vulnerabilities[*].notes[*].category (16)
- $.vulnerabilities[*].references[*].category (8)
- $.vulnerabilities[*].remediations[*].category (14)
- $.vulnerabilities[*].remediations[*].restart_required.category (20)
- $.vulnerabilities[*].threats[*].category (14)
- $.vulnerabilities[*].x_extensions[*].category (13)
- $.x_extensions[*].category (13)

C.6 Const

A string which is a const has a fixed maximum length given by its own value.

> Later versions of CSAF might modify the value which could change its length. Therefore, this sizes should not be implemented as fixed limits if forward compatibility is desired.

The value of $['$schema'] is a fixed URL, currently pointing to the JSON schema location. It seems to be safe to assume that the length of this value is not greater than 150. This applies to:

- $['$schema'] (59)

For all other values, it seems to be safe to assume that the length of each value is not greater than 50. This applies to:

- $.document.csaf_version (3)
- $.vulnerabilities[*].metrics[*].content.ssvc_v2.schemaVersion (5)

C.7 URI Length

A string with format uri should not have a length greater than 20000. This applies to:

- $.document.acknowledgments[*].urls[*]
- $.document.aggregate_severity.namespace
- $.document.distribution.tlp.url
- $.document.involvement.entities[*].contact.public_openpgp_key_url
- $.document.involvement.entities[*].contact.url
- $.document.publisher.contact.public_openpgp_key_url
- $.document.publisher.contact.url
- $.document.publisher.namespace
- $.document.references[*].url
- $.document.x_extensions[*]['$schema']
- $.product_tree..branches[*].product.product_identification_helper.sbom_urls[*]
- $.product_tree..branches[*].product.product_identification_helper.x_generic_uris[*].namespace
- $.product_tree..branches[*].product.product_identification_helper.x_generic_uris[*].uri
- $.product_tree..branches[*].product.x_extensions[*]['$schema']
- $.product_tree.full_product_names[*].product_identification_helper.sbom_urls[*]
- $.product_tree.full_product_names[*].product_identification_helper.x_generic_uris[*].namespace
- $.product_tree.full_product_names[*].product_identification_helper.x_generic_uris[*].uri
- $.product_tree.full_product_names[*].x_extensions[*]['$schema']
- $.product_tree.product_paths[*].full_product_name.product_identification_helper.sbom_urls[*]
- $.product_tree.product_paths[*].full_product_name.product_identification_helper.x_generic_uris[*].namespace
- $.product_tree.product_paths[*].full_product_name.product_identification_helper.x_generic_uris[*].uri
- $.product_tree.product_paths[*].full_product_name.x_extensions[*]['$schema']

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- $.vulnerabilities[*].acknowledgments[*].urls[*]
- $.vulnerabilities[*].metrics[*].content.ssvc_v2.decision_point_resources[*].uri
- $.vulnerabilities[*].metrics[*].content.ssvc_v2.references[*].uri
- $.vulnerabilities[*].metrics[*].content.x_extensions[*]['$schema']
- $.vulnerabilities[*].metrics[*].source
- $.vulnerabilities[*].references[*].url
- $.vulnerabilities[*].remediations[*].url
- $.vulnerabilities[*].x_extensions[*]['$schema']
- $.x_extensions[*]['$schema']

C.8 UUID Length

A string with format uuid should not have a length greater than 50. This applies to:

- $.document.distribution.sharing_group.id (36)

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Appendix D. Collapsing Product Paths

(This appendix does not form an integral part of this specification and is informational.)

The following examples are intended to aid in understanding under which circumstances product paths can be collapsed.

Example 1 (which is not collapsed but collapsible):

```
"product_tree": {
 "branches": [
  {
   "branches": [
    {
     "branches": [
      {
       "category": "product_version",
       "name": "1.0.0",
       "product": {
        "name": "Example Company Product A 1.0.0",
        "product_id": "CSAFPID-908070601"
       }
      },
      // ...
     ],
     "category": "product_name",
     "name": "Product A"
    },
    {
     "branches": [
      {
       "category": "product_version",
       "name": "2023",
       "product": {
        "name": "Example Company Product B 2023",
        "product_id": "CSAFPID-908070603"
       }
      },
      // ...
     ],
     "category": "product_name",
     "name": "Product B"
    },
    {
     "branches": [
      {
       "category": "product_version",
       "name": "EU",
       "product": {
        "name": "Example Company Product C EU",
        "product_id": "CSAFPID-908070605"
       }
      },
      // ...
     ],
     "category": "product_name",
     "name": "Product C"
    }
   ],
   "category": "vendor",
   "name": "Example Company"
  }
 ],
 "product_paths": [
  {
   "beginning_product_reference": "CSAFPID-908070601",
   "full_product_name": {
    "name": "Example Company Product A 1.0.0 installed on Example Company Product B 2023",
    "product_id": "CSAFPID-908070607"
   },
   "subpaths": [
    {
     "category": "installed_on",
     "next_product_reference": "CSAFPID-908070603"
    }
   ]
  },
  // ...
  {
   "beginning_product_reference": "CSAFPID-908070607",
```

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```
      "full_product_name": {
        "name": "Example Company Product A 1.0.0 installed on Example Company Product B 2023 installed on Example Company Product C EU",
        "product_id": "CSAFPID-908070611"
      },
      "subpaths": [
        {
          "category": "installed_on",
          "next_product_reference": "CSAFPID-908070605"
        }
      ]
    },
    // ...
  ]
}
```

Product paths in example 1 above can be collapse as shown in example 2 below.

Example 2 (which is collapsed):

```
"product_tree": {
  "branches": [
    {
      "branches": [
        {
          "branches": [
            {
              "category": "product_version",
              "name": "1.0.0",
              "product": {
                "name": "Example Company Product A 1.0.0",
                "product_id": "CSAFPID-908070601"
              }
            },
            // ...
          ],
          "category": "product_name",
          "name": "Product A"
        },
        {
          "branches": [
            {
              "category": "product_version",
              "name": "2023",
              "product": {
                "name": "Example Company Product B 2023",
                "product_id": "CSAFPID-908070603"
              }
            },
            // ...
          ],
          "category": "product_name",
          "name": "Product B"
        },
        {
          "branches": [
            {
              "category": "product_version",
              "name": "EU",
              "product": {
                "name": "Example Company Product C EU",
                "product_id": "CSAFPID-908070605"
              }
            },
            // ...
          ],
          "category": "product_name",
          "name": "Product C"
        }
      ],
      "category": "vendor",
      "name": "Example Company"
    }
  ],
  "product_paths": [
```

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```
    {
      "beginning_product_reference": "CSAFPID-908070601",
      "full_product_name": {
        "name": "Example Company Product A 1.0.0 installed on Example Company Product B 2023 installed on Example Company Product C EU",
        "product_id": "CSAFPID-908070611"
      },
      "subpaths": [
        {
          "category": "installed_on",
          "next_product_reference": "CSAFPID-908070603"
        },
        {
          "category": "installed_on",
          "next_product_reference": "CSAFPID-908070605"
        }
      ]
    },
    // ...
  ]
}
```

Example 3 (which is not collapsed nor collapsible without information loss):

```
"product_tree": {
  "branches": [
    {
      "branches": [
        {
          "branches": [
            {
              "category": "product_version",
              "name": "1.0.0",
              "product": {
                "name": "Example Company Product A 1.0.0",
                "product_id": "CSAFPID-908070601",
                "product_identification_helper": {
                  "cpe": "cpe:2.3:a:example_company:product_a:1.0.0:*:*:*:*:*:*:*"
                }
              }
            },
            {
              "category": "product_version",
              "name": "1.1.0",
              "product": {
                "name": "Example Company Product A 1.1.0",
                "product_id": "CSAFPID-908070602",
                "product_identification_helper": {
                  "cpe": "cpe:2.3:a:example_company:product_a:1.1.0:*:*:*:*:*:*:*"
                }
              }
            }
          ],
          "category": "product_name",
          "name": "Product A"
        },
        {
          "branches": [
            {
              "category": "product_version",
              "name": "2023",
              "product": {
                "name": "Example Company Product B 2023",
                "product_id": "CSAFPID-908070603",
                "product_identification_helper": {
                  "cpe": "cpe:2.3:o:example_company:product_b:2023:*:*:*:*:*:*:*"
                }
              }
            },
            {
              "category": "product_version",
              "name": "2024",
              "product": {
                "name": "Example Company Product B 2024",
                "product_id": "CSAFPID-908070604",
                "product_identification_helper": {
                  "cpe": "cpe:2.3:o:example_company:product_b:2024:*:*:*:*:*:*:*"
                }
              }
            }
```

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```
          }
         ],
         "category": "product_name",
         "name": "Product B"
        },
        {
         "branches": [
          {
           "category": "product_version",
           "name": "EU",
           "product": {
            "name": "Example Company Product C EU",
            "product_id": "CSAFPID-908070605",
            "product_identification_helper": {
             "cpe": "cpe:2.3:h:example_company:product_c:eu:*:*:*:*:*:*:*"
            }
           }
          },
          {
           "category": "product_version",
           "name": "US",
           "product": {
            "name": "Example Company Product C US",
            "product_id": "CSAFPID-908070606",
            "product_identification_helper": {
             "cpe": "cpe:2.3:h:example_company:product_c:us:*:*:*:*:*:*:*"
            }
           }
          }
         ],
         "category": "product_name",
         "name": "Product C"
        }
       ],
       "category": "vendor",
       "name": "Example Company"
      }
     ],
     "product_paths": [
      {
       "beginning_product_reference": "CSAFPID-908070601",
       "full_product_name": {
        "name": "Example Company Product A 1.0.0 installed on Example Company Product B 2023",
        "product_id": "CSAFPID-908070607",
        "product_identification_helper": {
         "cpe": "cpe:2.3:a:example_company:product_a:1.0.0:*:*:*:*:product_b_2023:*:*"
        }
       },
       "subpaths": [
        {
         "category": "installed_on",
         "next_product_reference": "CSAFPID-908070603"
        }
       ]
      },
      // ...
      {
       "beginning_product_reference": "CSAFPID-908070607",
       "full_product_name": {
        "name": "Example Company Product A 1.0.0 installed on Example Company Product B 2023 installed on Example Company Product C EU",
        "product_id": "CSAFPID-908070611",
        "product_identification_helper": {
         "cpe": "cpe:2.3:a:example_company:product_a:1.0.0:*:*:*:*:product_b_2023:product_c_eu:*"
        }
       },
       "subpaths": [
        {
         "category": "installed_on",
         "next_product_reference": "CSAFPID-908070605"
        }
       ]
      },
      // ...
     ]
    }
```

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Product paths in example 3 cannot be collapsed without information loss. For example, the cpe of the product identified by CSAFPID-908070607 would be lost.

Example 4 (which is not collapsed nor collapsible as products are referenced):

```
{
  // ...
  "product_tree": {
    "branches": [
      {
        "branches": [
          {
            "branches": [
              {
                "category": "product_version",
                "name": "1.0.0",
                "product": {
                  "name": "Example Company Product A 1.0.0",
                  "product_id": "CSAFPID-908070601"
                }
              },
              // ...
            ],
            "category": "product_name",
            "name": "Product A"
          },
          {
            "branches": [
              {
                "category": "product_version",
                "name": "2023",
                "product": {
                  "name": "Example Company Product B 2023",
                  "product_id": "CSAFPID-908070603"
                }
              },
              // ...
            ],
            "category": "product_name",
            "name": "Product B"
          },
          {
            "branches": [
              {
                "category": "product_version",
                "name": "EU",
                "product": {
                  "name": "Example Company Product C EU",
                  "product_id": "CSAFPID-908070605"
                }
              },
              // ...
            ],
            "category": "product_name",
            "name": "Product C"
          }
        ],
        "category": "vendor",
        "name": "Example Company"
      }
    ],
    // ...
    "product_paths": [
      {
        "beginning_product_reference": "CSAFPID-908070601",
        "full_product_name": {
          "name": "Example Company Product A 1.0.0 installed on Example Company Product B 2023",
          "product_id": "CSAFPID-908070607"
        },
        "subpaths": [
          {
            "category": "installed_on",
            "next_product_reference": "CSAFPID-908070603"
          }
        ]
      },
```

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```
      // ...
      {
        "beginning_product_reference": "CSAFPID-908070607",
        "full_product_name": {
          "name": "Example Company Product A 1.0.0 installed on Example Company Product B 2023 installed on Example Company Product C EU",
          "product_id": "CSAFPID-908070611"
        },
        "subpaths": [
          {
            "category": "installed_on",
            "next_product_reference": "CSAFPID-908070605"
          }
        ]
      },
      // ...
    ]
  },
  "vulnerabilities": [
    {
      "product_status": {
        "under_investigation": [
          "CSAFPID-908070607",
          // ...
        ]
      }
    },
    {
      "threats": [
        {
          "category": "target_set",
          "details": "These products are known to be specifically targeted by APT-00.",
          "product_ids": [
            "CSAFPID-908070607",
            // ...
          ]
        }
      ]
    }
  ]
}
```

Product paths in example 4 cannot be collapsed as the products inside the path are referenced elsewhere in the document. For example, CSAFPID-908070607 is also referenced in the product status and threats.

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